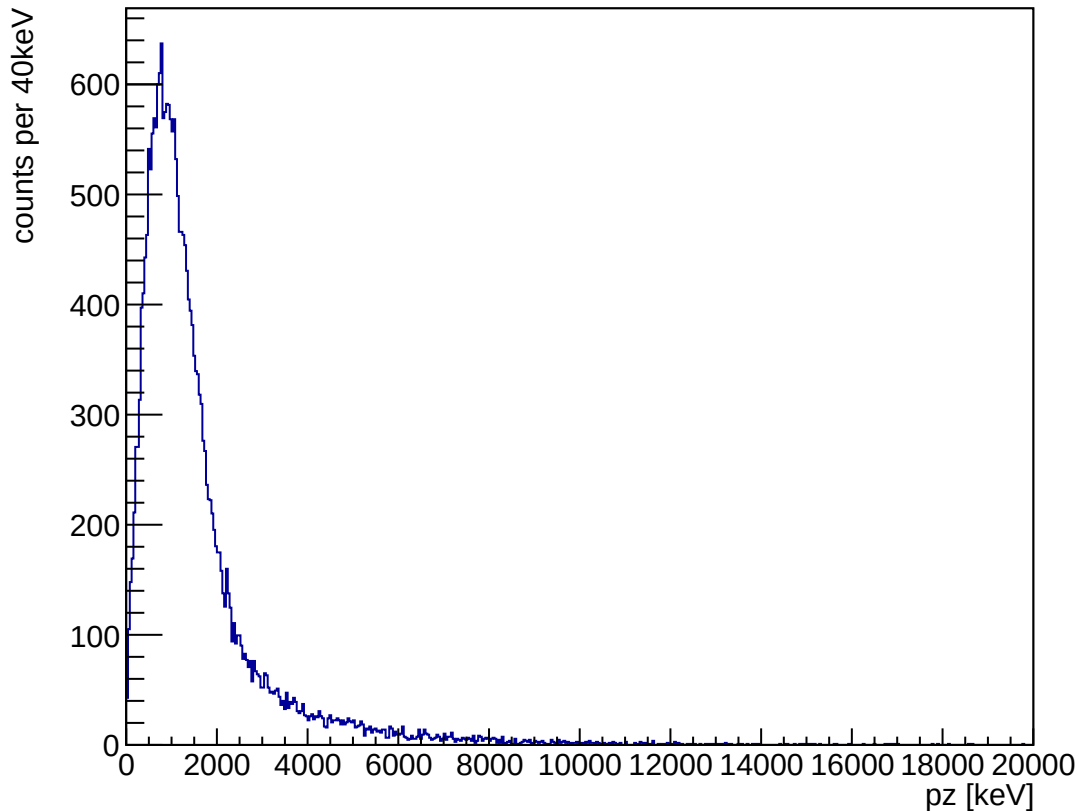
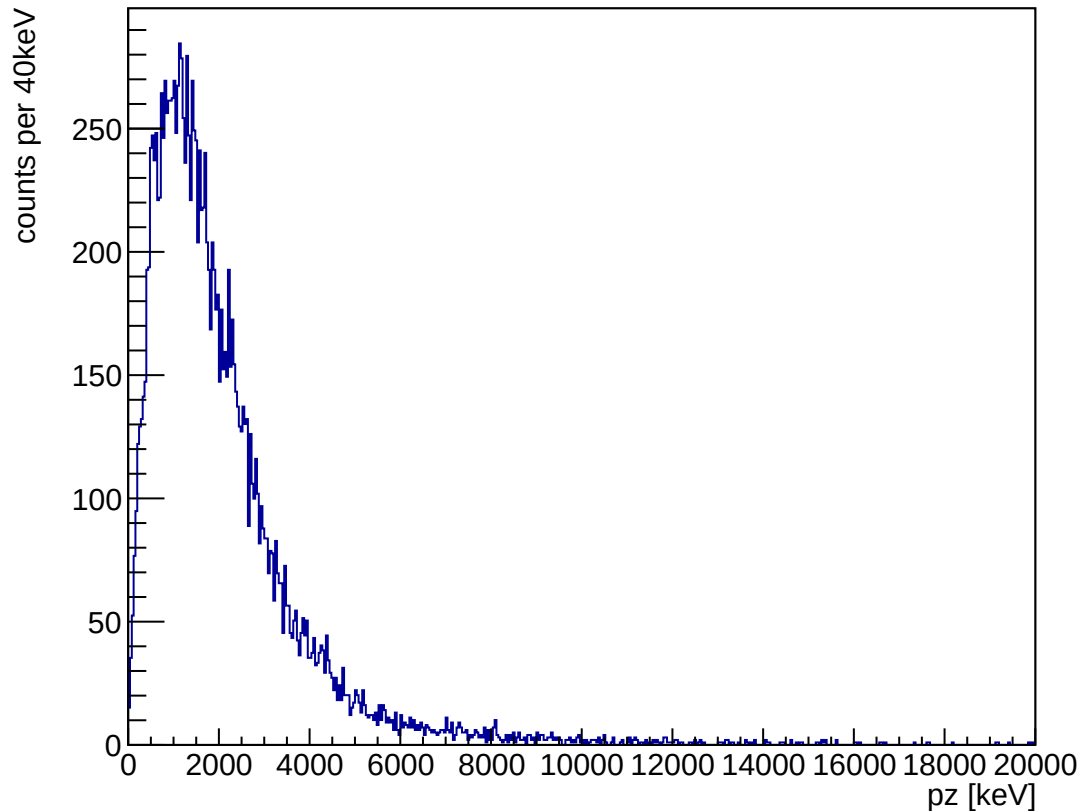


Stock G4: positron momentum pz



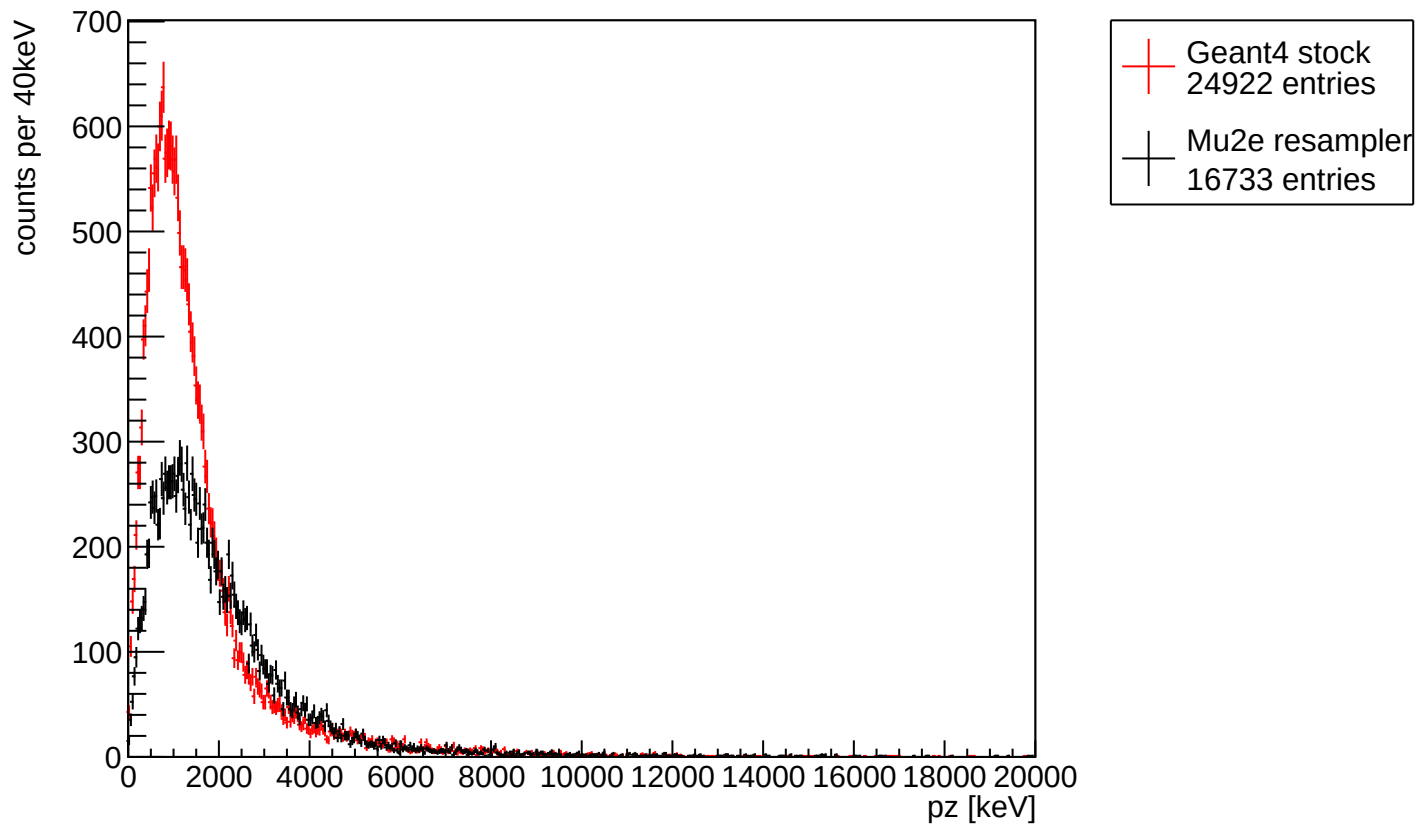
Entries	26791
Mean	1547
Std Dev	1538
Underflow	0
Overflow	11.16
Integral	2.491e+04

Mu2e mu resampler: positron momentum pz

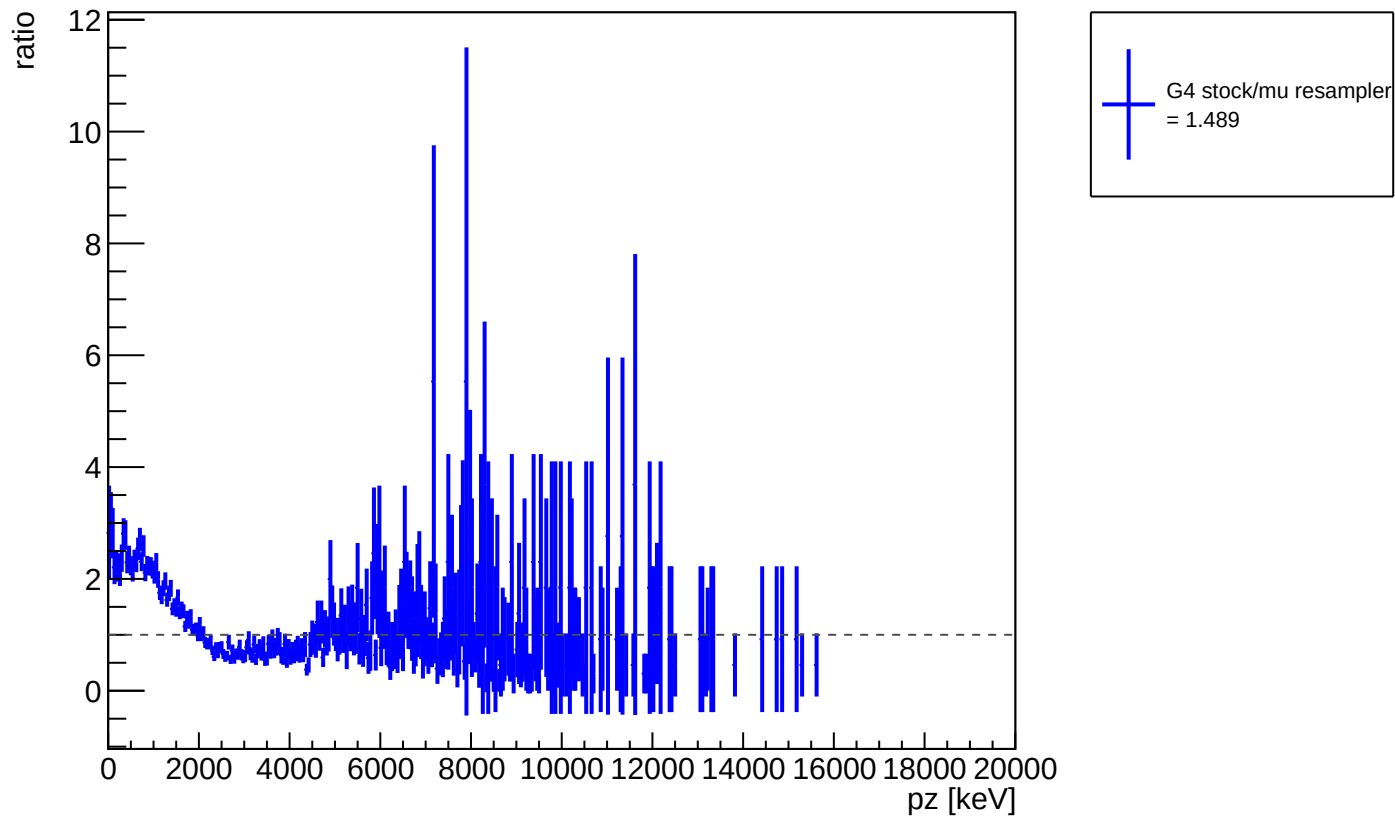


Entries	16582
Mean	2046
Std Dev	1778
Underflow	0
Overflow	10.09
Integral	1.672e+04

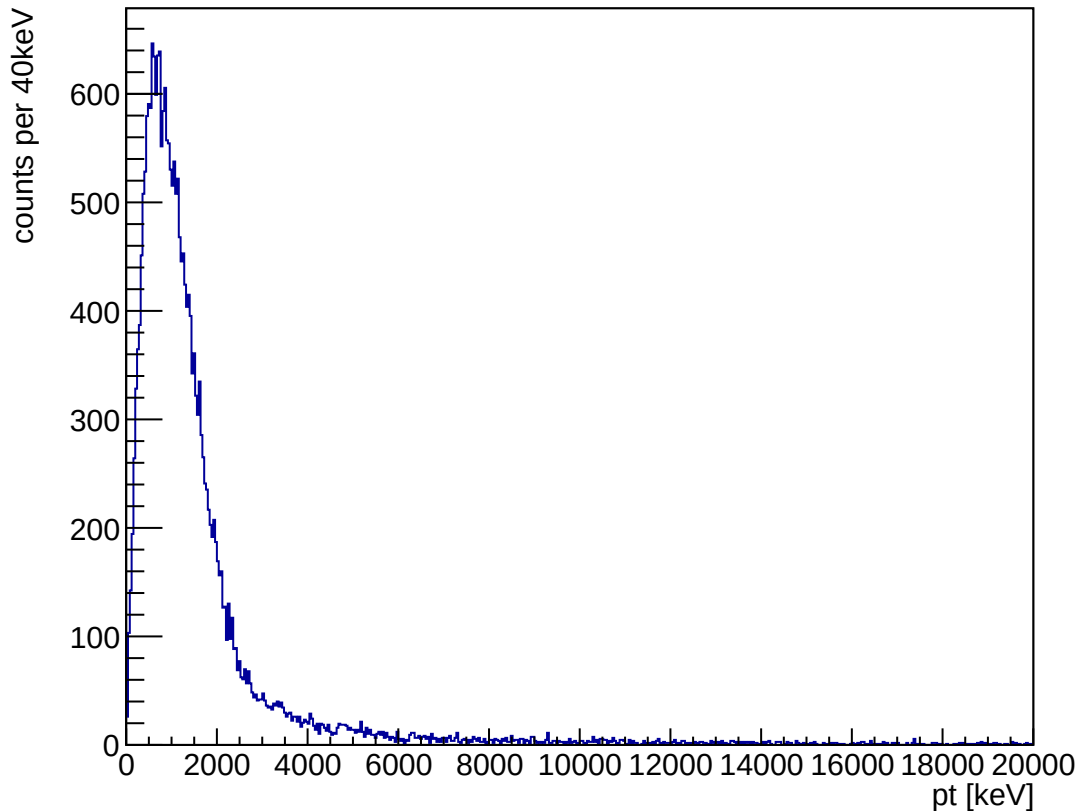
positron momentum pz



ratio of G4 stock / muon resampler: positron momentum pz

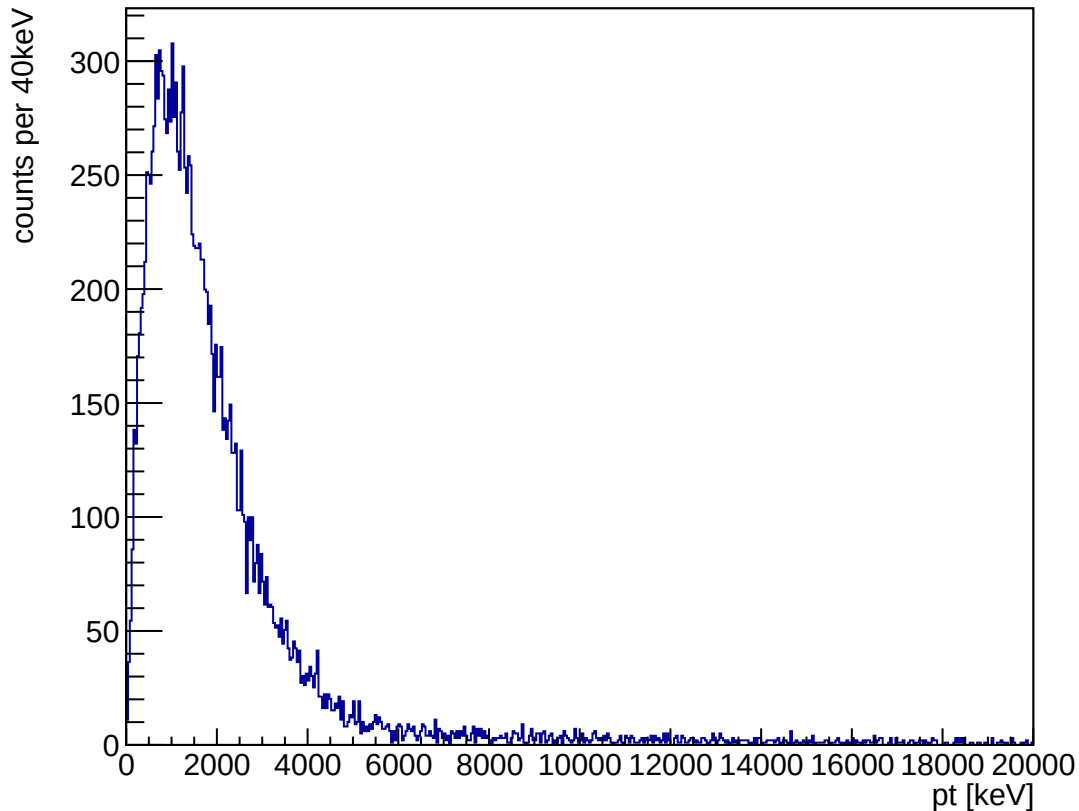


Stock G4: positron transverse momentum pT



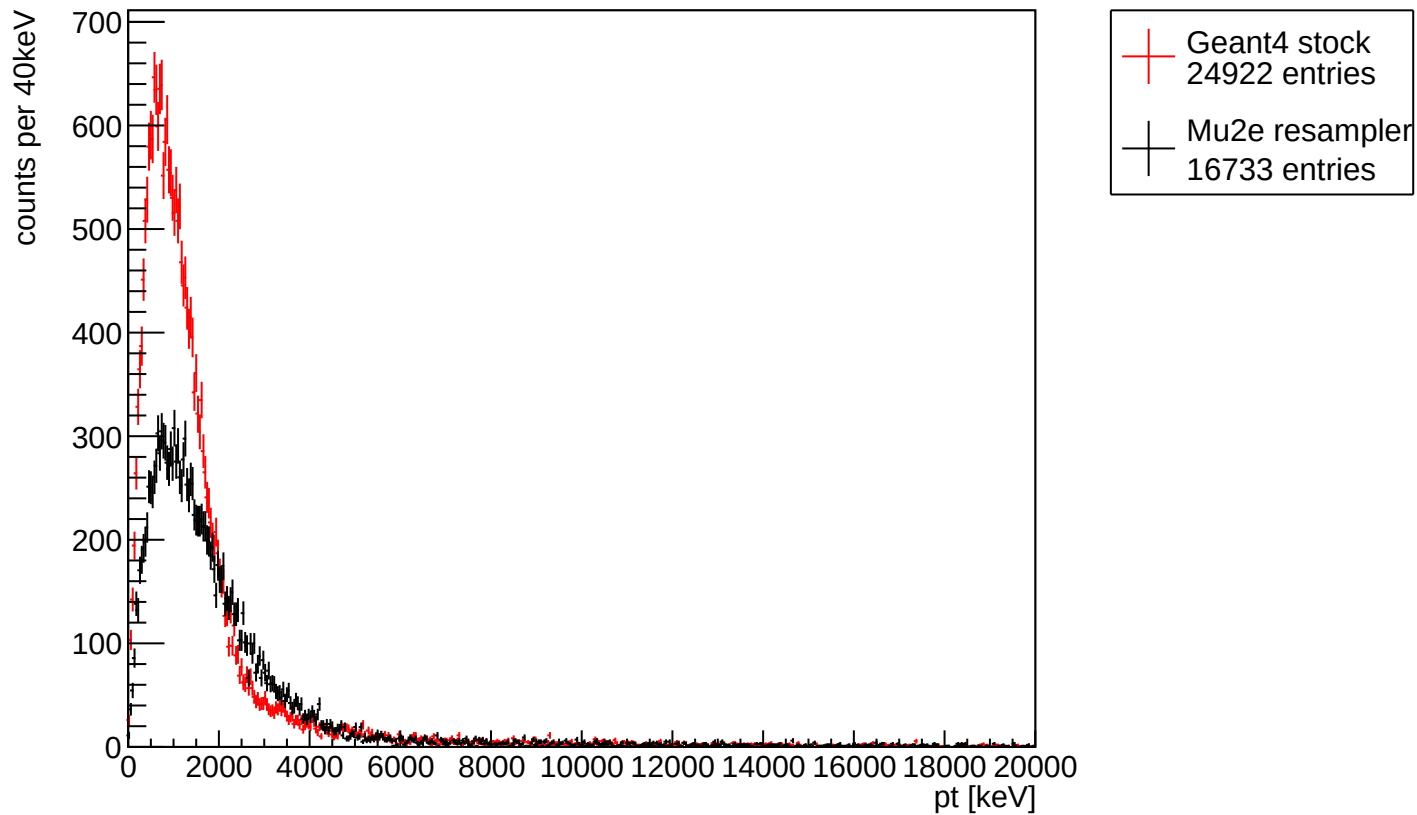
Entries	26791
Mean	1570
Std Dev	2013
Underflow	0
Overflow	126.5
Integral	2.48e+04

Mu2e mu resampler: positron transverse momentum pT

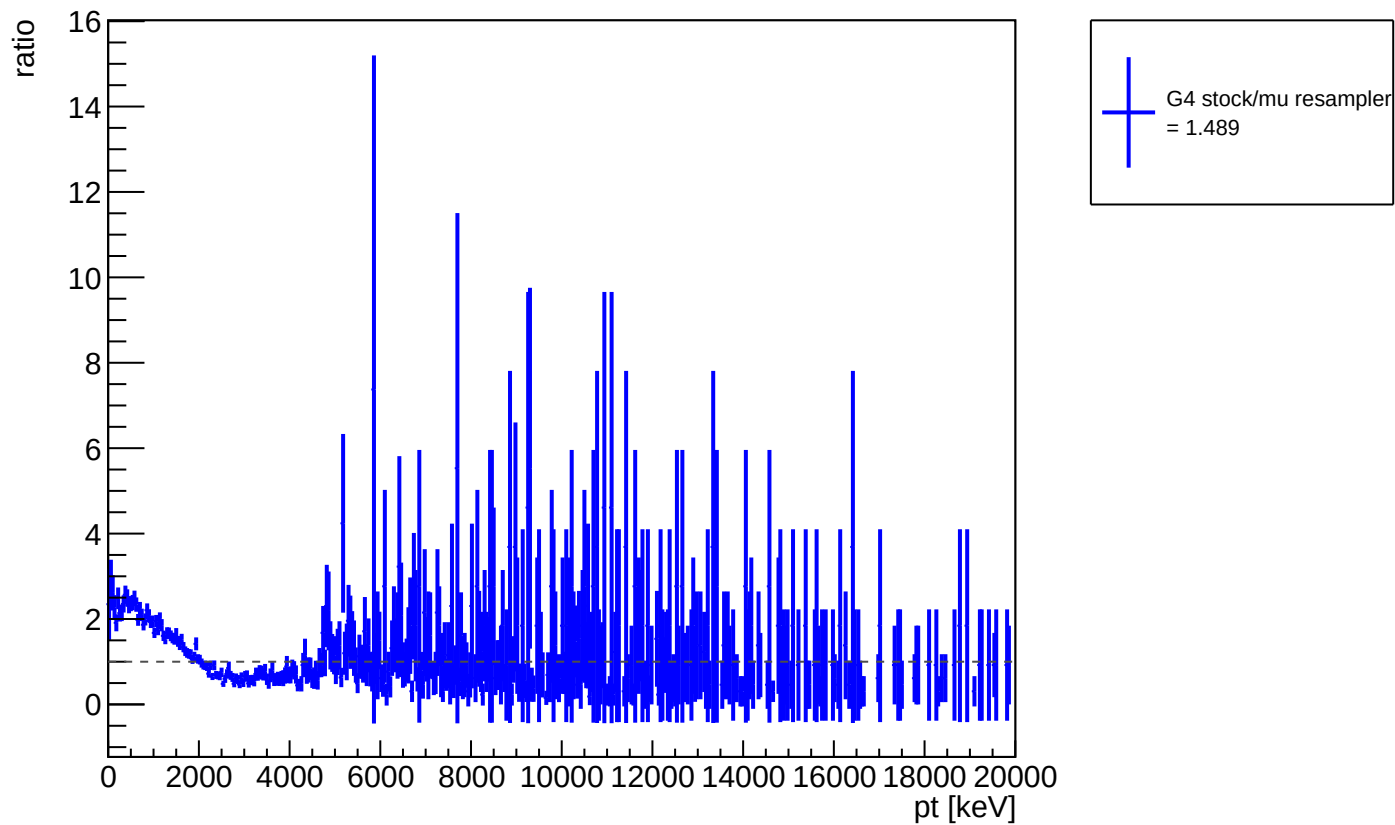


Entries	16582
Mean	2076
Std Dev	2359
Underflow	0
Overflow	120.1
Integral	1.661e+04

positron transverse momentum pT

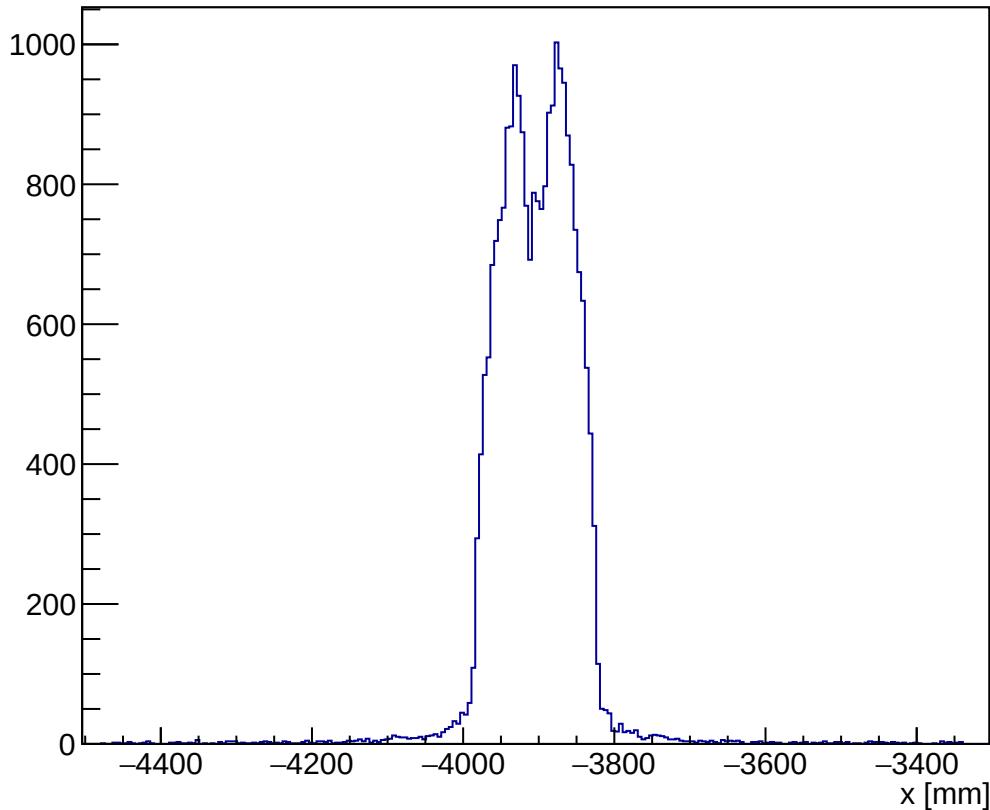


ratio of G4 stock / muon resampler: positron transverse momentum pT



Stock G4: positron x position

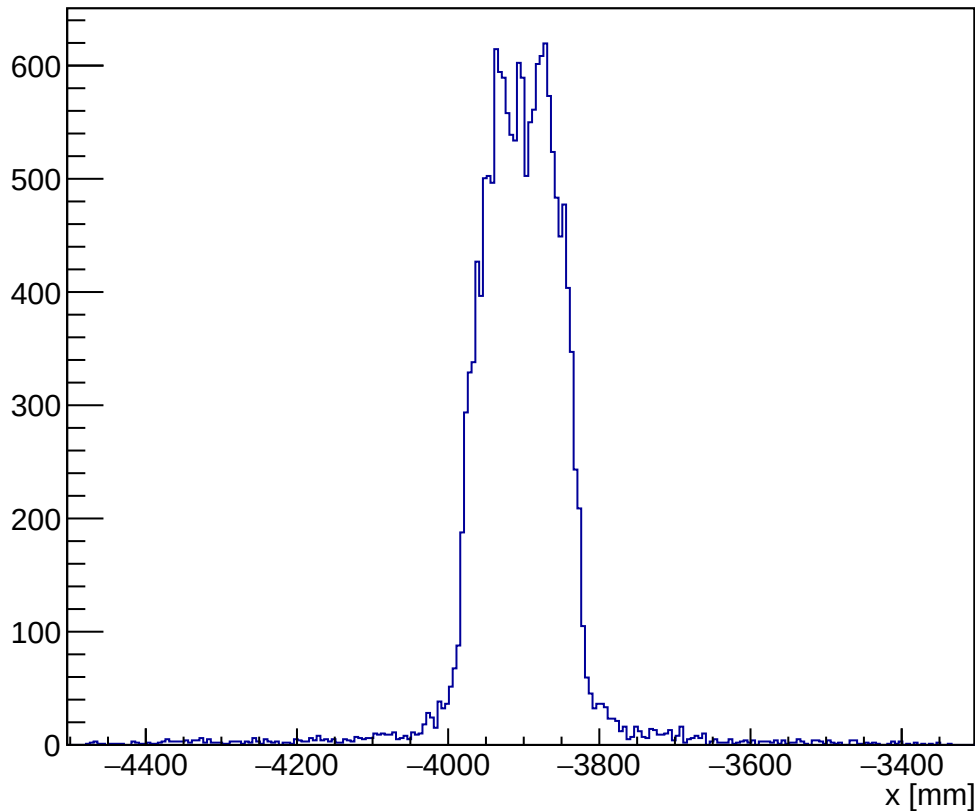
counts per 5mm



Entries	26791
Mean	-3903
Std Dev	61.56
Underflow	0
Overflow	0
Integral	2.492e+04

Mu2e mu resampler: positron x position

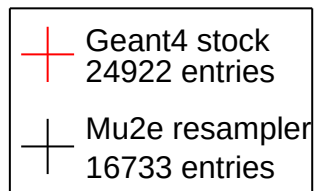
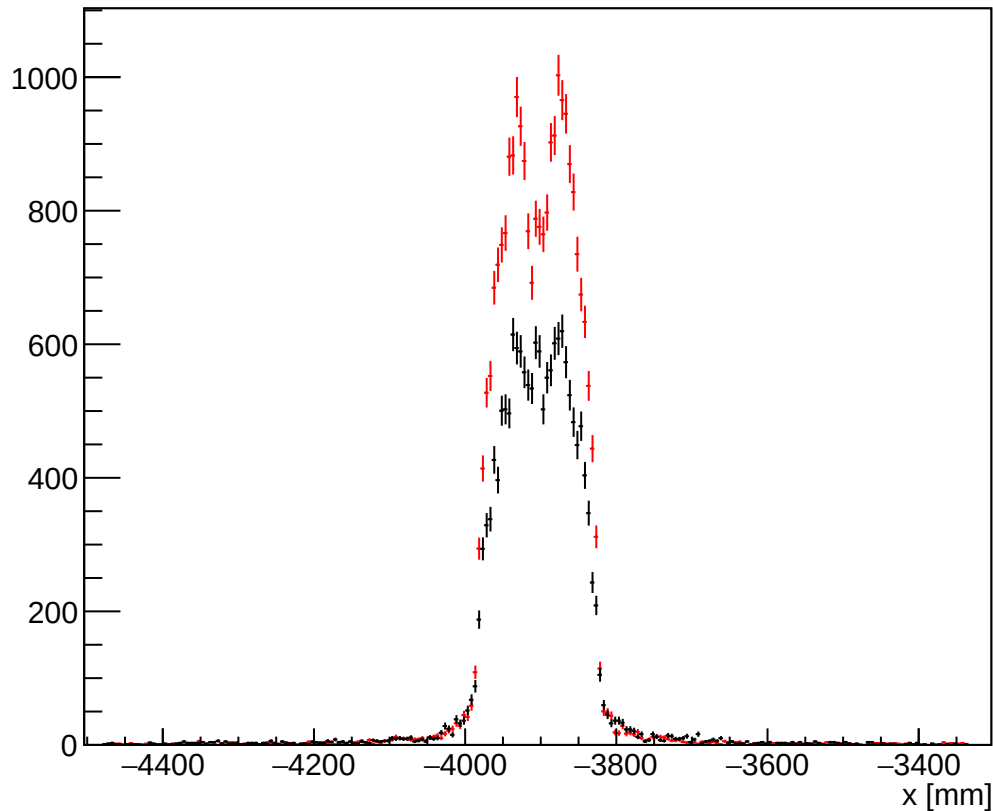
counts per 5mm



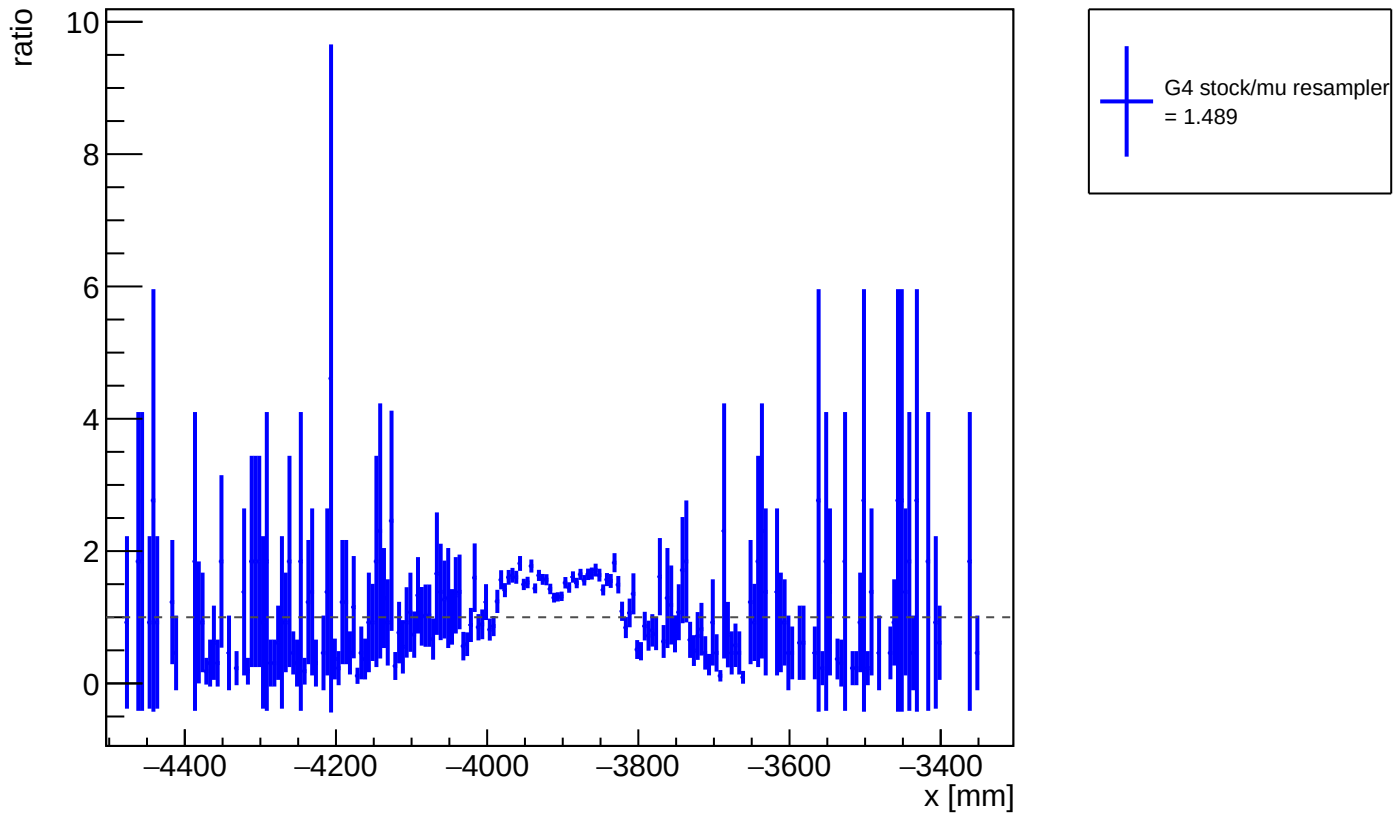
Entries	16582
Mean	-3904
Std Dev	73.55
Underflow	0
Overflow	0
Integral	1.673e+04

positron x position

counts per 5mm

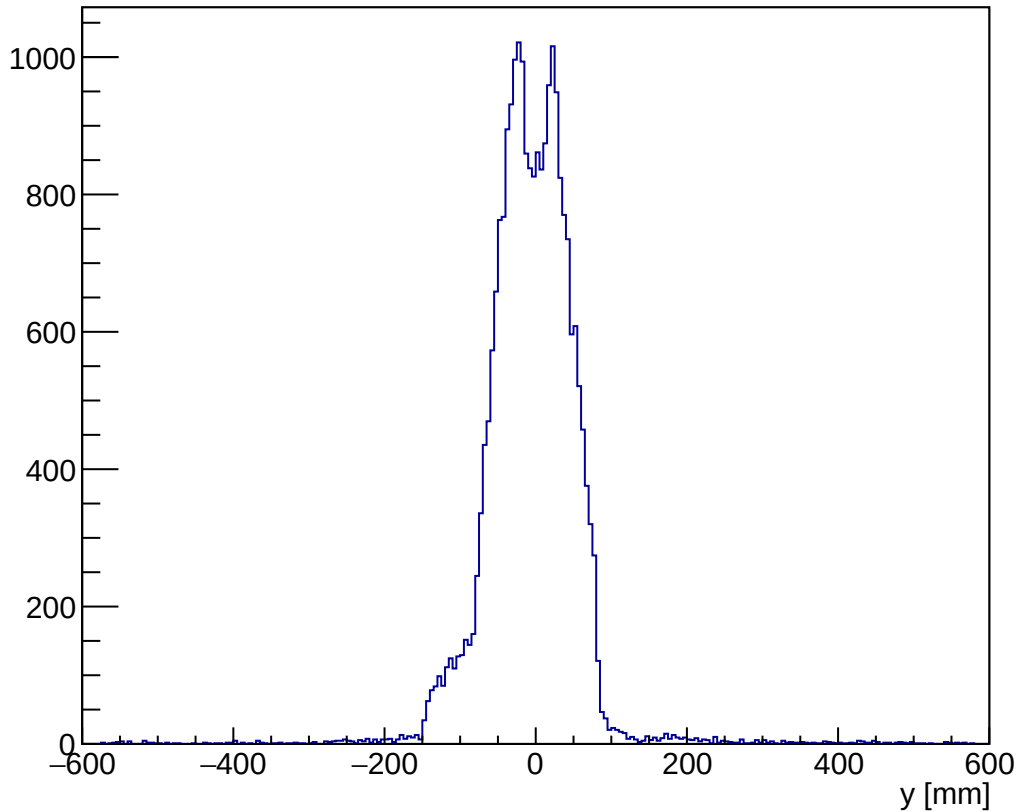


ratio of G4 stock / muon resampler: positron x position



Stock G4: positron y position

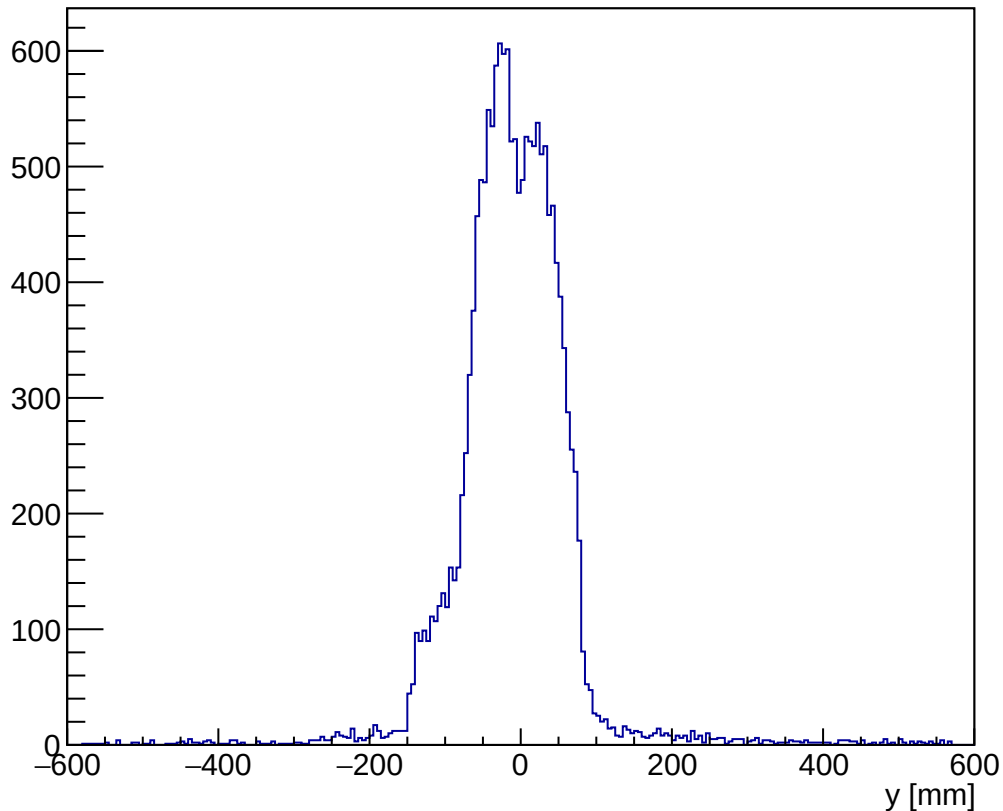
counts per 5mm



Entries	26791
Mean	-5.652
Std Dev	63.43
Underflow	0
Overflow	0
Integral	2.492e+04

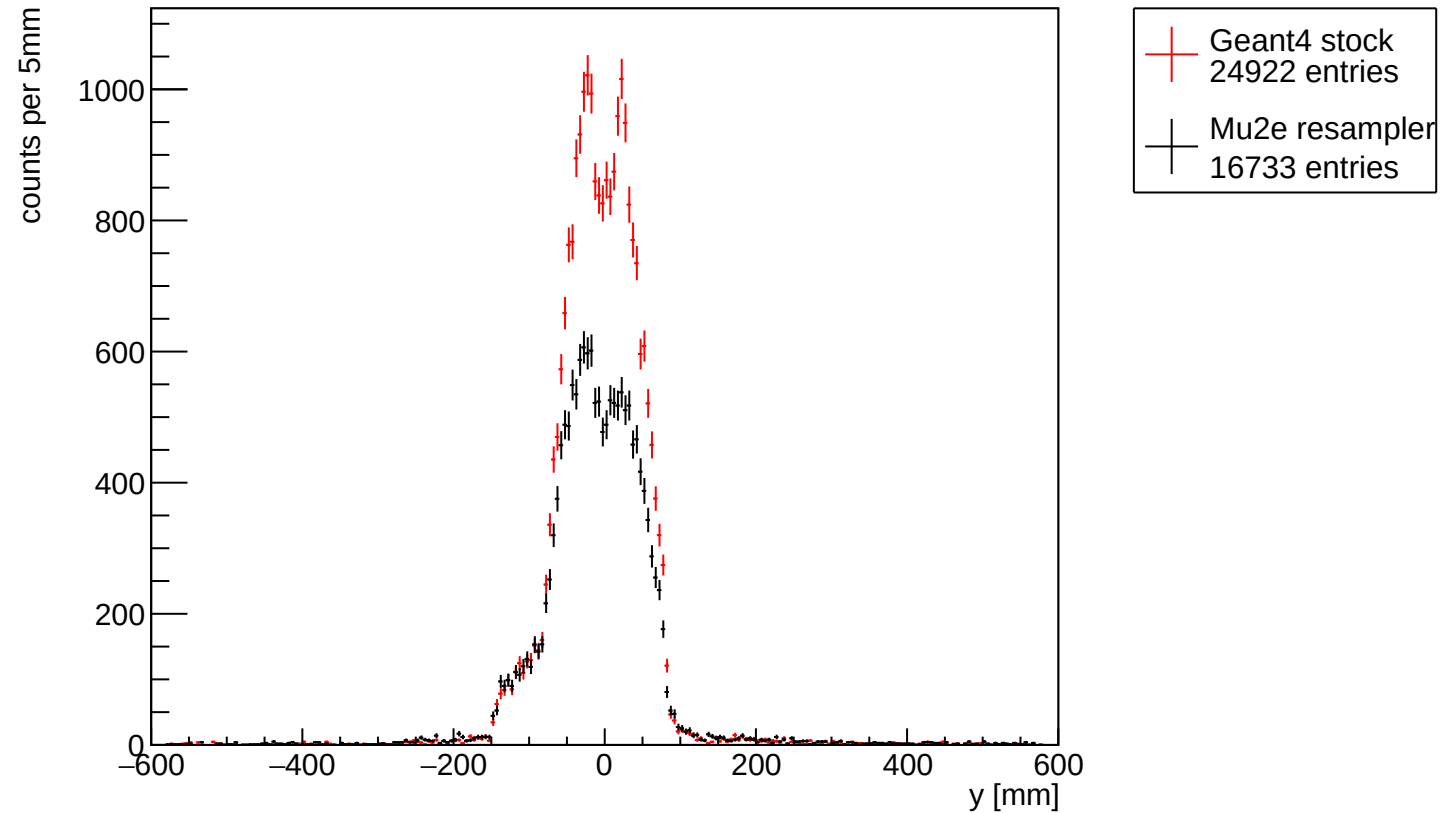
Mu2e mu resampler: positron y position

counts per 5mm

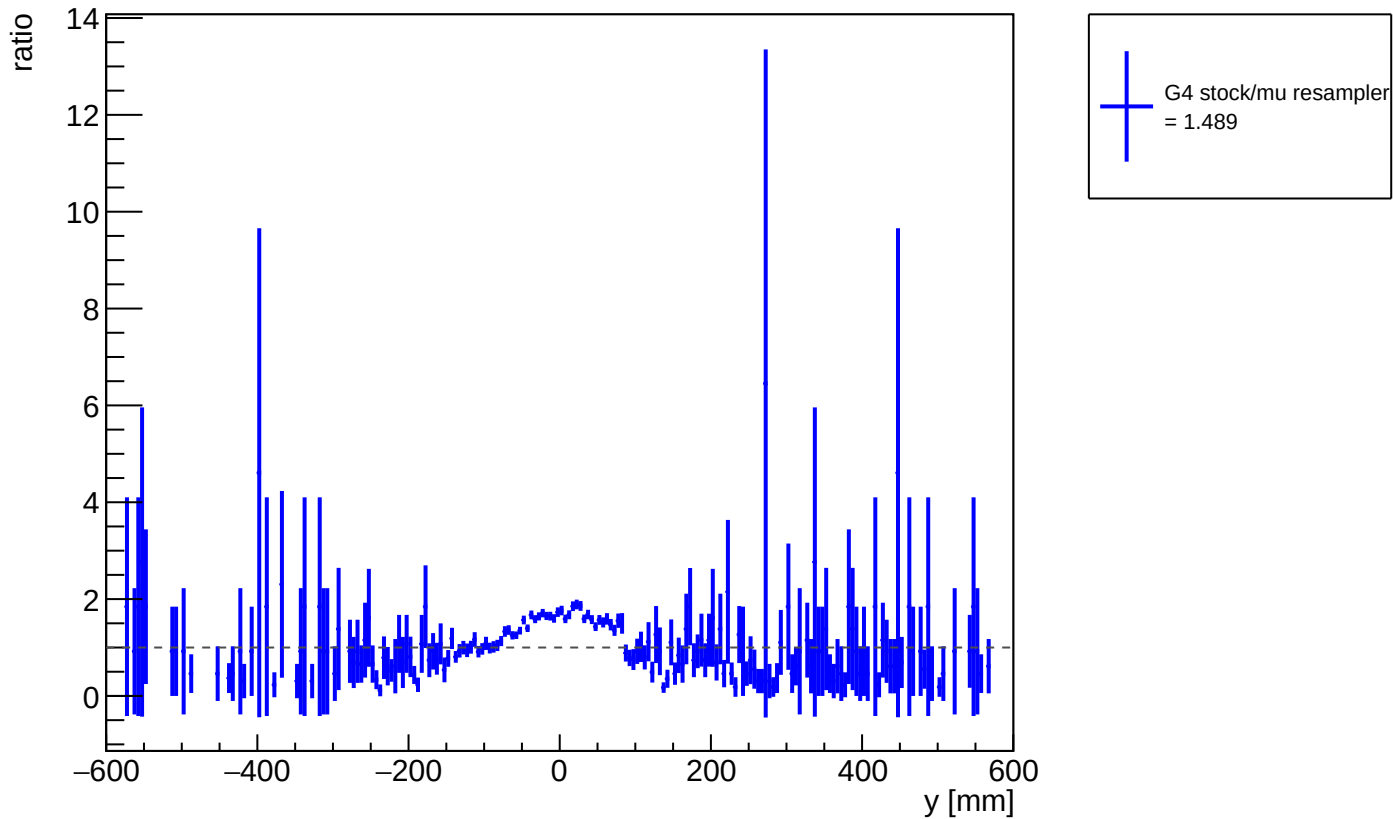


Entries	16582
Mean	-9.193
Std Dev	78.05
Underflow	0
Overflow	0
Integral	1.673e+04

positron y position

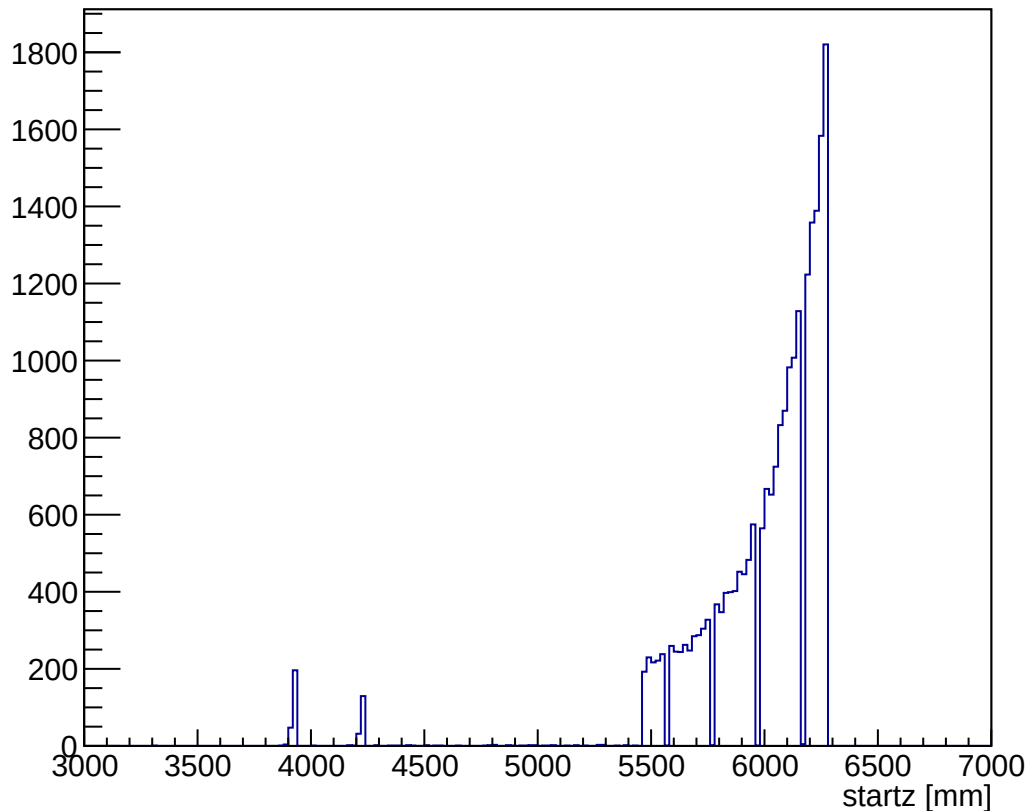


ratio of G4 stock / muon resampler: positron y position



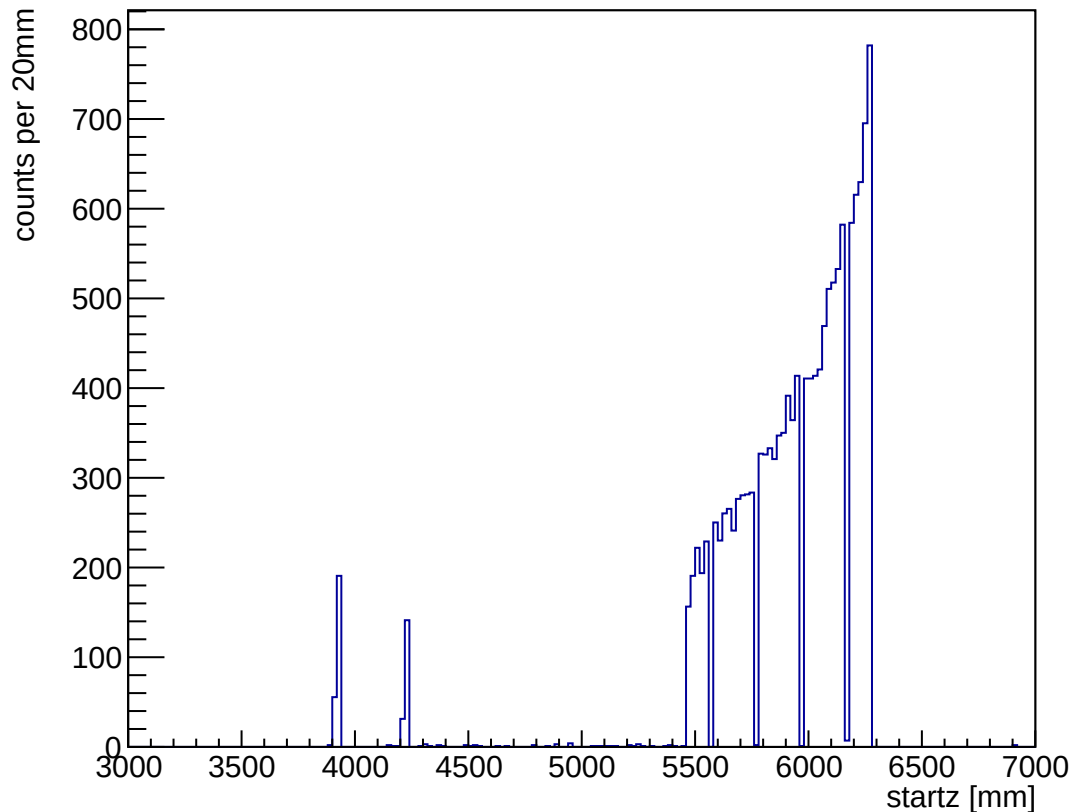
Stock G4: positron start z

counts per 20mm



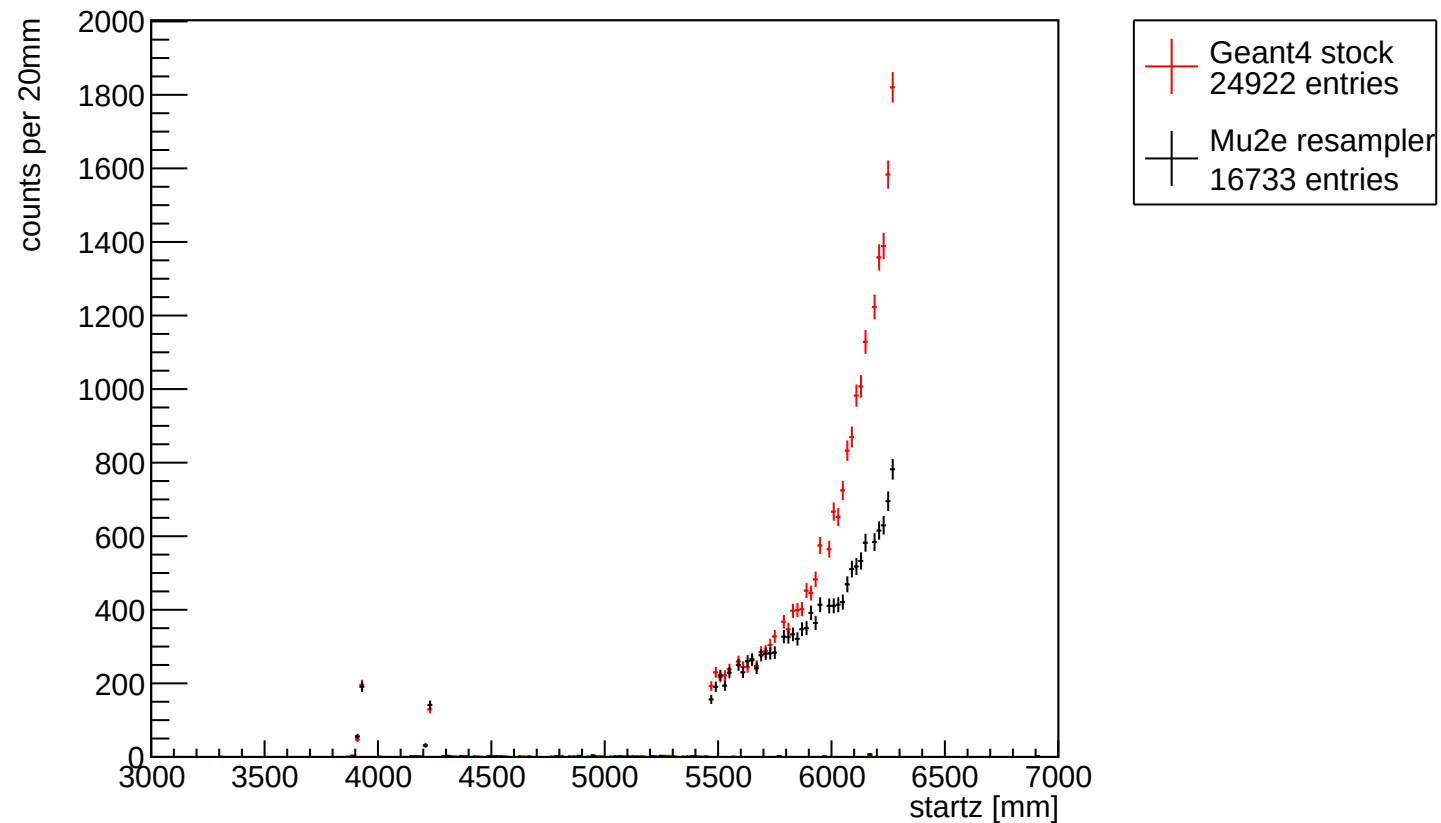
Entries	26791
Mean	5985
Std Dev	346.2
Underflow	2204
Overflow	22.33
Integral	2.27e+04

Mu2e mu resampler: positron start z

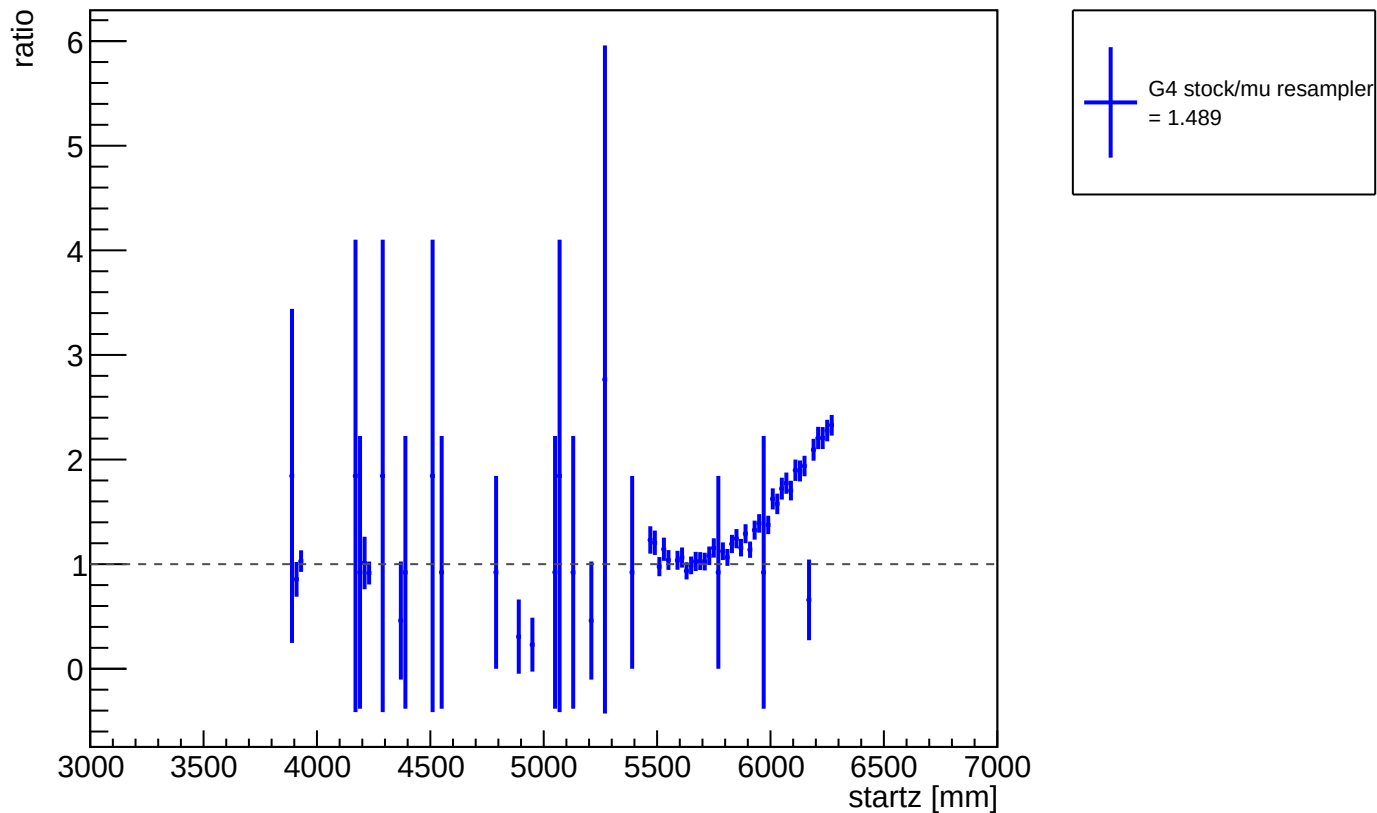


Entries	16582
Mean	5902
Std Dev	398.9
Underflow	2112
Overflow	28.25
Integral	1.459e+04

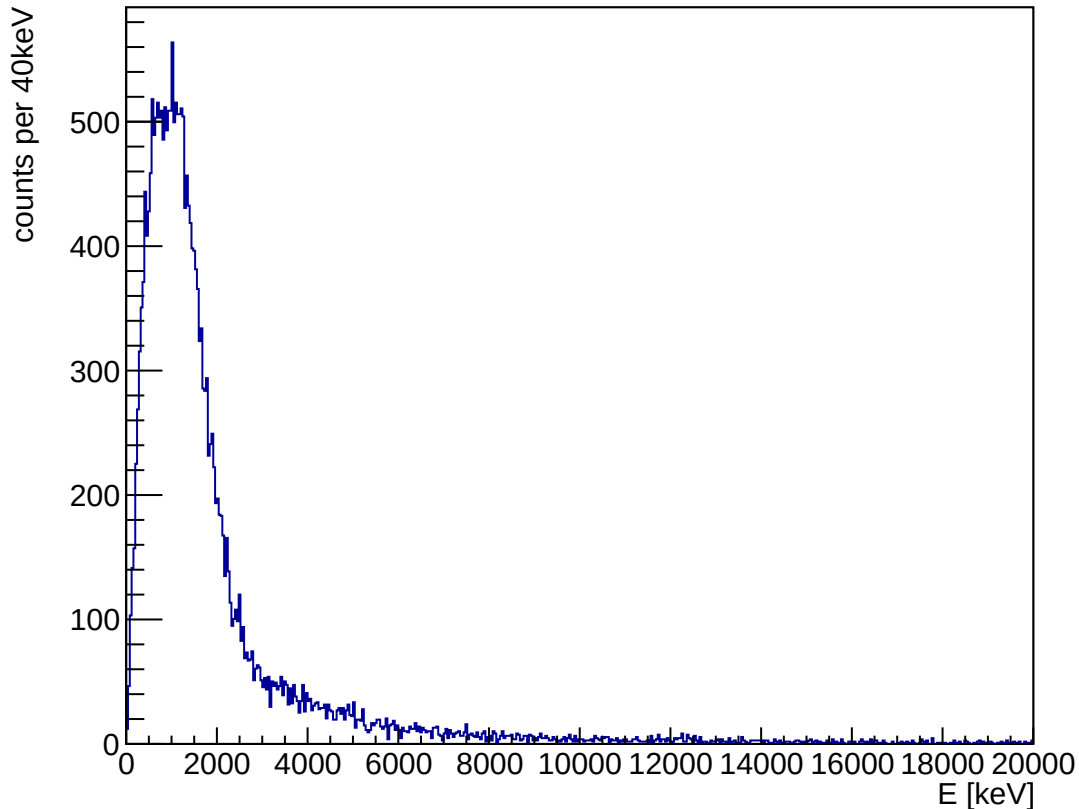
positron start z



ratio of G4 stock / muon resampler: positron start z

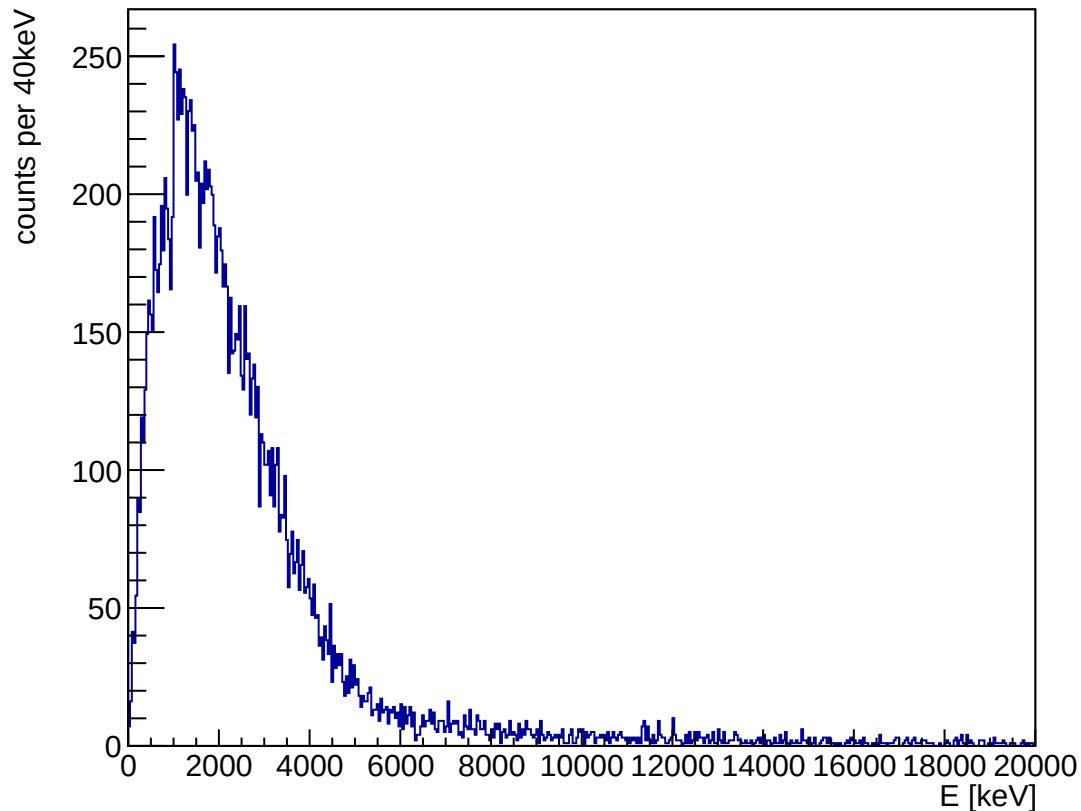


Stock G4: positron kin. energy



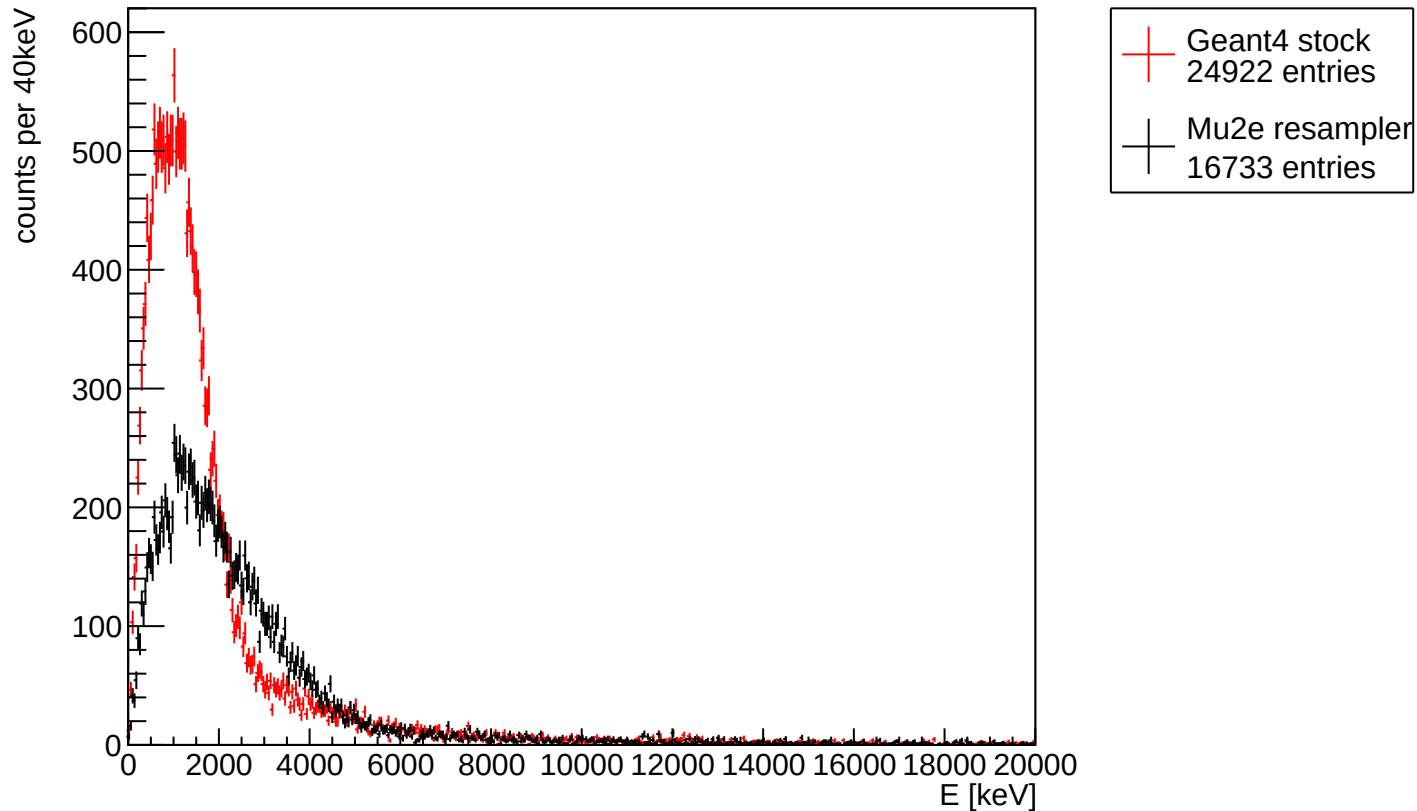
Entries	26791
Mean	1870
Std Dev	2252
Underflow	0
Overflow	152.6
Integral	2.477e+04

Mu2e mu resampler: positron kin. energy

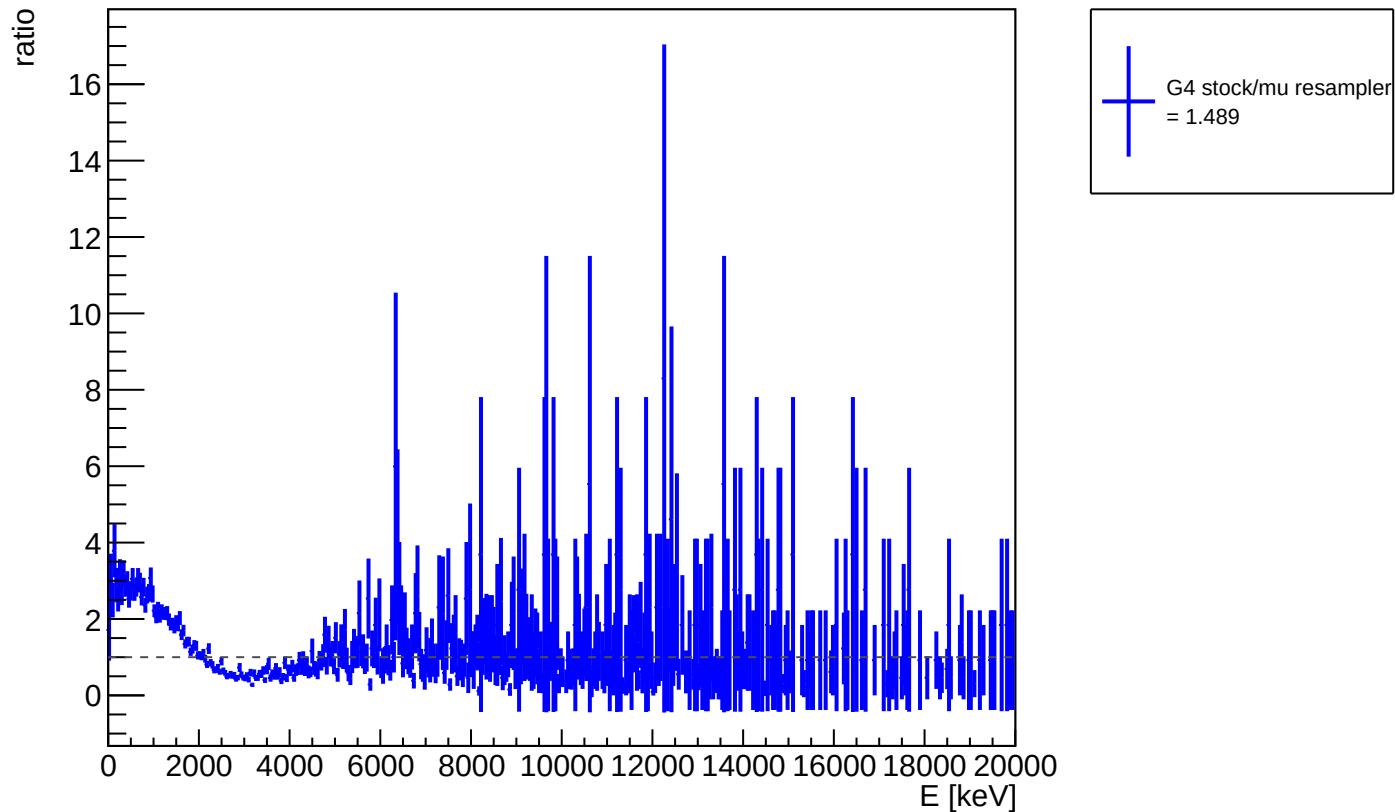


Entries	16582
Mean	2575
Std Dev	2538
Underflow	0
Overflow	155.4
Integral	1.658e+04

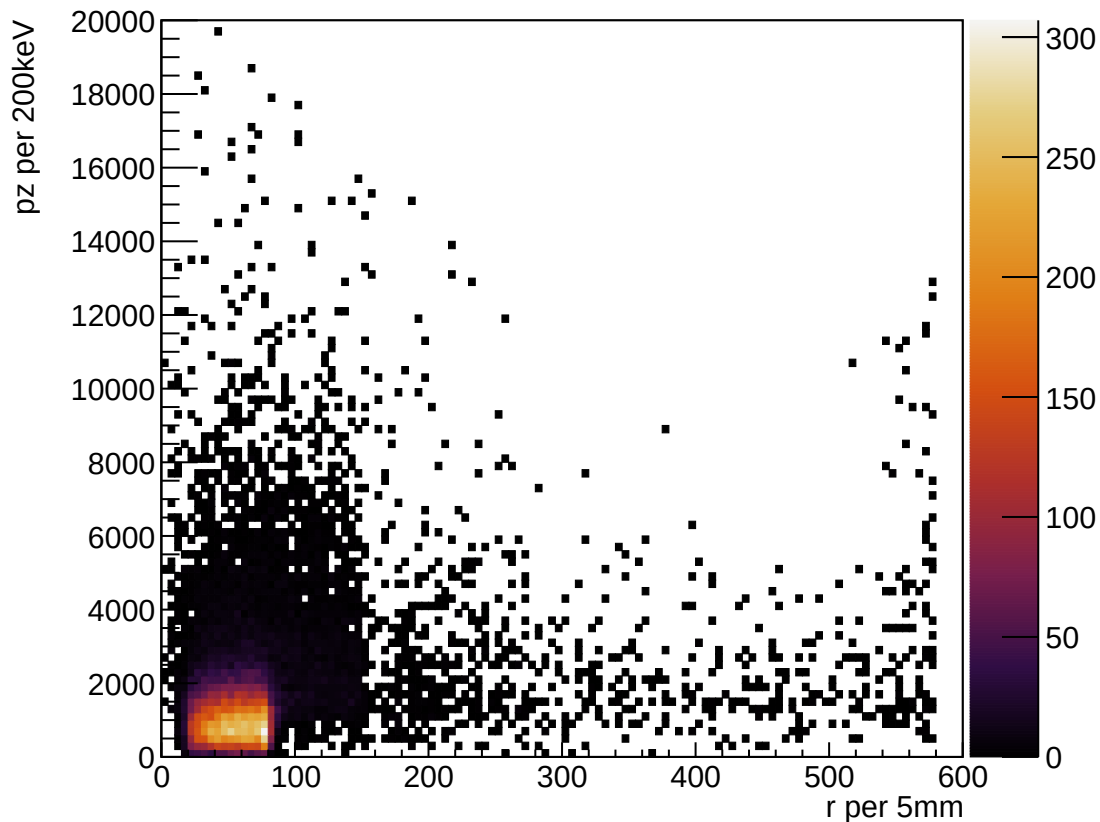
positron kin. energy



ratio of G4 stock / muon resampler: positron kin. energy

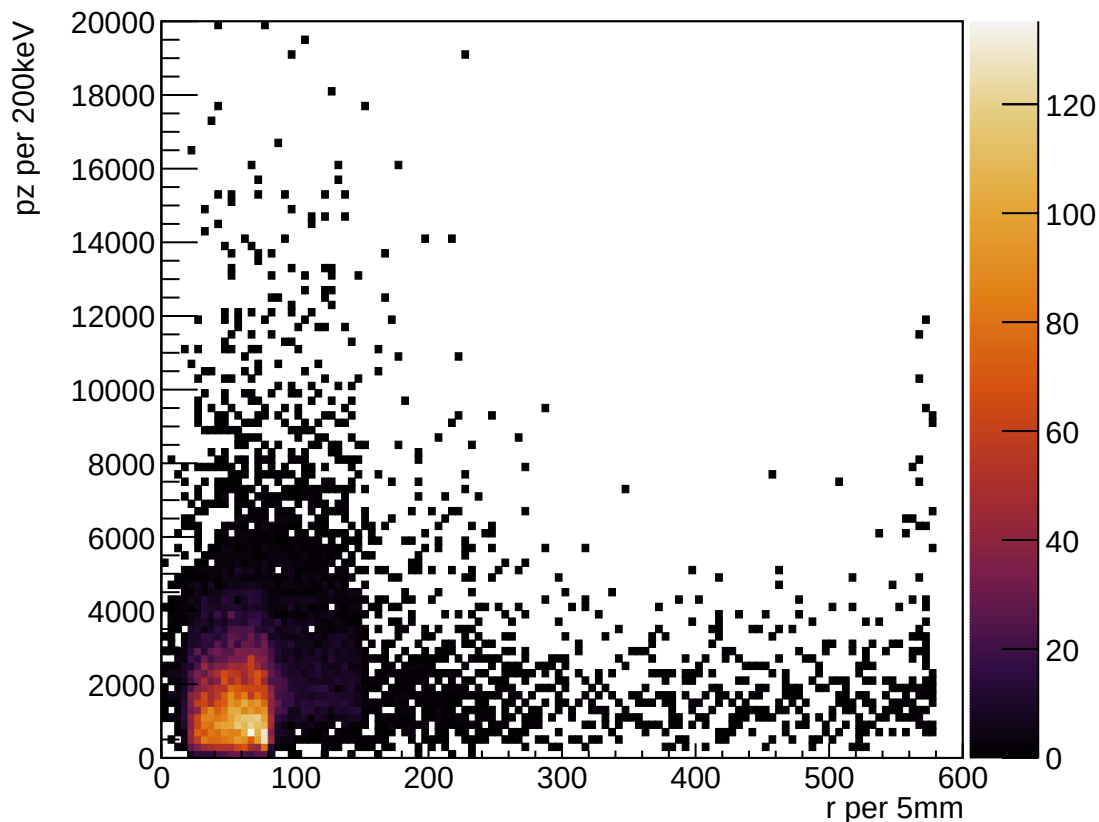


Stock G4: positron pz vs r



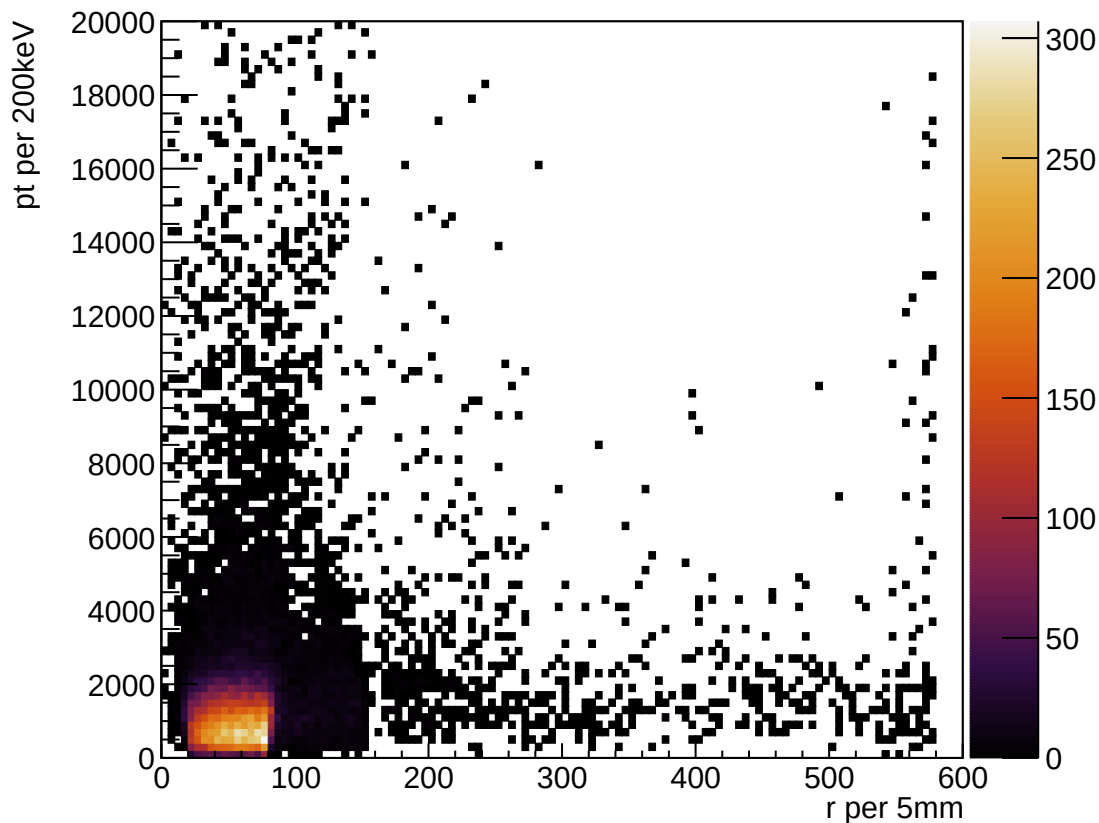
Entries	26791	
Mean x	67.69	
Mean y	1547	
Std Dev x	57.1	
Std Dev y	1538	
Integral	2.491e+04	
0	11.1626	0
0	24910.4	0
0	0	0

Mu2e mu resampler: positron pz vs r



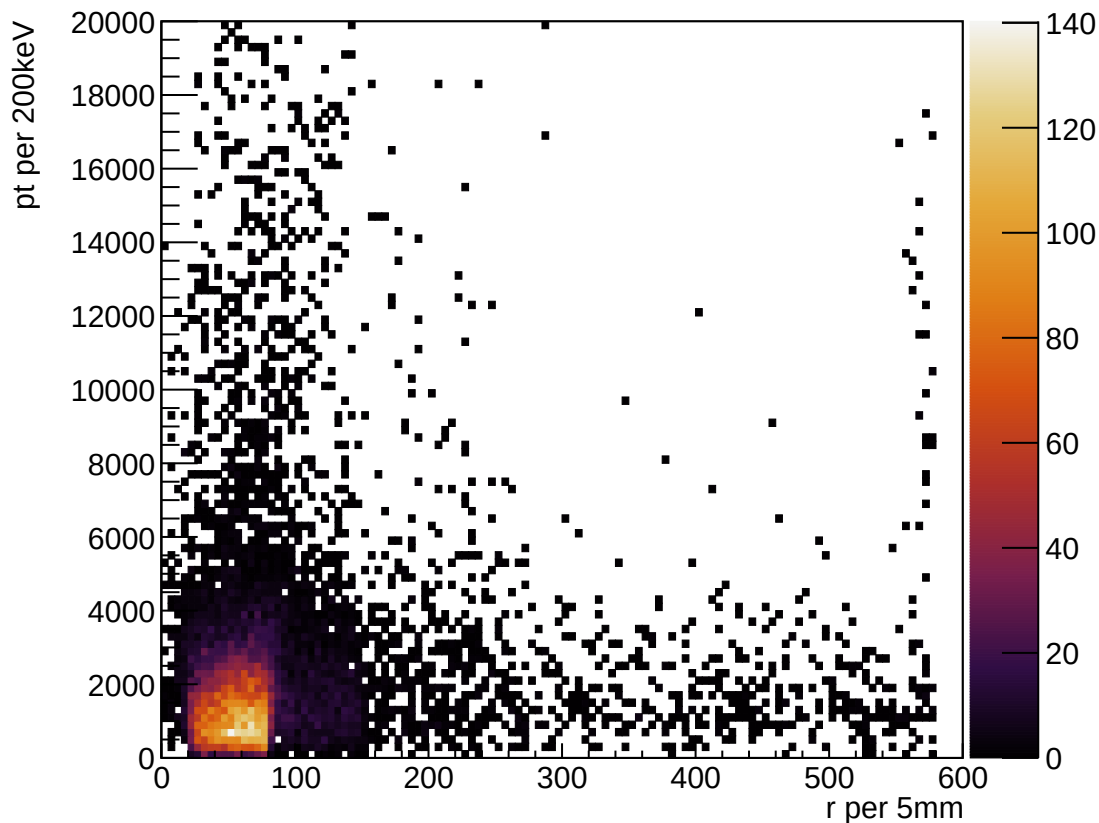
Entries	16582	
Mean x	78.48	
Mean y	2046	
Std Dev x	73.69	
Std Dev y	1778	
Integral	1.672e+04	
0	10.0908	0
0	16722.5	0
0	0	0

Stock G4: positron pt vs r



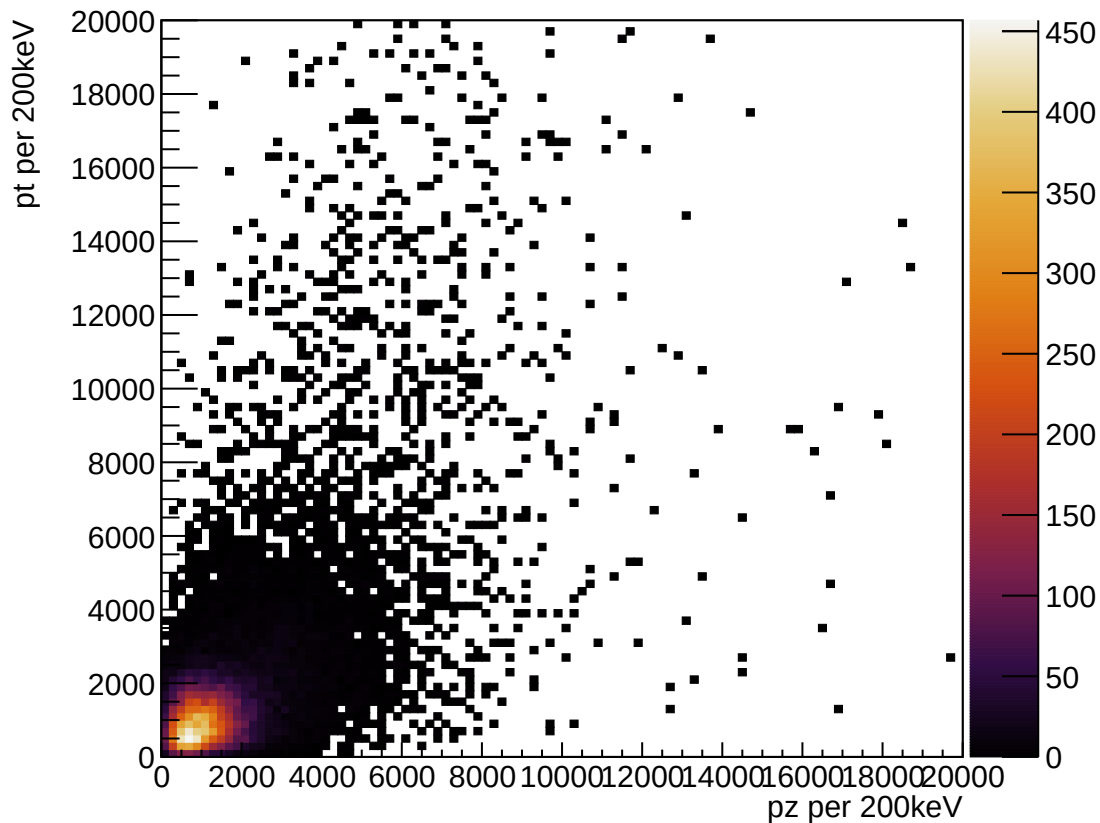
Entries	26791	
Mean x	67.31	
Mean y	1570	
Std Dev x	56.25	
Std Dev y	2013	
Integral	2.48e+04	
0	126.51	0
0	24795	0
0	0	0

Mu2e mu resampler: positron pt vs r



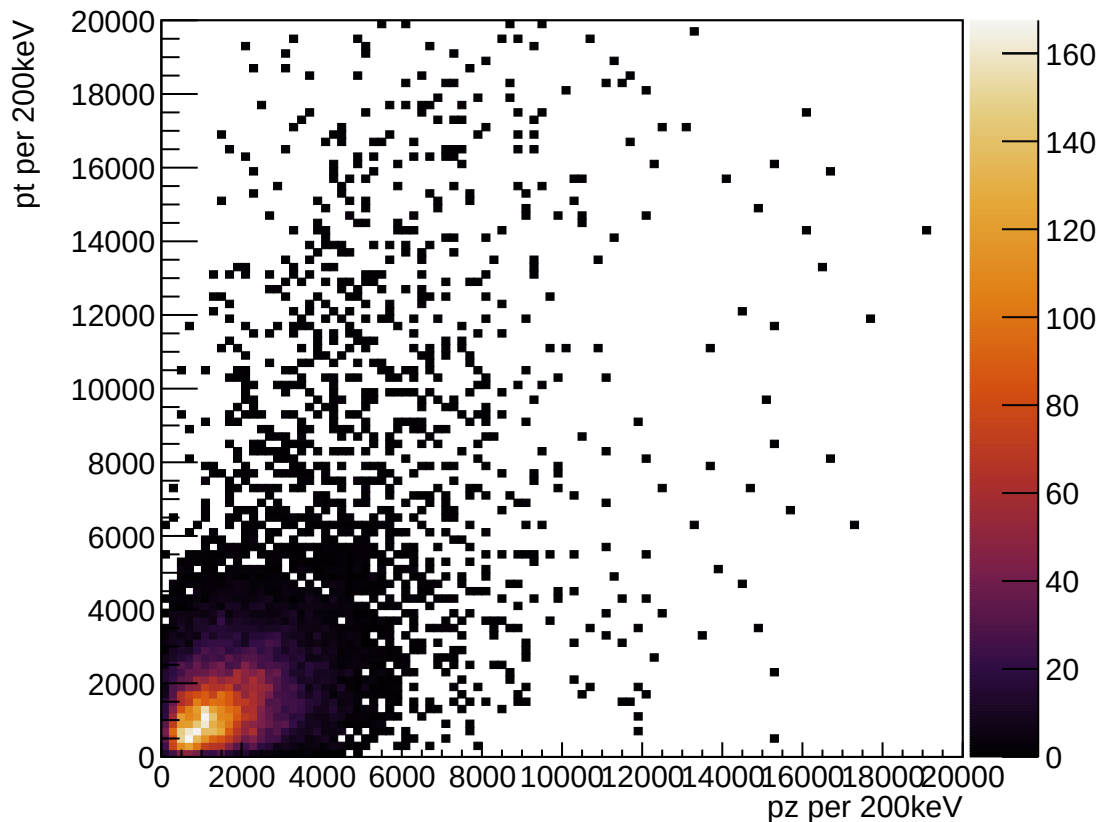
Entries	16582	
Mean x	78.13	
Mean y	2076	
Std Dev x	73.37	
Std Dev y	2359	
Integral	1.661e+04	
0	120.081	0
0	16612.5	0
0	0	0

Stock G4: positron pt vs pz



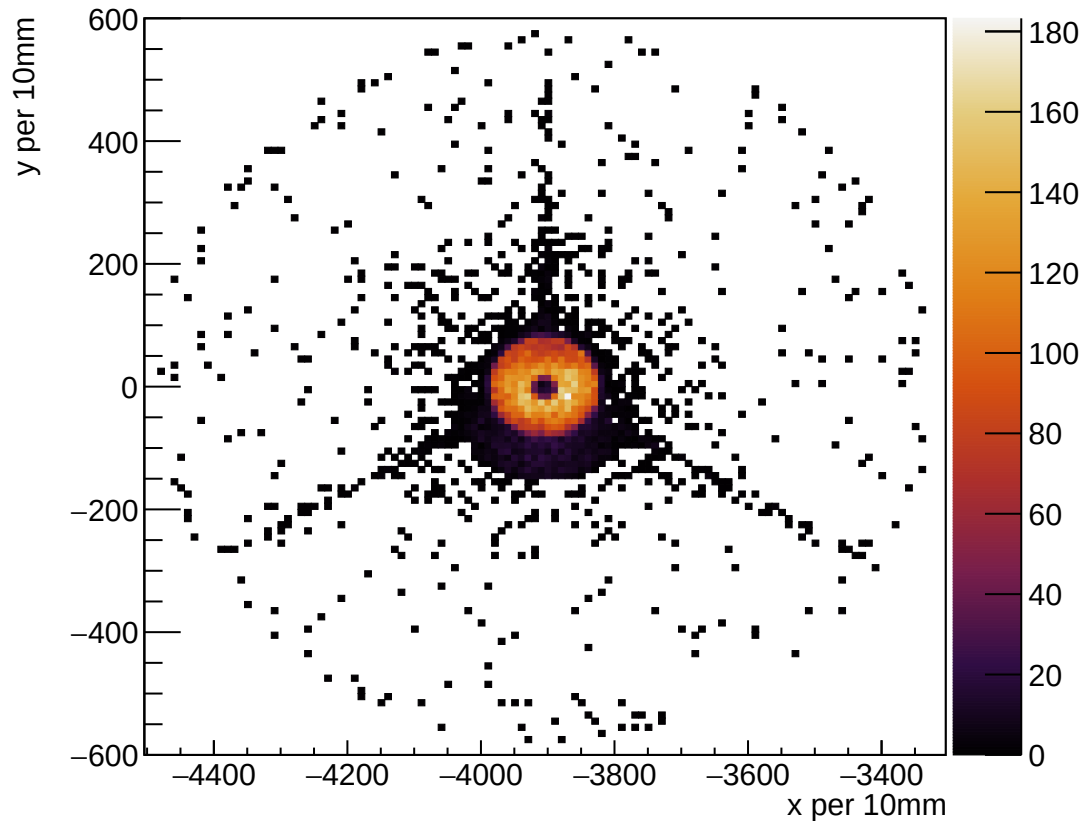
Entries	26791	
Mean x	1510	
Mean y	1567	
Std Dev x	1432	
Std Dev y	2002	
Integral	2.479e+04	
0	124.649	1.86044
0	24785.7	9.3022
0	0	0

Mu2e mu resampler: positron pt vs pz



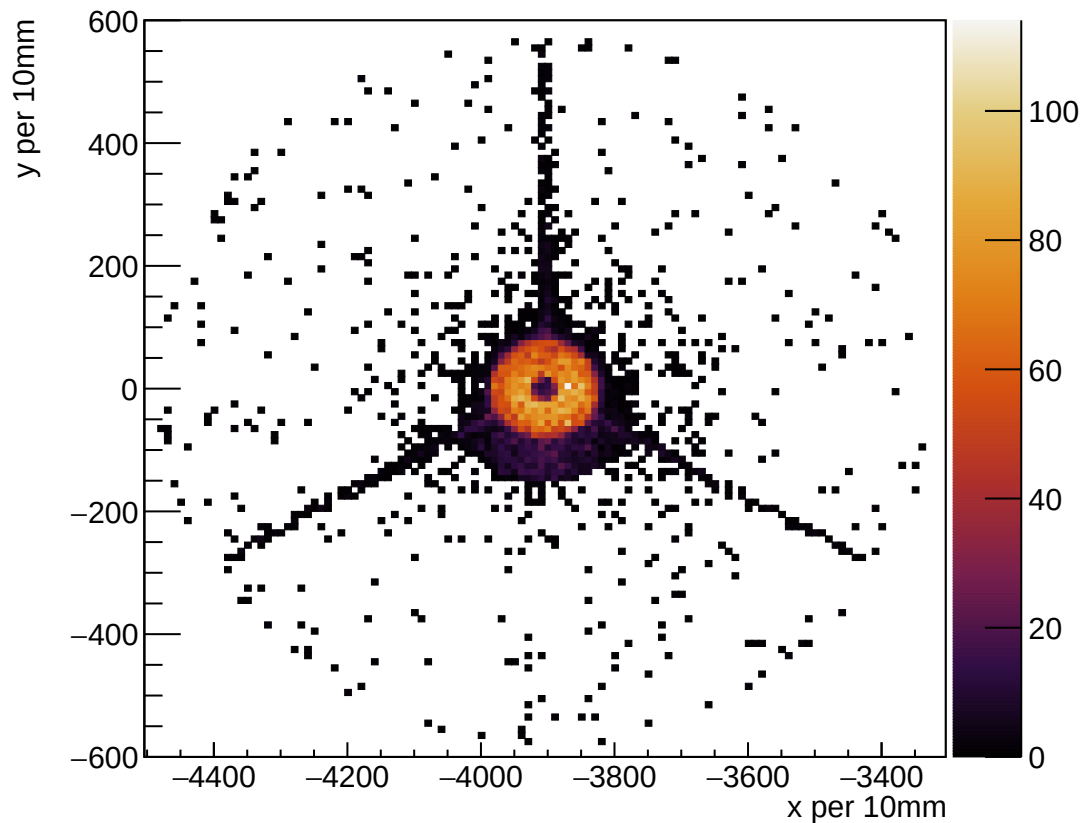
Entries	16582	
Mean x	1993	
Mean y	2073	
Std Dev x	1637	
Std Dev y	2352	
Integral	1.661e+04	
0	117.053	3.02725
0	16605.4	47.06357
0	0	0

Stock G4: positron y vs x



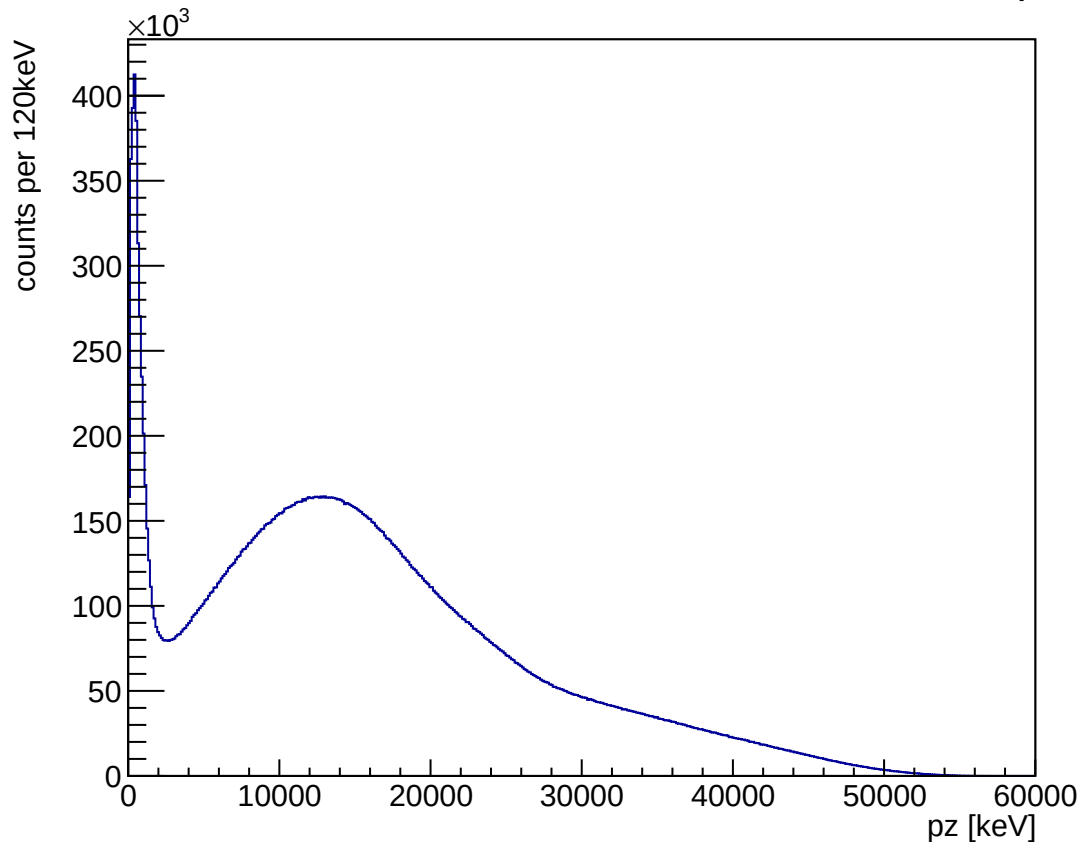
Entries	26791		
Mean x	-3903		
Mean y	-5.652		
Std Dev x	61.56		
Std Dev y	63.43		
Integral	2.492e+04		
0	0	0	
0	24921.5	0	
0	0	0	

Mu2e mu resampler: positron y vs x



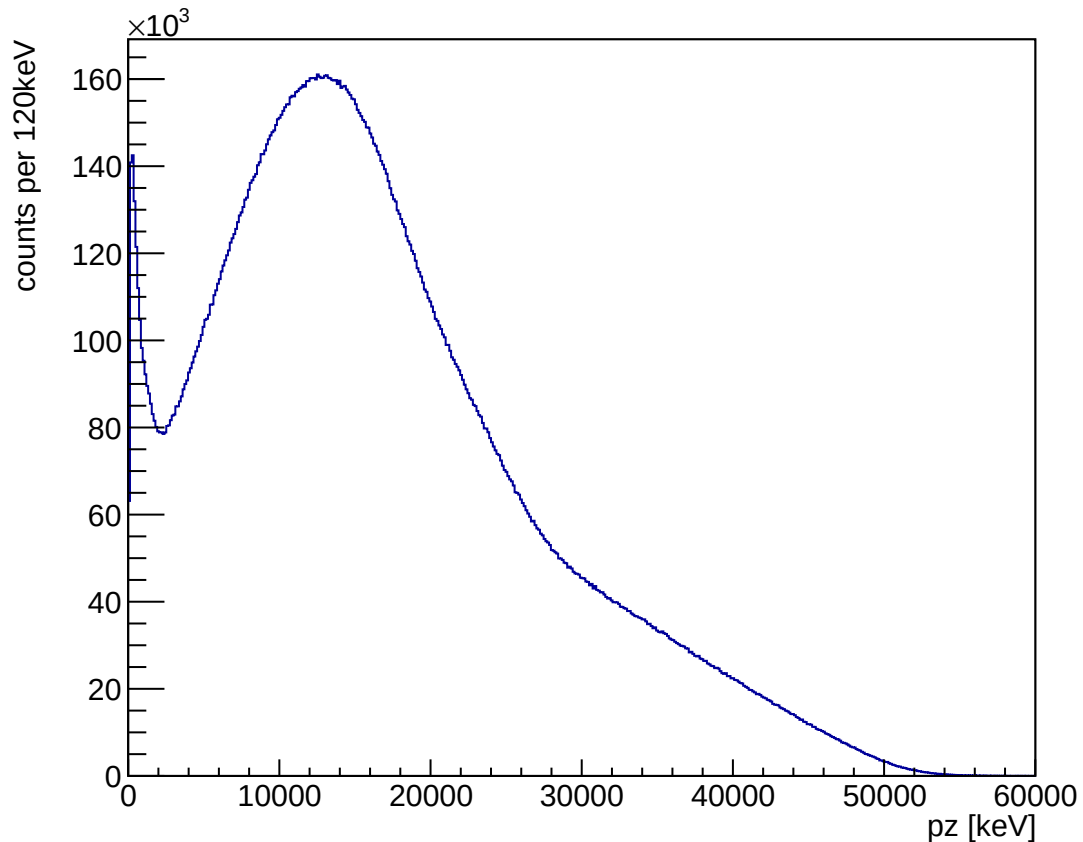
Entries	16582		
Mean x	-3904		
Mean y	-9.193		
Std Dev x	73.55		
Std Dev y	78.05		
Integral	1.673e+04		
0	0	0	
0	16732.6	0	
0	0	0	

Stock G4: electron momentum pz



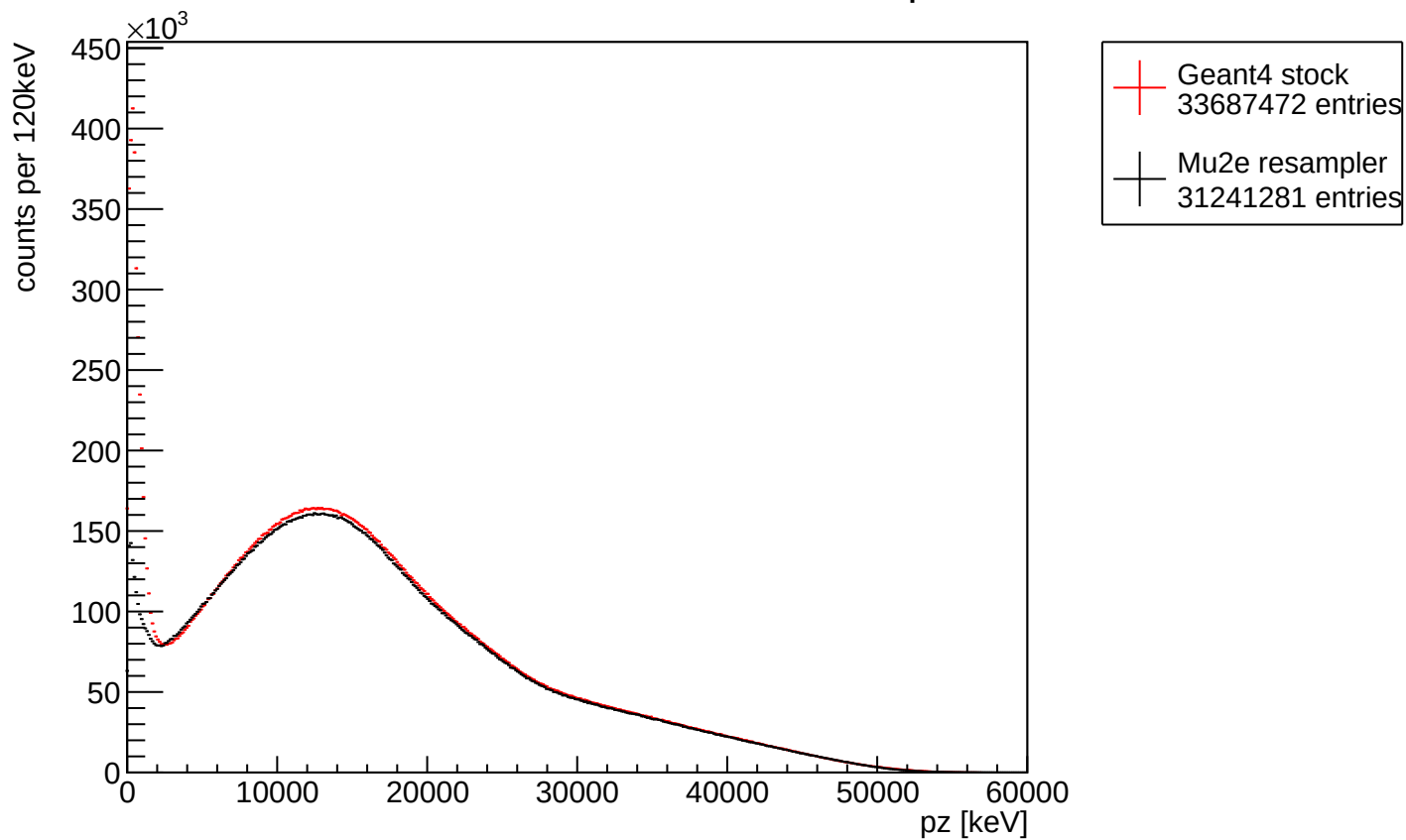
Entries	3.621453e+07
Mean	1.572e+04
Std Dev	1.092e+04
Underflow	0
Overflow	0
Integral	3.369e+07

Mu2e mu resampler: electron momentum pz

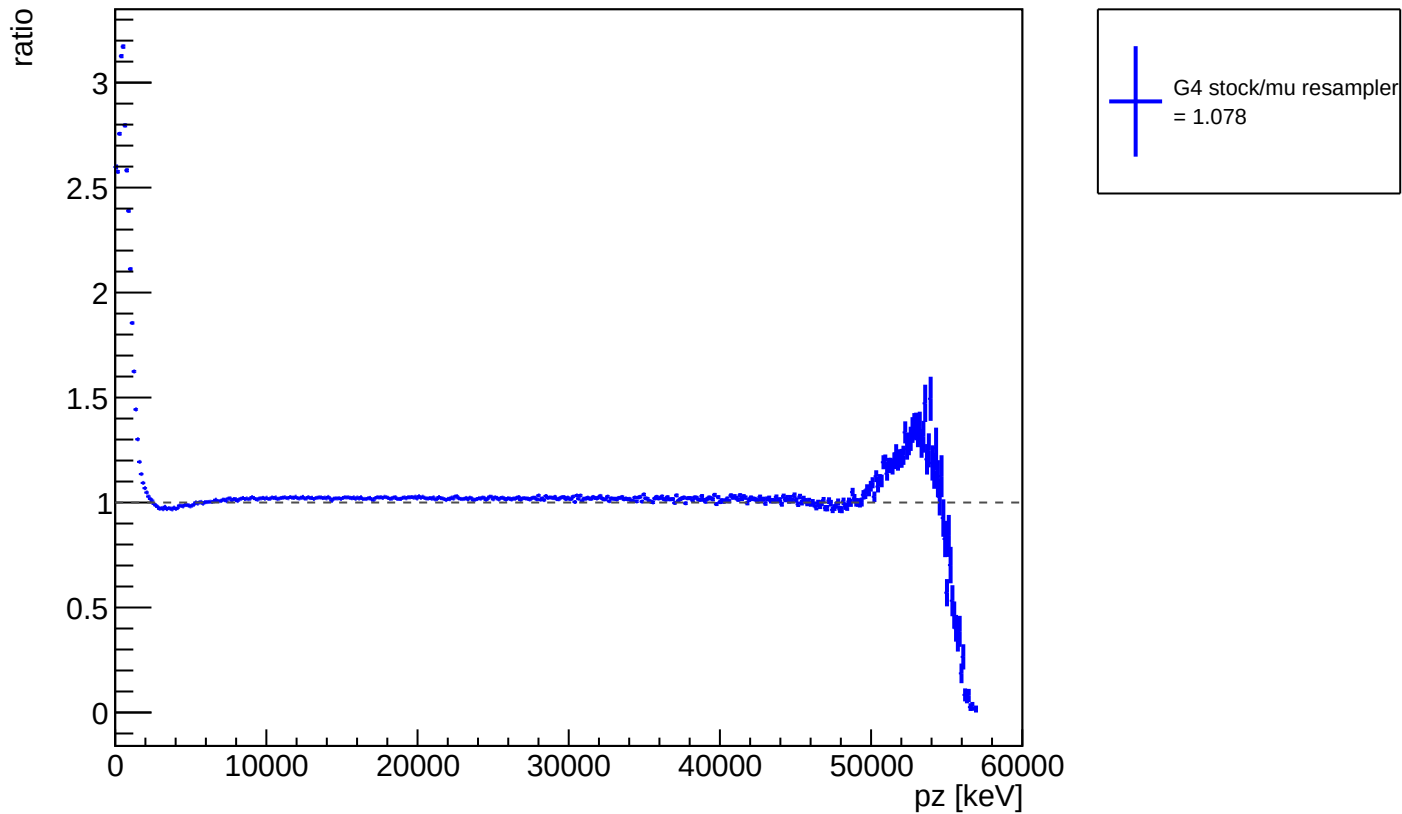


Entries	3.096011e+07
Mean	1.66e+04
Std Dev	1.059e+04
Underflow	0
Overflow	125.1
Integral	3.124e+07

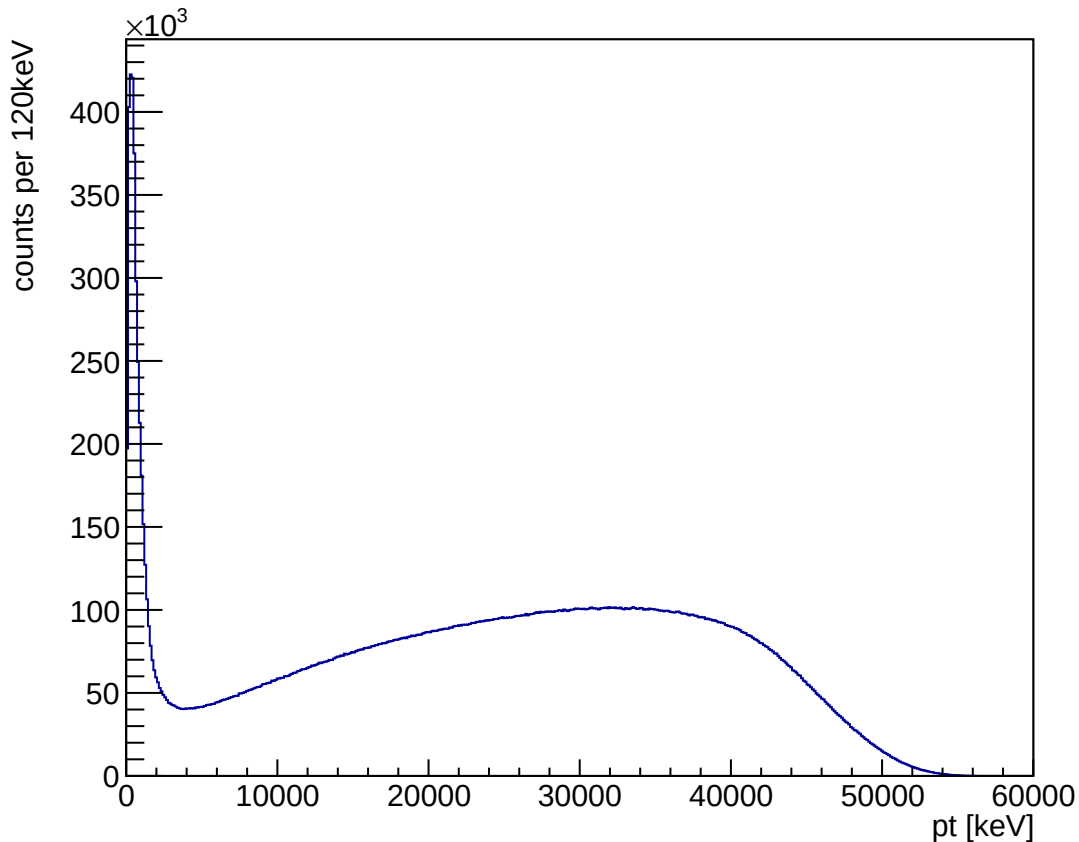
electron momentum pz



ratio of G4 stock / muon resampler: electron momentum pz

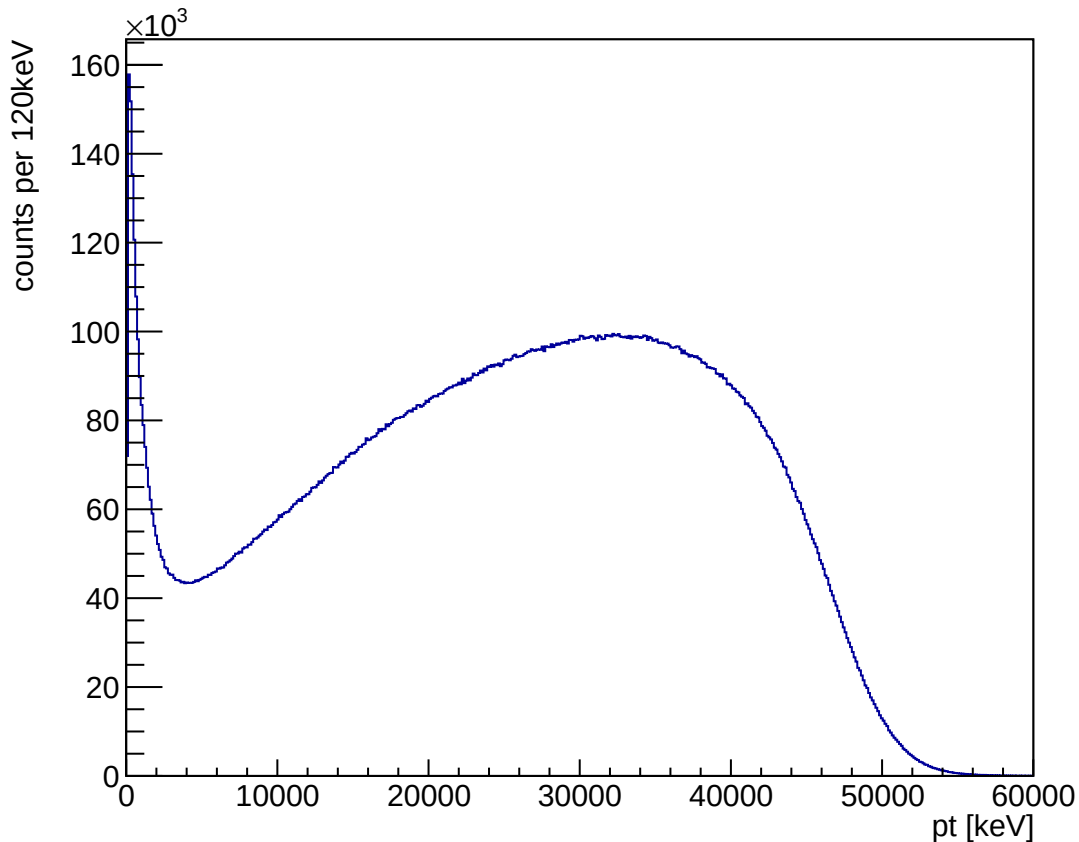


Stock G4: electron transverse momentum pT



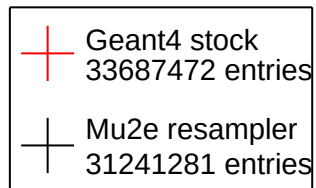
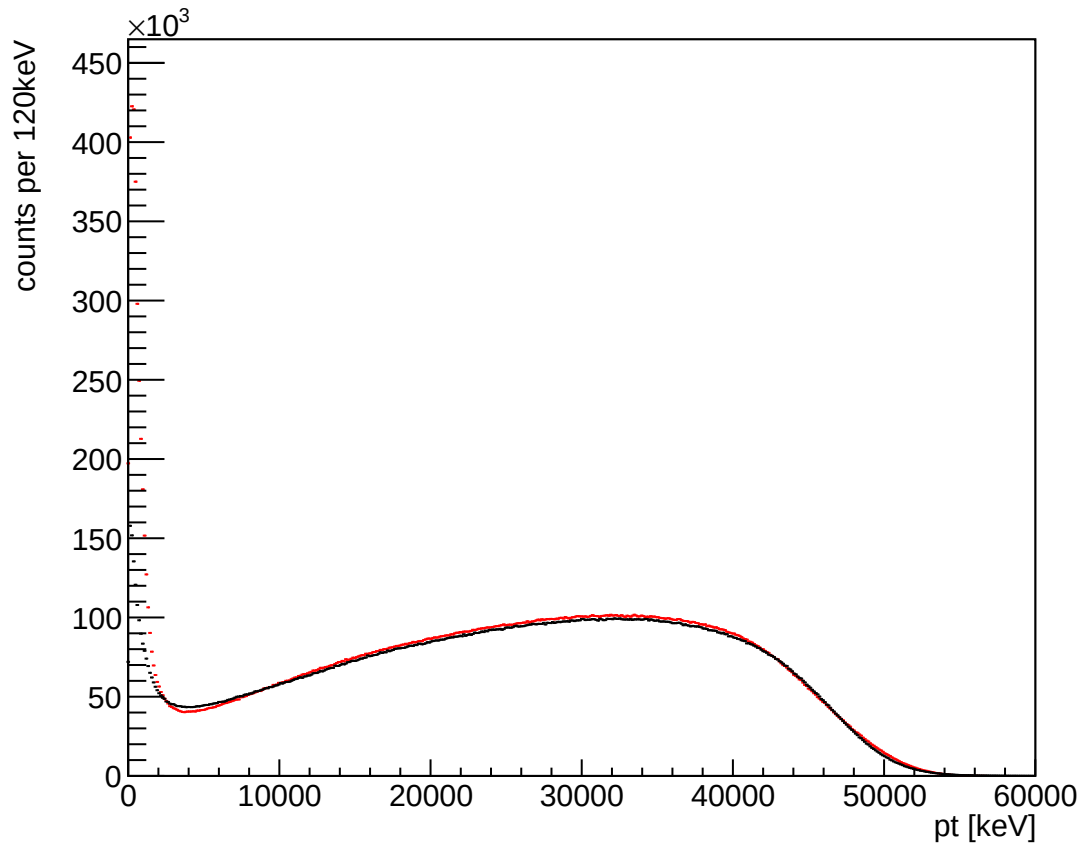
Entries	3.621453e+07
Mean	2.413e+04
Std Dev	1.393e+04
Underflow	0
Overflow	0
Integral	3.369e+07

Mu2e mu resampler: electron transverse momentum pT

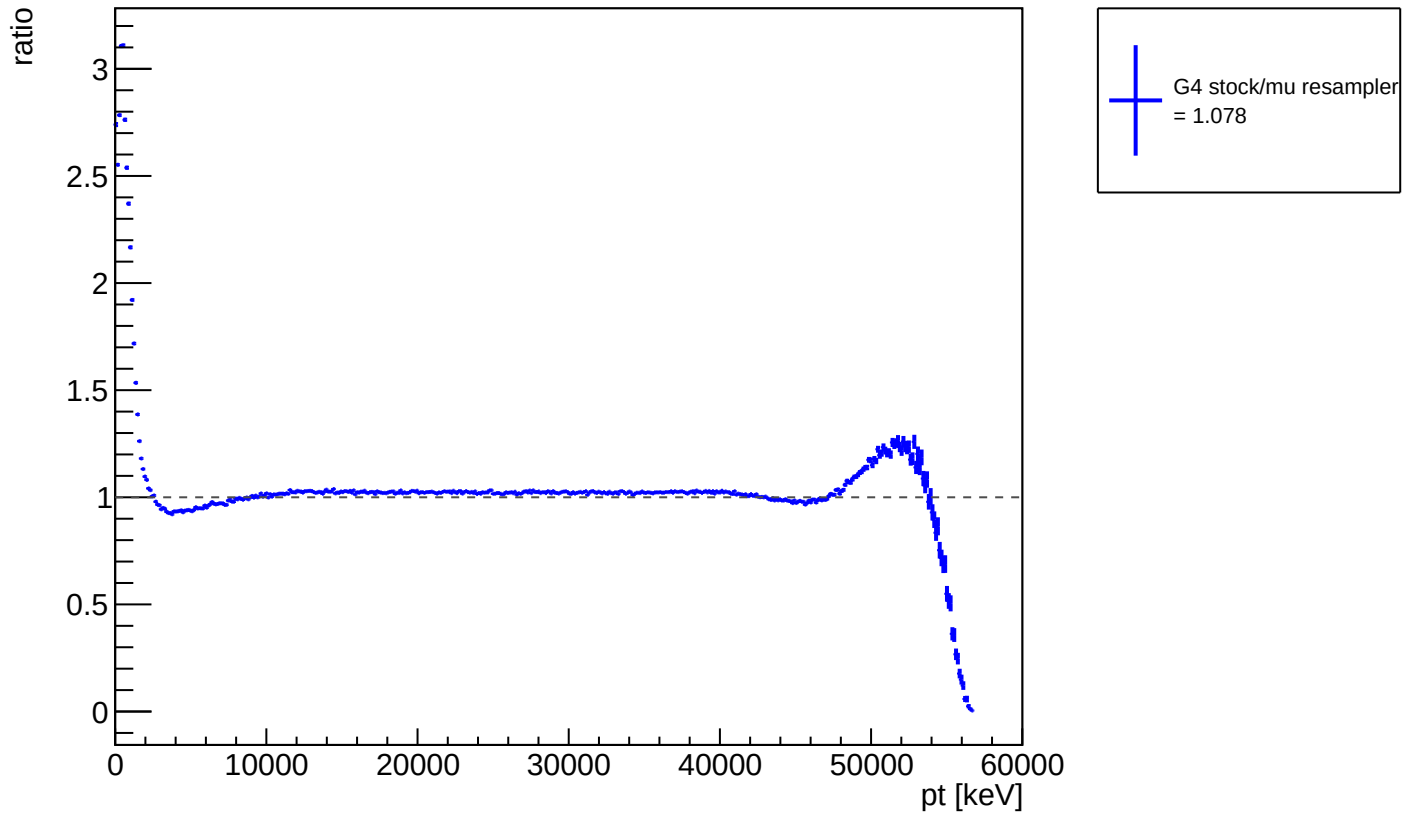


Entries	3.096011e+07
Mean	2.548e+04
Std Dev	1.308e+04
Underflow	0
Overflow	517.7
Integral	3.124e+07

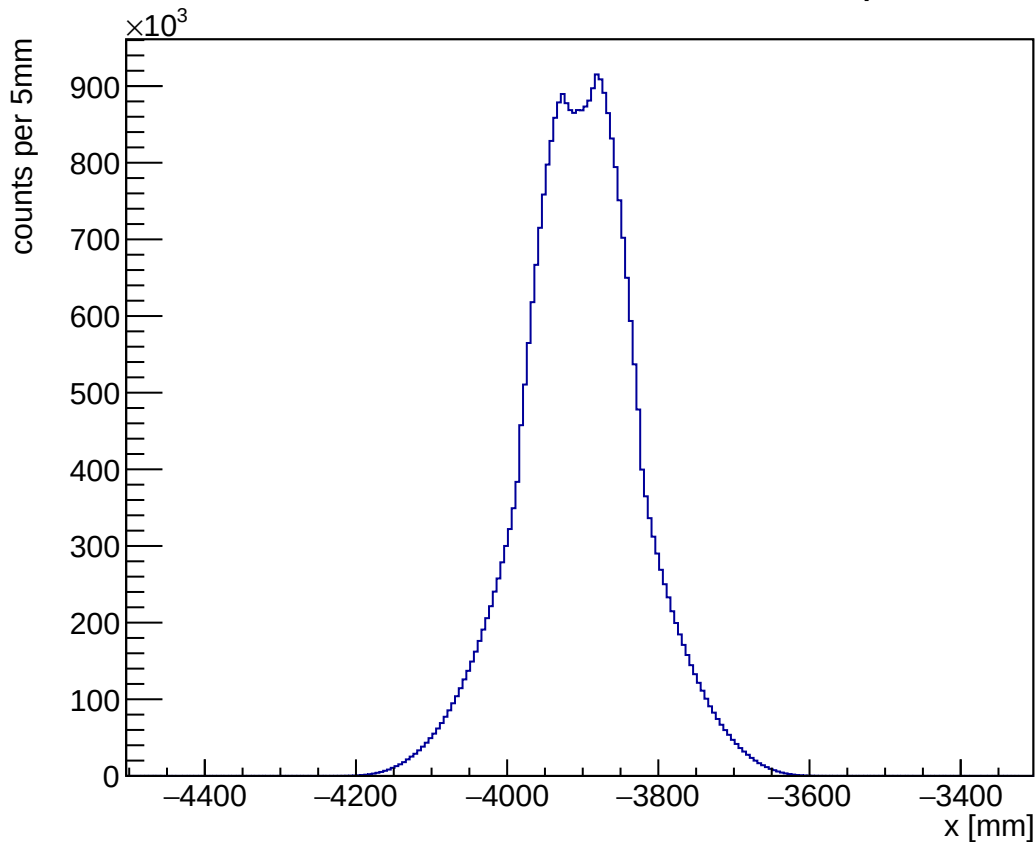
electron transverse momentum pT



ratio of G4 stock / muon resampler: electron transverse momentum pT

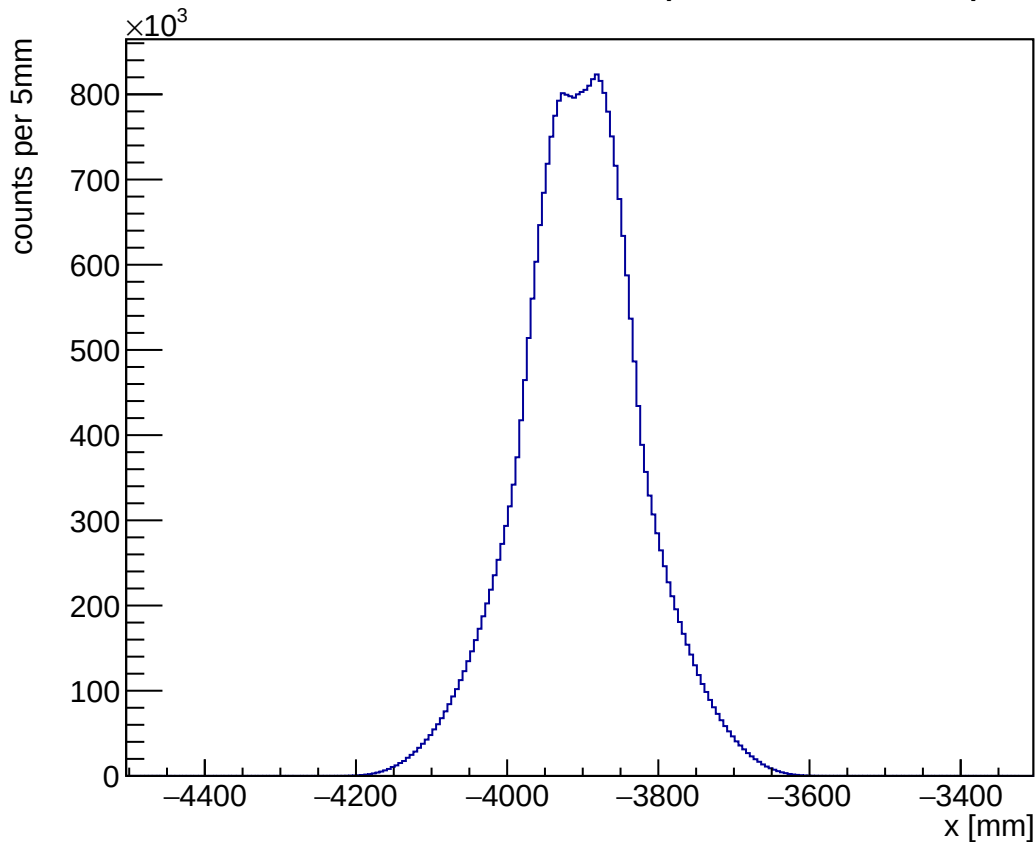


Stock G4: electron x position



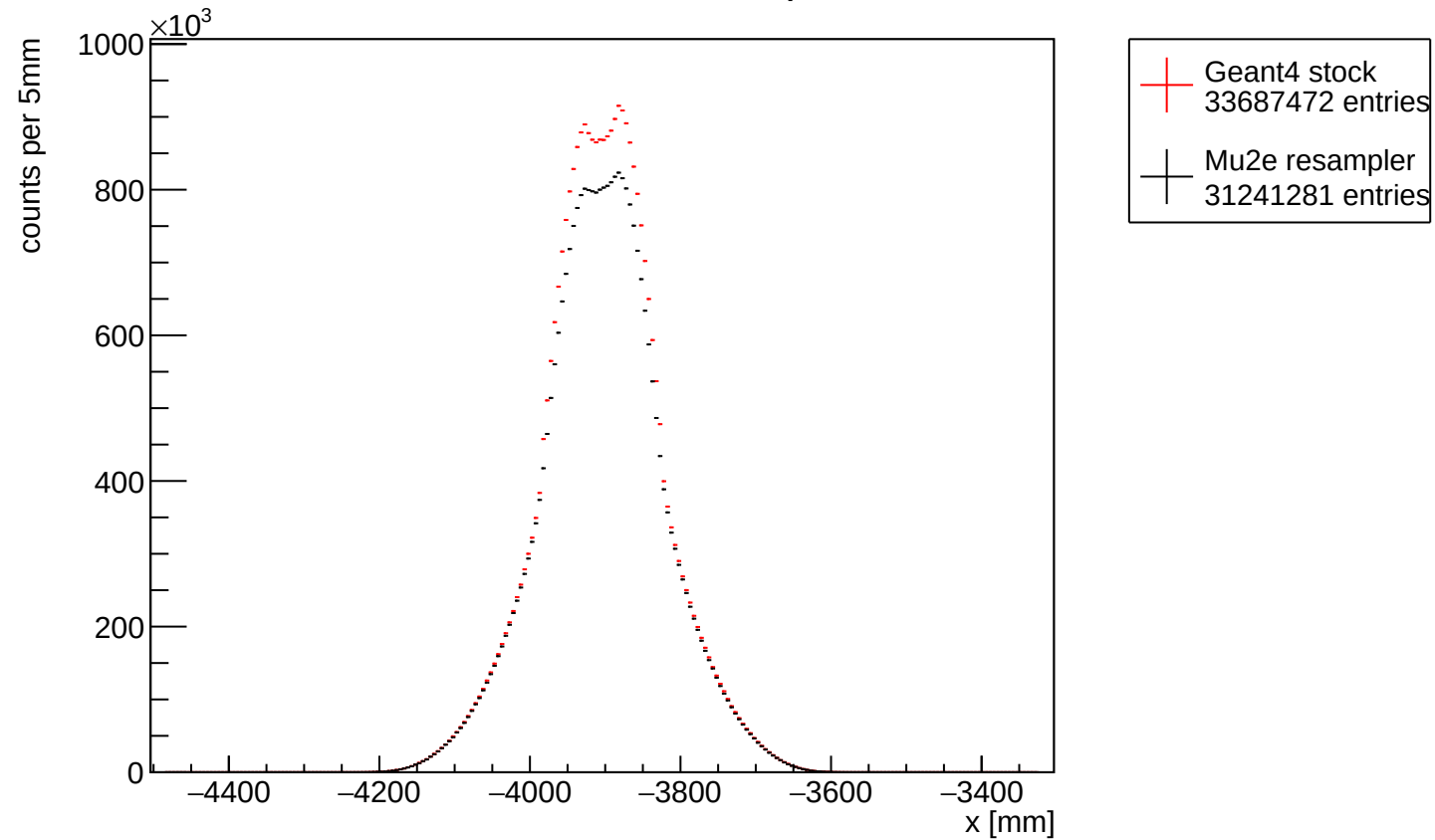
Entries	3.621453e+07
Mean	-3902
Std Dev	78.87
Underflow	0
Overflow	0
Integral	3.369e+07

Mu2e mu resampler: electron x position

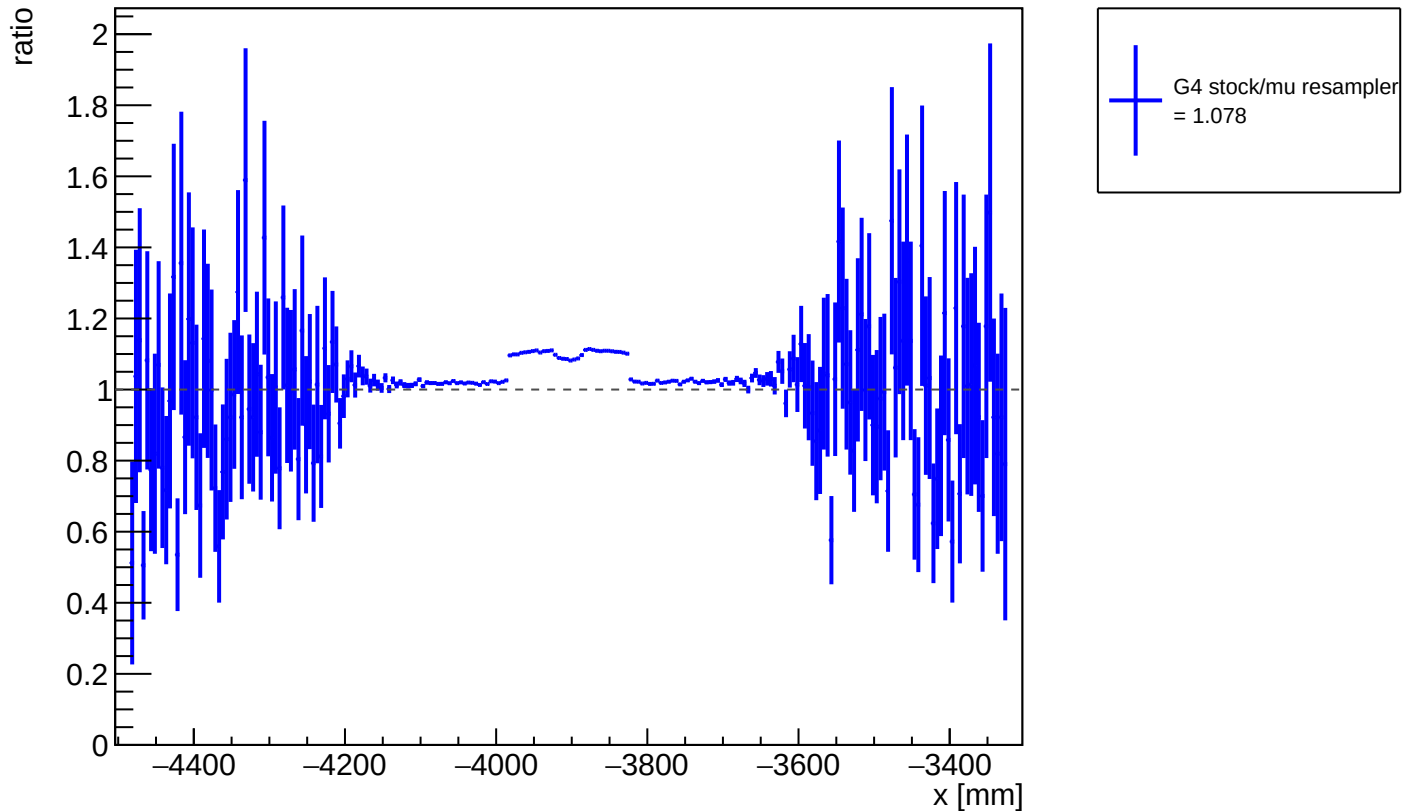


Entries	3.096011e+07
Mean	-3903
Std Dev	80.44
Underflow	0
Overflow	0
Integral	3.124e+07

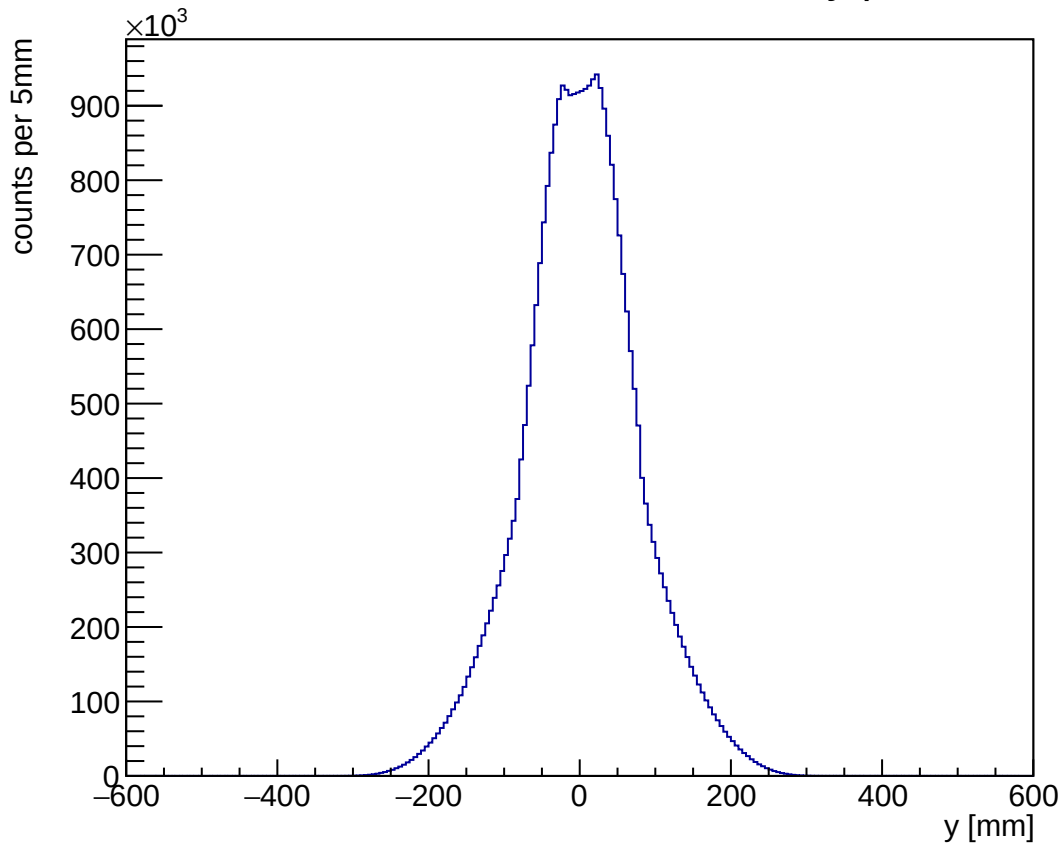
electron x position



ratio of G4 stock / muon resampler: electron x position

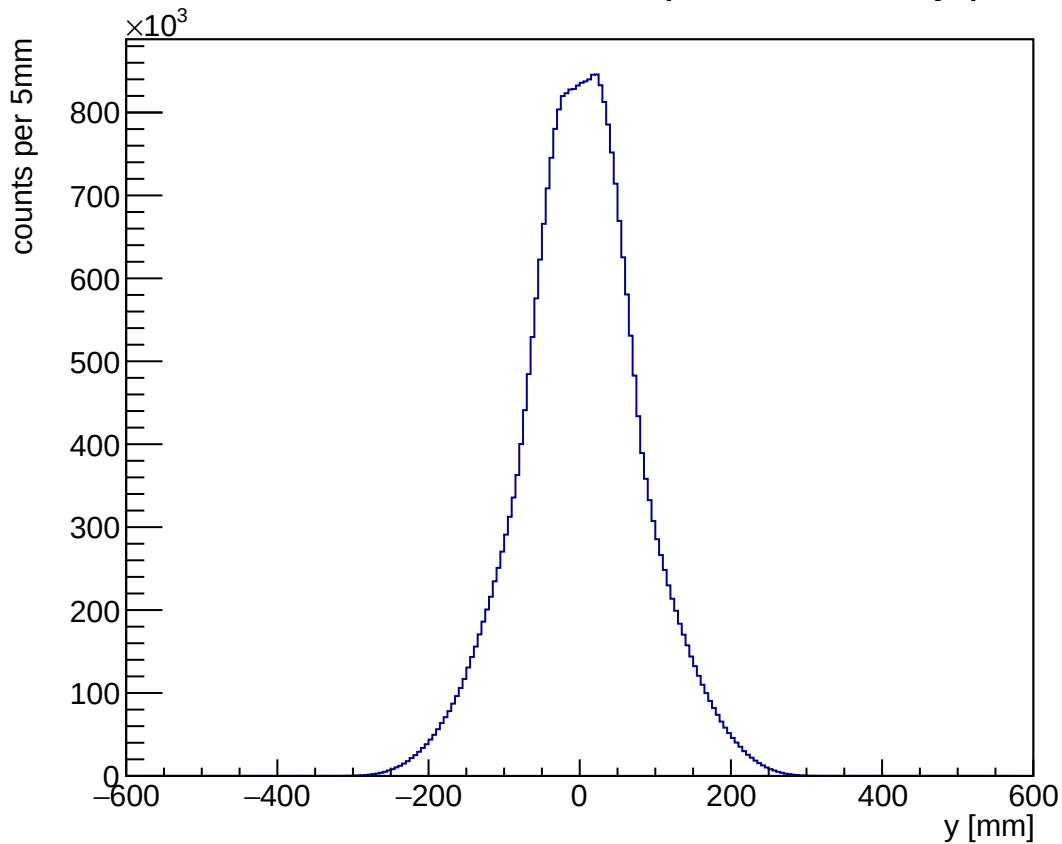


Stock G4: electron y position



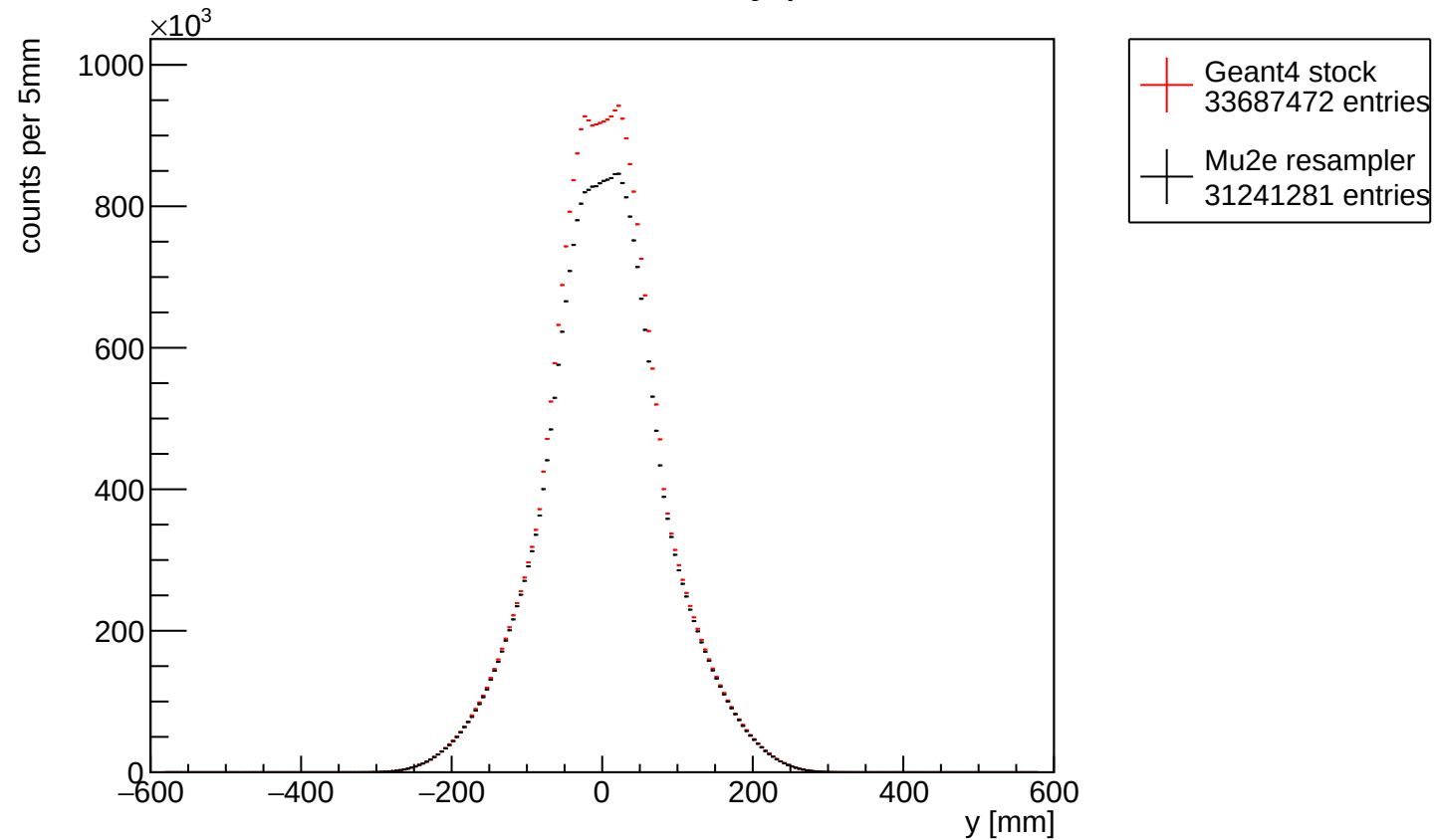
Entries	3.621453e+07
Mean	2.382
Std Dev	77.71
Underflow	0
Overflow	0
Integral	3.369e+07

Mu2e mu resampler: electron y position

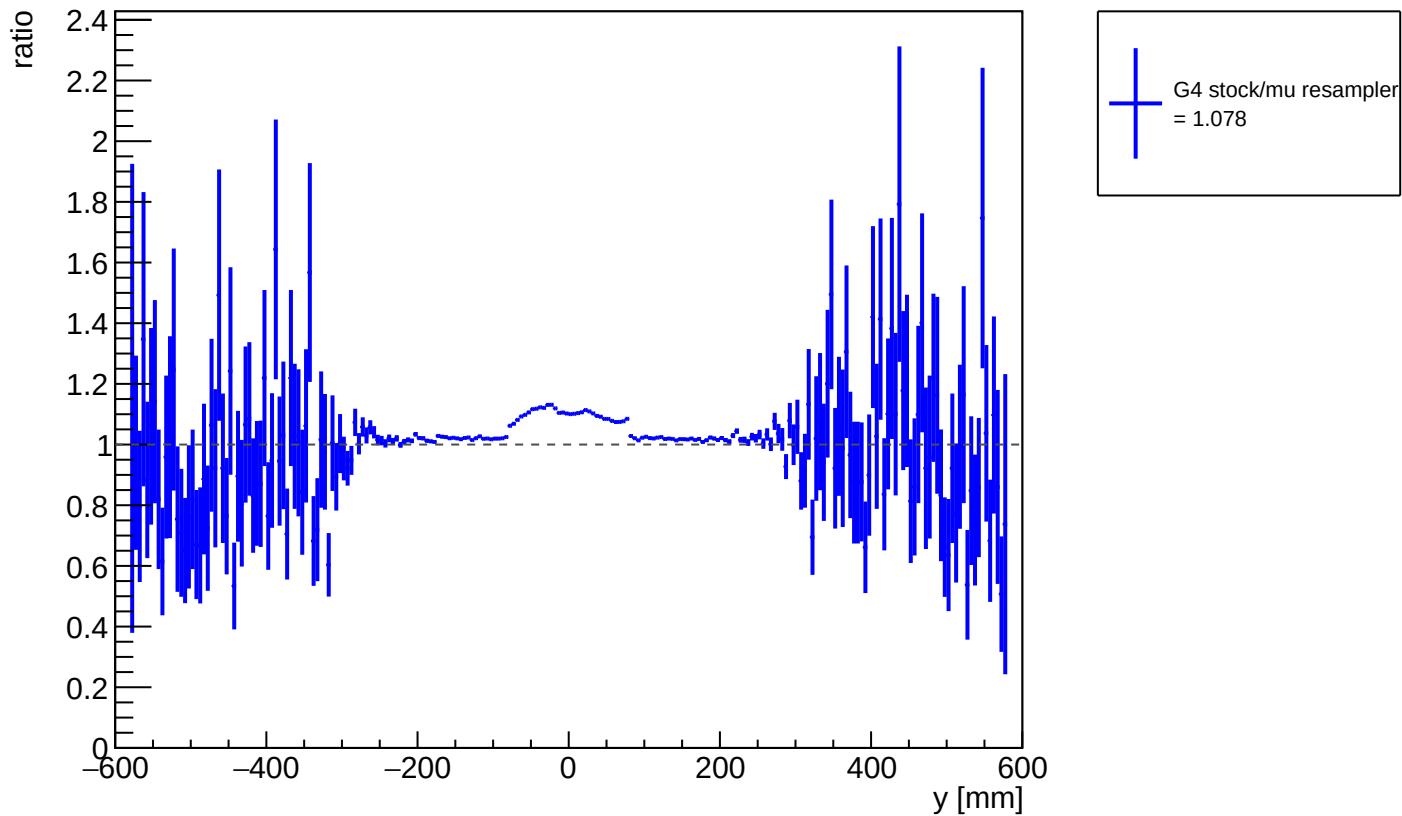


Entries	3.096011e+07
Mean	2.634
Std Dev	79.38
Underflow	0
Overflow	0
Integral	3.124e+07

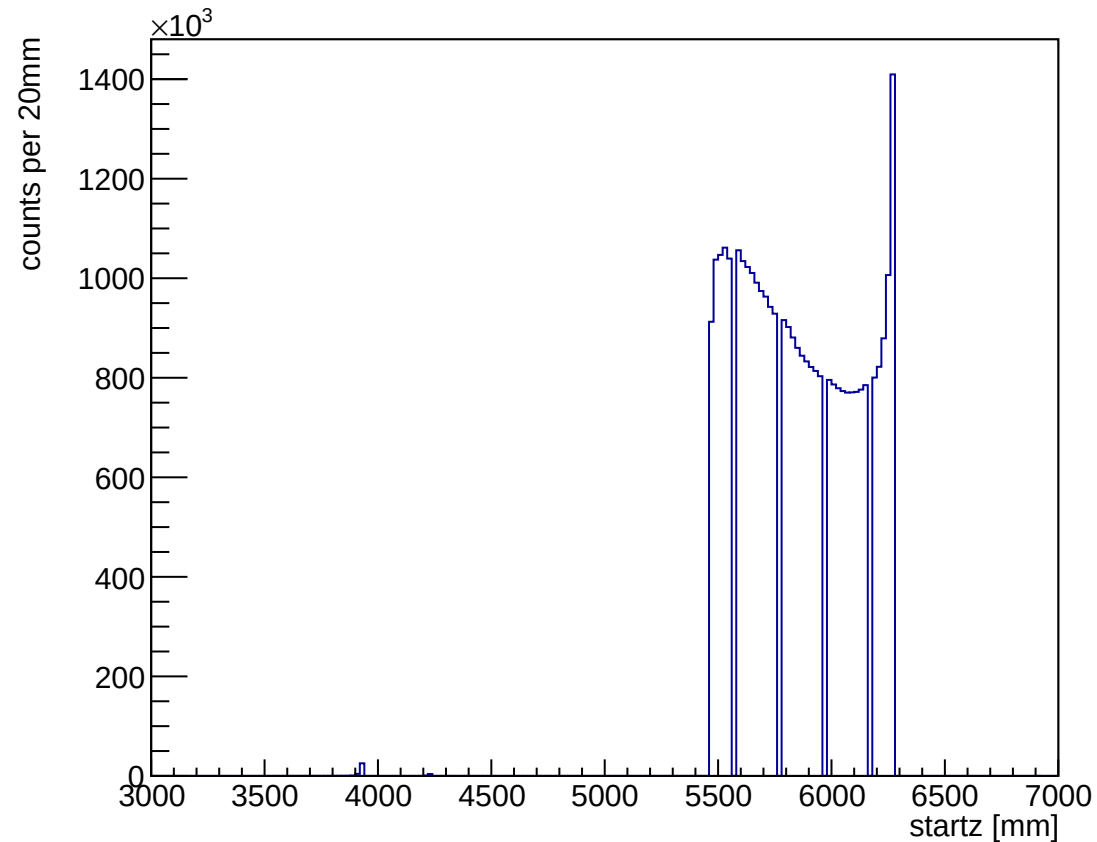
electron y position



ratio of G4 stock / muon resampler: electron y position

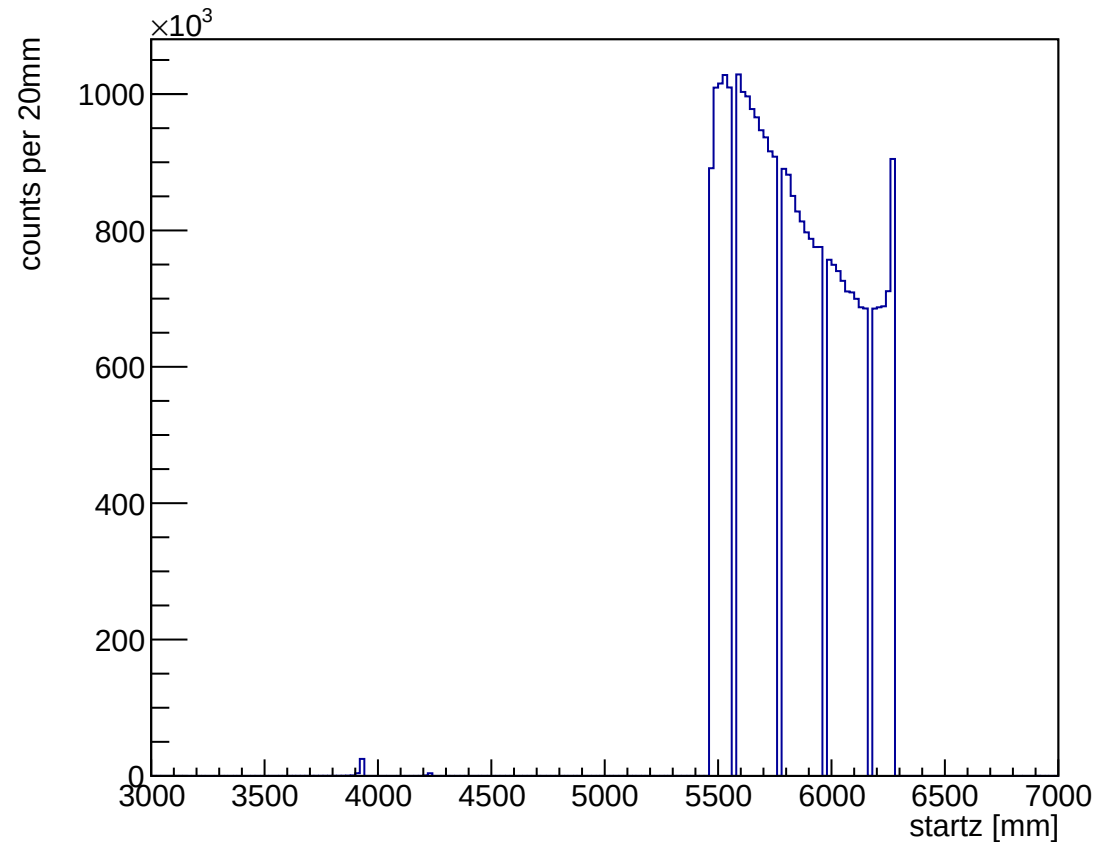


Stock G4: electron start z



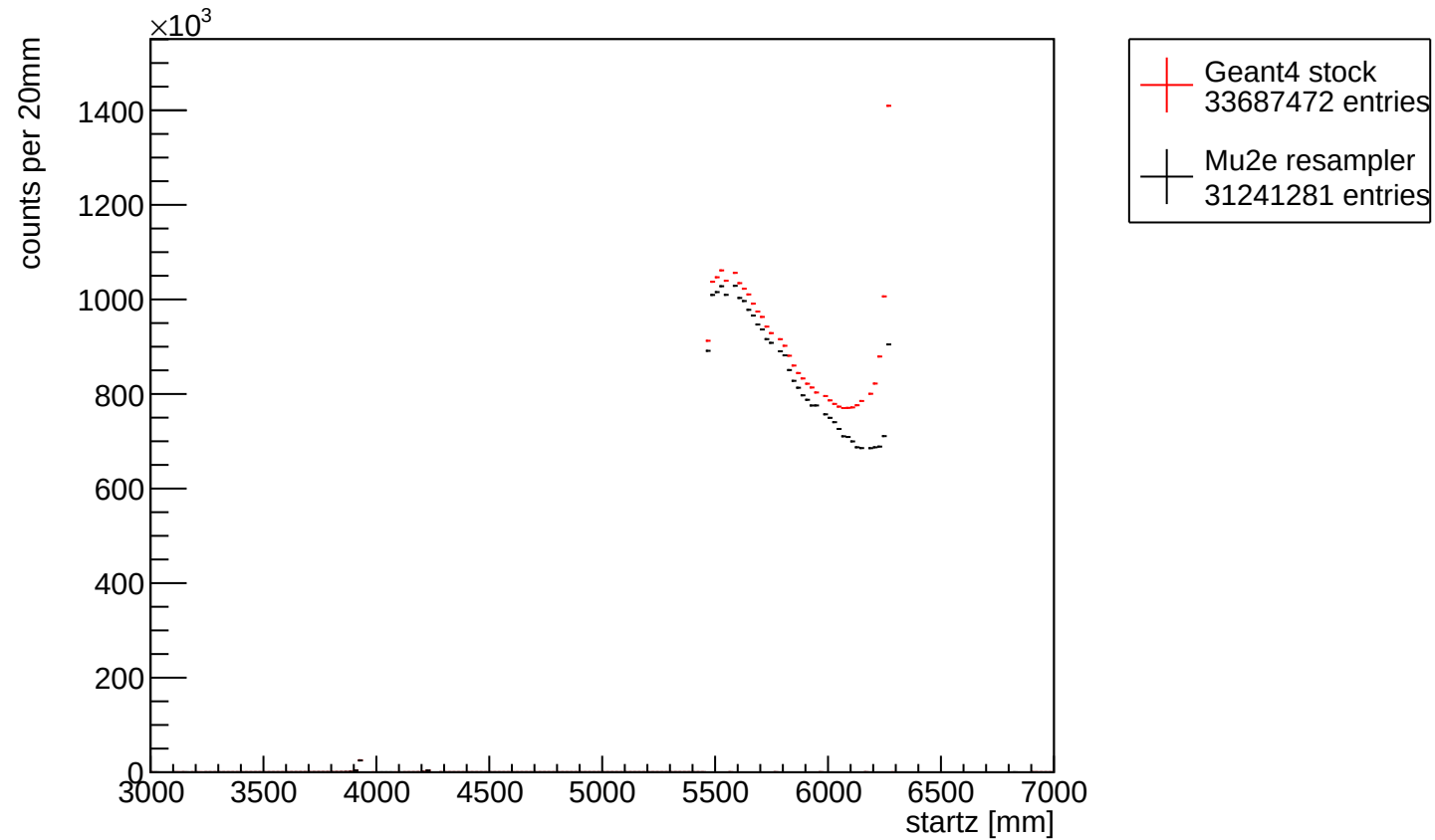
Entries	3.621453e+07
Mean	5855
Std Dev	252.6
Underflow	2.534e+04
Overflow	1575
Integral	3.366e+07

Mu2e mu resampler: electron start z

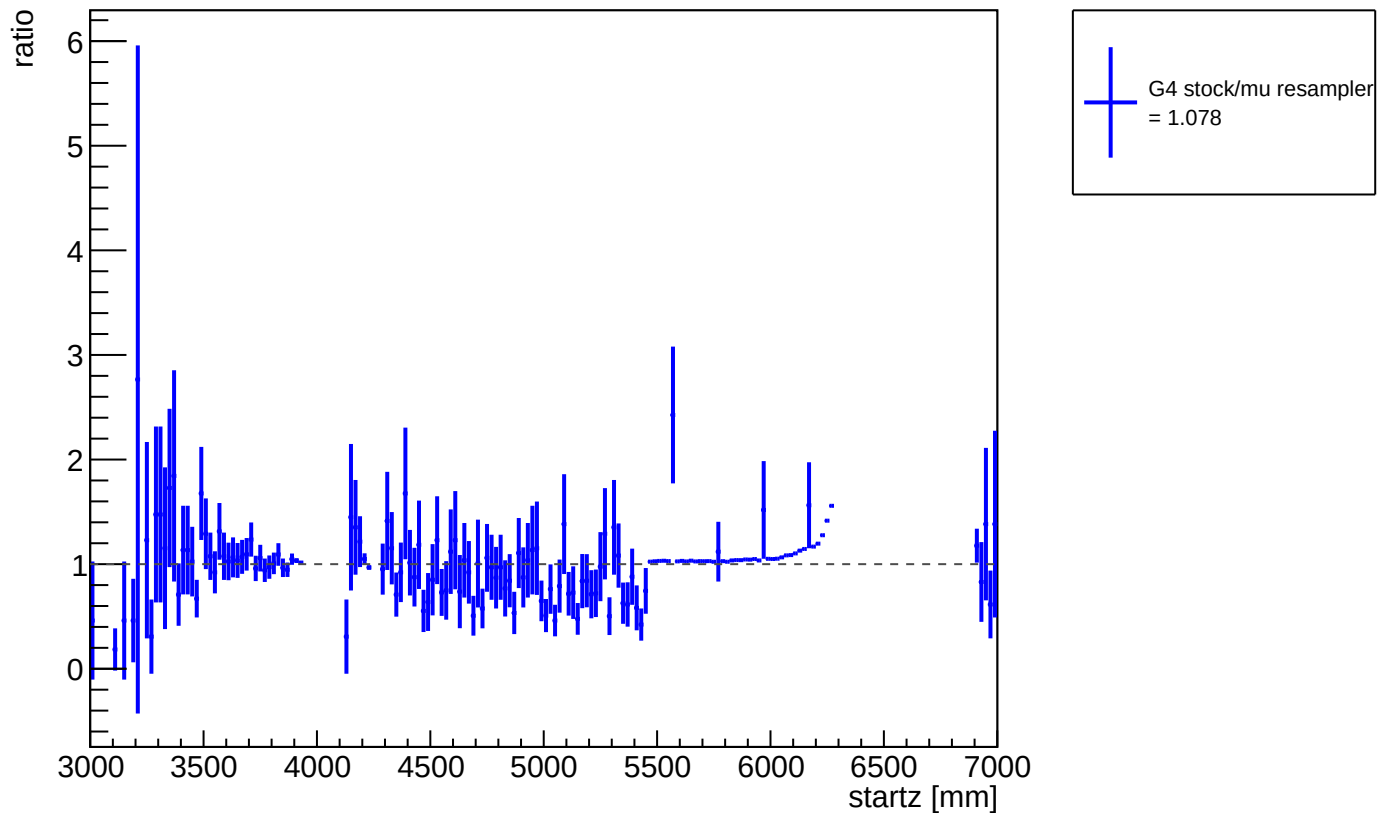


Entries	3.096011e+07
Mean	5839
Std Dev	246.1
Underflow	2.489e+04
Overflow	1498
Integral	3.121e+07

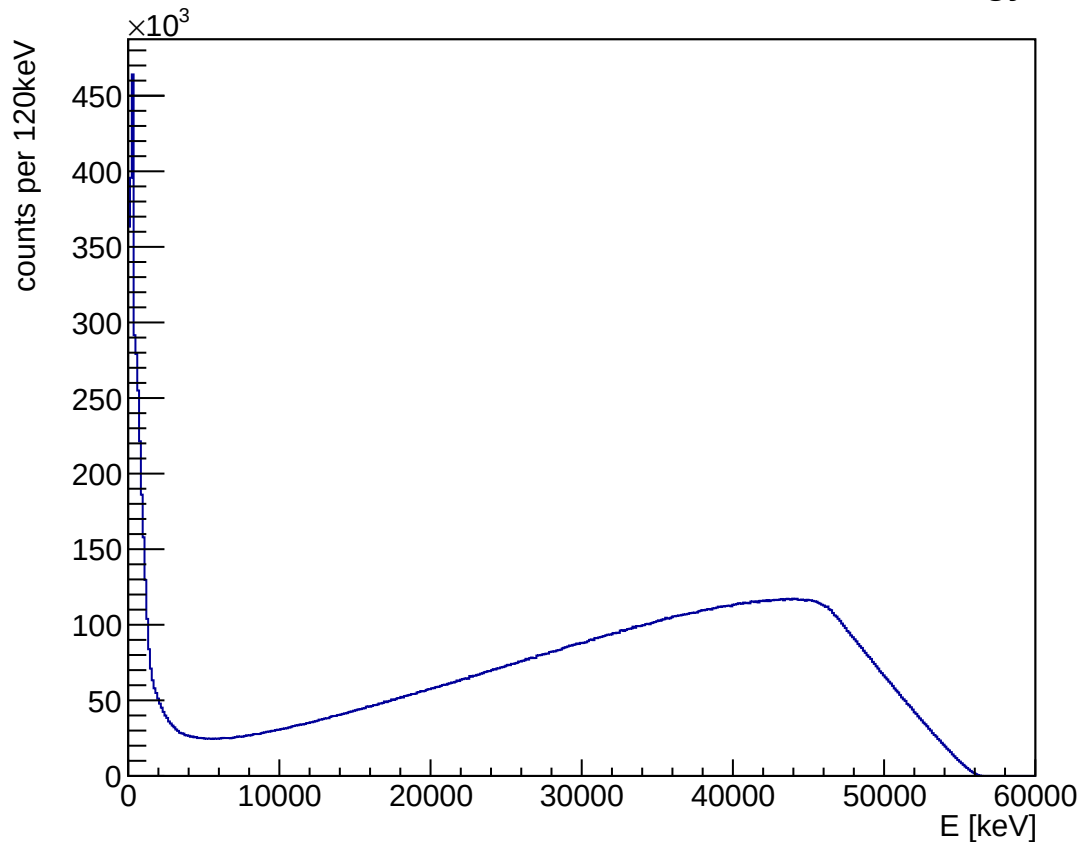
electron start z



ratio of G4 stock / muon resampler: electron start z

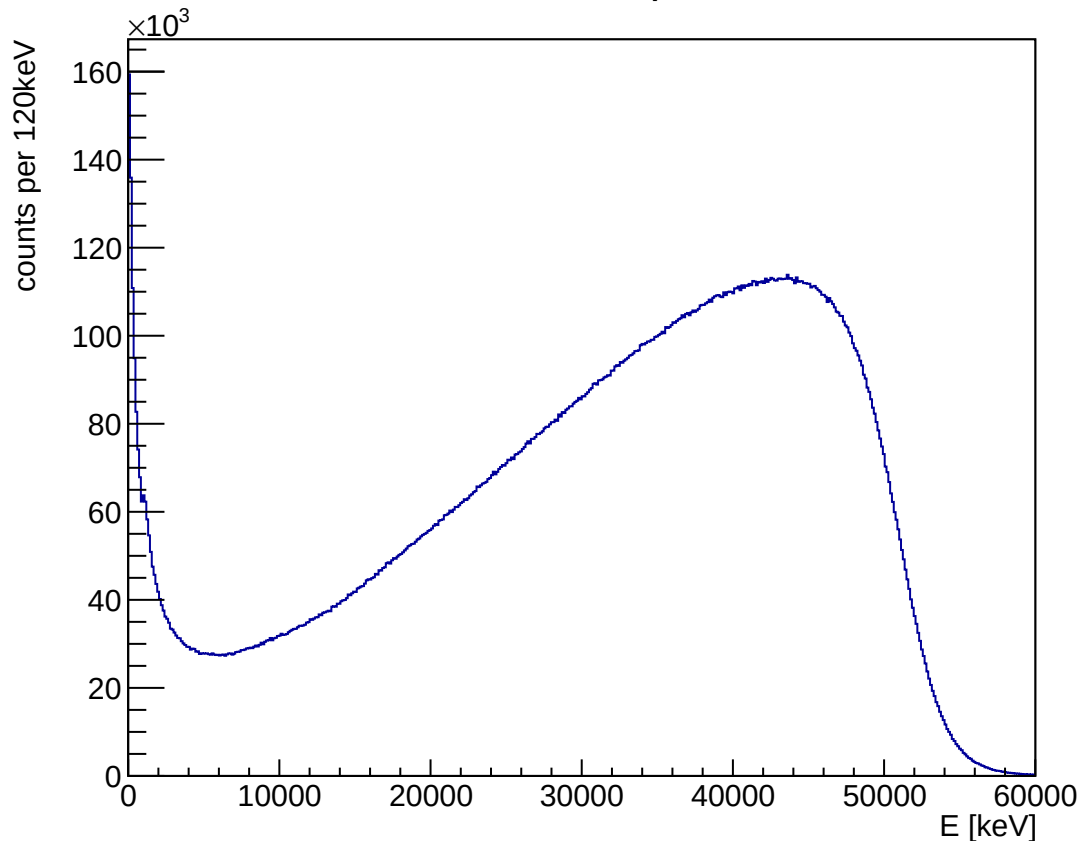


Stock G4: electron kin. energy



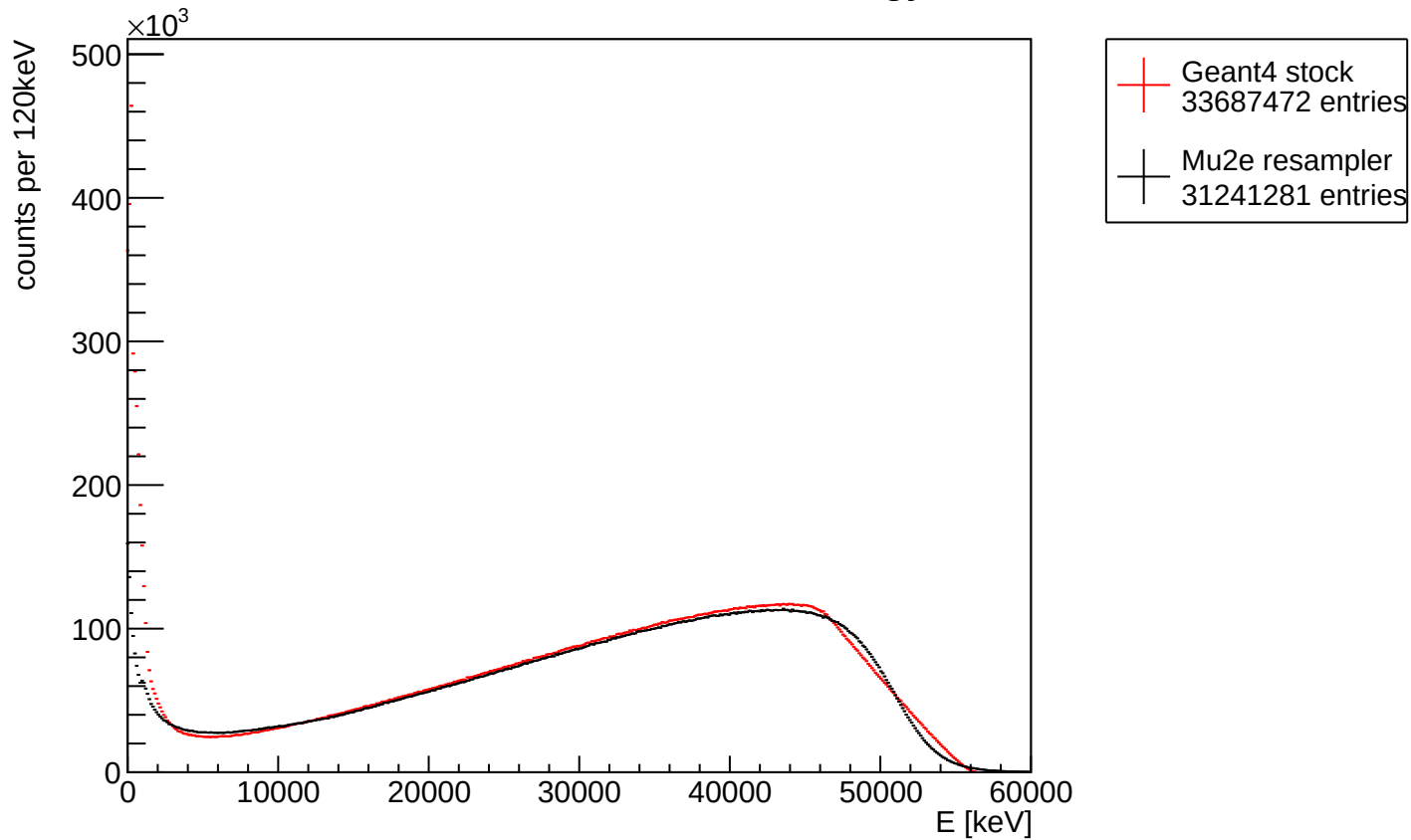
Entries	3.621453e+07
Mean	2.963e+04
Std Dev	1.531e+04
Underflow	0
Overflow	0
Integral	3.369e+07

Mu2e mu resampler: electron kin. energy

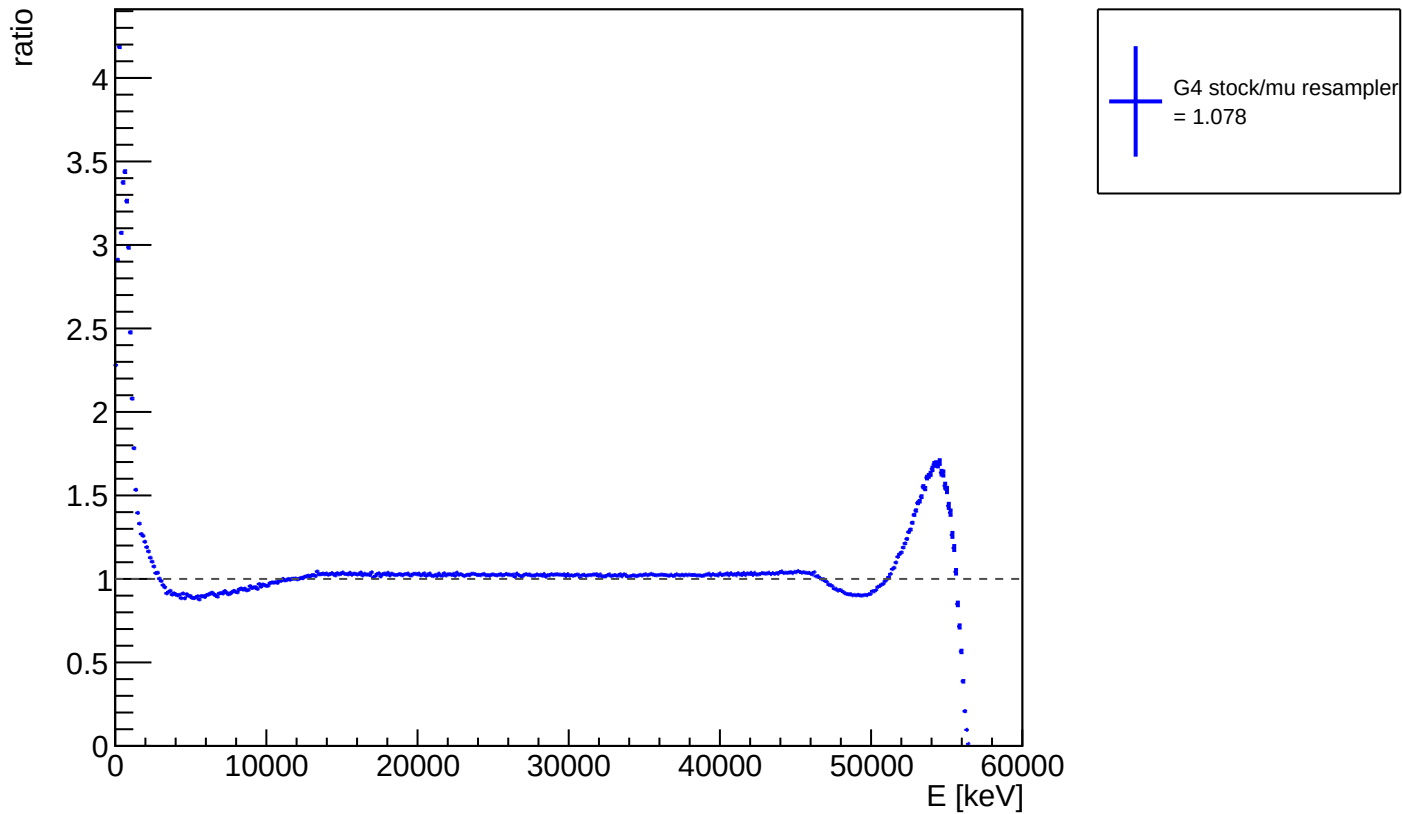


Entries	3.096011e+07
Mean	3.131e+04
Std Dev	1.398e+04
Underflow	0
Overflow	4313
Integral	3.124e+07

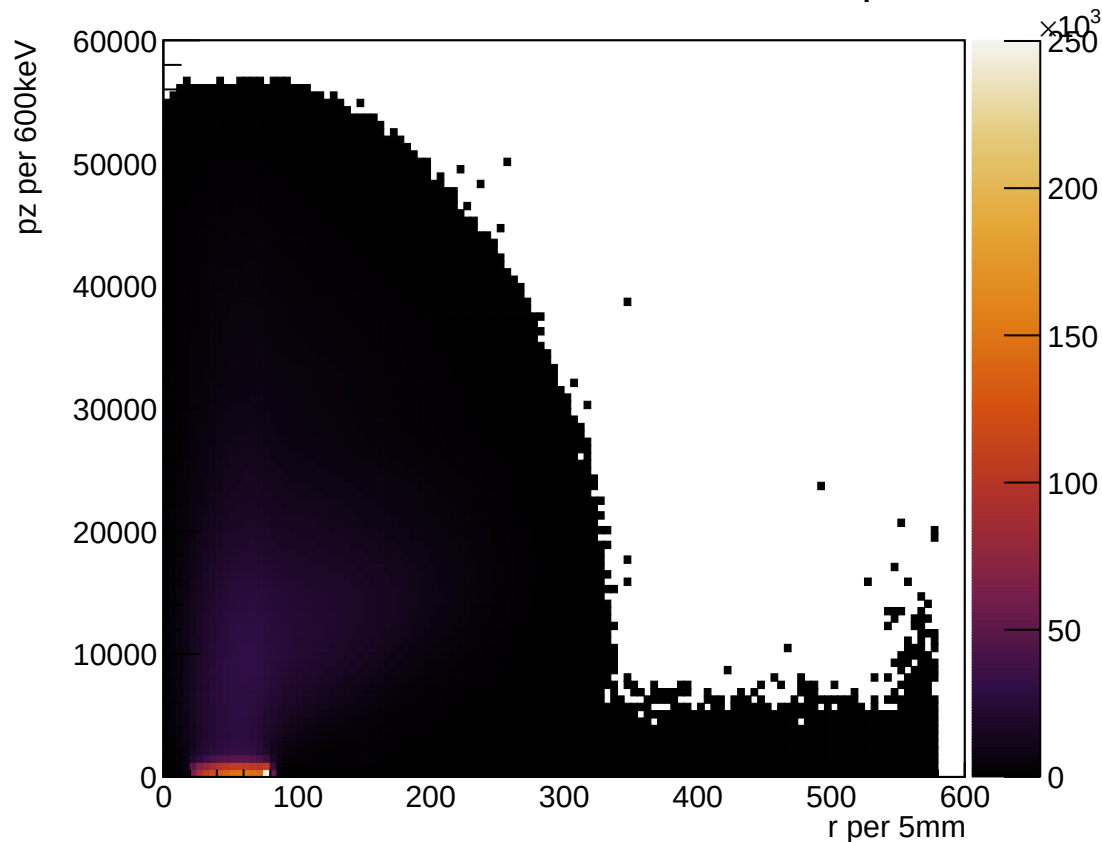
electron kin. energy



ratio of G4 stock / muon resampler: electron kin. energy

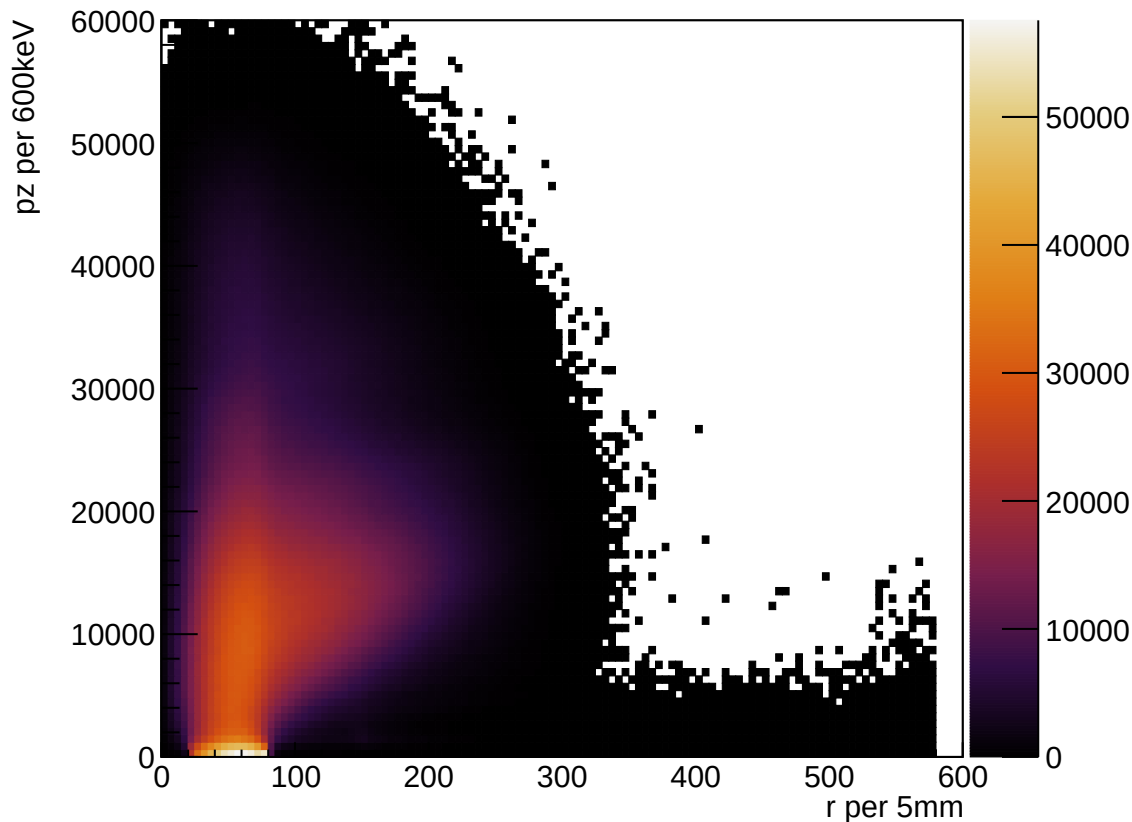


Stock G4: electron pz vs r



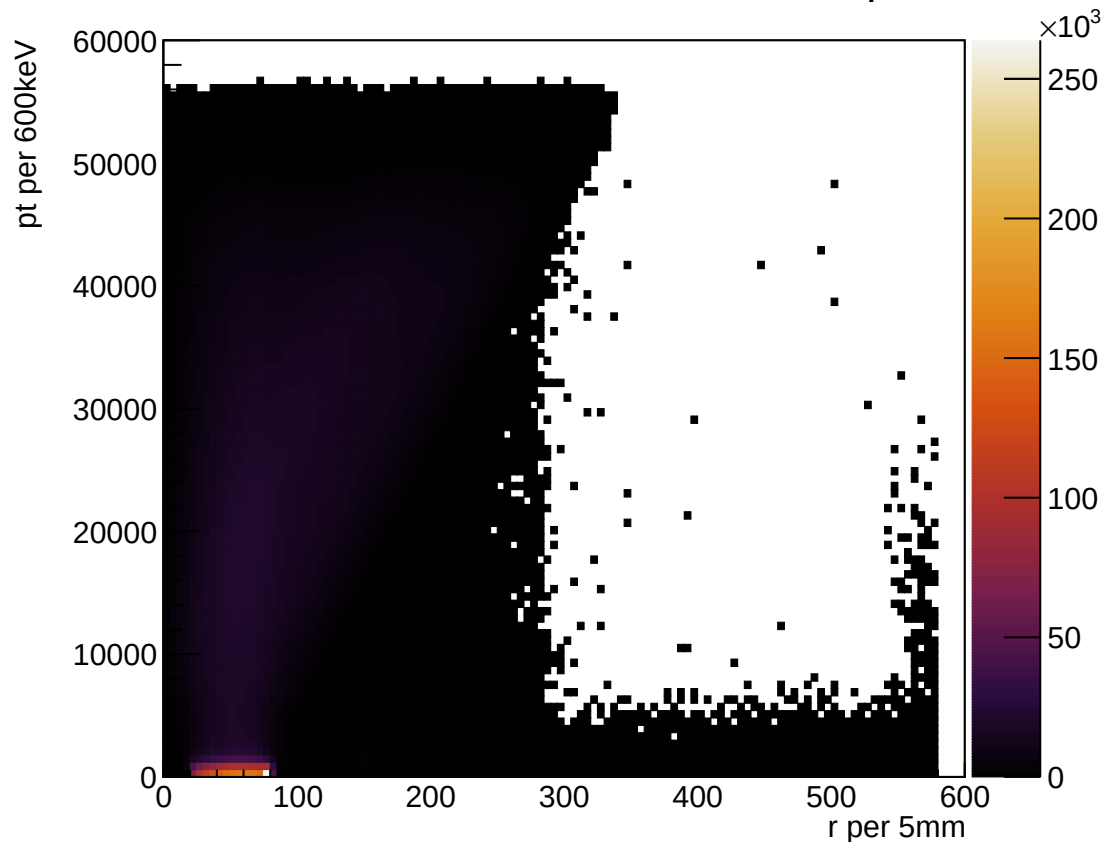
Entries	3.621453e+07		
Mean x	95.38		
Mean y	1.572e+04		
Std Dev x	56.3		
Std Dev y	1.092e+04		
Integral	3.369e+07		
0	0	0	
0	3.36875e+07	0	
0	0	0	

Mu2e mu resampler: electron pz vs r



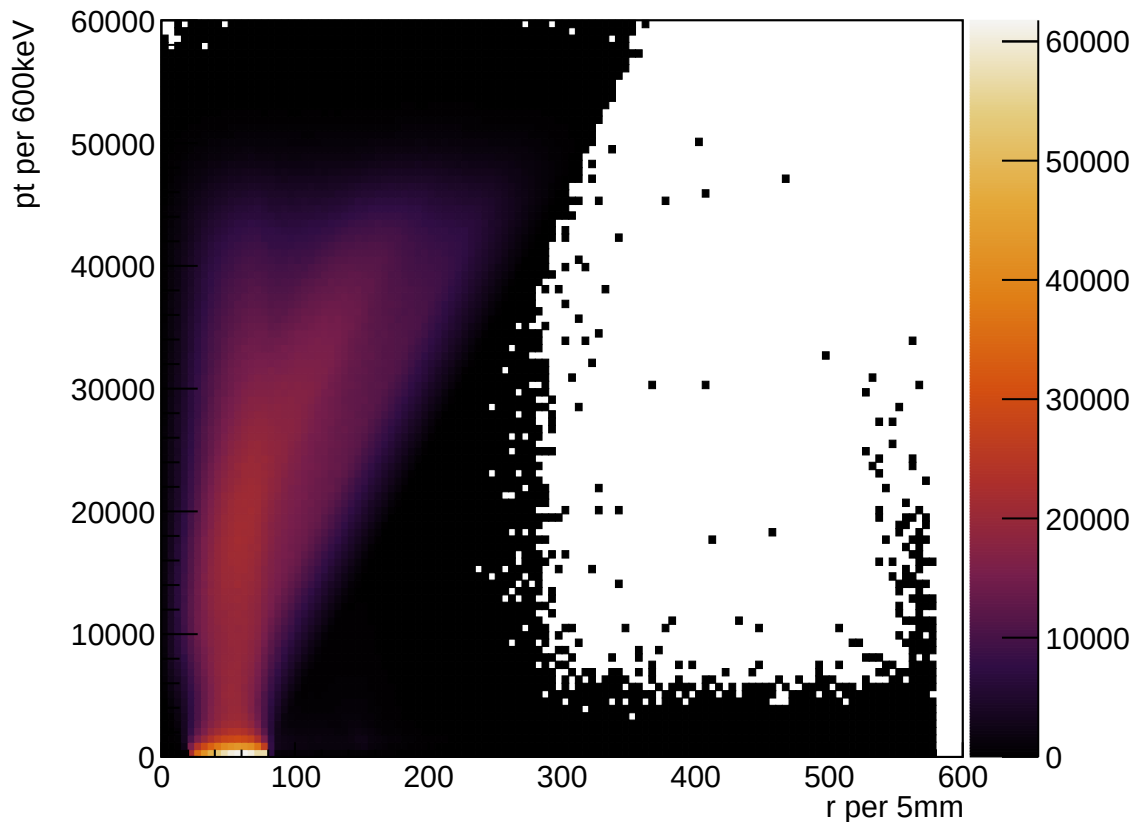
Entries	3.096011e+07	
Mean x	97.69	
Mean y	1.66e+04	
Std Dev x	56.91	
Std Dev y	1.059e+04	
Integral	3.124e+07	
0	125.126	0
0	3.12412e+07	0
0	0	0

Stock G4: electron pt vs r



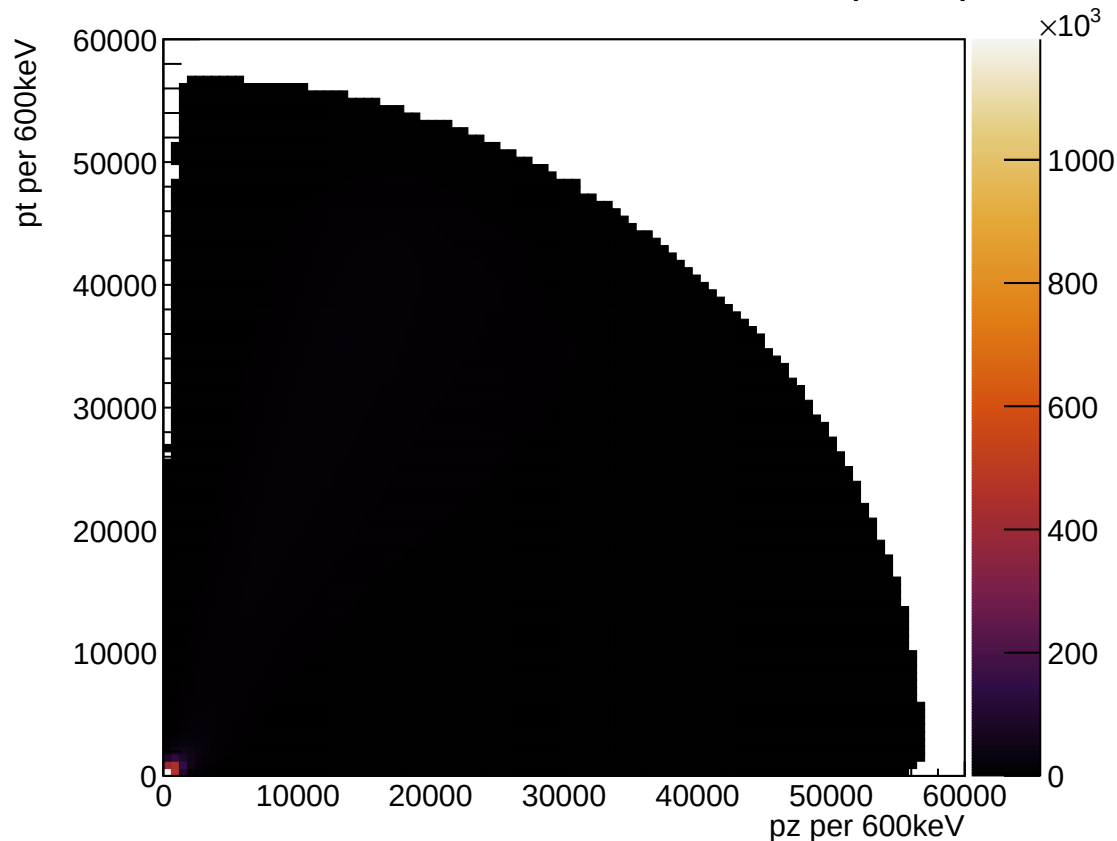
Entries	3.621453e+07		
Mean x	95.38		
Mean y	2.413e+04		
Std Dev x	56.3		
Std Dev y	1.393e+04		
Integral	3.369e+07		
0	0	0	
0	3.36875e+07	0	
0	0	0	

Mu2e mu resampler: electron pt vs r



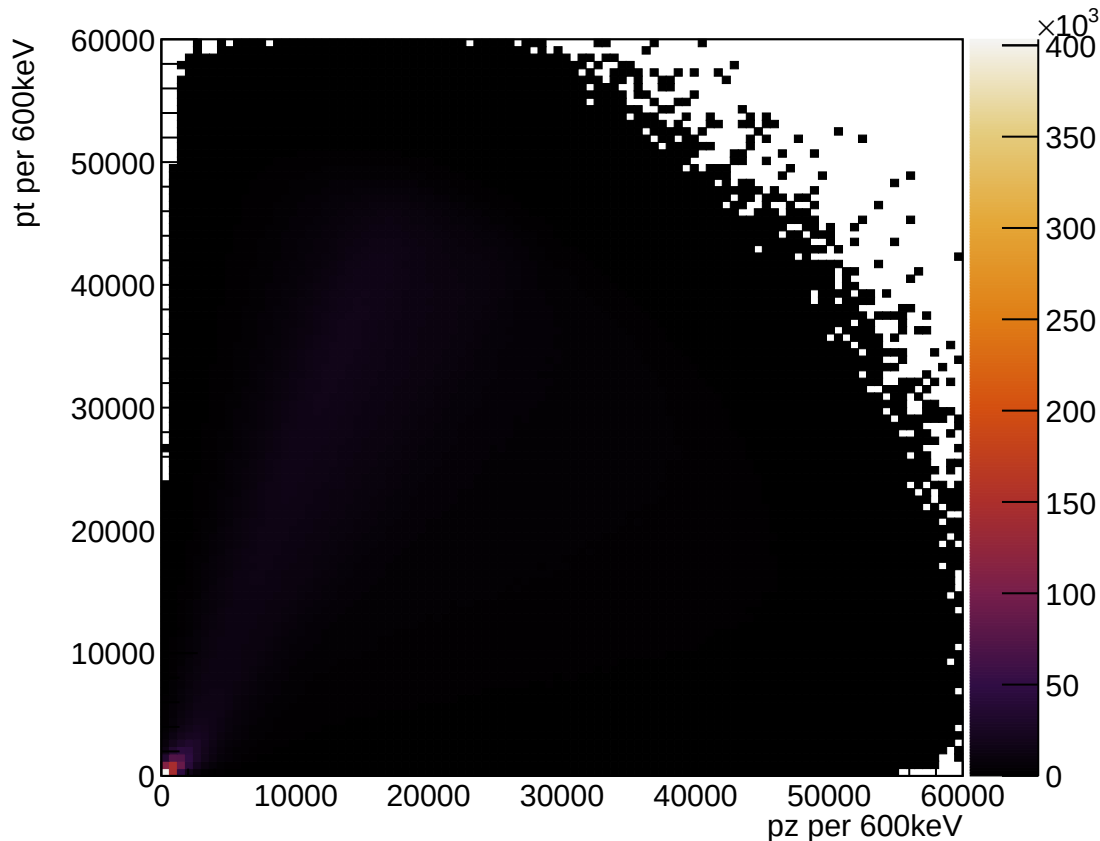
Entries	3.096011e+07	
Mean x	97.69	
Mean y	2.548e+04	
Std Dev x	56.91	
Std Dev y	1.308e+04	
Integral	3.124e+07	
0	517.659	0
0	3.12408e+07	0
0	0	0

Stock G4: electron pt vs pz



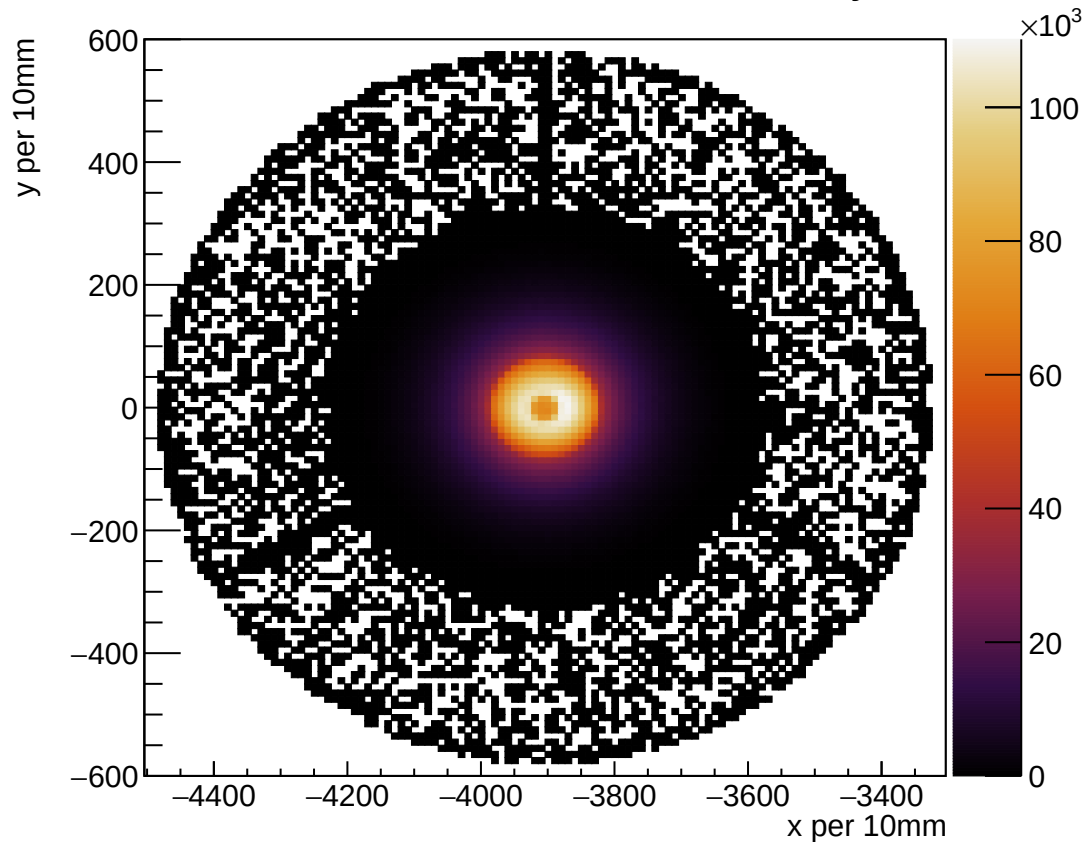
Entries	3.621453e+07		
Mean x	1.572e+04		
Mean y	2.413e+04		
Std Dev x	1.092e+04		
Std Dev y	1.393e+04		
Integral	3.369e+07		
0	0	0	
0	3.36875e+07	0	0
0	0	0	

Mu2e mu resampler: electron pt vs pz



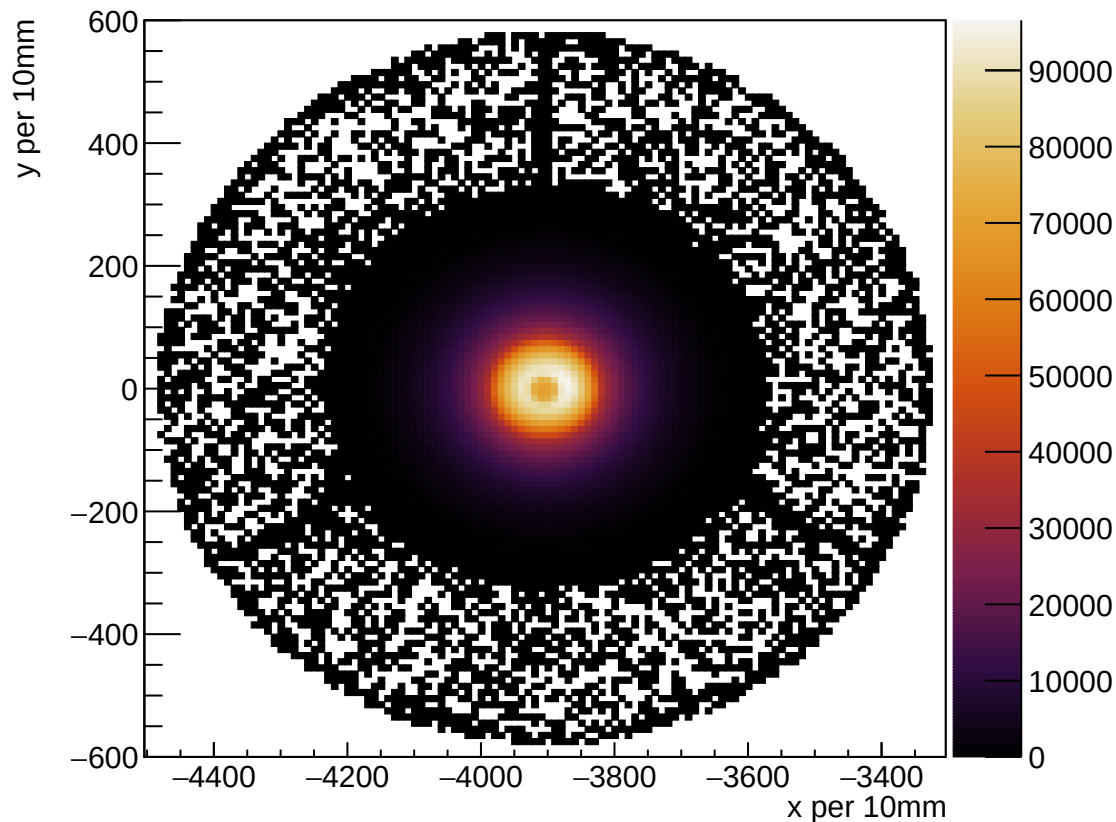
Entries	3.096011e+07	
Mean x	1.66e+04	
Mean y	2.548e+04	
Std Dev x	1.059e+04	
Std Dev y	1.308e+04	
Integral	3.124e+07	
0	517.659	0
0	3.12406e+07	125.126
0	0	0

Stock G4: electron y vs x



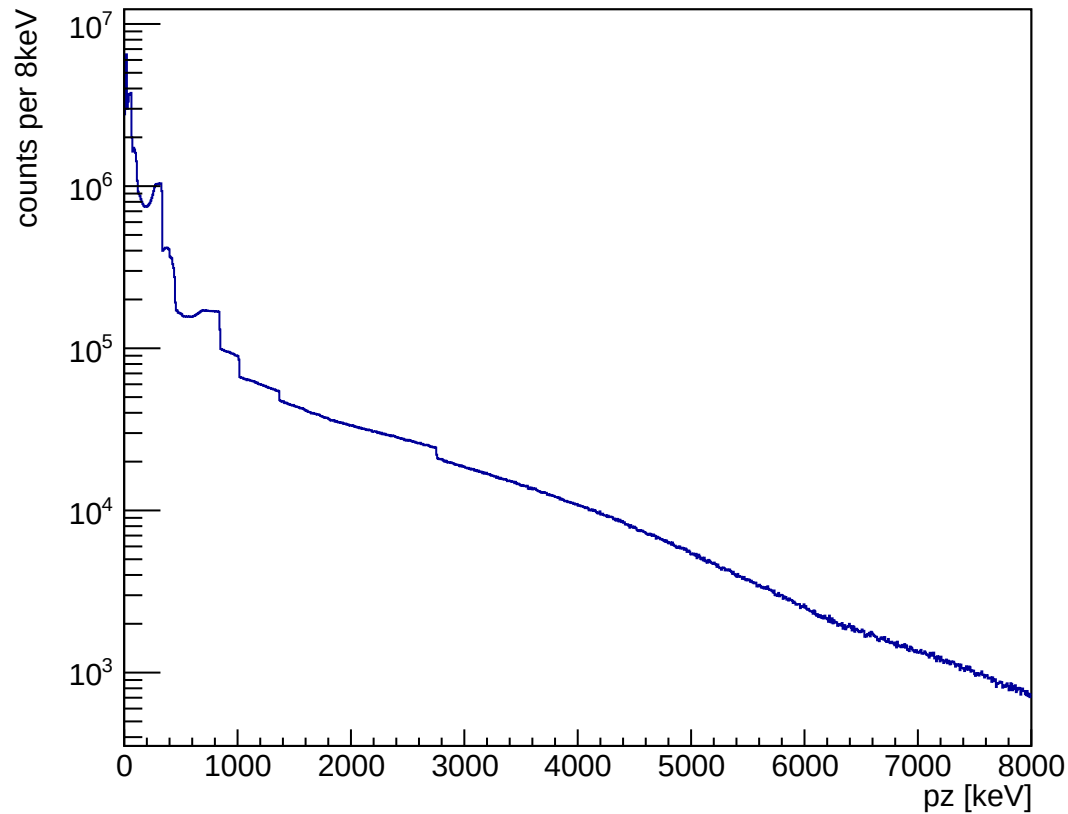
Entries	3.621453e+07		
Mean x	-3902		
Mean y	2.382		
Std Dev x	78.87		
Std Dev y	77.71		
Integral	3.369e+07		
0	0	0	
0	3.36875e+07	0	
0	0	0	

Mu2e mu resampler: electron y vs x



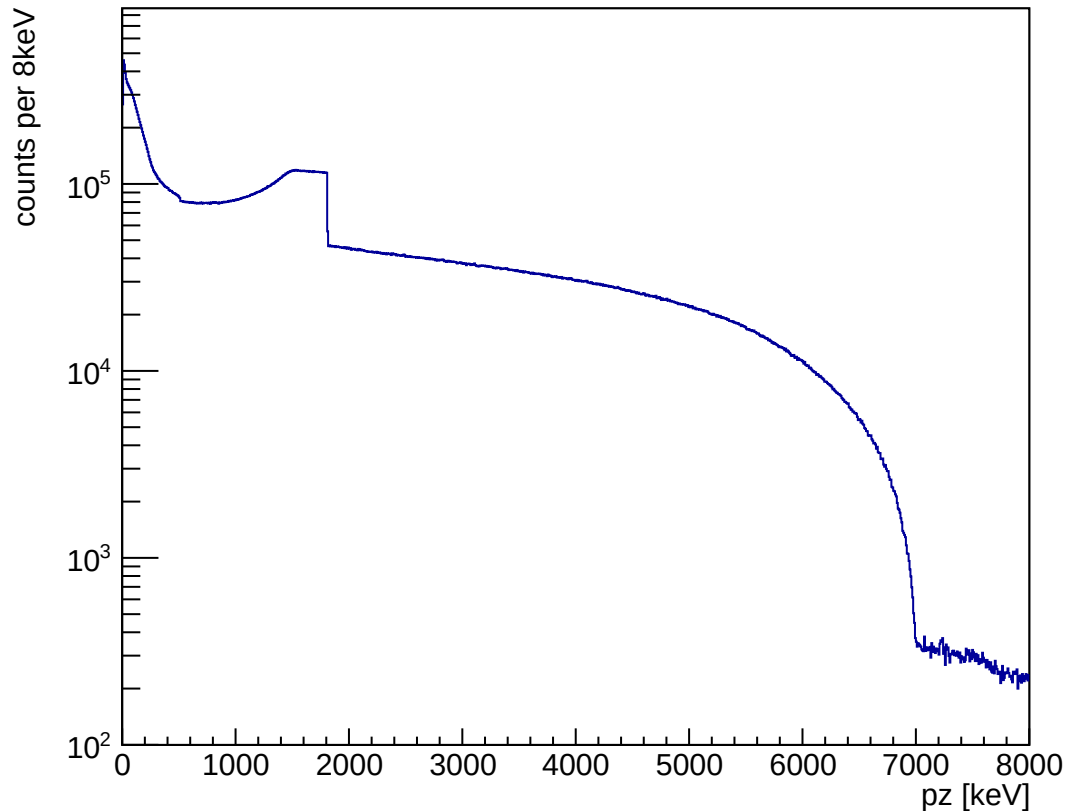
Entries	3.096011e+07		
Mean x	-3903		
Mean y	2.634		
Std Dev x	80.44		
Std Dev y	79.38		
Integral	3.124e+07		
0	0	0	
0	3.12413e+07	0	0
0	0	0	

Stock G4: photon momentum pz



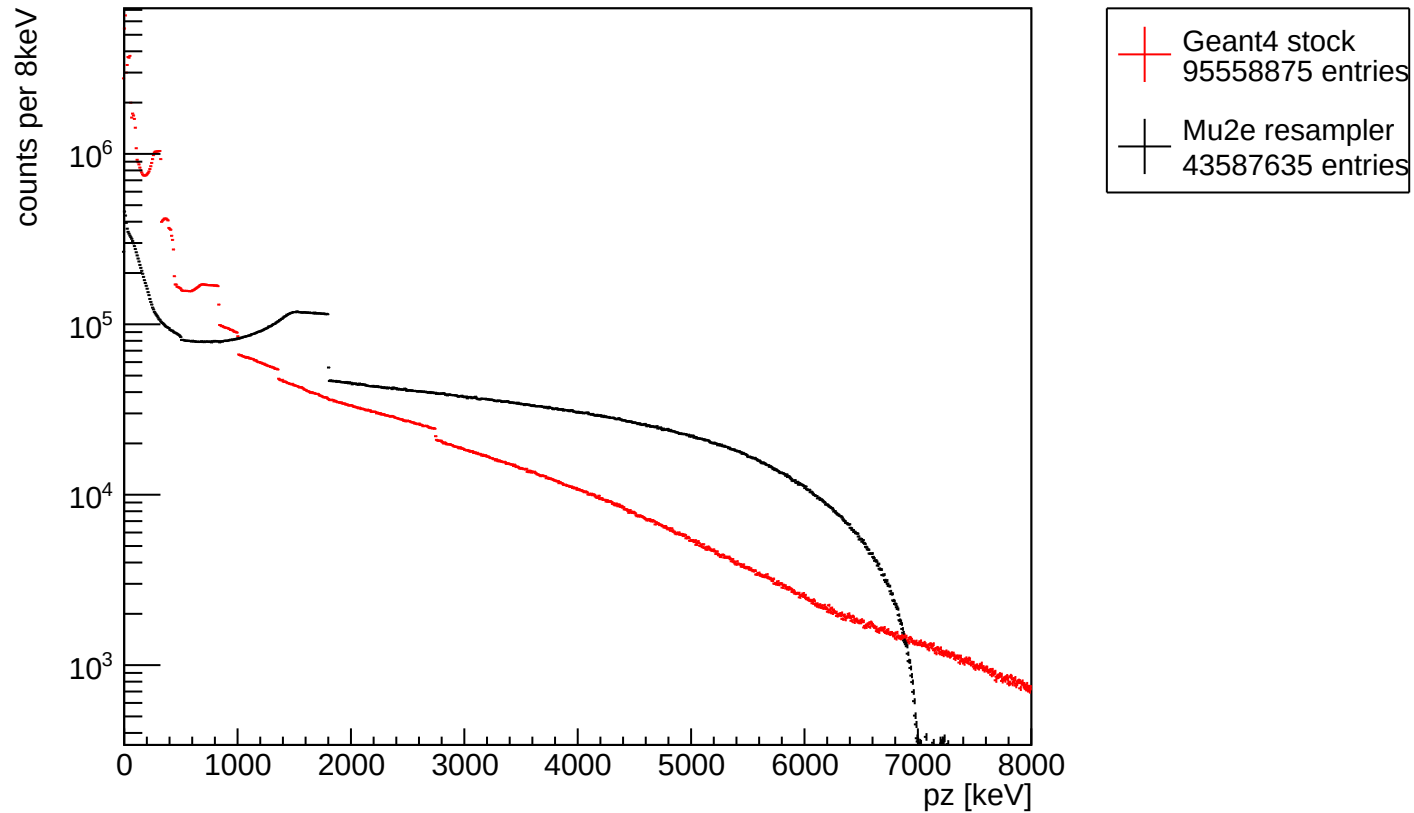
Entries	1.027272e+08
Mean	513.5
Std Dev	957.7
Underflow	0
Overflow	2.205e+05
Integral	9.534e+07

Mu2e mu resampler: photon momentum pz

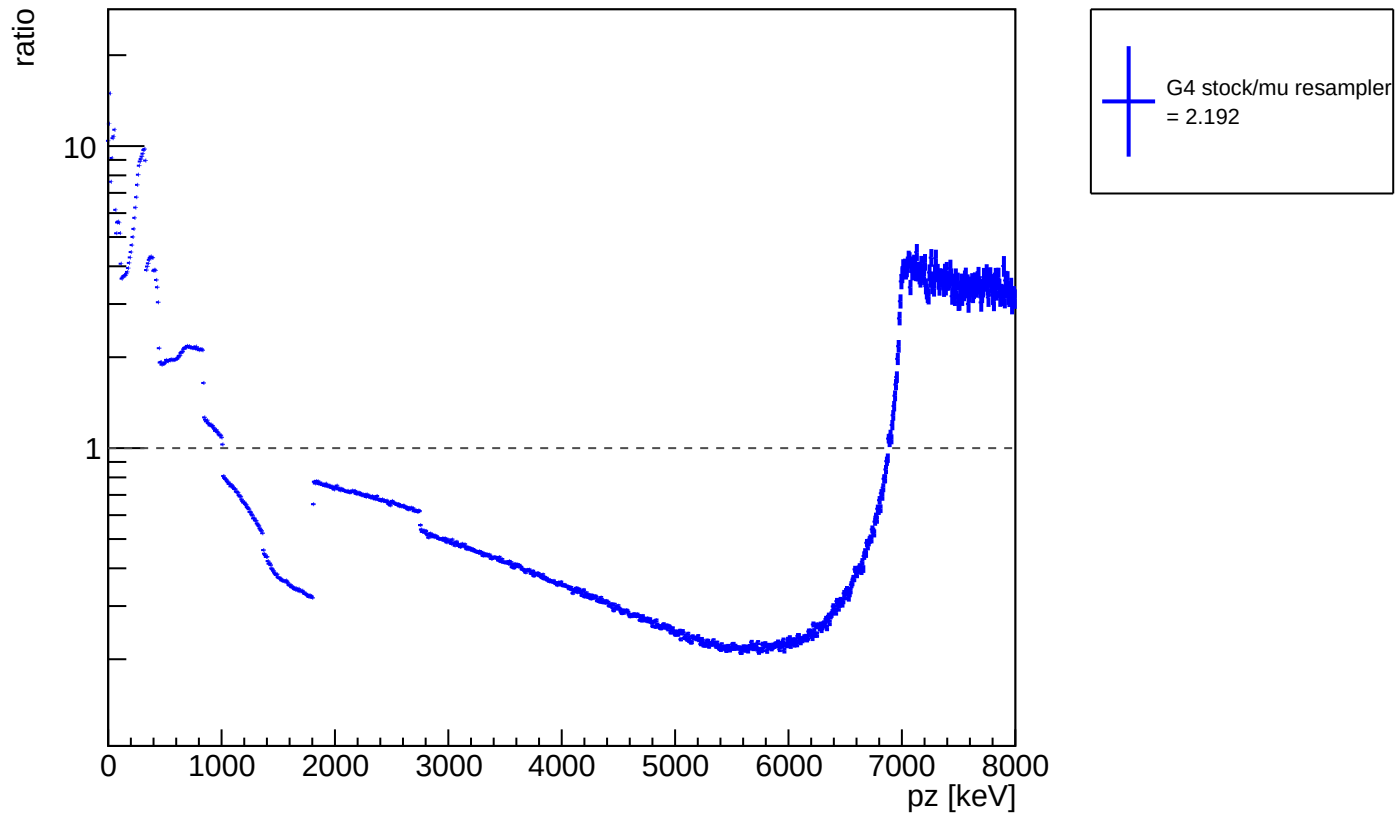


Entries	4.319534e+07
Mean	1879
Std Dev	1664
Underflow	0
Overflow	1.49e+05
Integral	4.344e+07

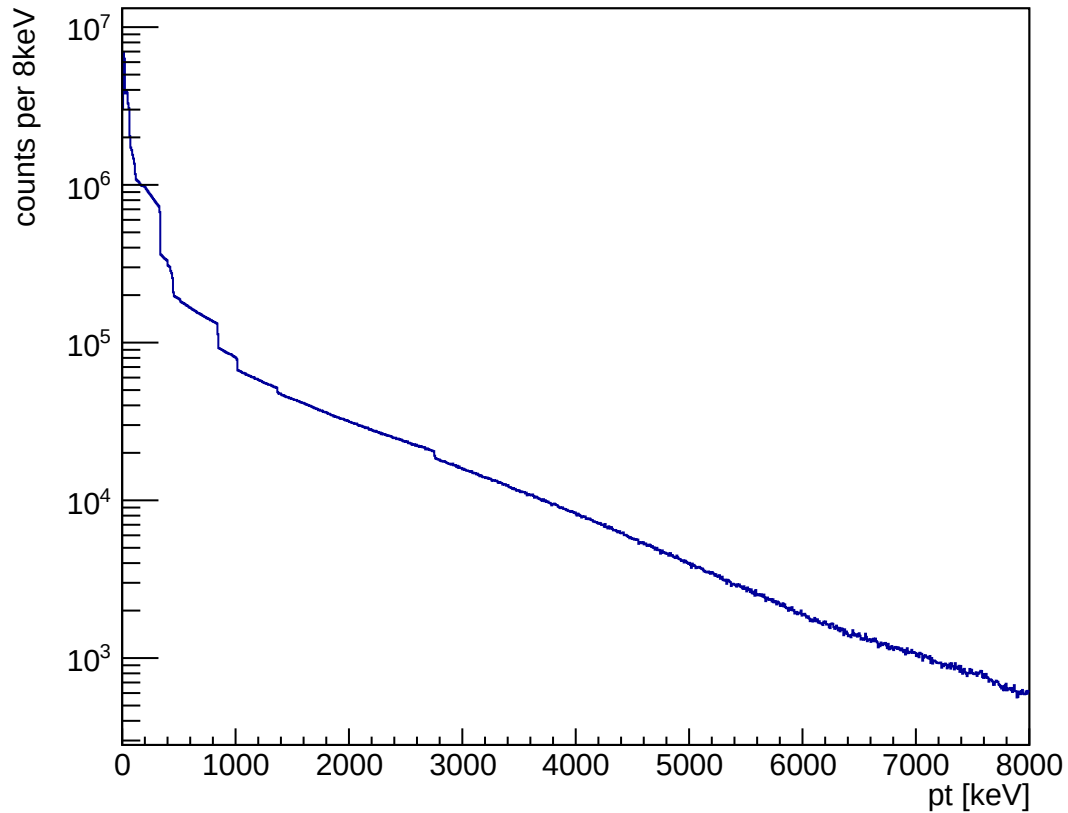
photon momentum pz



ratio of G4 stock / muon resampler: photon momentum pz

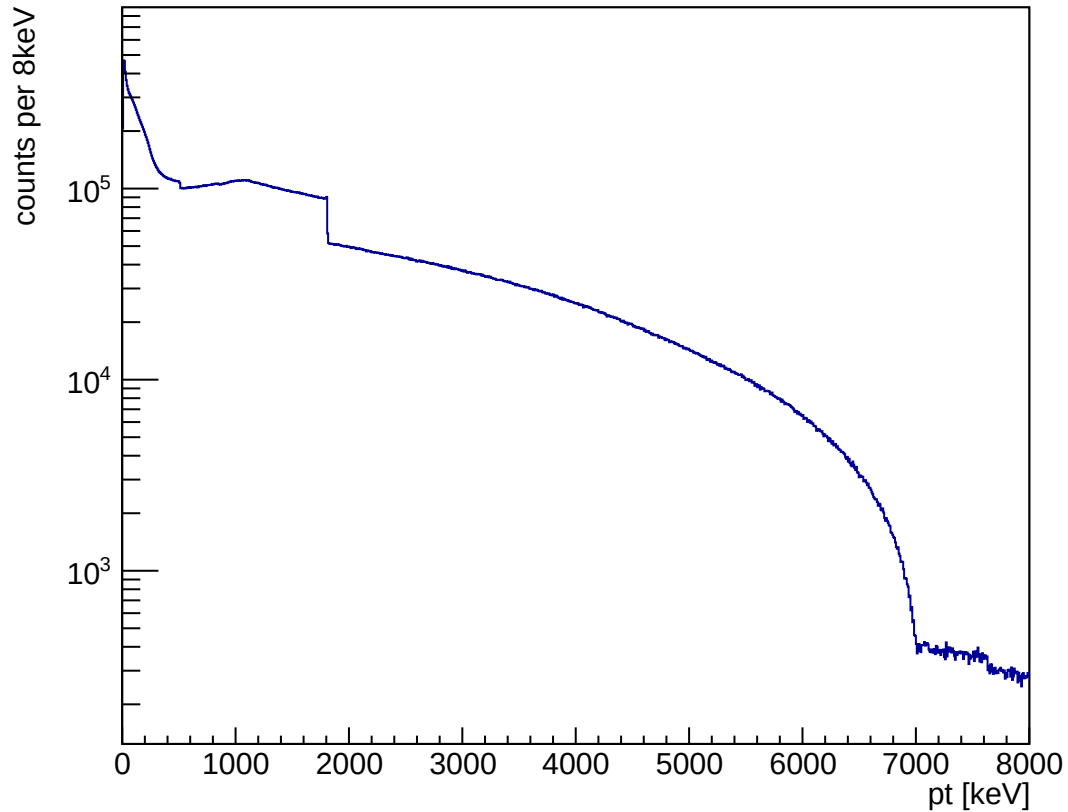


Stock G4: photon transverse momentum pT



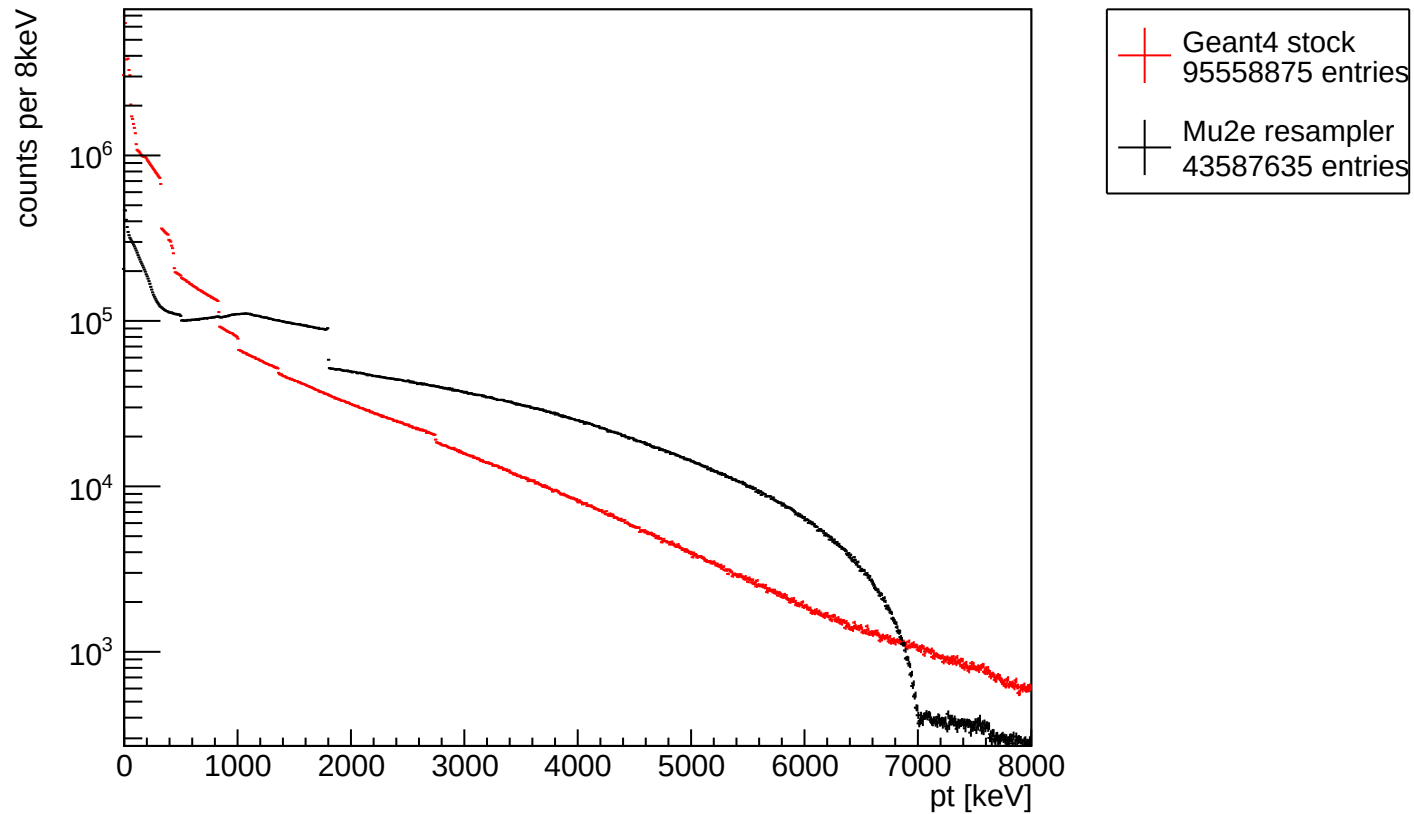
Entries	1.027272e+08
Mean	454.4
Std Dev	873.6
Underflow	0
Overflow	2.611e+05
Integral	9.53e+07

Mu2e mu resampler: photon transverse momentum pT

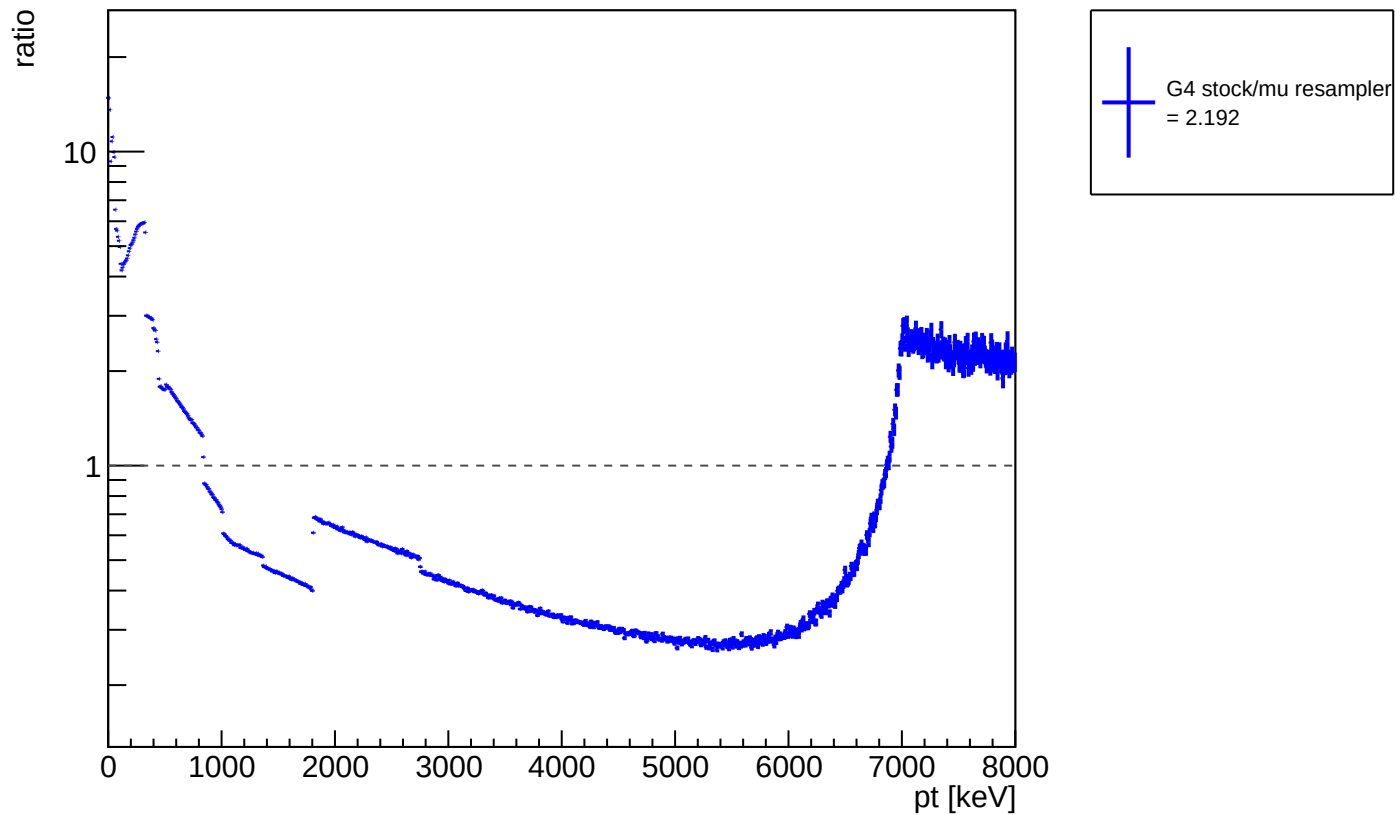


Entries	4.319534e+07
Mean	1641
Std Dev	1506
Underflow	0
Overflow	2.179e+05
Integral	4.337e+07

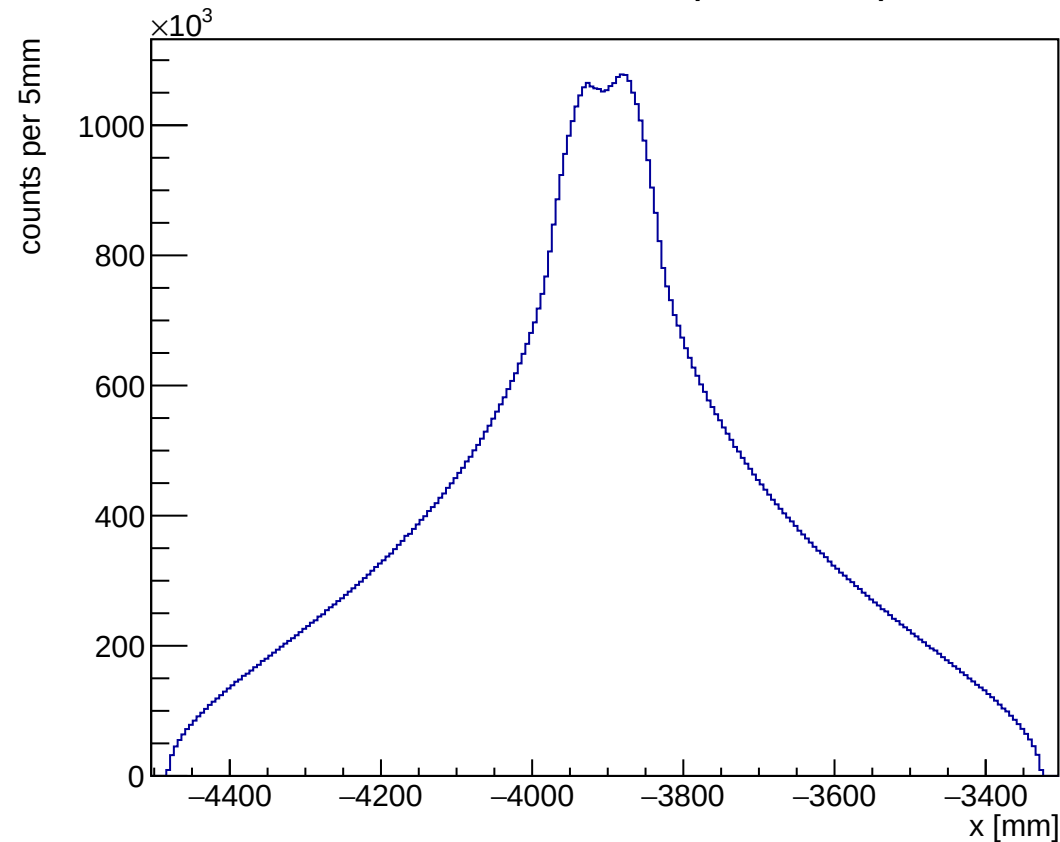
photon transverse momentum pT



ratio of G4 stock / muon resampler: photon transverse momentum pT

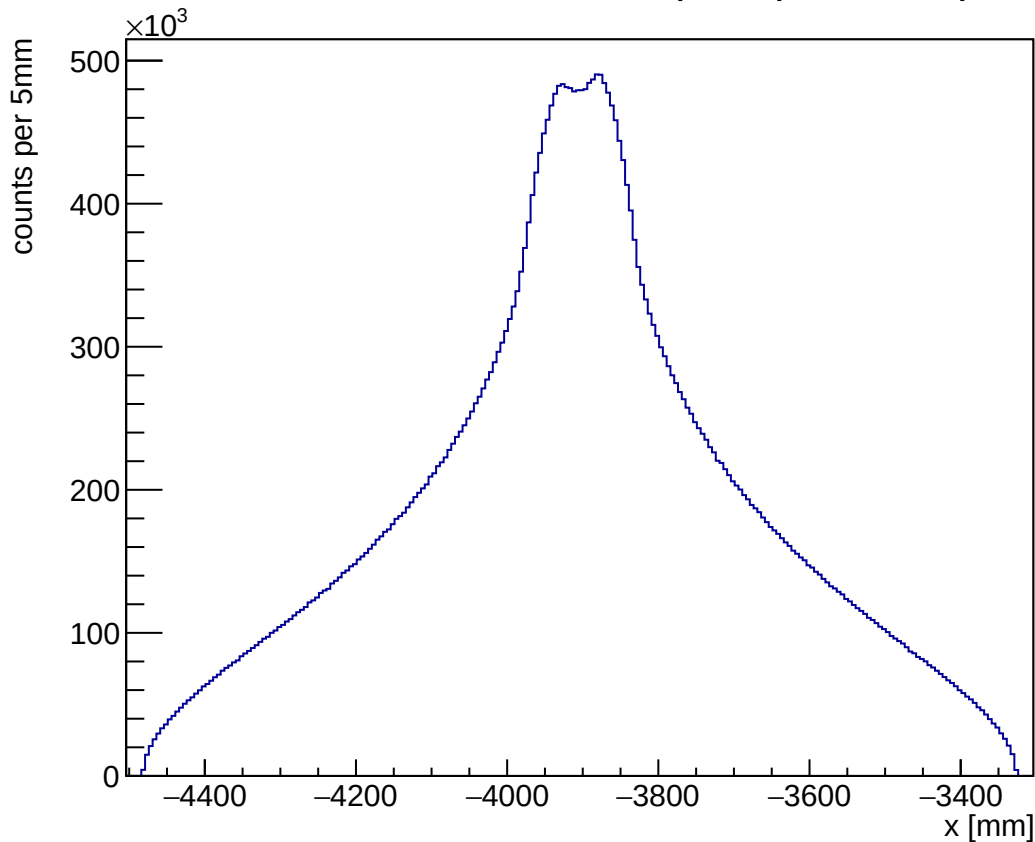


Stock G4: photon x position



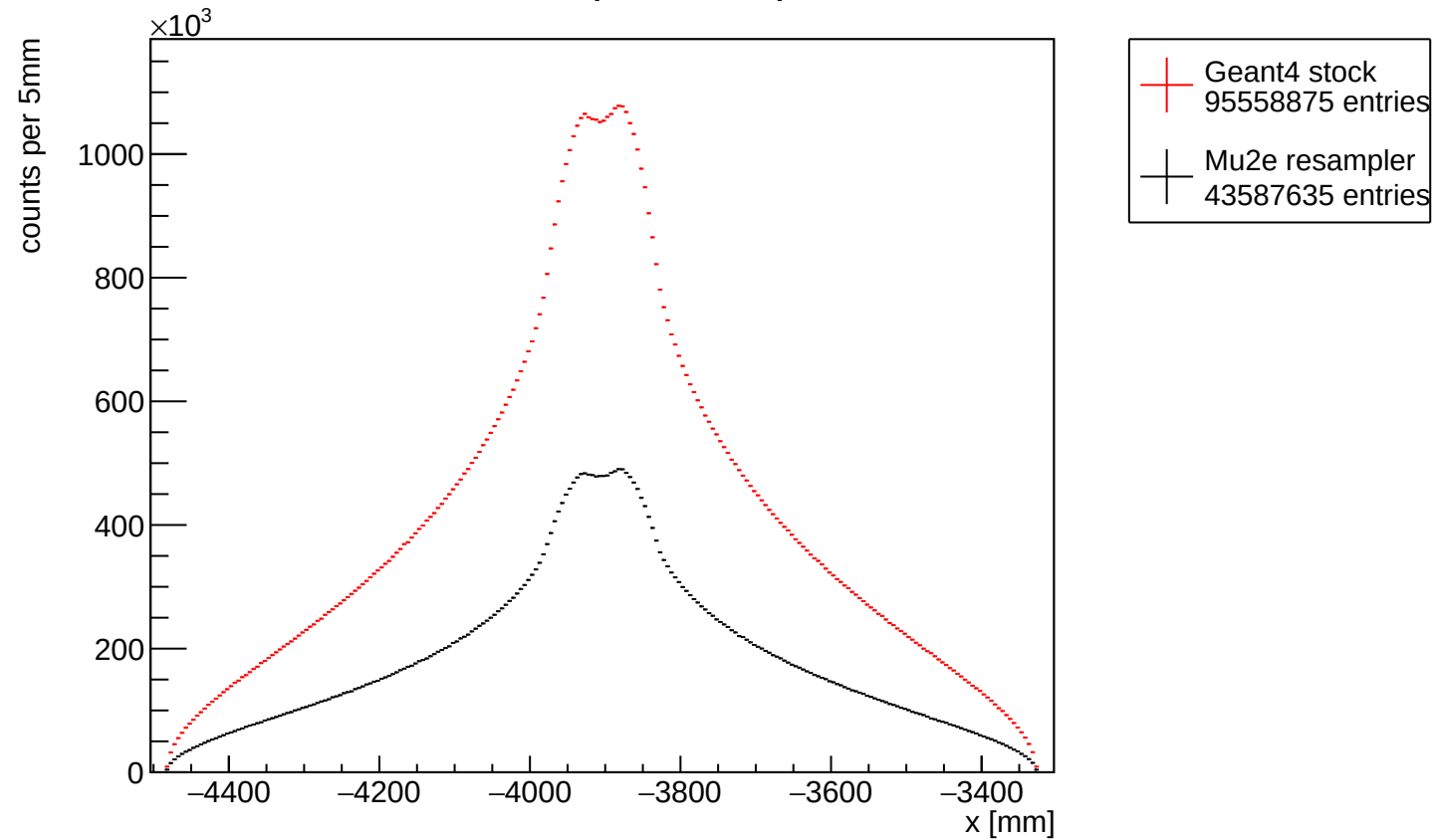
Entries	1.027272e+08
Mean	-3903
Std Dev	230.3
Underflow	0
Overflow	0
Integral	9.556e+07

Mu2e mu resampler: photon x position

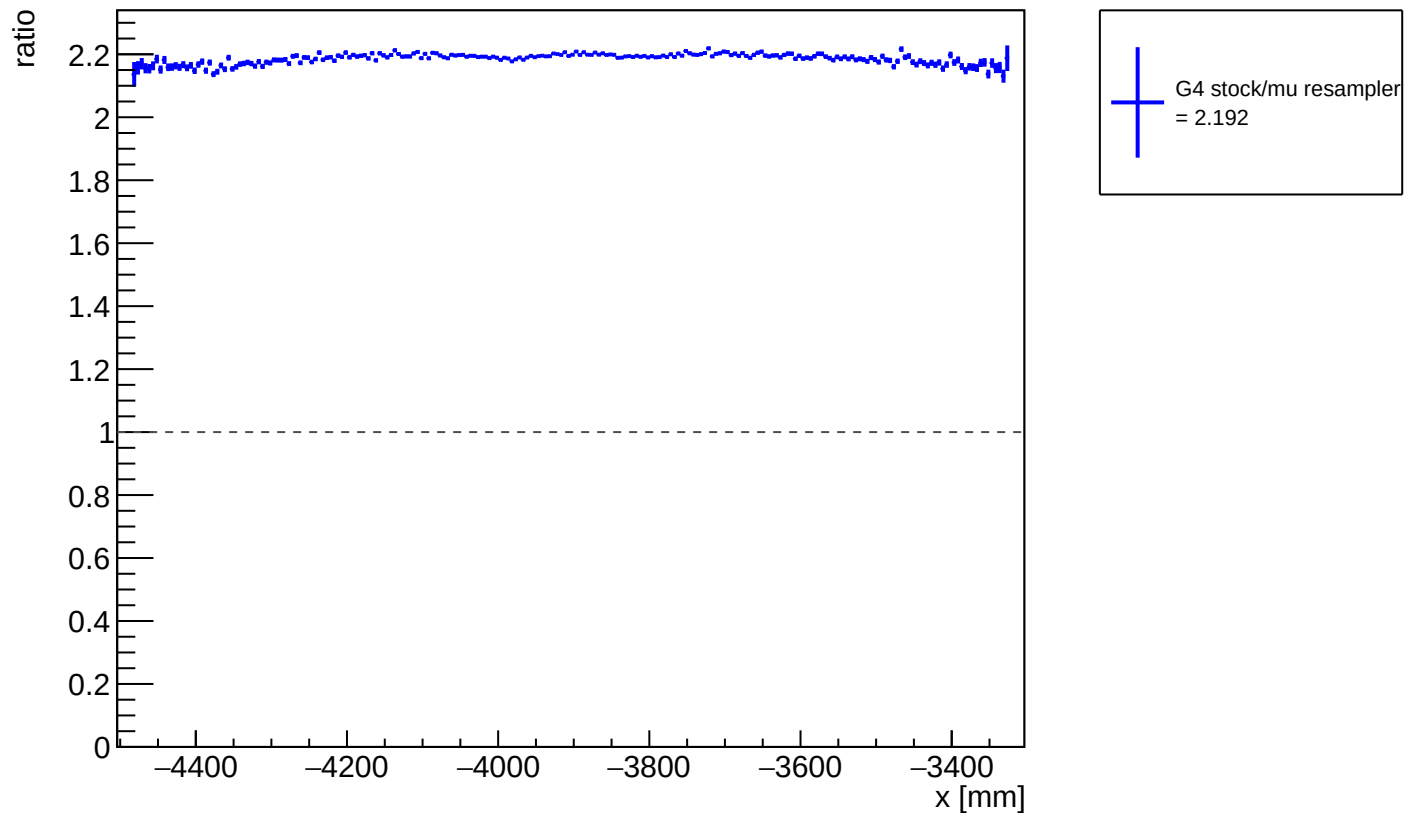


Entries	4.319534e+07
Mean	-3903
Std Dev	230.9
Underflow	0
Overflow	0
Integral	4.359e+07

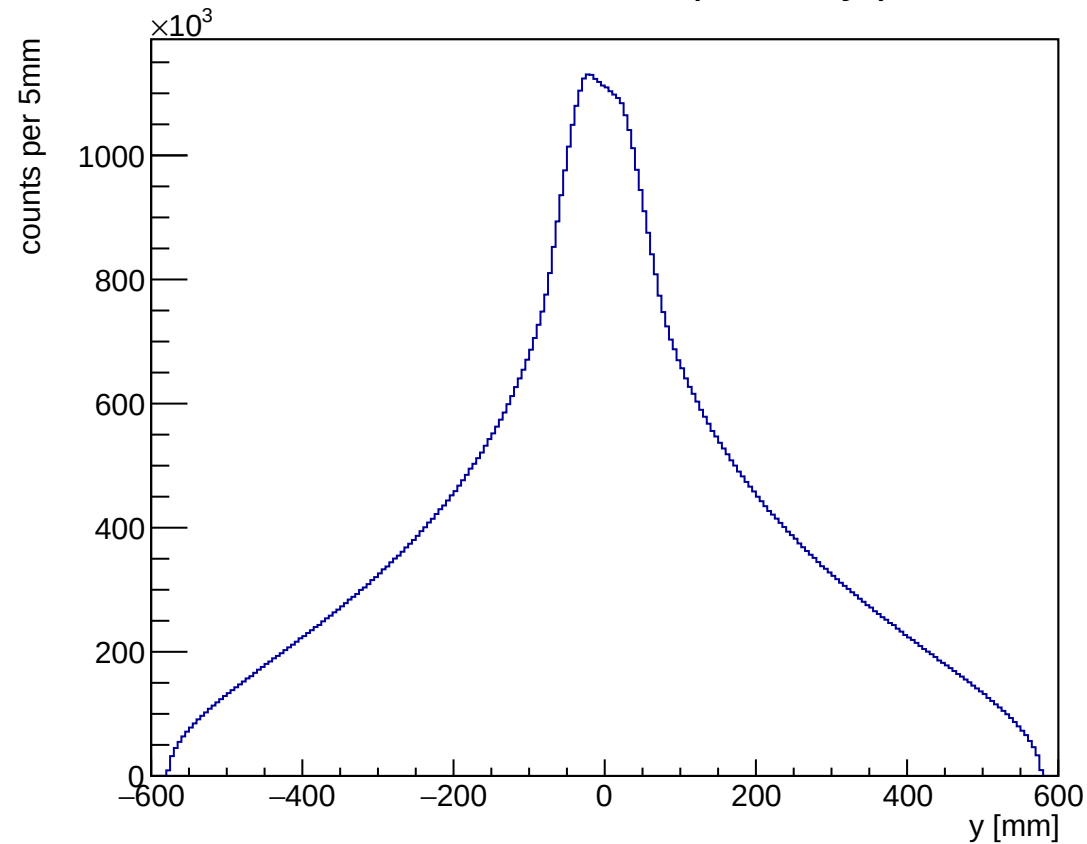
photon x position



ratio of G4 stock / muon resampler: photon x position

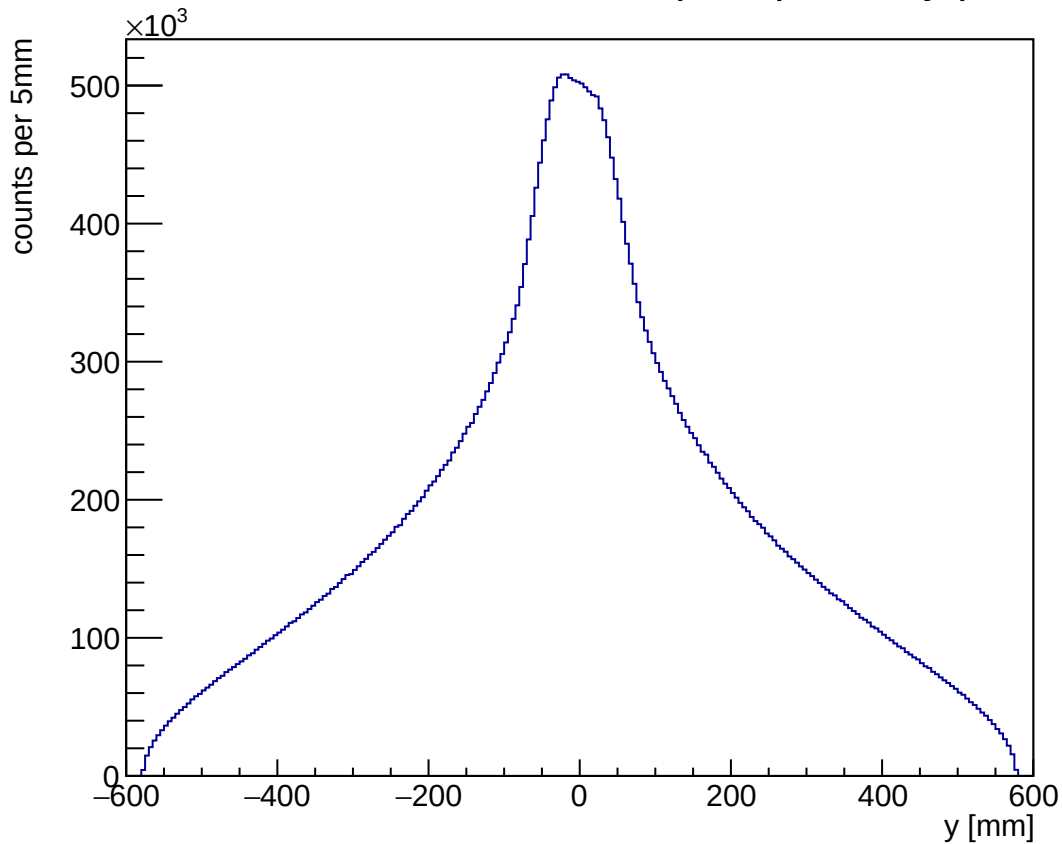


Stock G4: photon y position



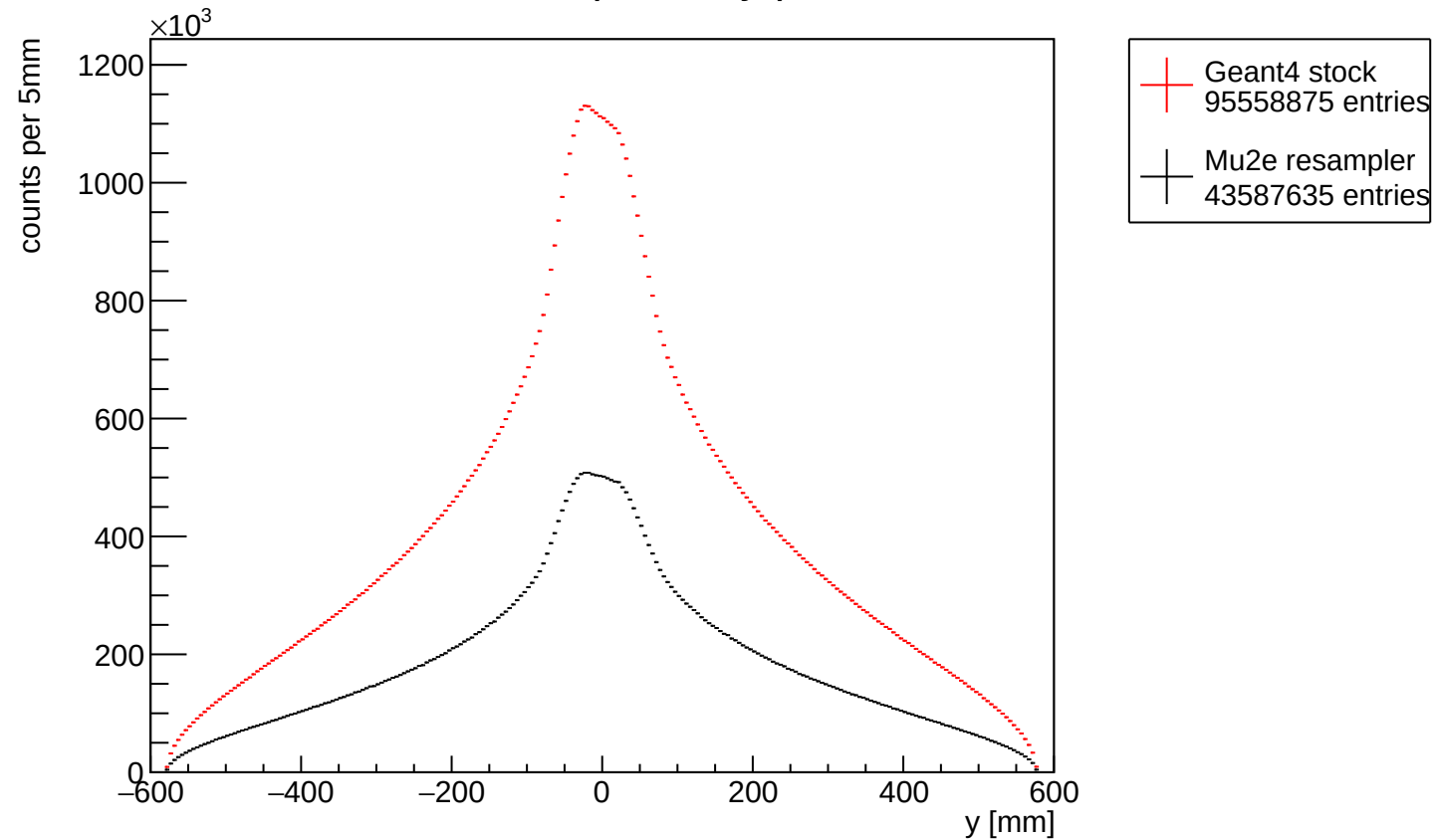
Entries	1.027272e+08
Mean	-0.005513
Std Dev	230.1
Underflow	0
Overflow	0
Integral	9.556e+07

Mu2e mu resampler: photon y position

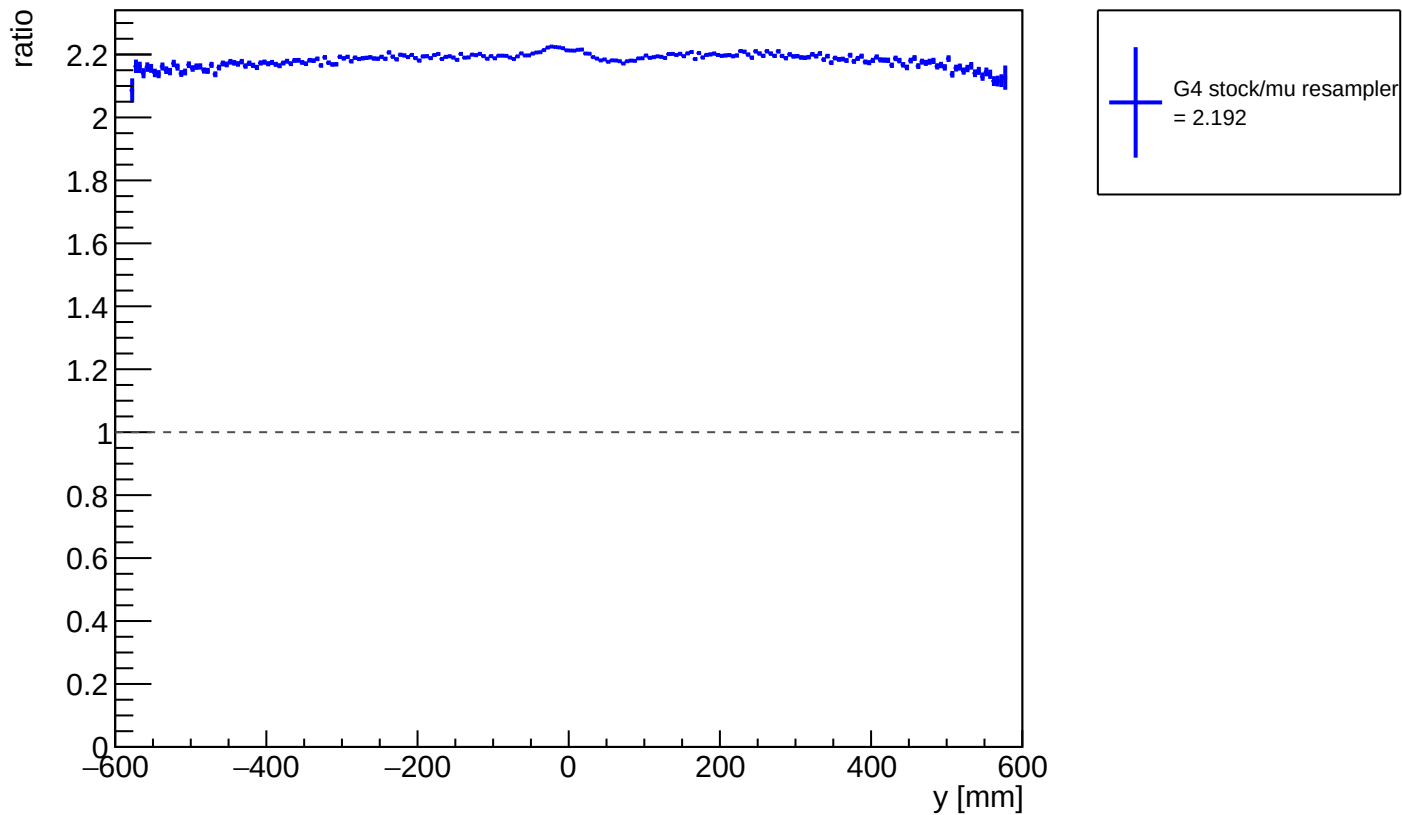


Entries	4.319534e+07
Mean	-0.2385
Std Dev	230.9
Underflow	0
Overflow	0
Integral	4.359e+07

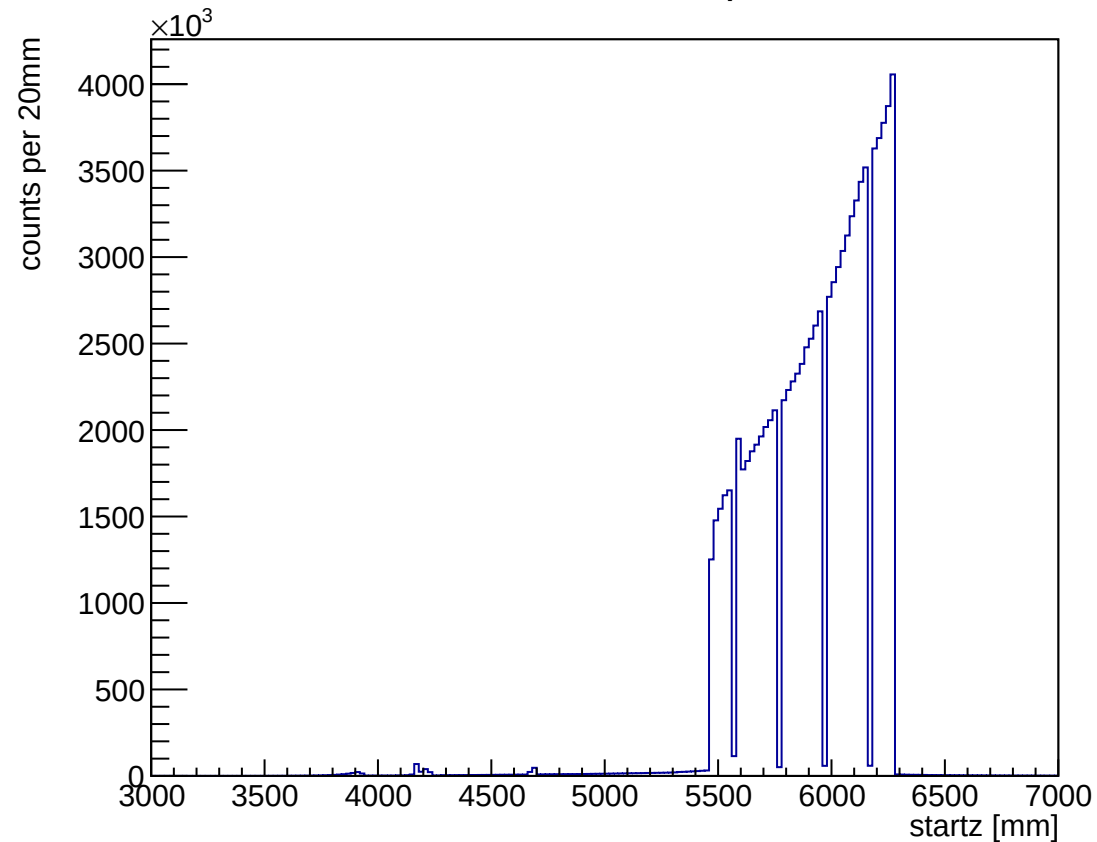
photon y position



ratio of G4 stock / muon resampler: photon y position

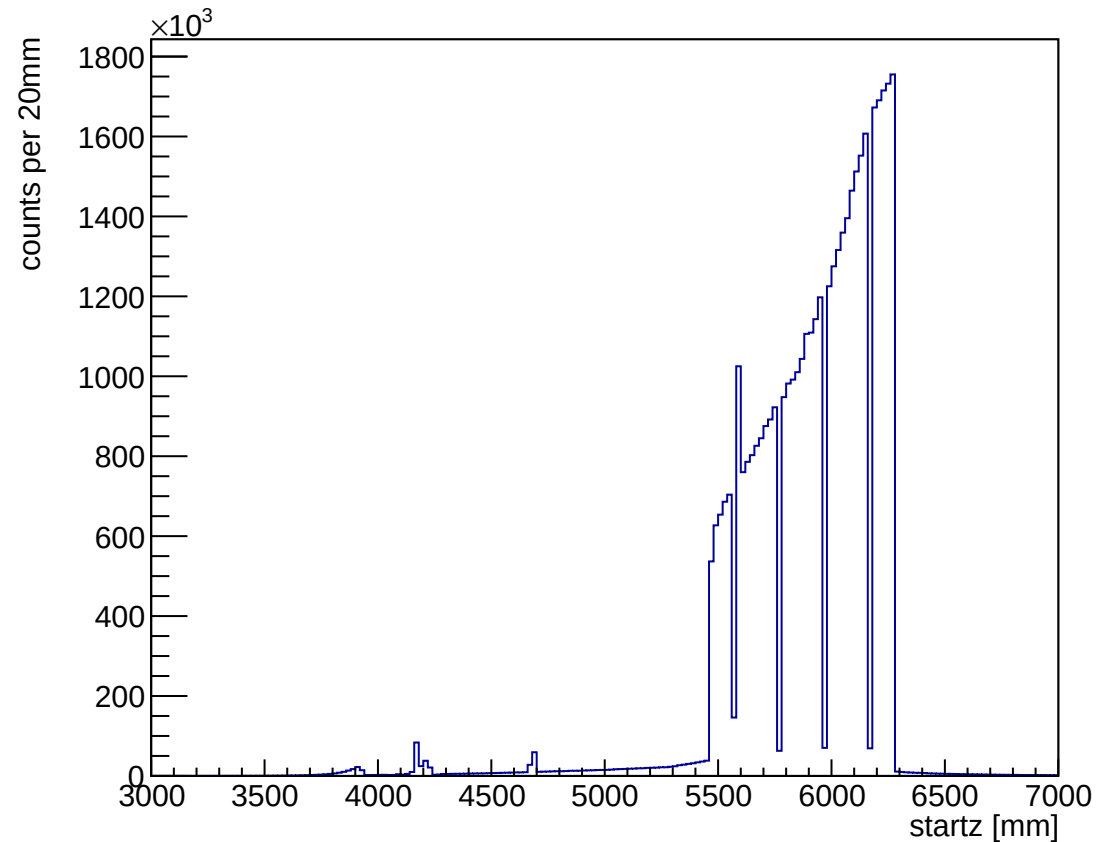


Stock G4: photon start z



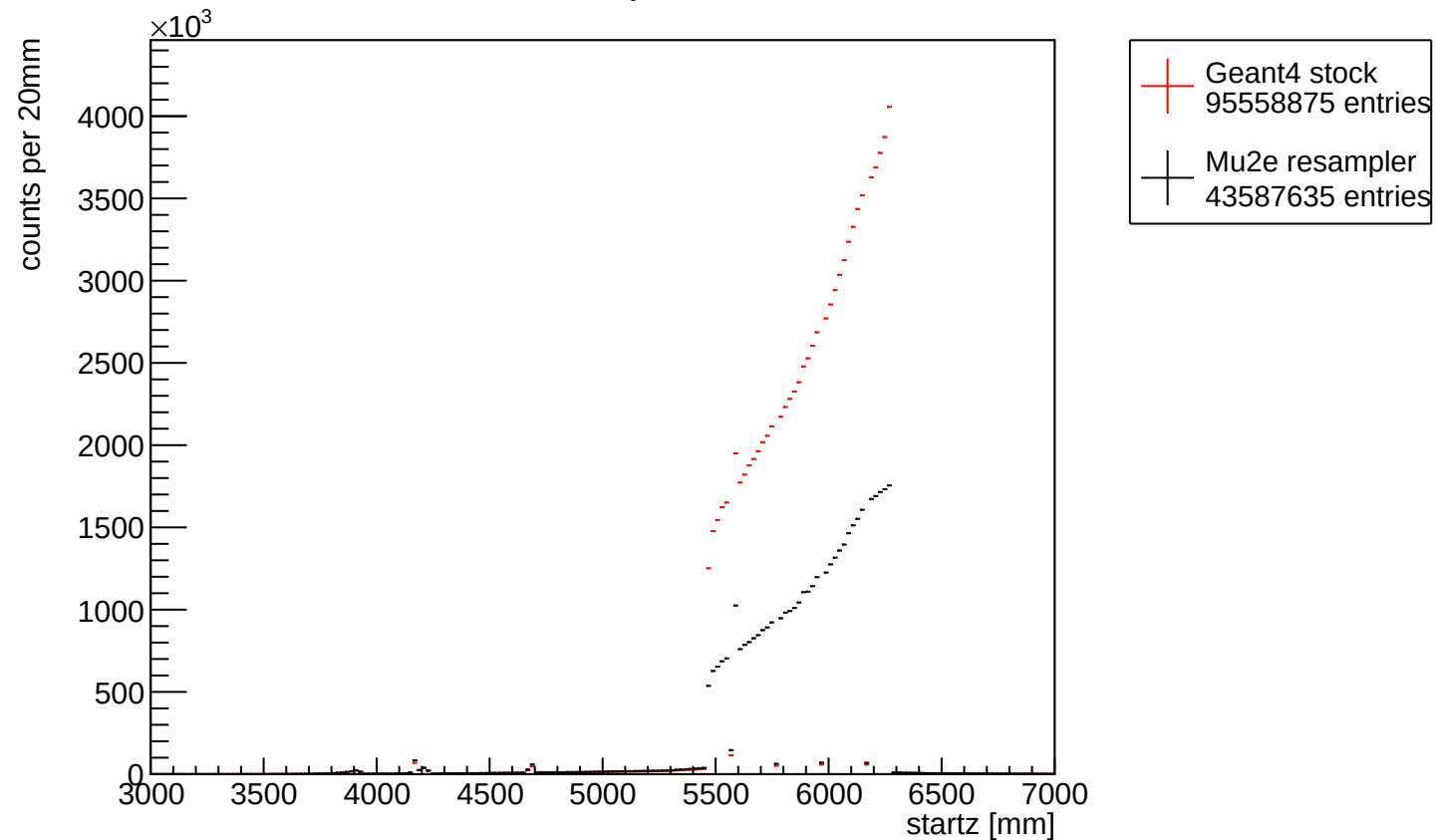
Entries	1.027272e+08
Mean	5927
Std Dev	270
Underflow	961.8
Overflow	4.609e+04
Integral	9.551e+07

Mu2e mu resampler: photon start z

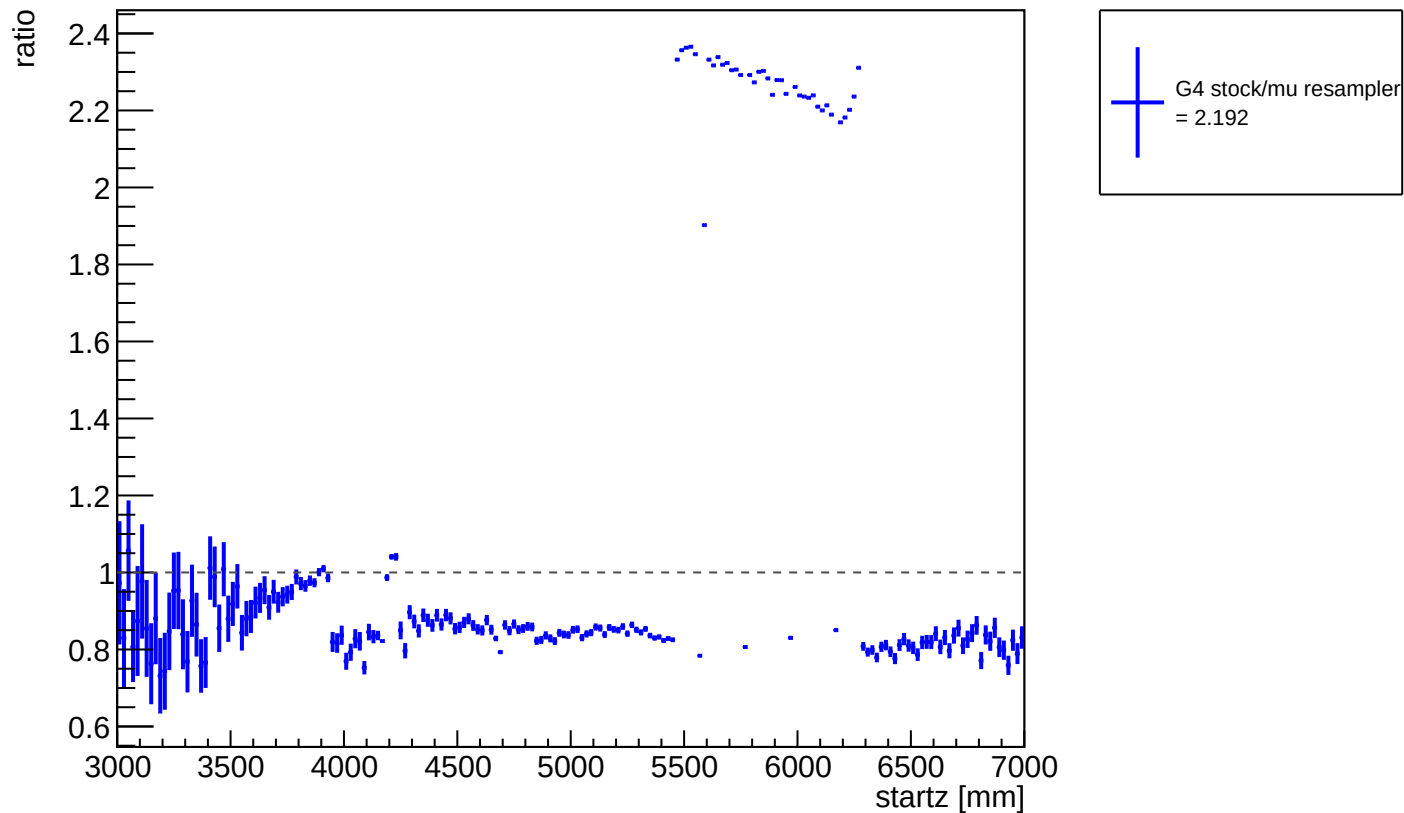


Entries	4.319534e+07
Mean	5910
Std Dev	317.2
Underflow	1036
Overflow	5.311e+04
Integral	4.353e+07

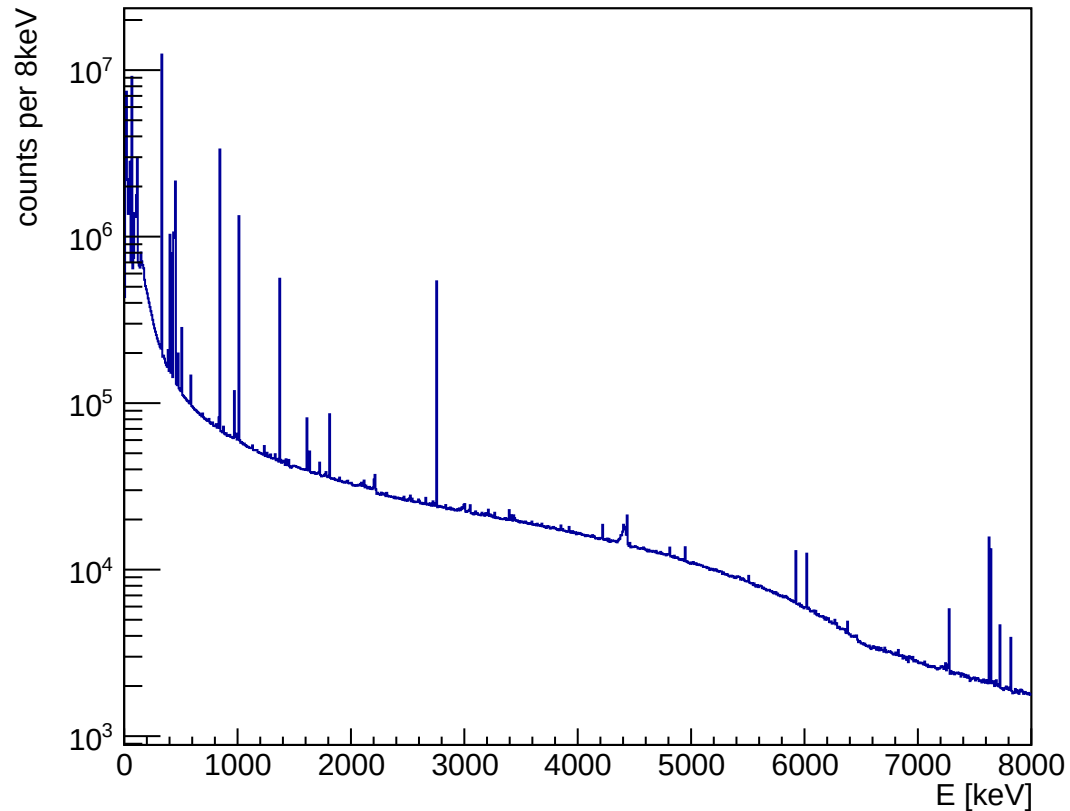
photon start z



ratio of G4 stock / muon resampler: photon start z

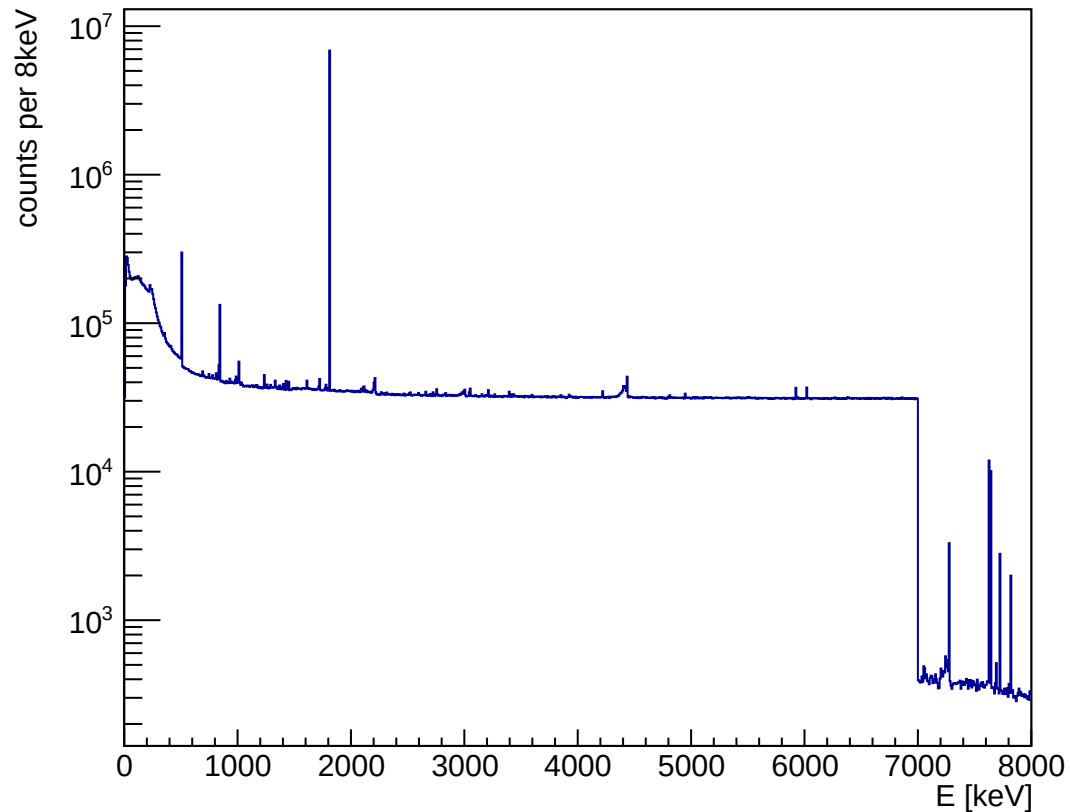


Stock G4: photon kin. energy



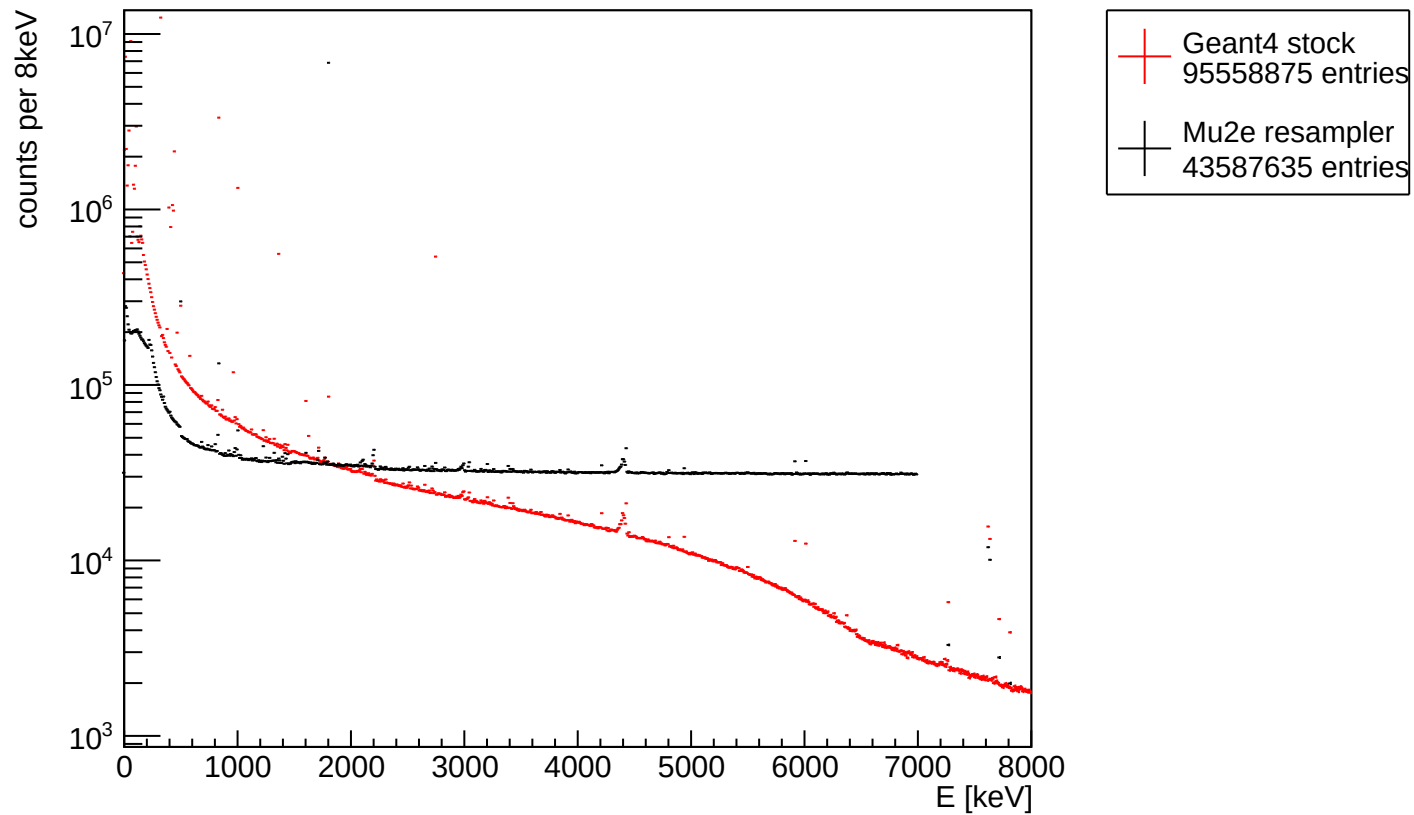
Entries	1.027272e+08
Mean	697.2
Std Dev	1201
Underflow	0
Overflow	5.845e+05
Integral	9.497e+07

Mu2e mu resampler: photon kin. energy

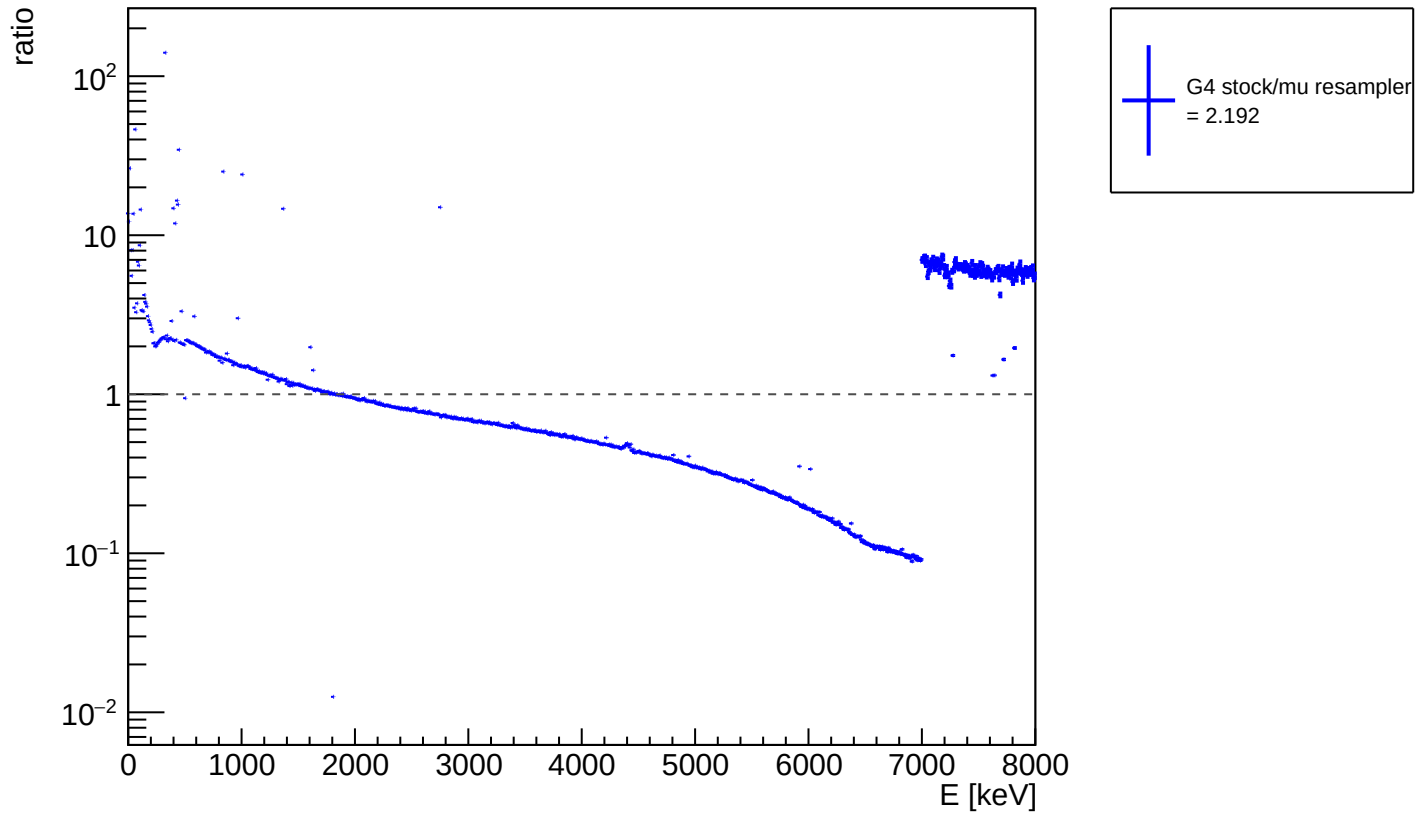


Entries	4.319534e+07
Mean	2614
Std Dev	2067
Underflow	0
Overflow	3.168e+05
Integral	4.327e+07

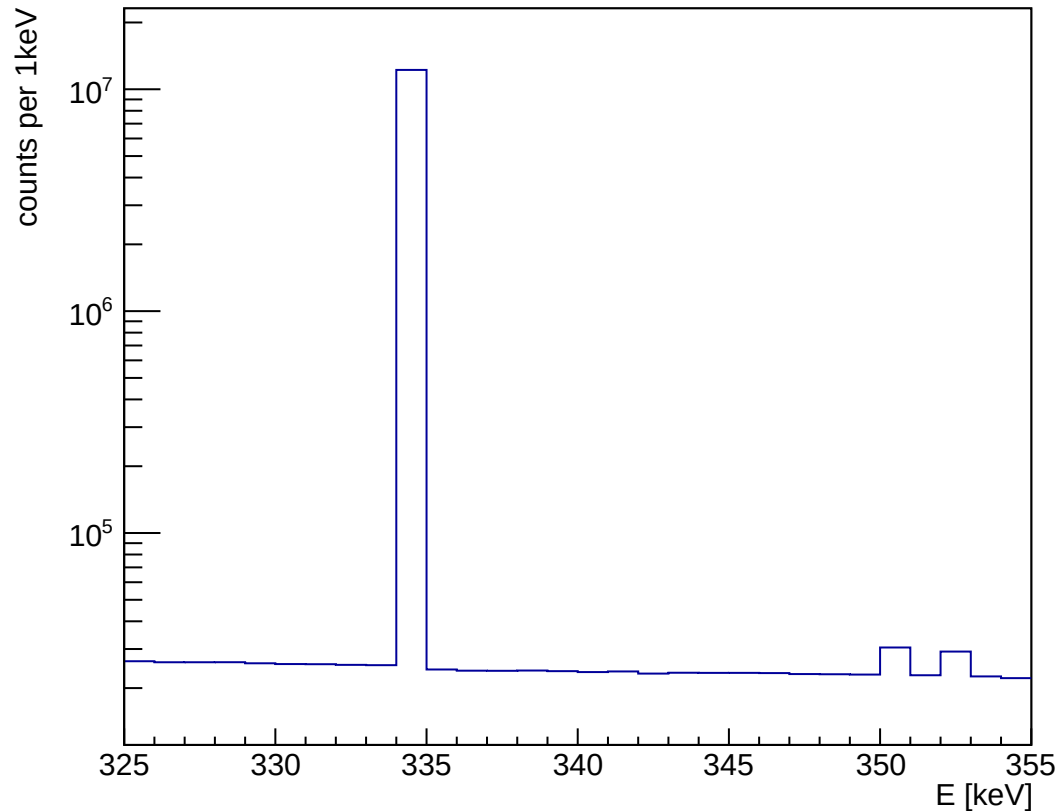
photon kin. energy



ratio of G4 stock / muon resampler: photon kin. energy



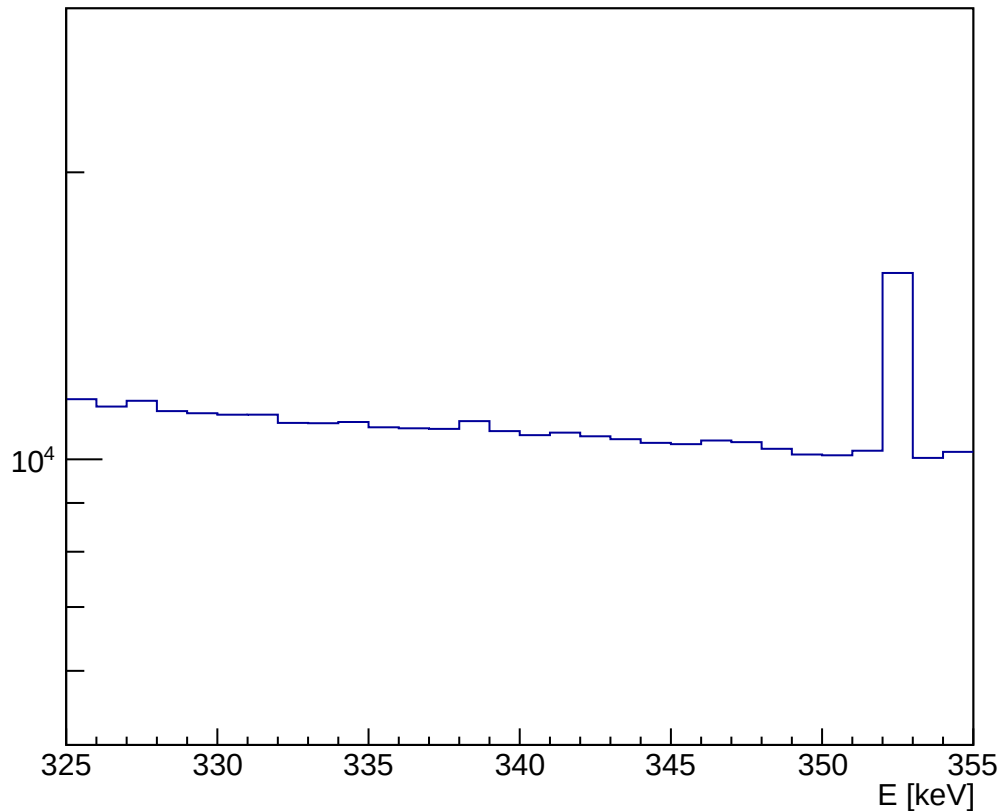
Stock G4: photon E (325-355 keV)



Entries	1.027272e+08
Mean	335
Std Dev	2.406
Underflow	4.81e+07
Overflow	3.452e+07
Integral	1.294e+07

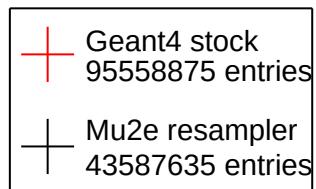
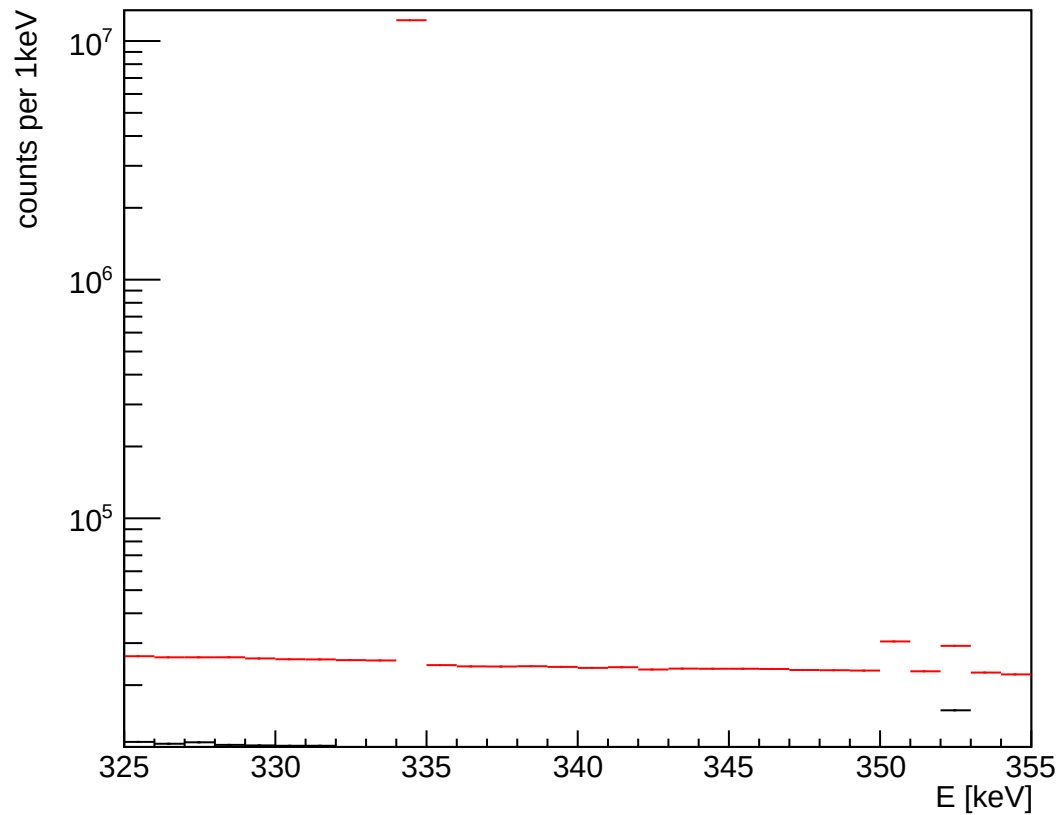
Mu2e mu resampler: photon E (325-355 keV)

counts per 1keV

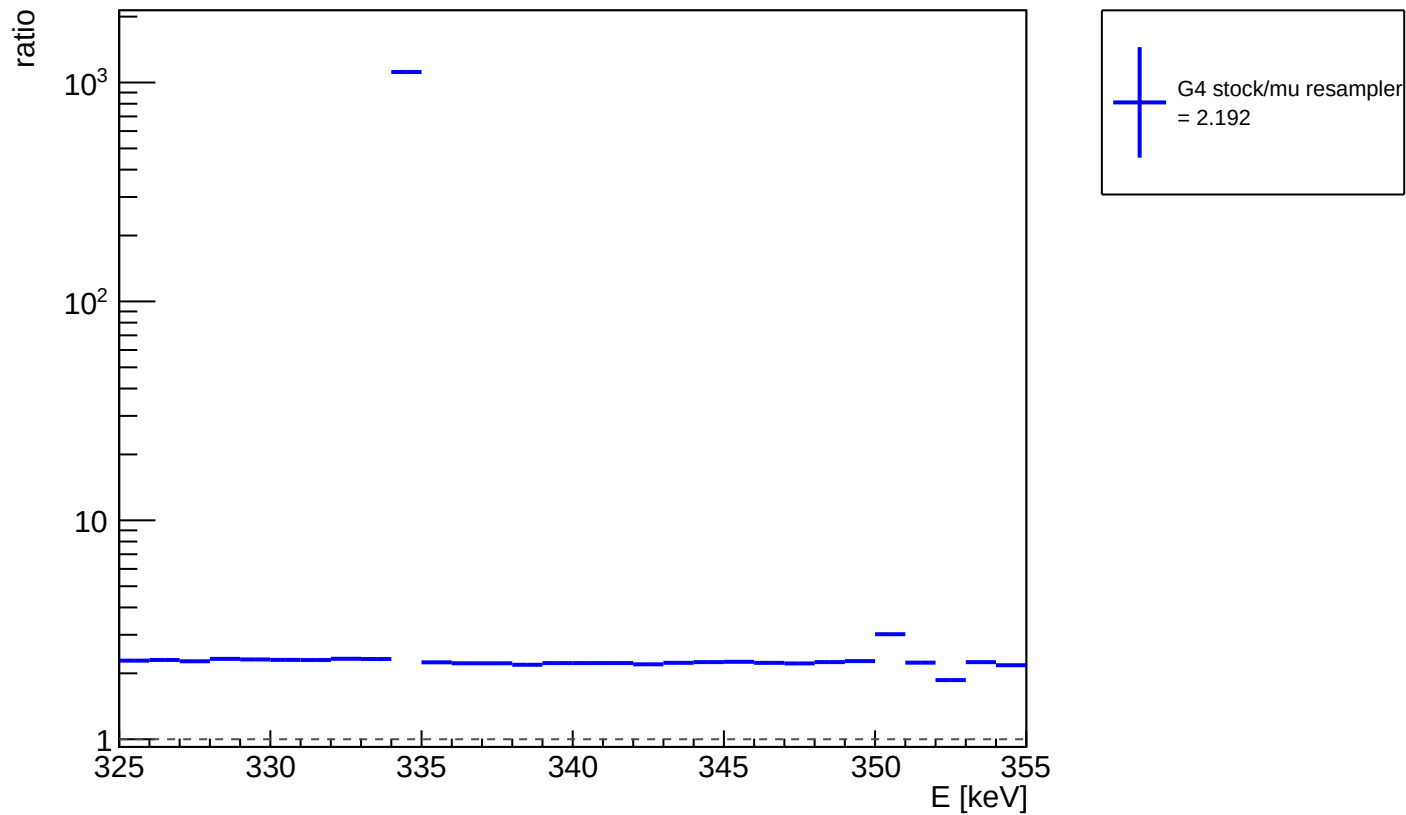


Entries	4.319534e+07
Mean	339.9
Std Dev	8.749
Underflow	7.085e+06
Overflow	3.618e+07
Integral	3.265e+05

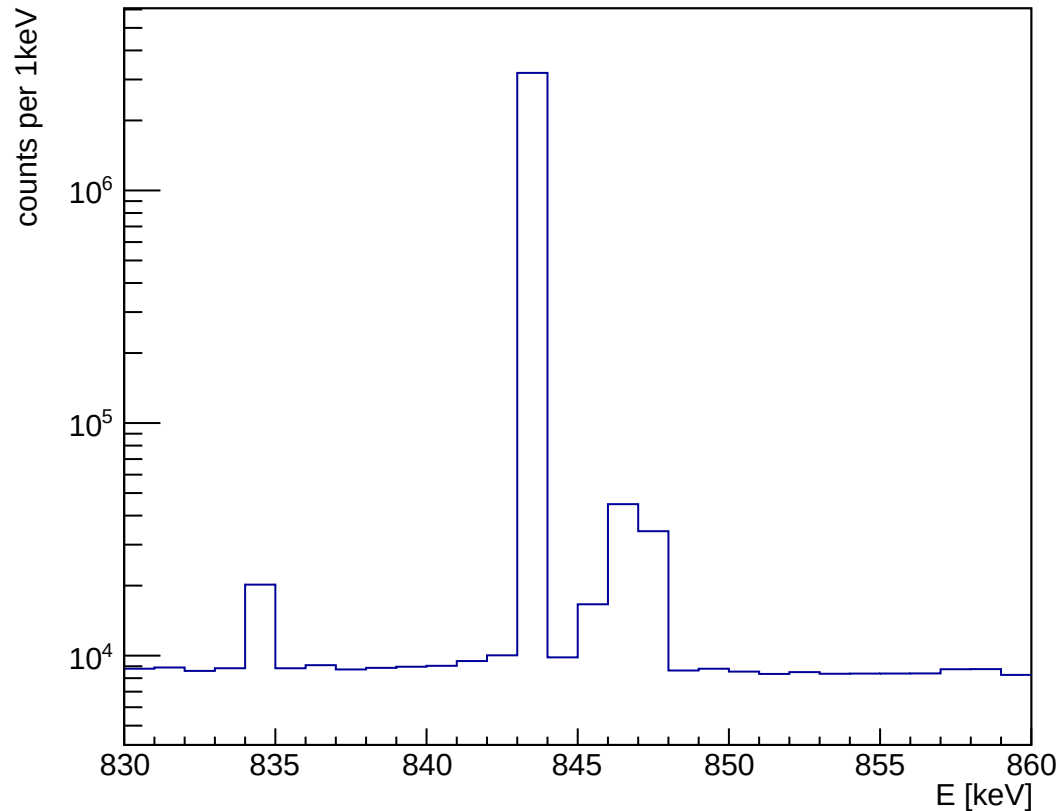
photon E (325-355 keV)



ratio of G4 stock / muon resampler: photon E (325-355 keV)



Stock G4: photon E (830-860 keV)



Entries	1.027272e+08
Mean	843.9
Std Dev	2.451
Underflow	7.297e+07
Overflow	1.905e+07
Integral	3.538e+06

Mu2e mu resampler: photon E (830-860 keV)

counts per 1keV

10^4

830

835

840

845

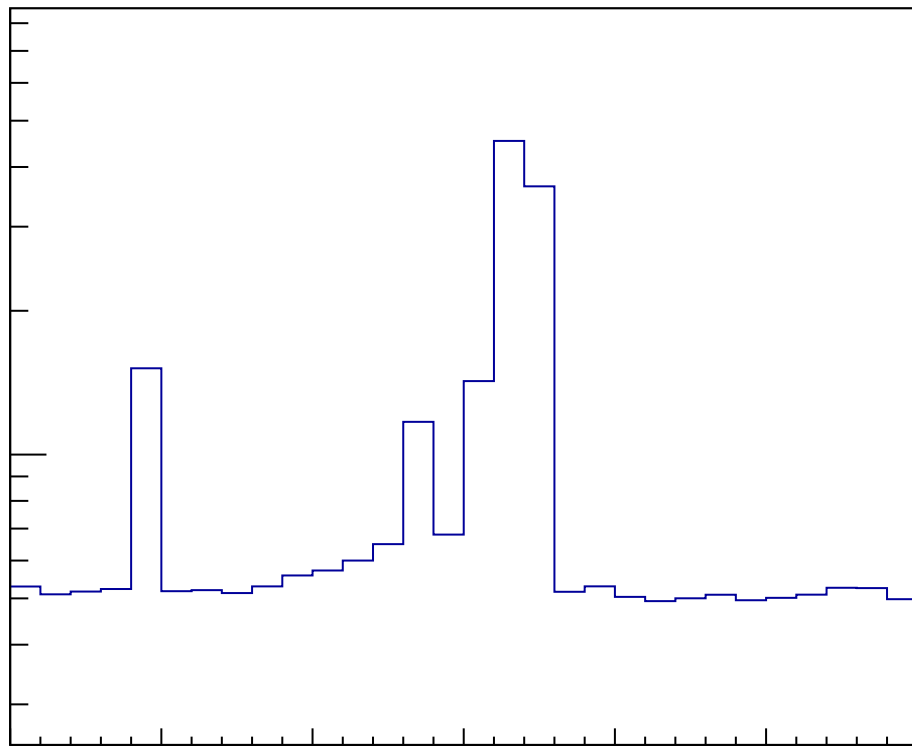
850

855

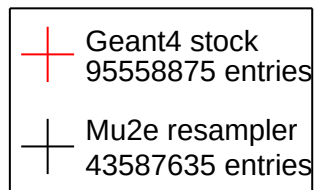
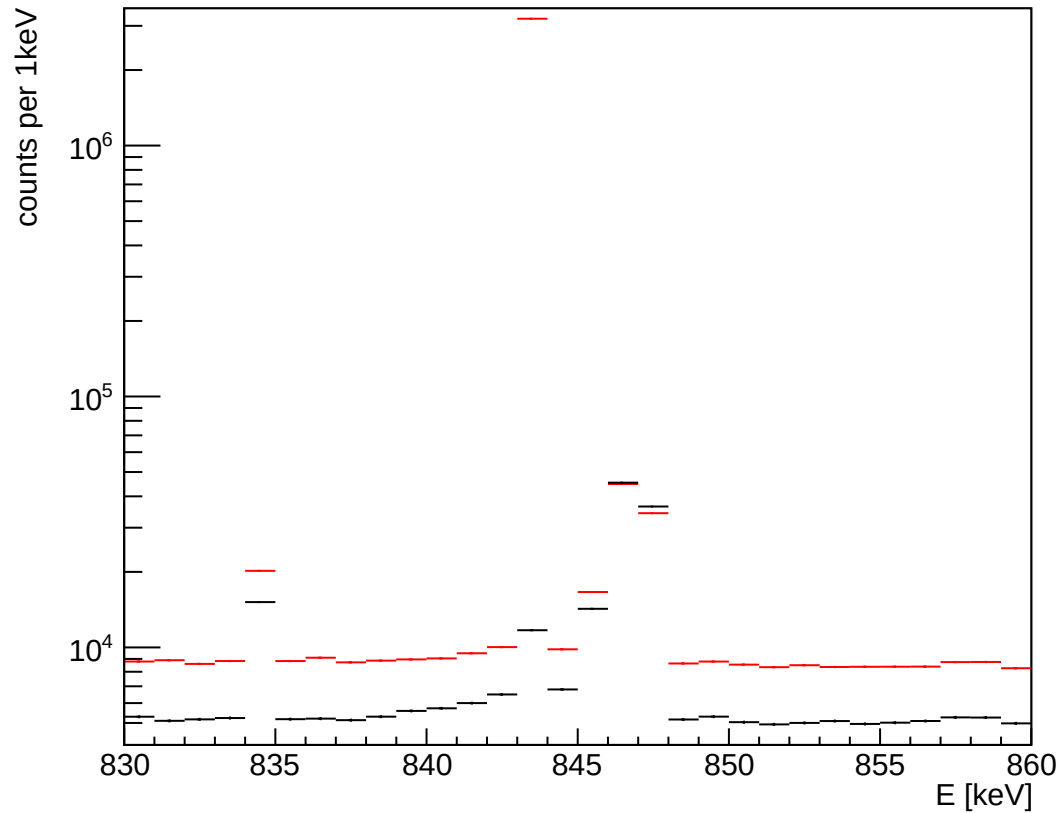
860

E [keV]

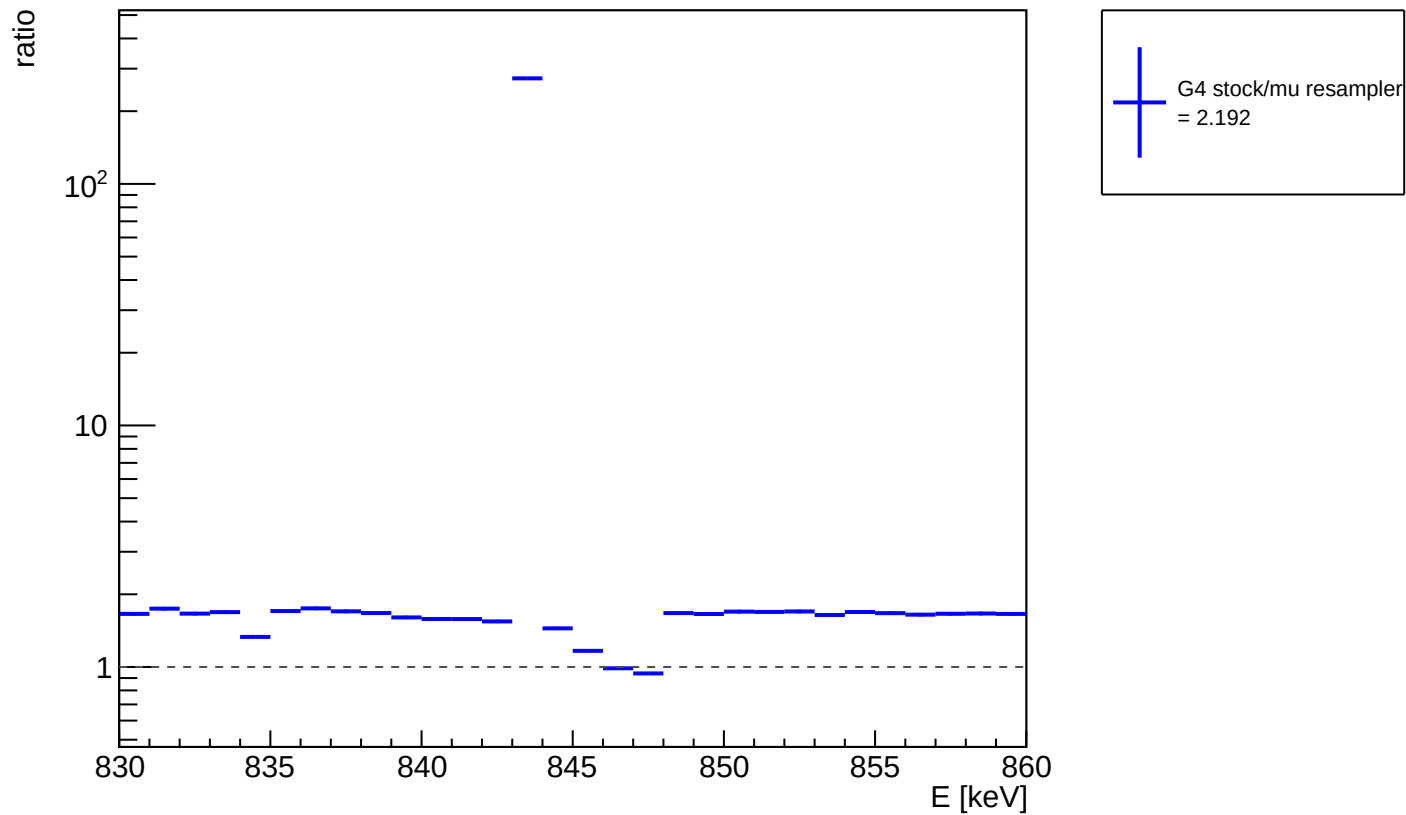
Entries	4.319534e+07
Mean	845
Std Dev	7.097
Underflow	1.076e+07
Overflow	3.257e+07
Integral	2.562e+05



photon E (830-860 keV)

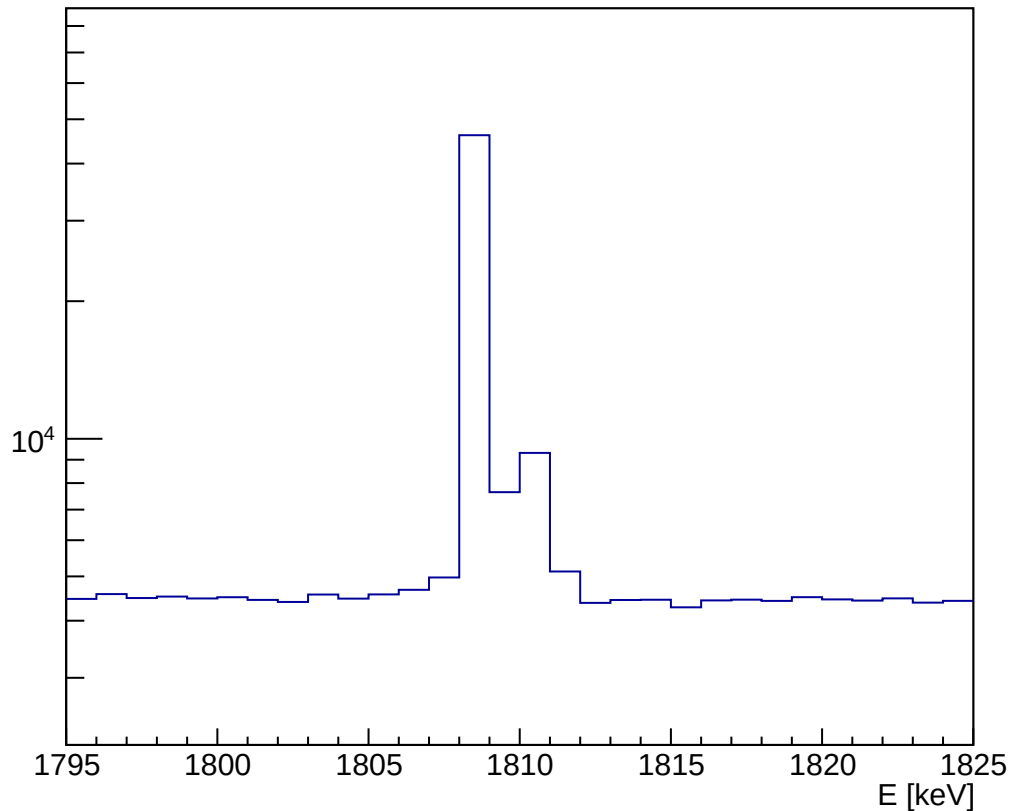


ratio of G4 stock / muon resampler: photon E (830-860 keV)



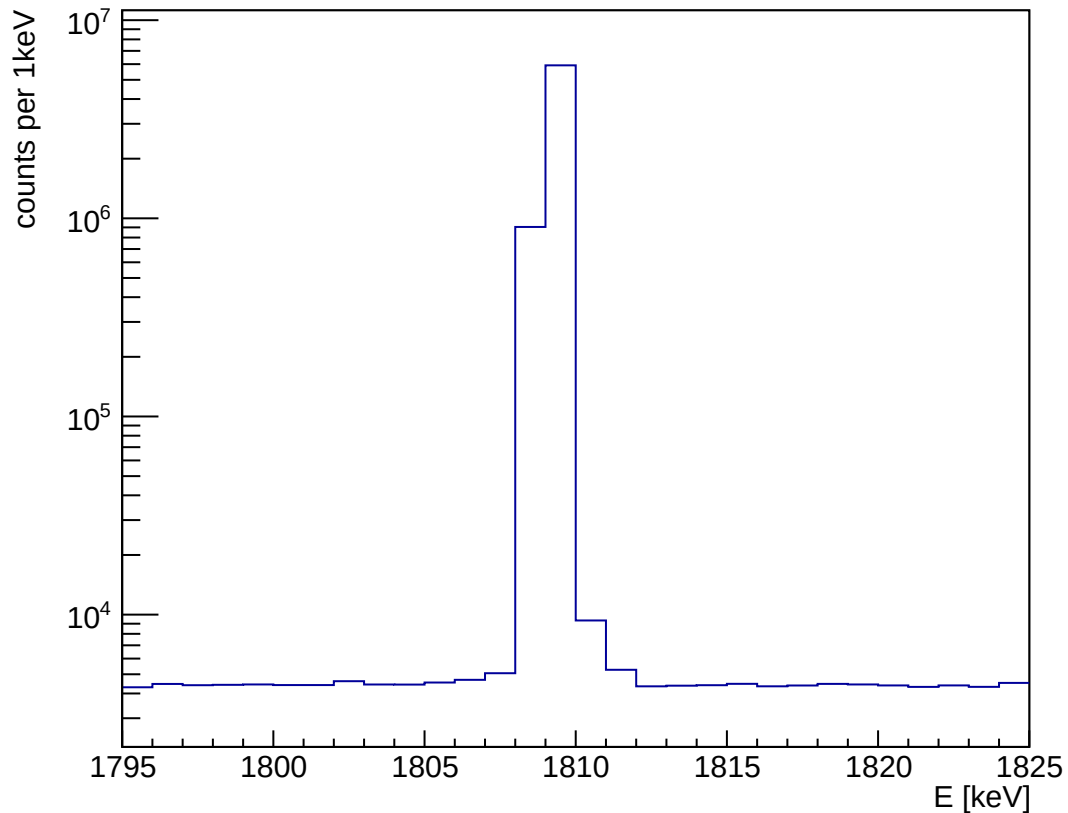
Stock G4: photon E (1795-1825 keV)

counts per 1keV



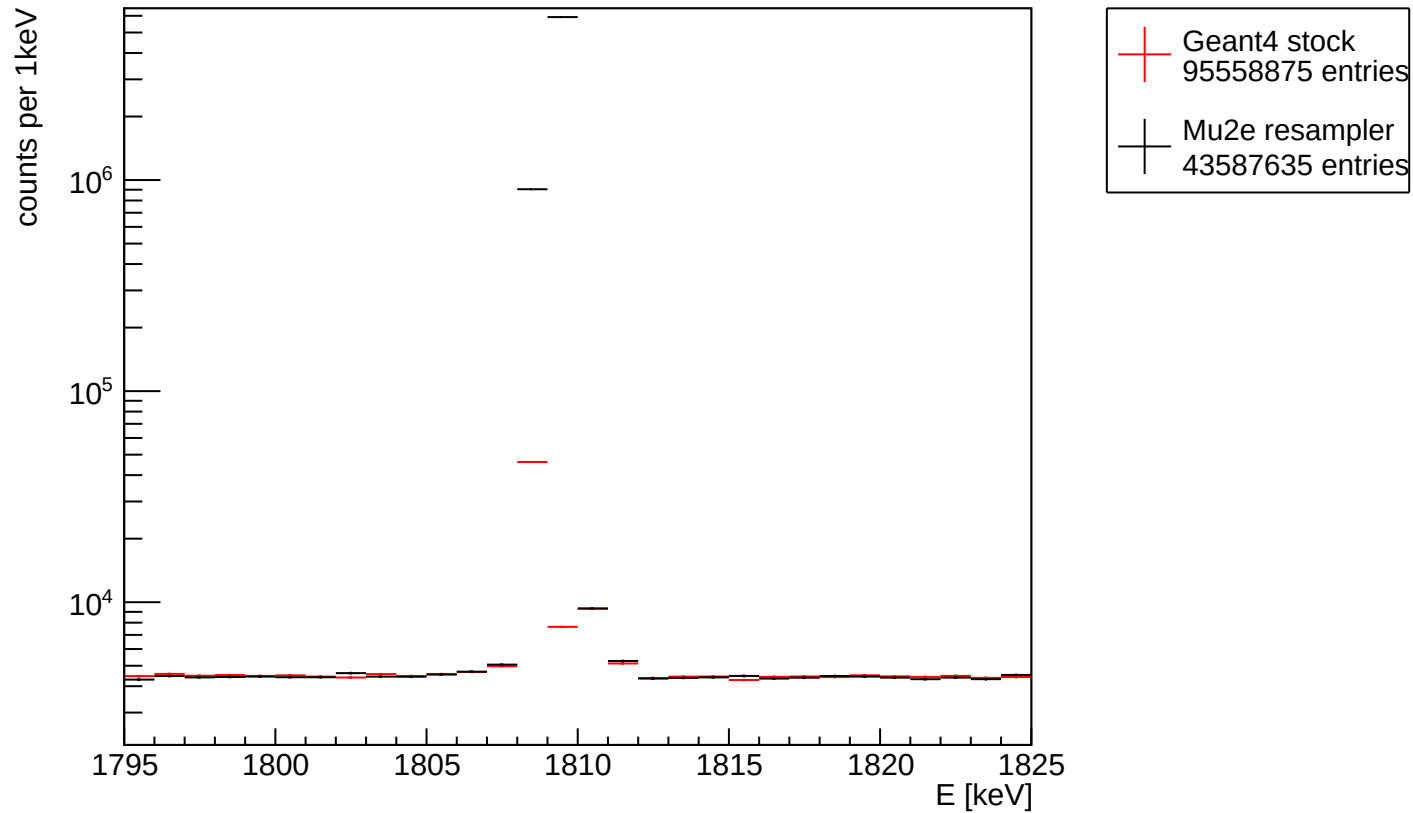
Entries	1.027272e+08
Mean	1810
Std Dev	7.395
Underflow	8.407e+07
Overflow	1.131e+07
Integral	1.848e+05

Mu2e mu resampler: photon E (1795-1825 keV)

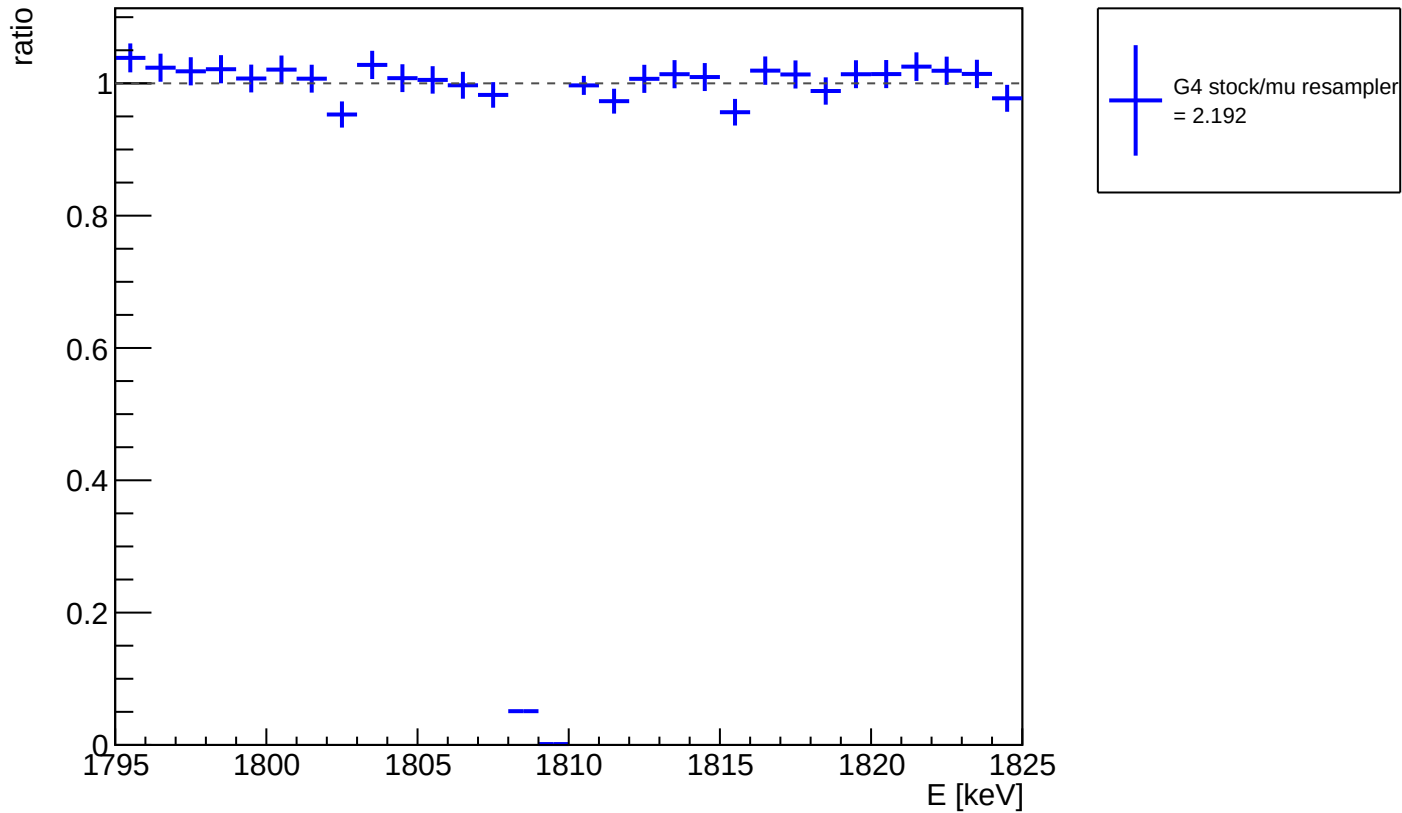


Entries	4.319534e+07
Mean	1809
Std Dev	1.203
Underflow	1.543e+07
Overflow	2.12e+07
Integral	6.952e+06

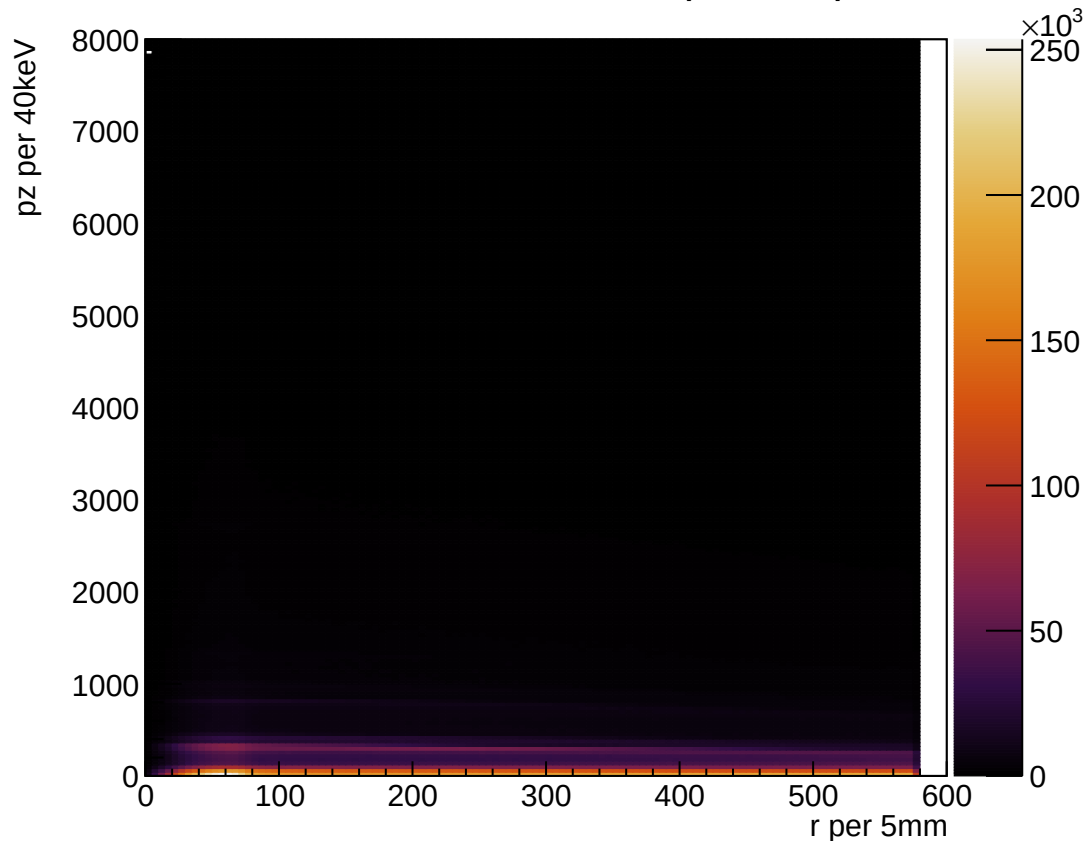
photon E (1795-1825 keV)



ratio of G4 stock / muon resampler: photon E (1795-1825 keV)

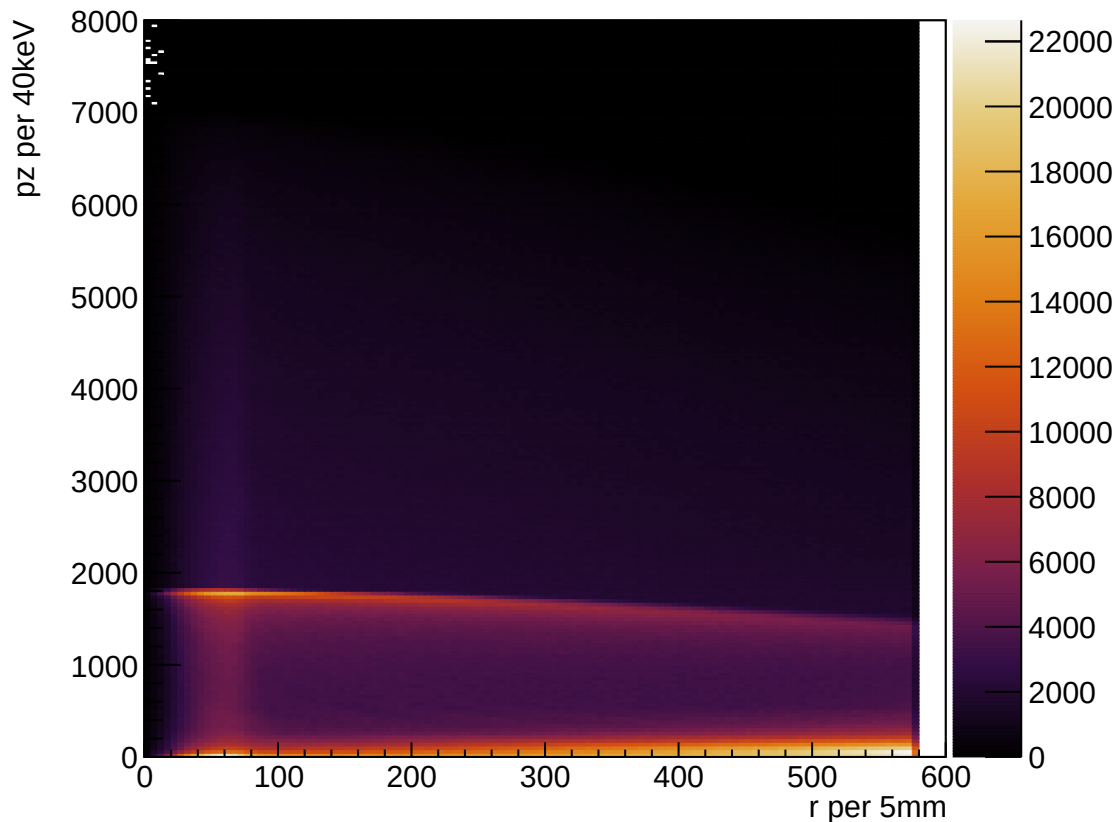


Stock G4: photon pz vs r



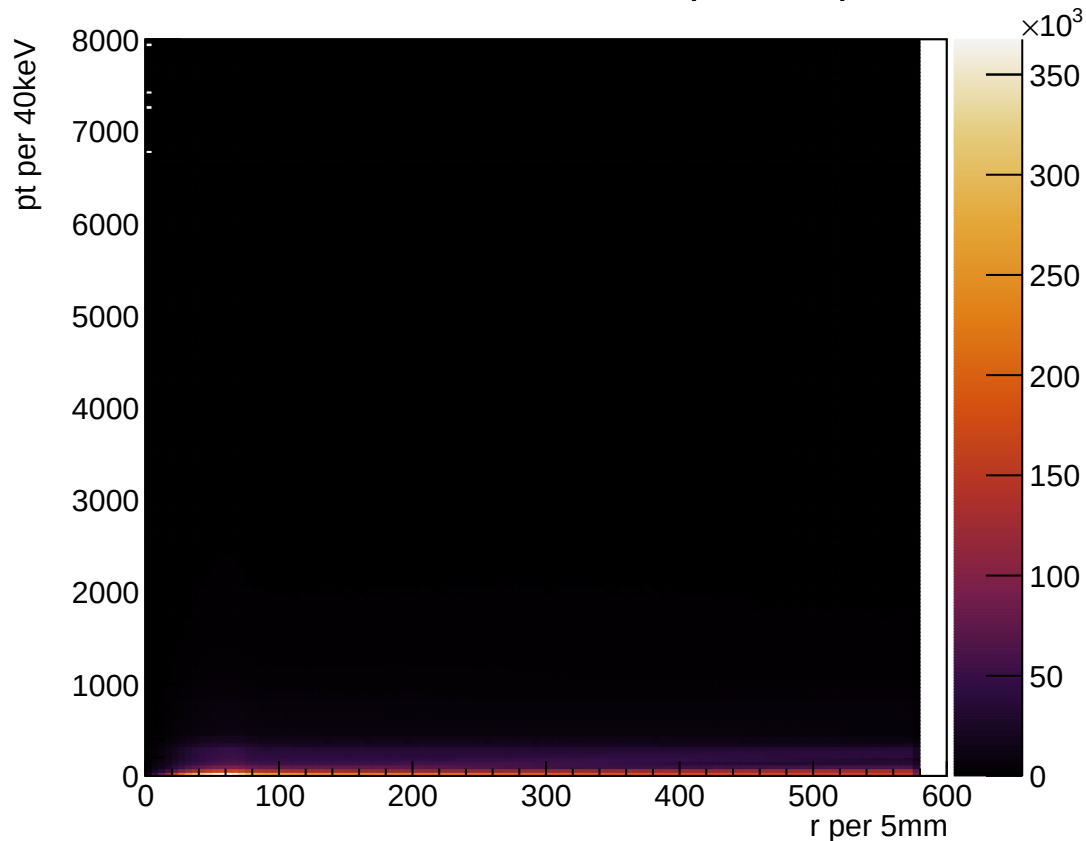
Entries	1.027272e+08	
Mean x	282.2	
Mean y	513.5	
Std Dev x	162.6	
Std Dev y	957.7	
Integral	9.534e+07	
0	220537	0
0	9.53383e+07	0
0	0	0

Mu2e mu resampler: photon pz vs r



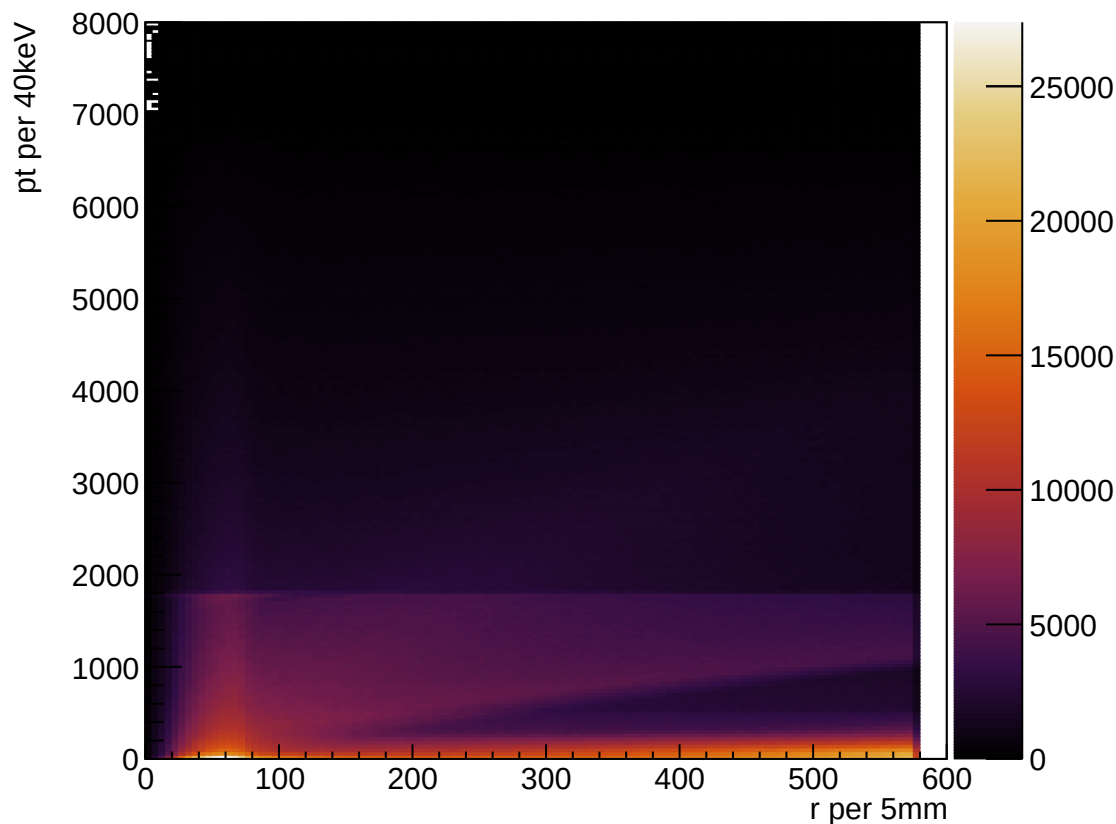
Entries	4.319534e+07	
Mean x	283.2	
Mean y	1879	
Std Dev x	162.8	
Std Dev y	1664	
Integral	4.344e+07	
0	148996	0
0	4.34386e+07	0
0	0	0

Stock G4: photon pt vs r



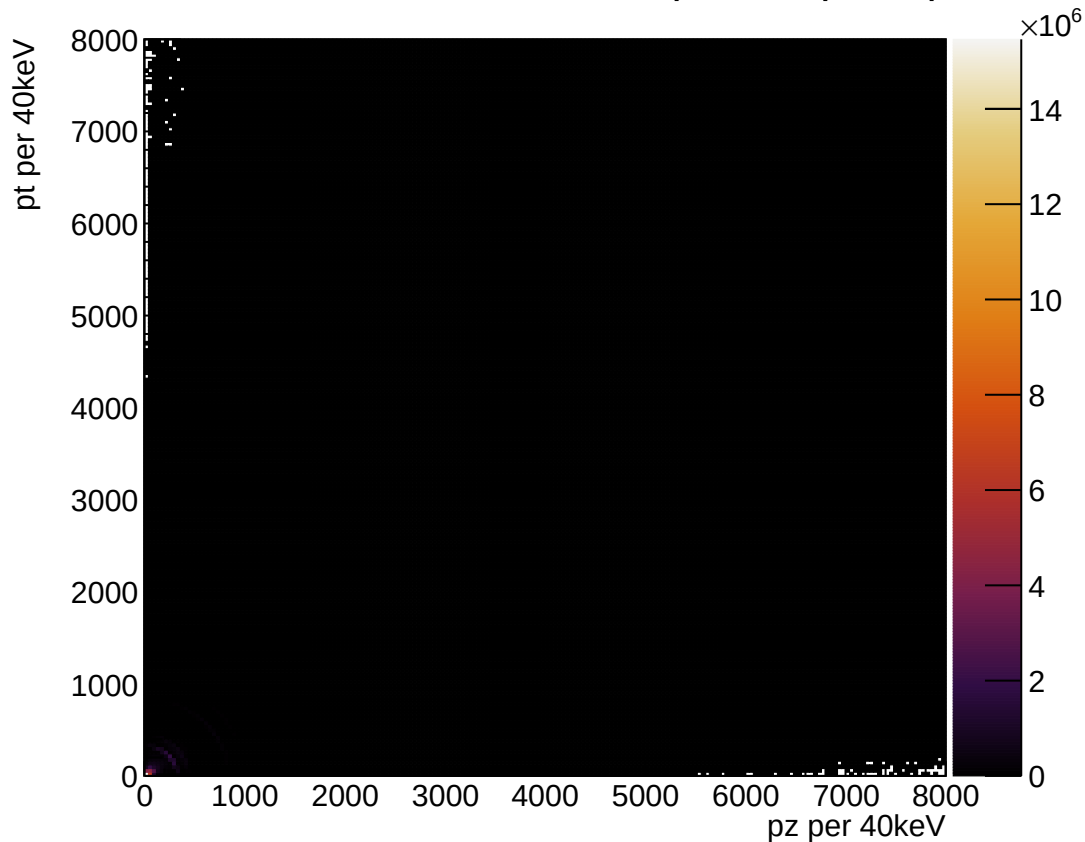
Entries	1.027272e+08	
Mean x	282.1	
Mean y	454.4	
Std Dev x	162.6	
Std Dev y	873.6	
Integral	9.53e+07	
0	261129	0
0	9.52977e+07	0
0	0	0

Mu2e mu resampler: photon pt vs r



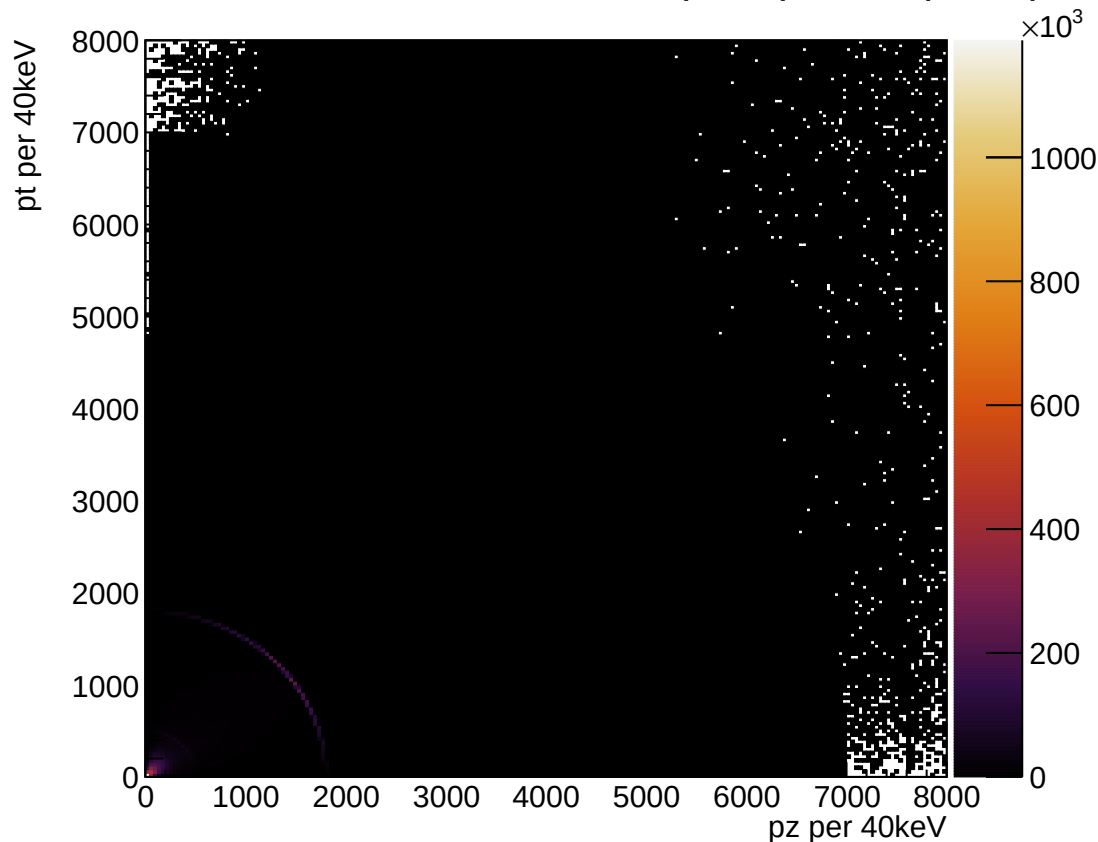
Entries	4.319534e+07	
Mean x	283.1	
Mean y	1641	
Std Dev x	162.9	
Std Dev y	1506	
Integral	4.337e+07	
0	217858	0
0	4.33698e+07	0
0	0	0

Stock G4: photon pt vs pz



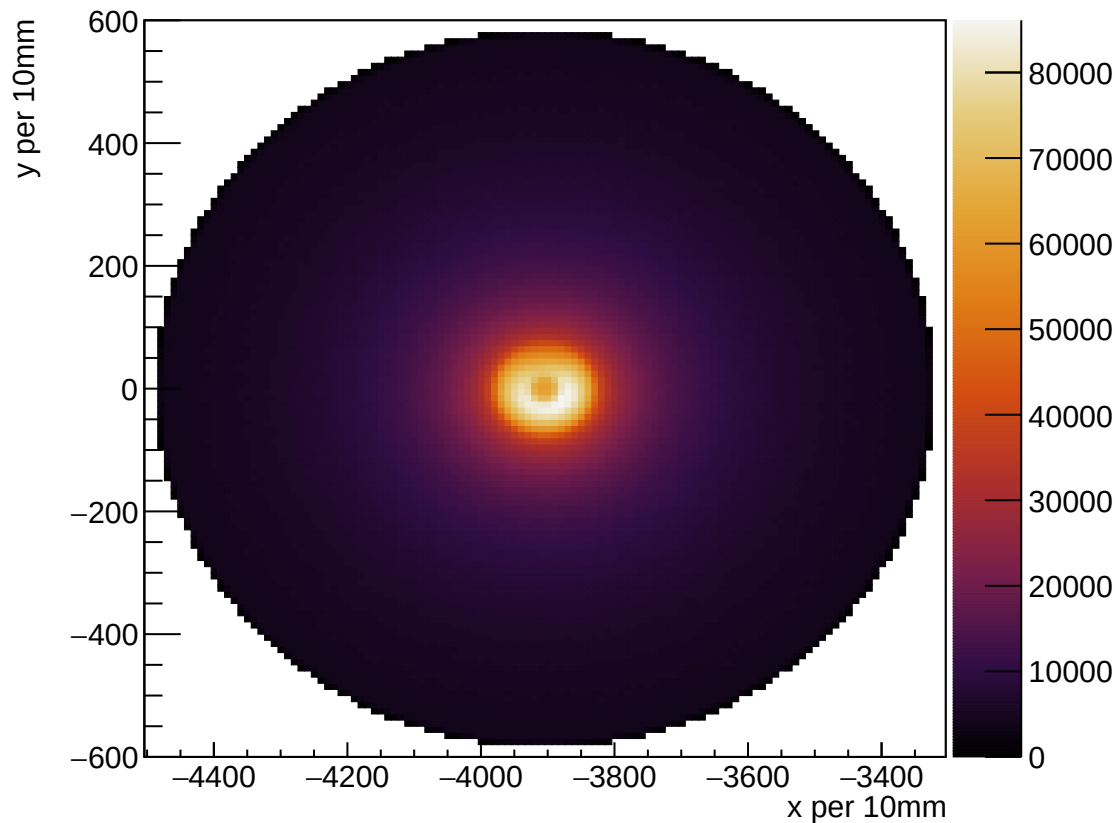
Entries	1.027272e+08	
Mean x	506.7	
Mean y	449.8	
Std Dev x	941	
Std Dev y	861.8	
Integral	9.518e+07	
0	161335	99793
0	9.5177e+07	7120743
0	0	0

Mu2e mu resampler: photon pt vs pz



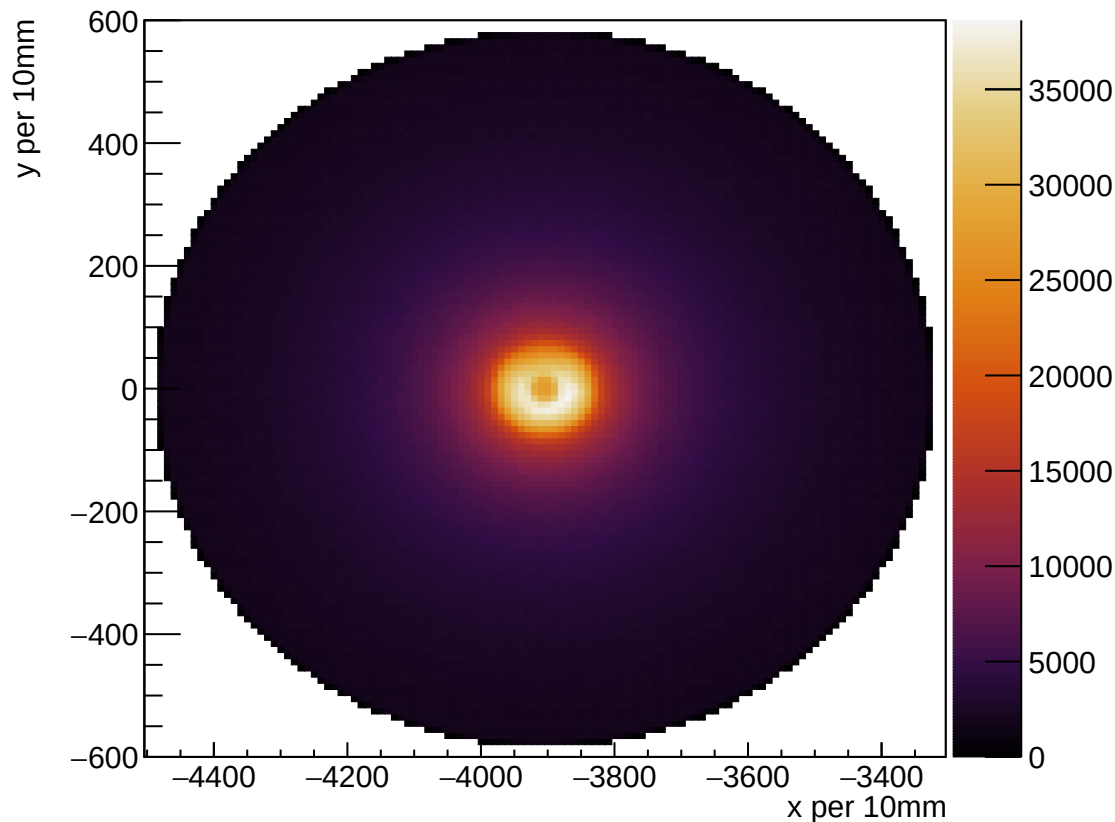
Entries	4.319534e+07	
Mean x	1871	
Mean y	1637	
Std Dev x	1656	
Std Dev y	1502	
Integral	4.332e+07	
0	119817	98040.4
0	4.33188e+07	70955.6
0	0	0

Stock G4: photon y vs x



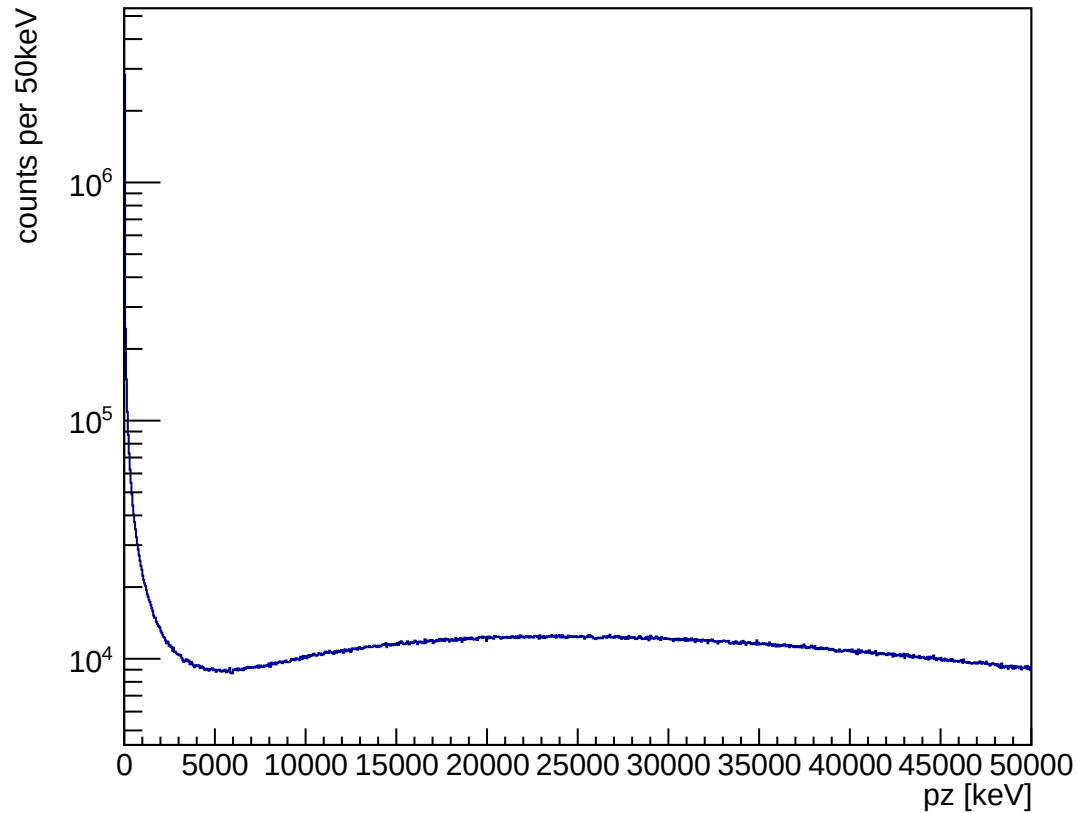
Entries	1.027272e+08		
Mean x	-3903		
Mean y	-0.005513		
Std Dev x	230.3		
Std Dev y	230.1		
Integral	9.556e+07		
0	0	0	
0	9.5589e+07	0	0
0	0	0	

Mu2e mu resampler: photon y vs x



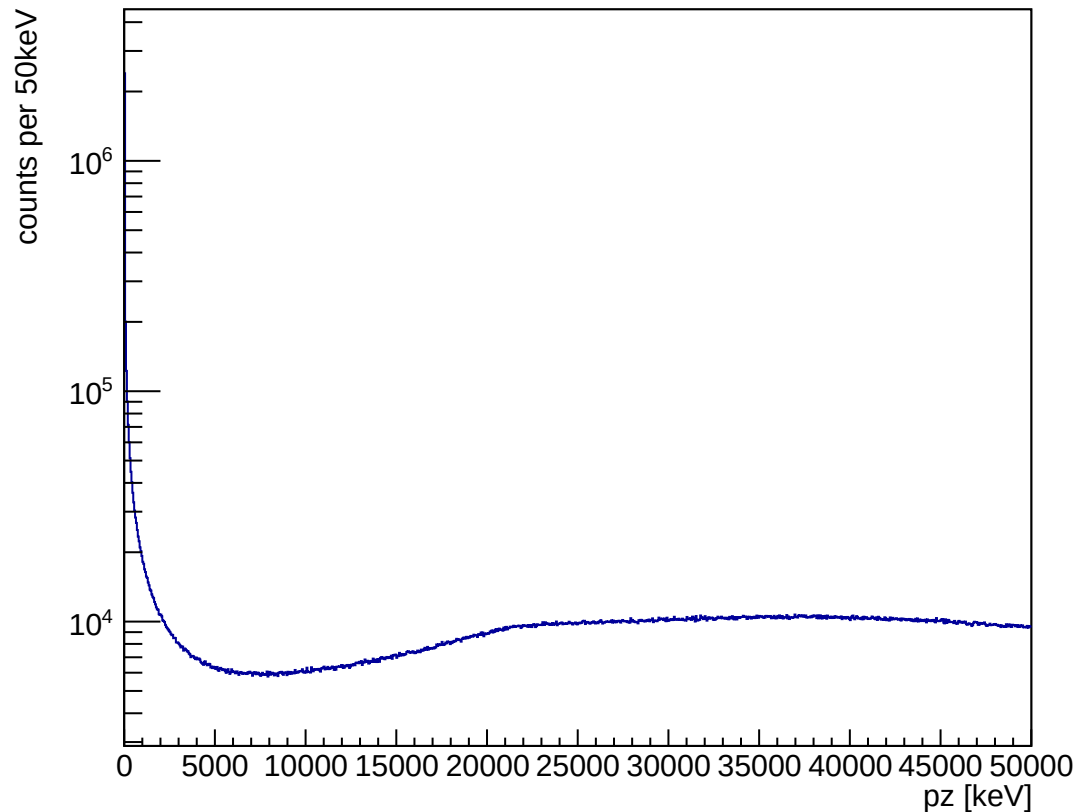
Entries	4.319534e+07		
Mean x	-3903		
Mean y	-0.2385		
Std Dev x	230.9		
Std Dev y	230.9		
Integral	4.359e+07		
0	0	0	
0	4.35876e+07	0	
0	0	0	

Stock G4: neutron momentum pz



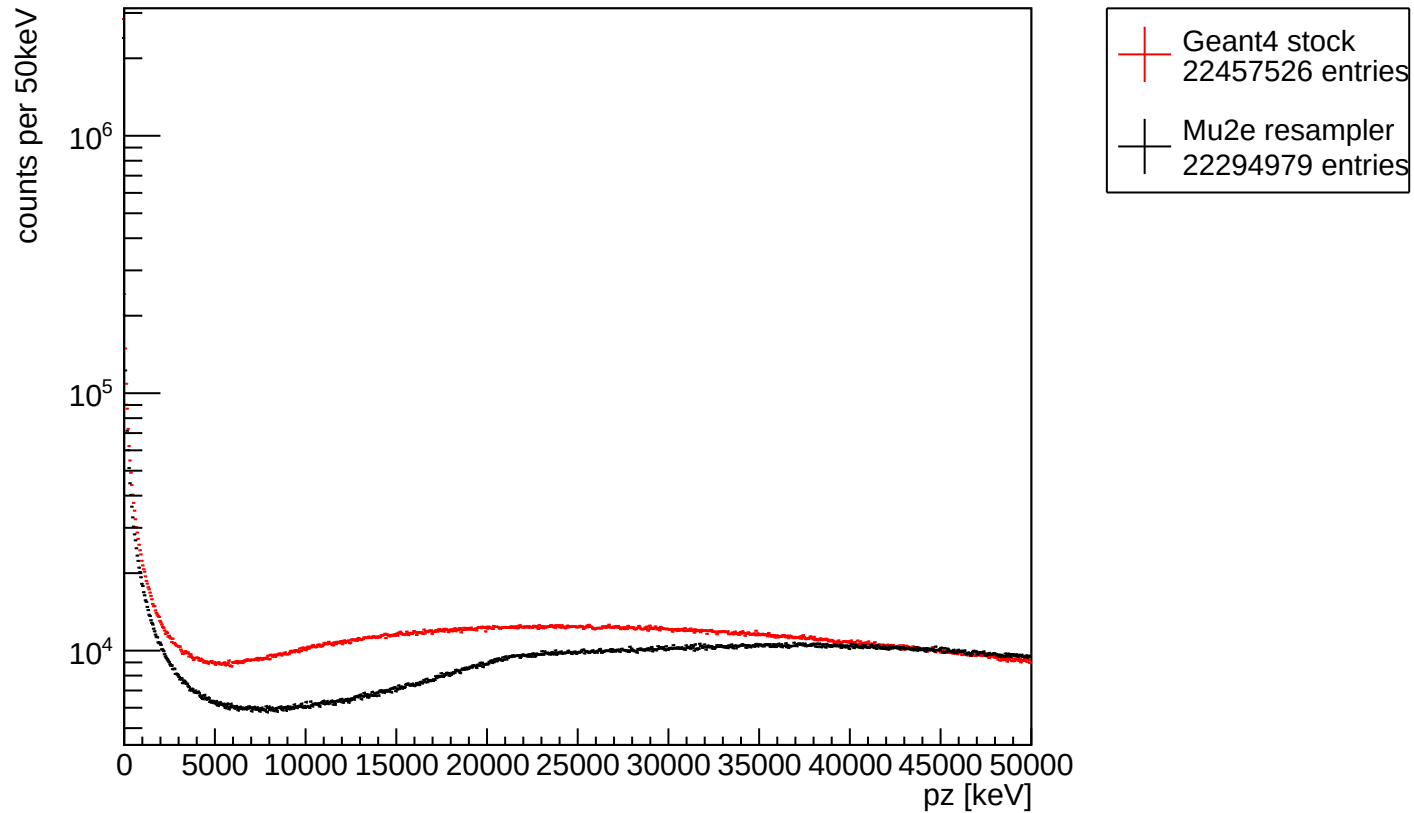
Entries	2.414217e+07
Mean	1.846e+04
Std Dev	1.619e+04
Underflow	0
Overflow	7.514e+06
Integral	1.494e+07

Mu2e mu resampler: neutron momentum pz

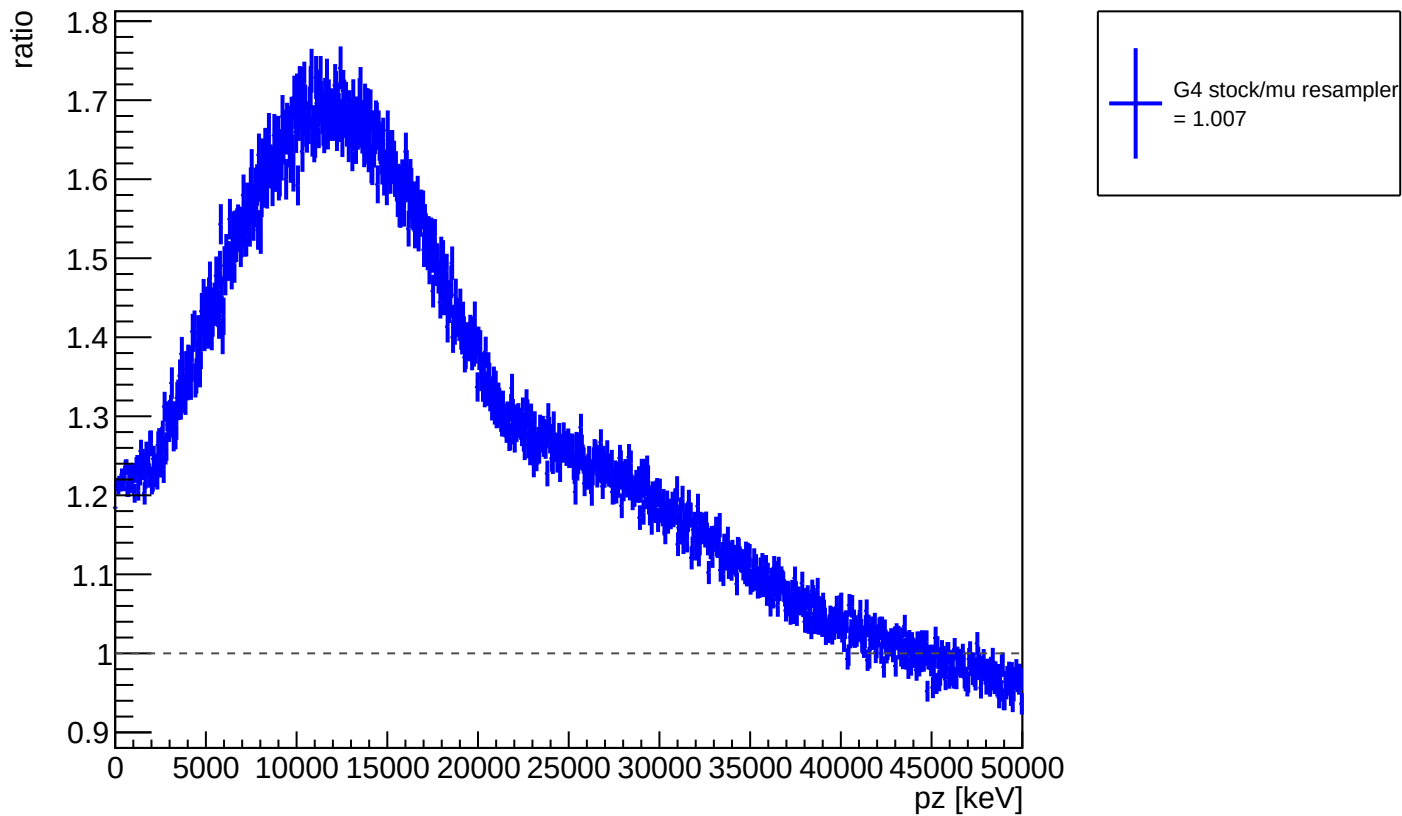


Entries	2.209432e+07
Mean	1.975e+04
Std Dev	1.695e+04
Underflow	0
Overflow	1.01e+07
Integral	1.219e+07

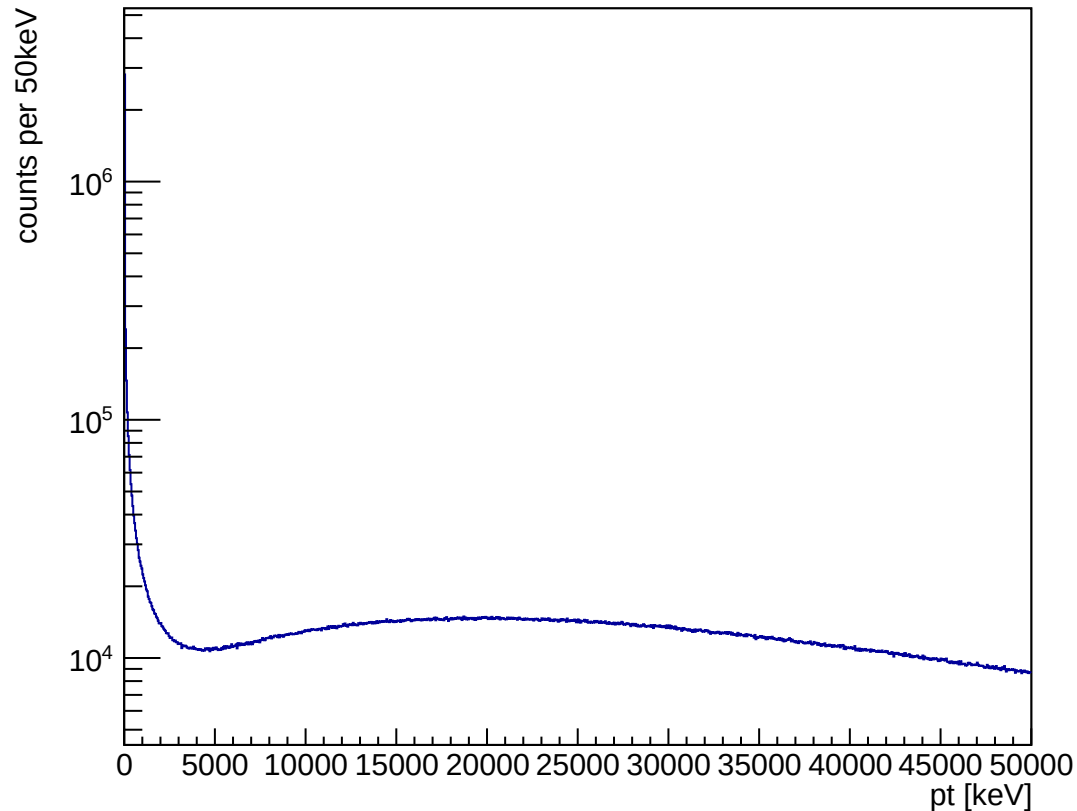
neutron momentum pz



ratio of G4 stock / muon resampler: neutron momentum pz



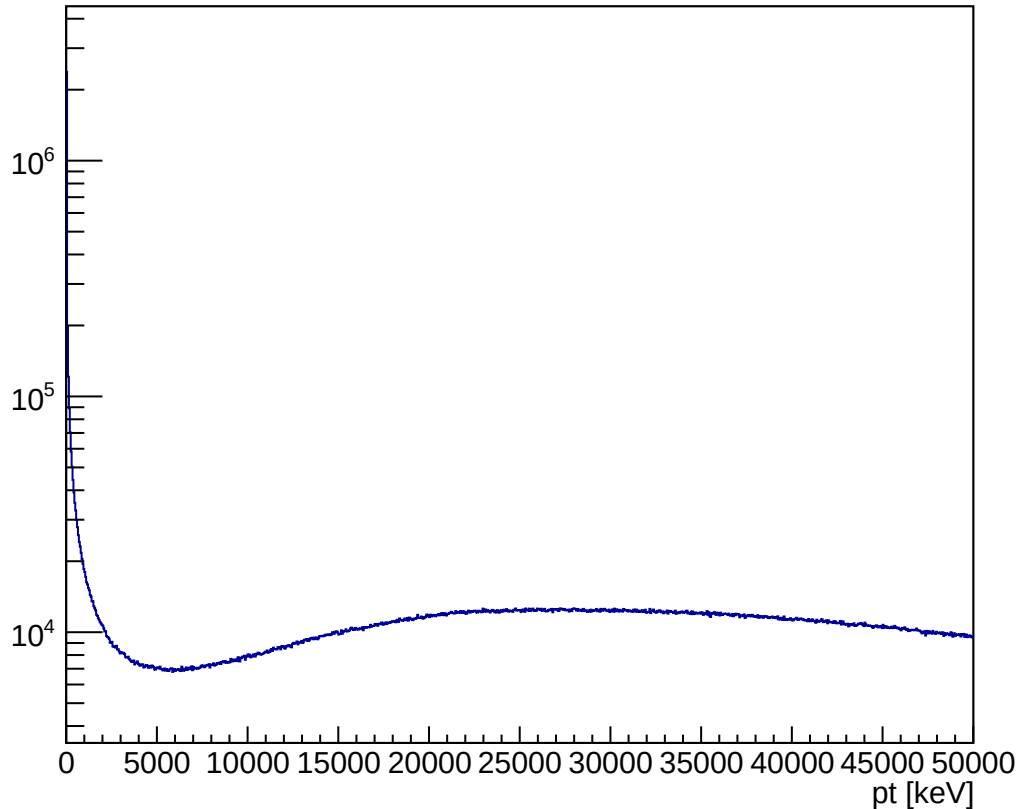
Stock G4: neutron transverse momentum pT



Entries	2.414217e+07
Mean	1.838e+04
Std Dev	1.566e+04
Underflow	0
Overflow	6.173e+06
Integral	1.628e+07

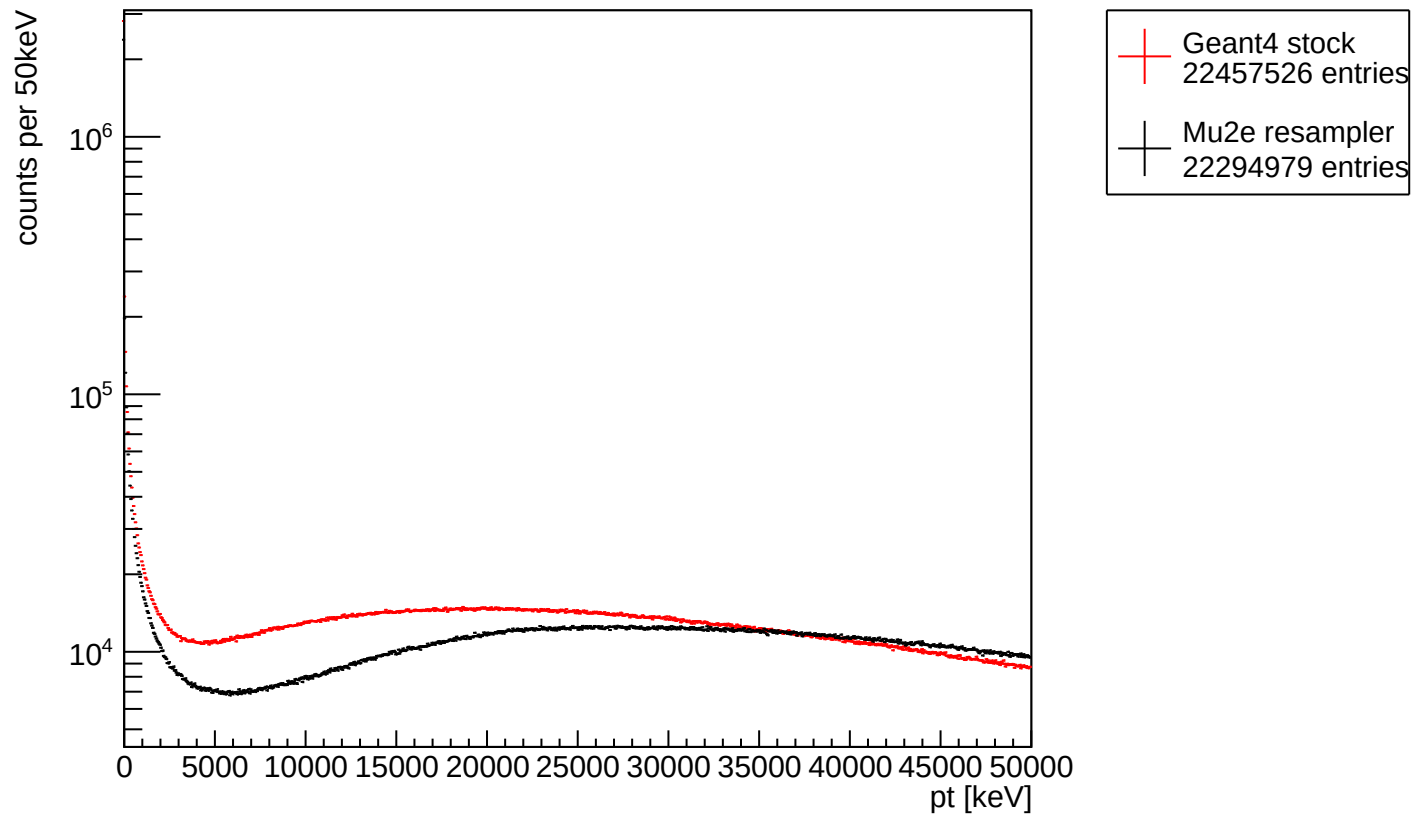
Mu2e mu resampler: neutron transverse momentum pT

counts per 50keV

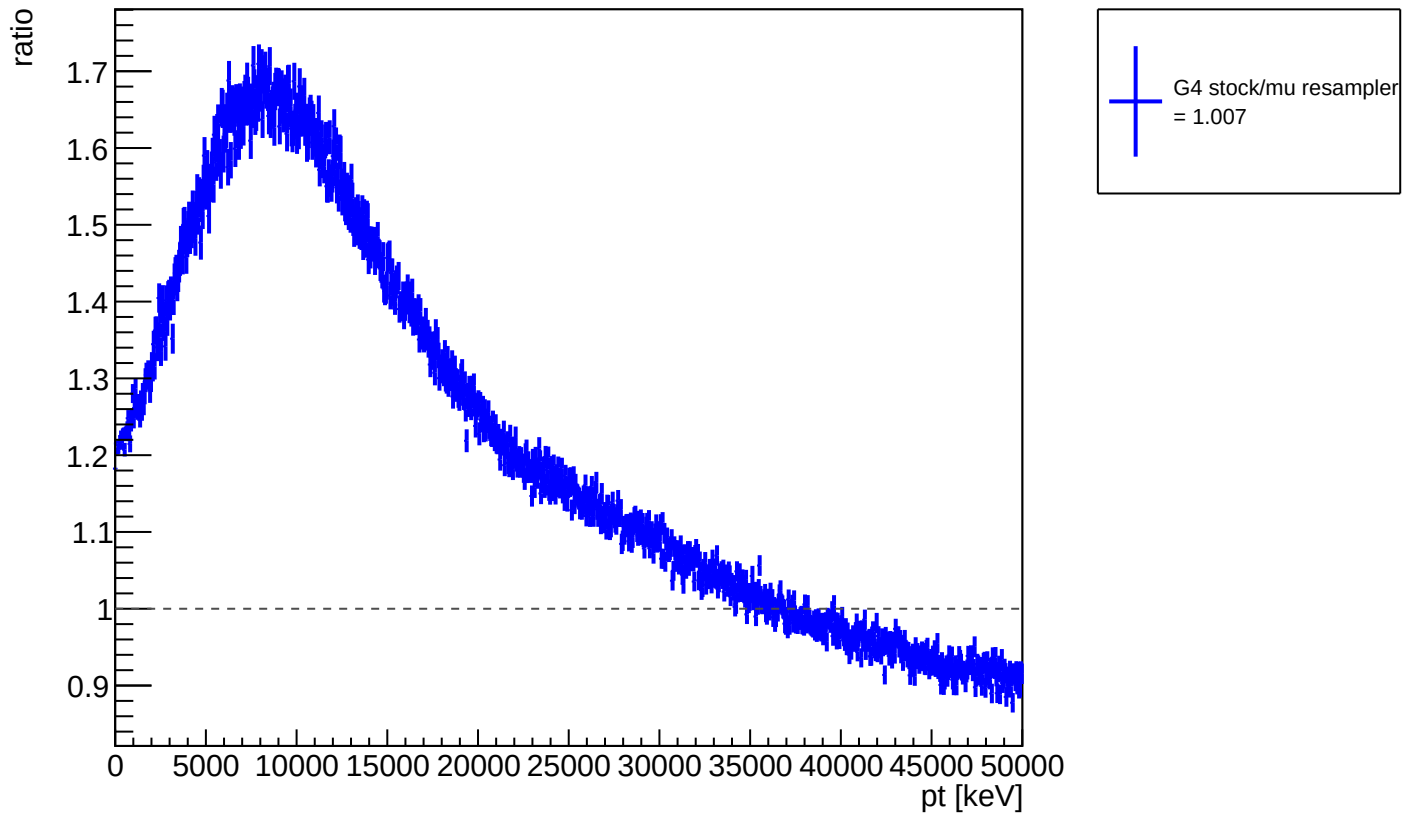


Entries	2.209432e+07
Mean	2.021e+04
Std Dev	1.634e+04
Underflow	0
Overflow	8.556e+06
Integral	1.374e+07

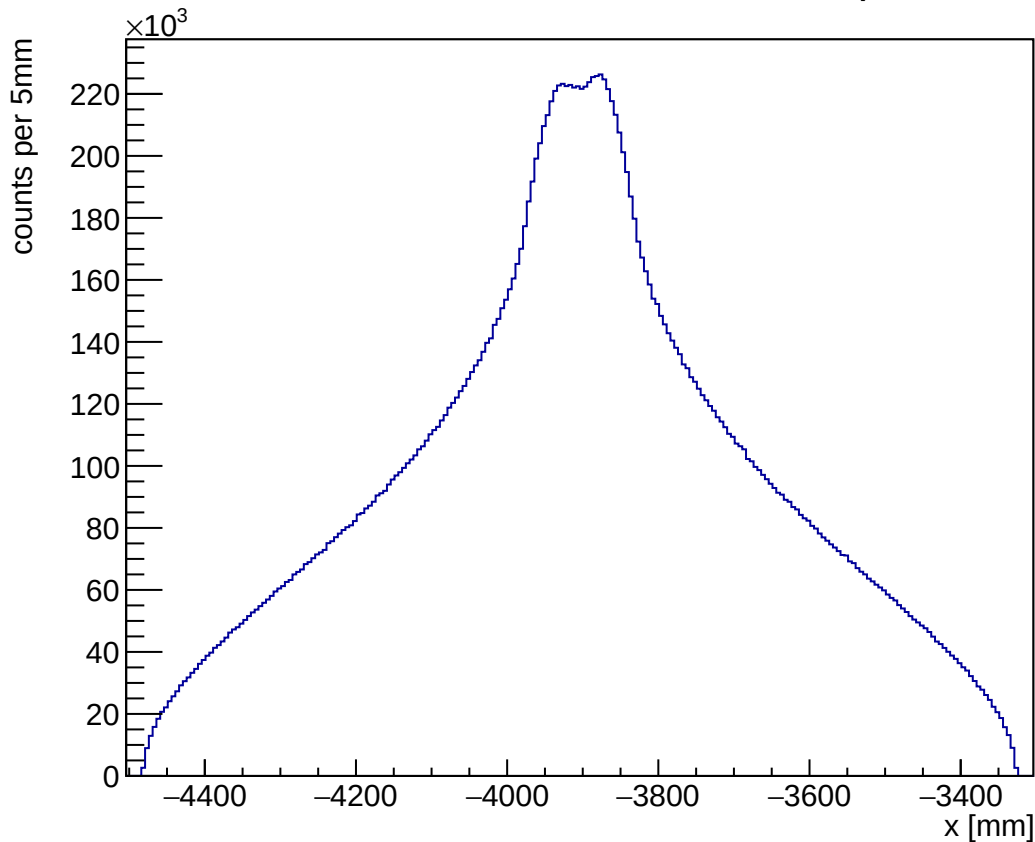
neutron transverse momentum pT



ratio of G4 stock / muon resampler: neutron transverse momentum pT

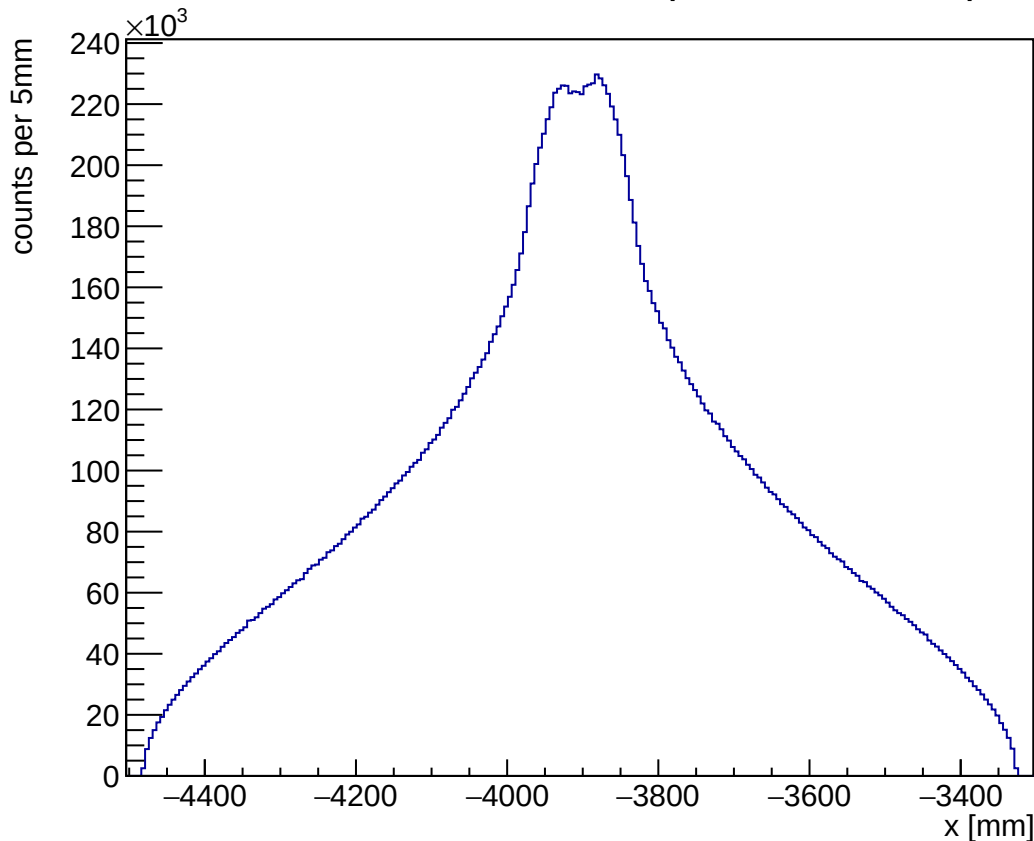


Stock G4: neutron x position



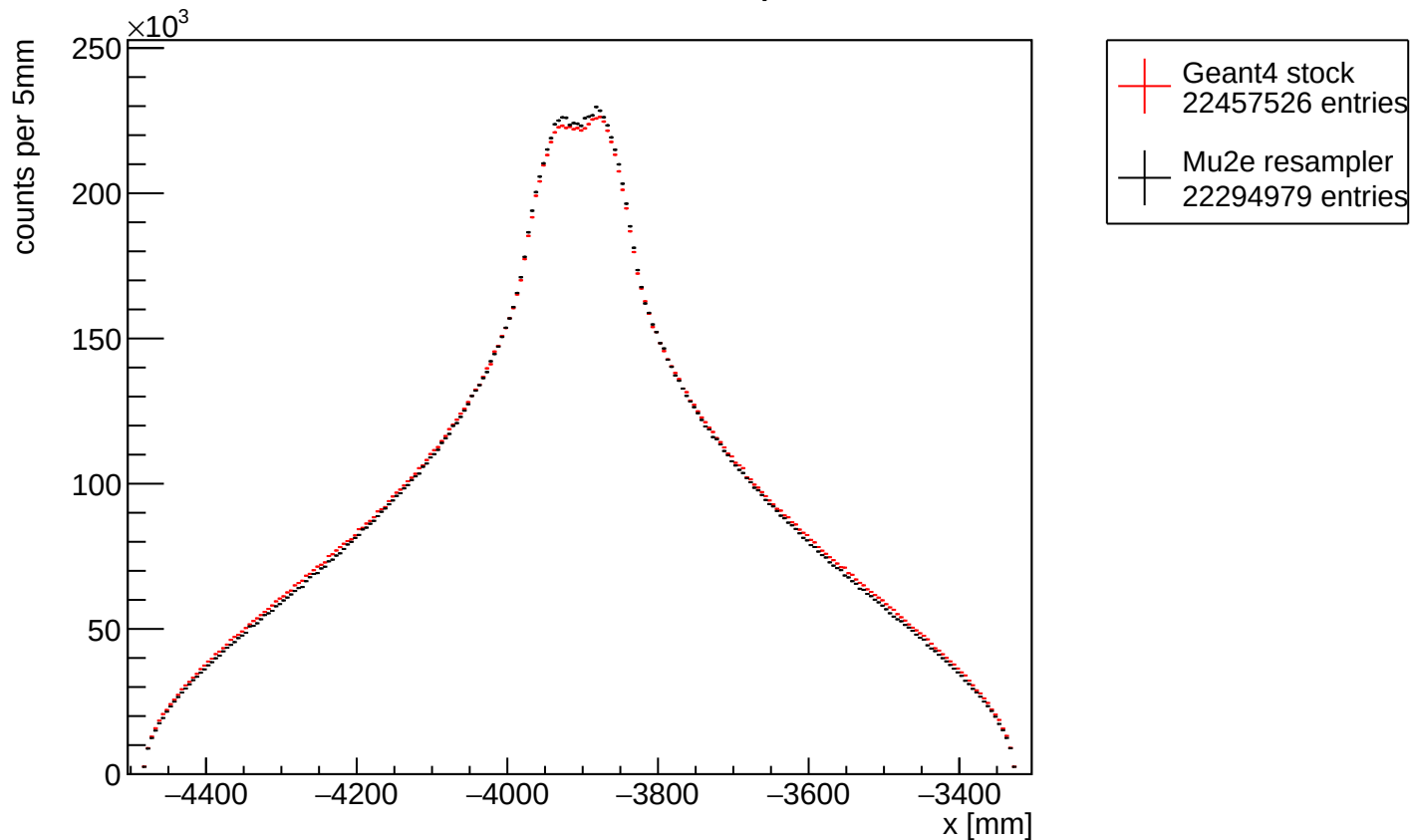
Entries	2.414217e+07
Mean	-3903
Std Dev	242.6
Underflow	0
Overflow	0
Integral	2.246e+07

Mu2e mu resampler: neutron x position

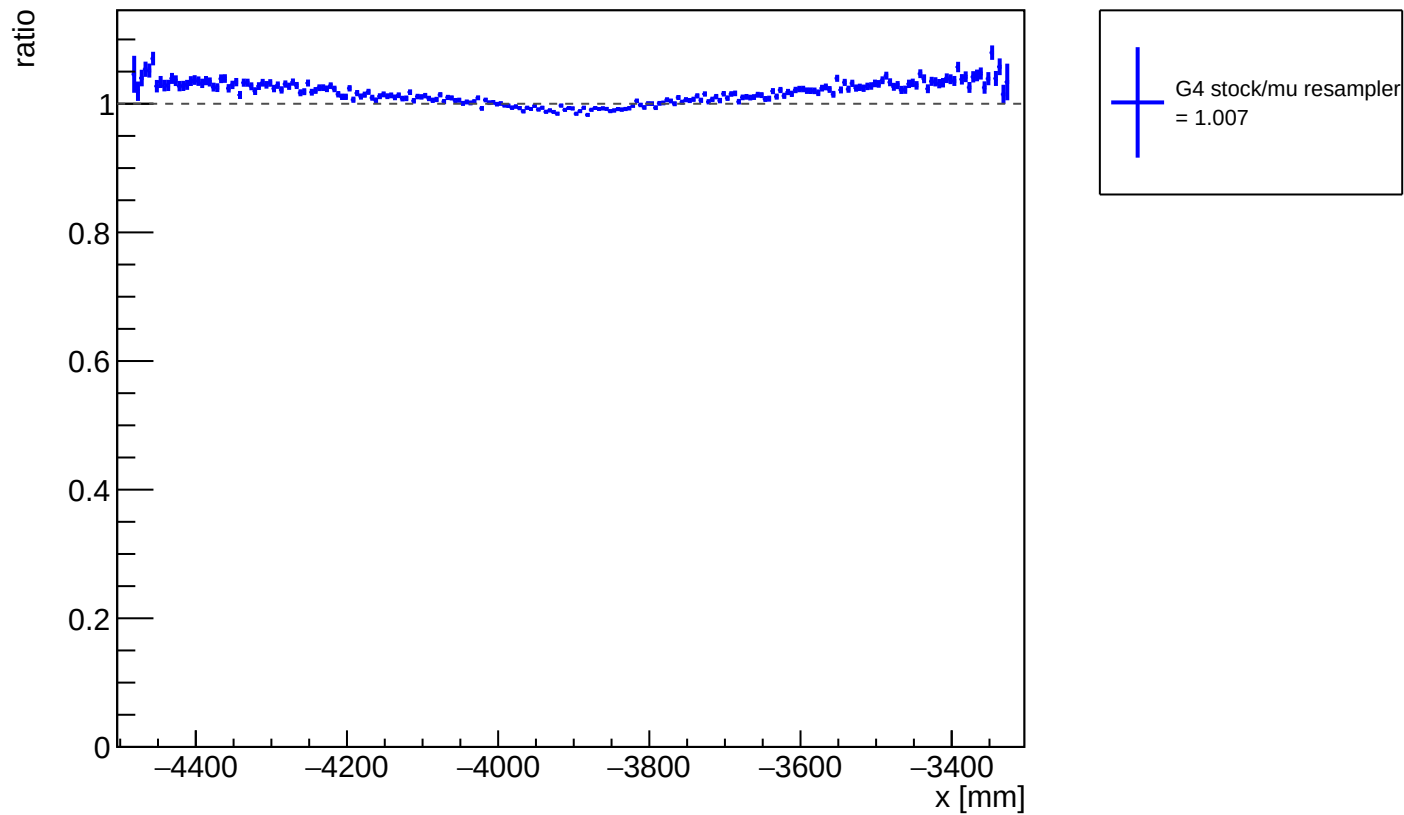


Entries	2.209432e+07
Mean	-3903
Std Dev	240.6
Underflow	0
Overflow	0
Integral	2.229e+07

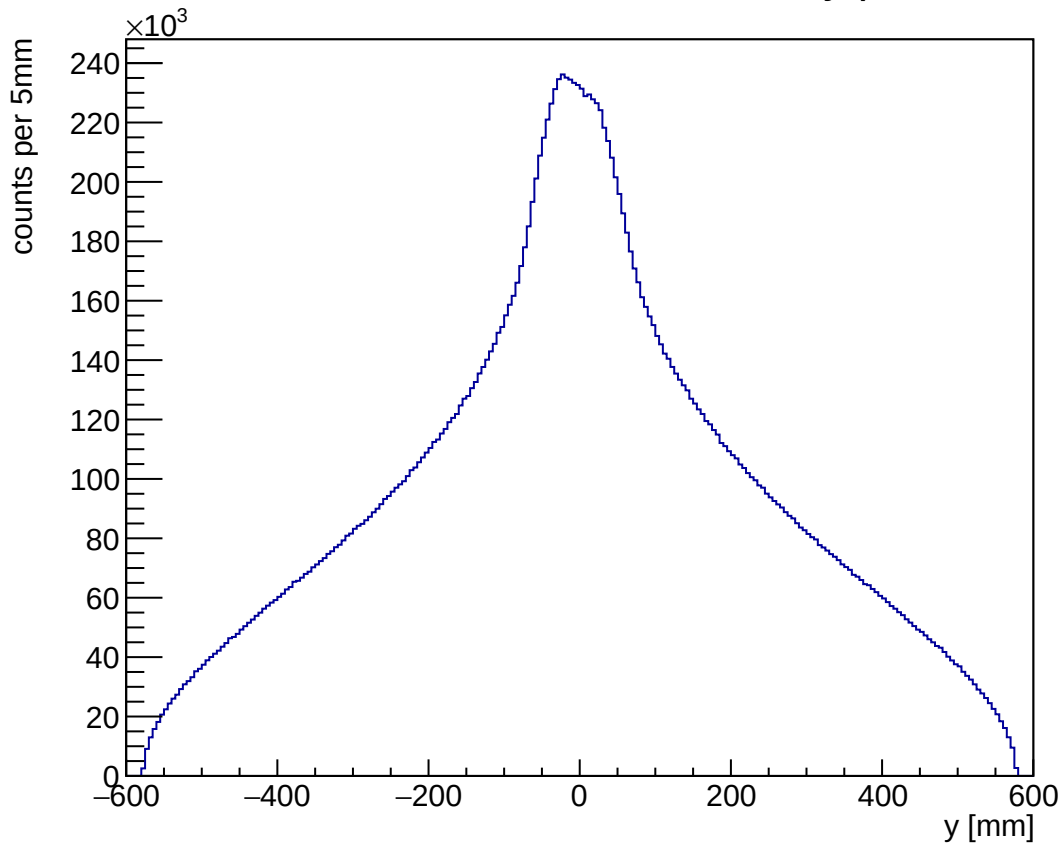
neutron x position



ratio of G4 stock / muon resampler: neutron x position

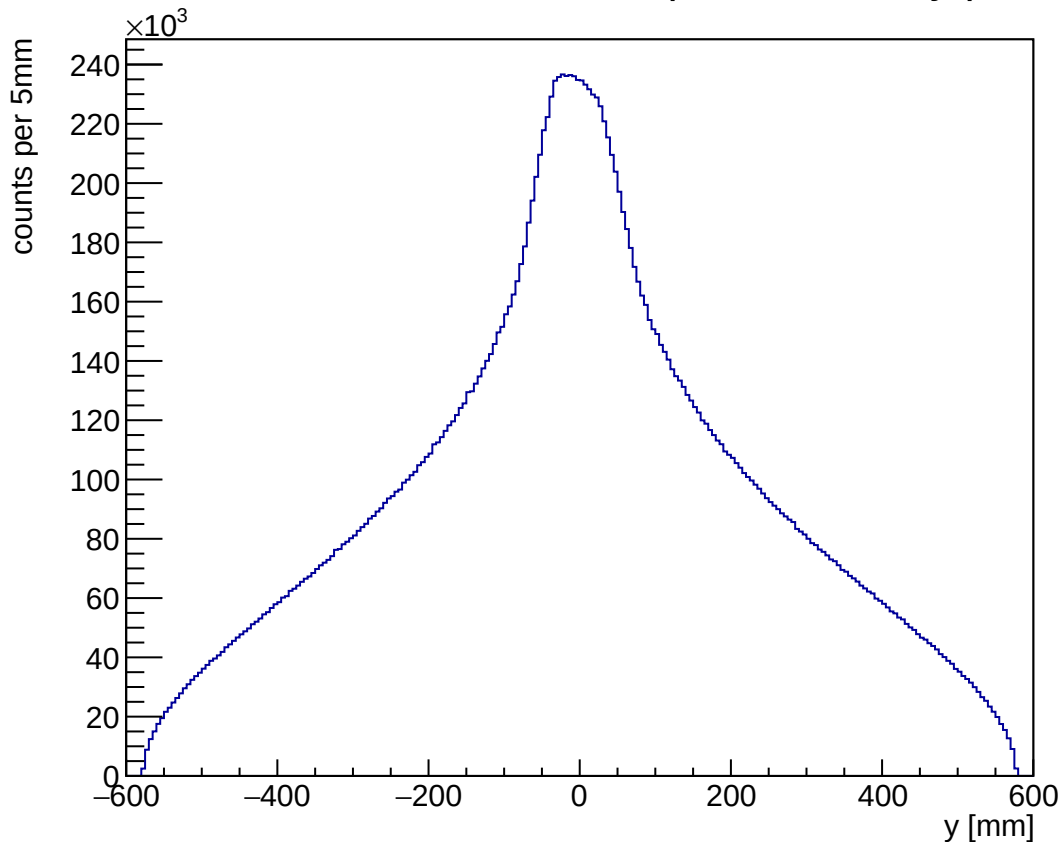


Stock G4: neutron y position



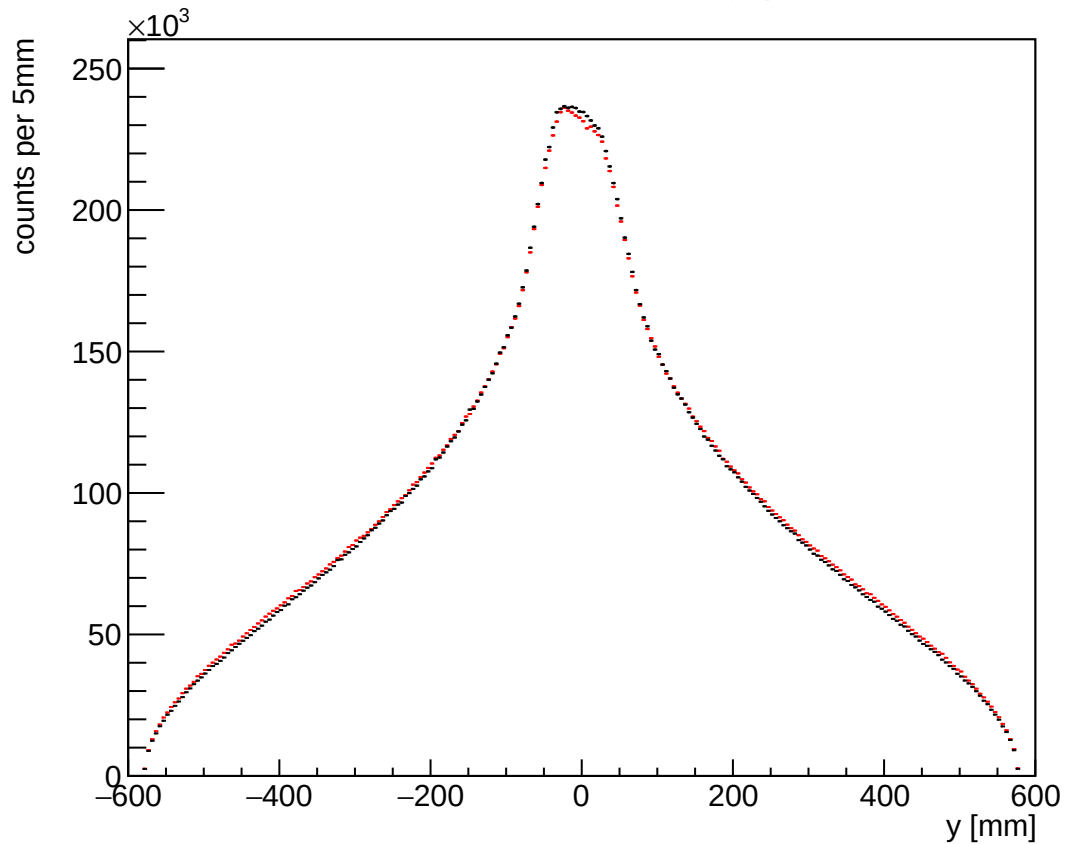
Entries	2.414217e+07
Mean	-0.3705
Std Dev	242.7
Underflow	0
Overflow	0
Integral	2.246e+07

Mu2e mu resampler: neutron y position

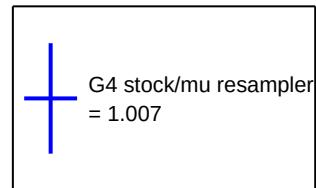
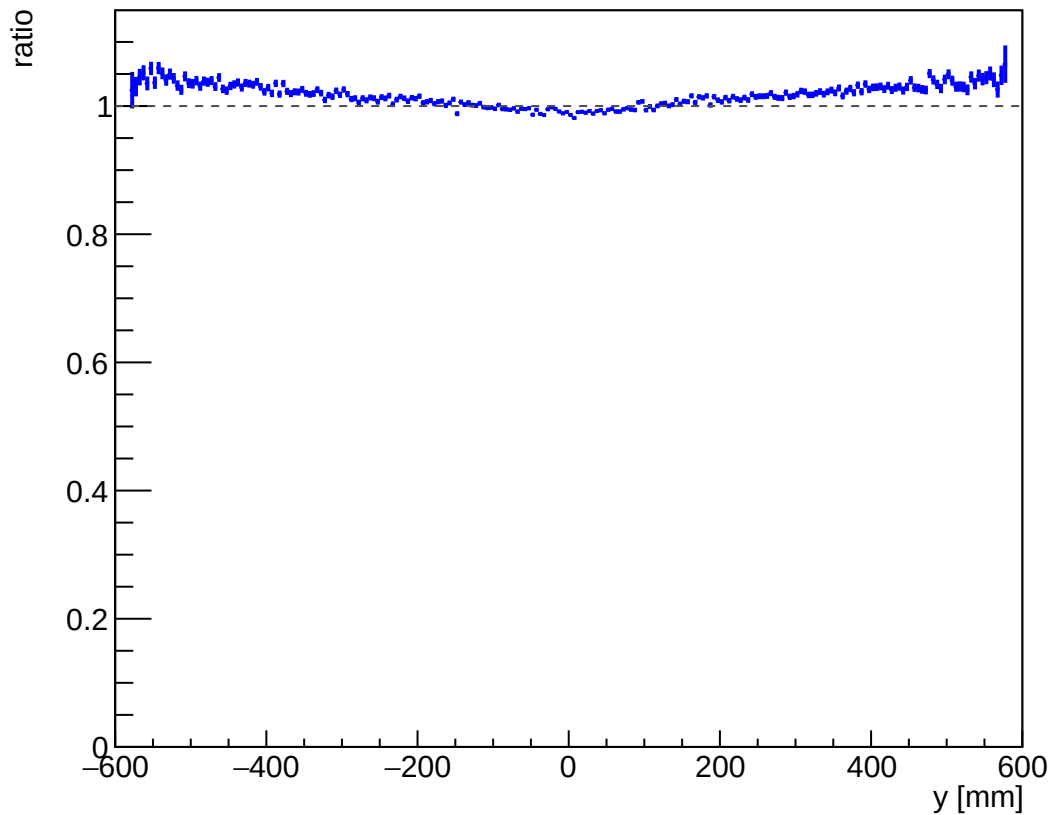


Entries	2.209432e+07
Mean	-0.4154
Std Dev	240.6
Underflow	0
Overflow	0
Integral	2.229e+07

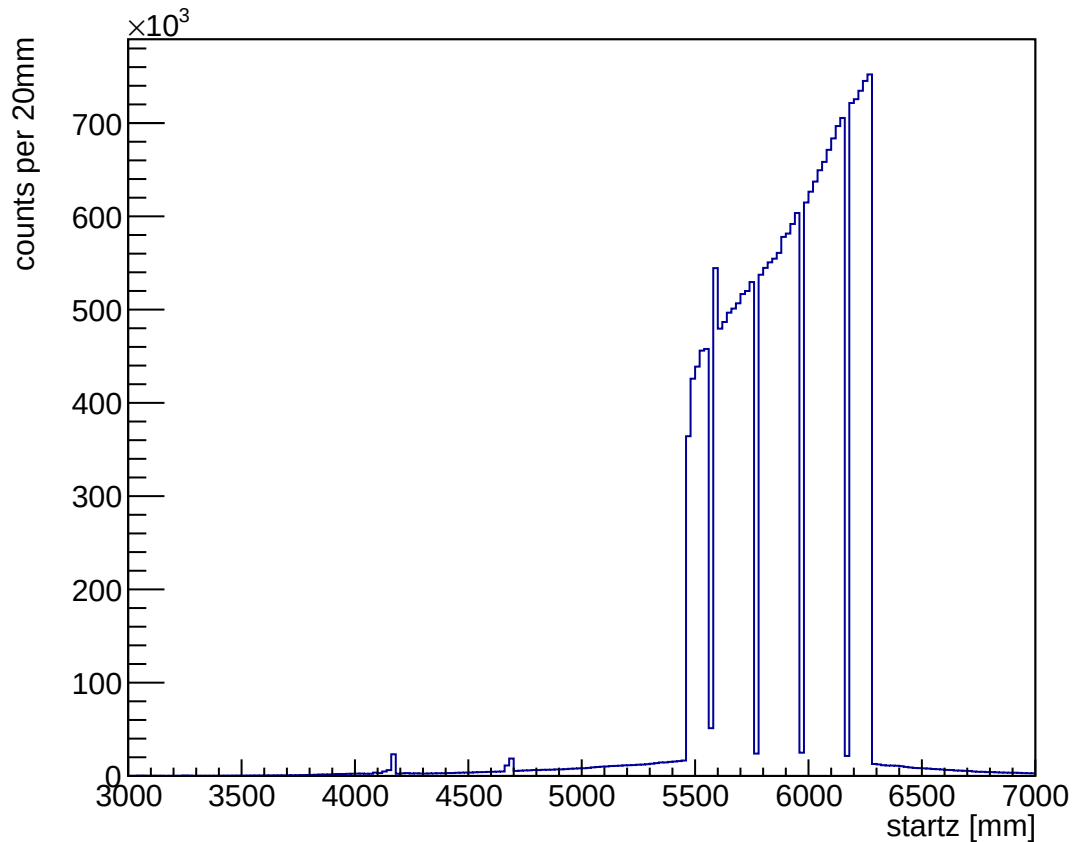
neutron y position



ratio of G4 stock / muon resampler: neutron y position

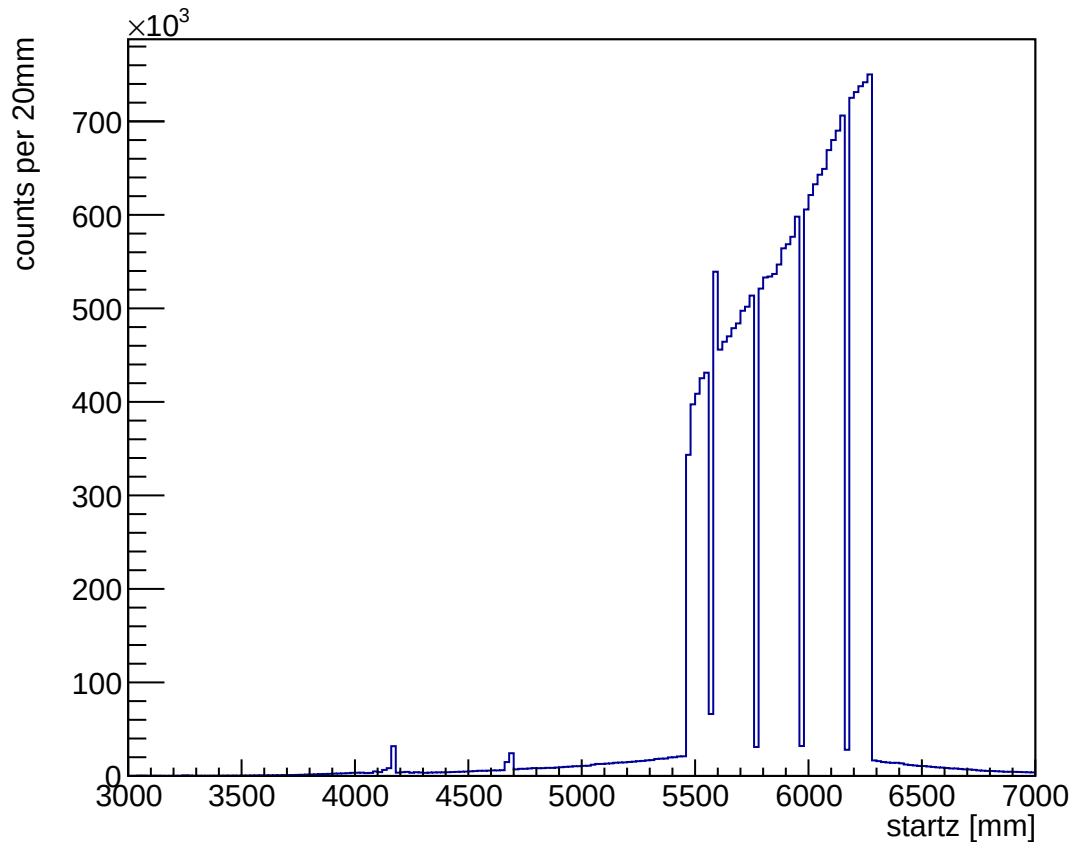


Stock G4: neutron start z



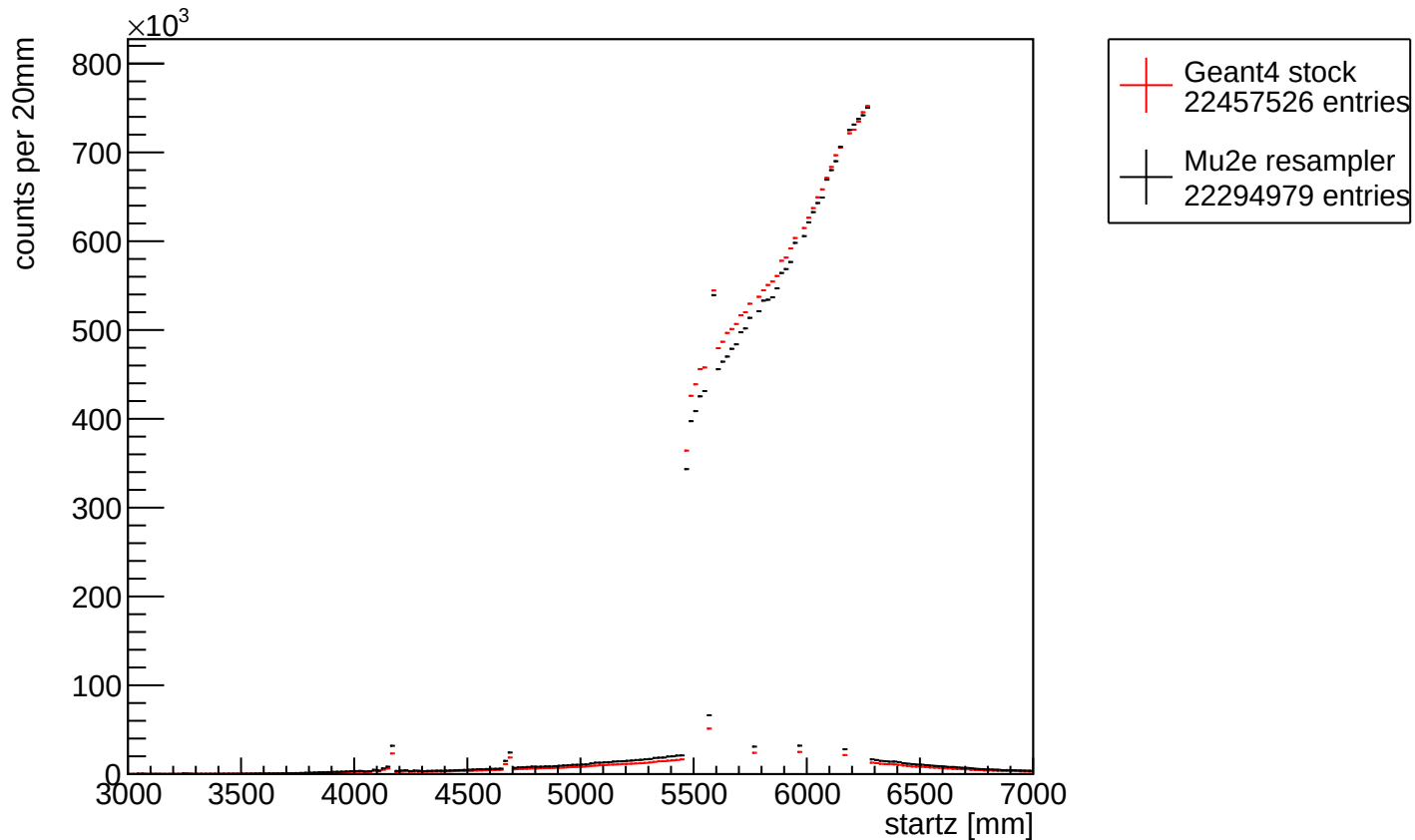
Entries	2.414217e+07
Mean	5890
Std Dev	302.2
Underflow	1123
Overflow	7.594e+04
Integral	2.238e+07

Mu2e mu resampler: neutron start z

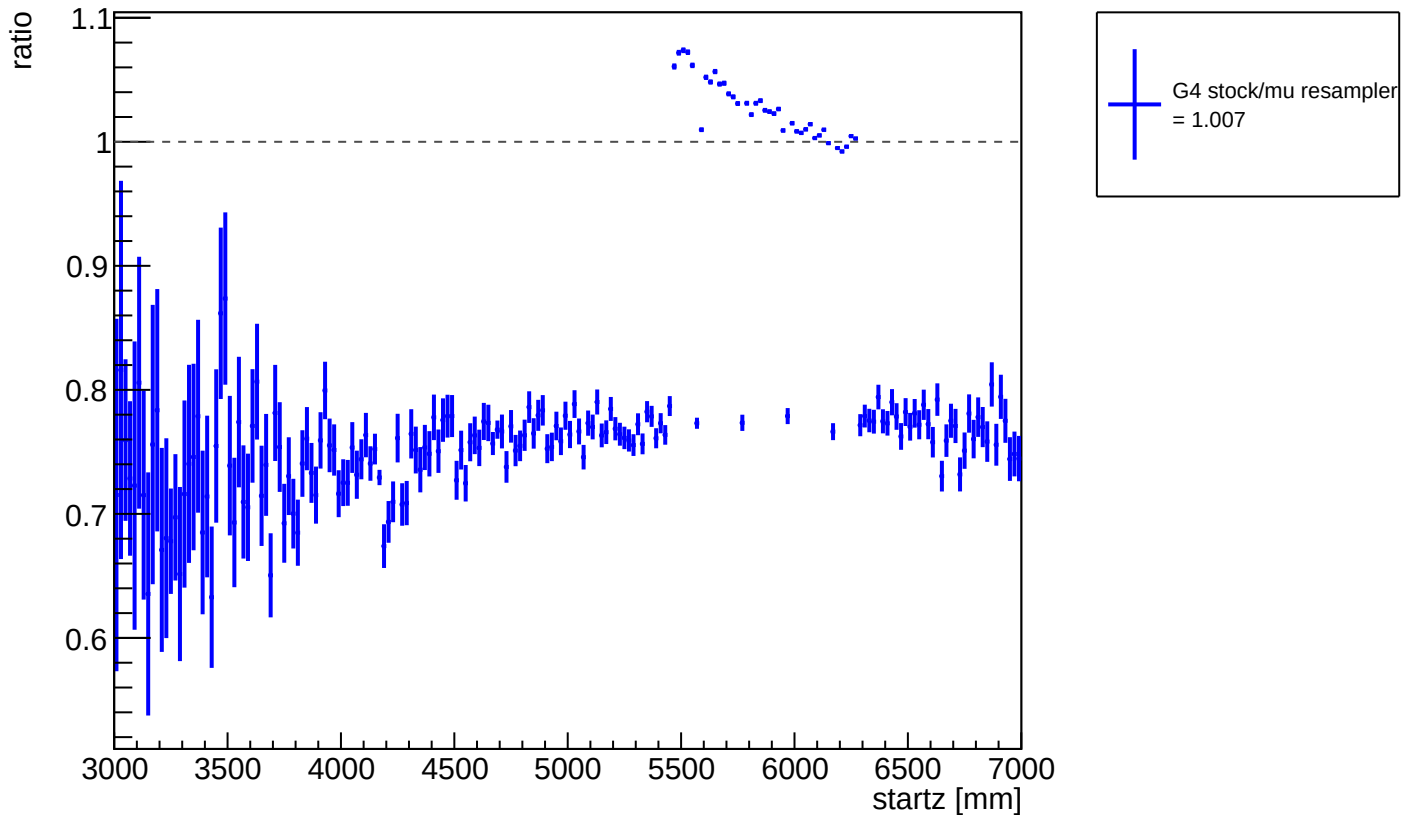


Entries	2.209432e+07
Mean	5888
Std Dev	321.7
Underflow	1517
Overflow	1.034e+05
Integral	2.219e+07

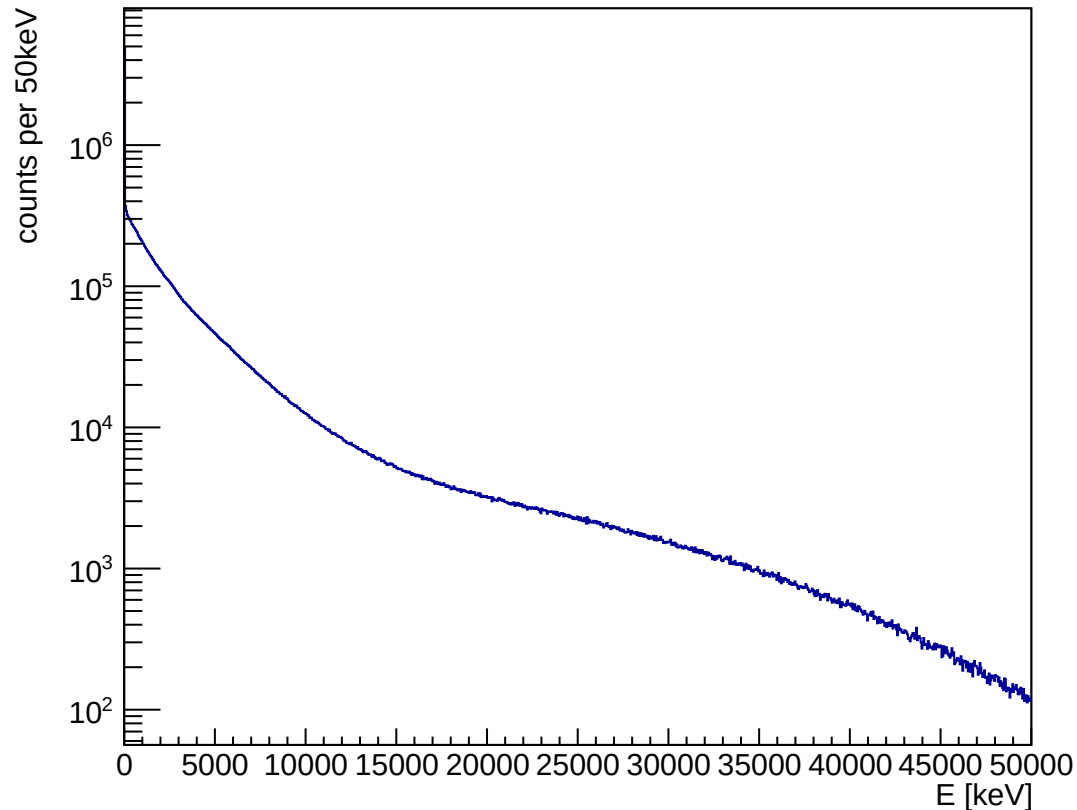
neutron start z



ratio of G4 stock / muon resampler: neutron start z

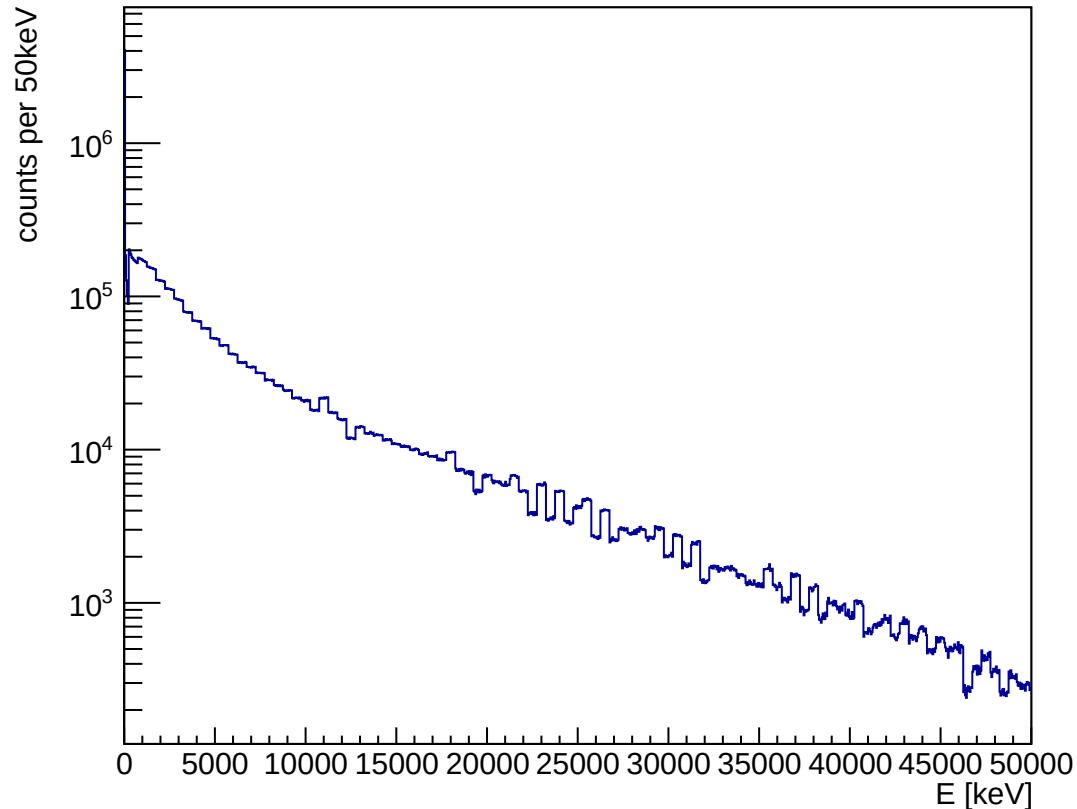


Stock G4: neutron kin. energy



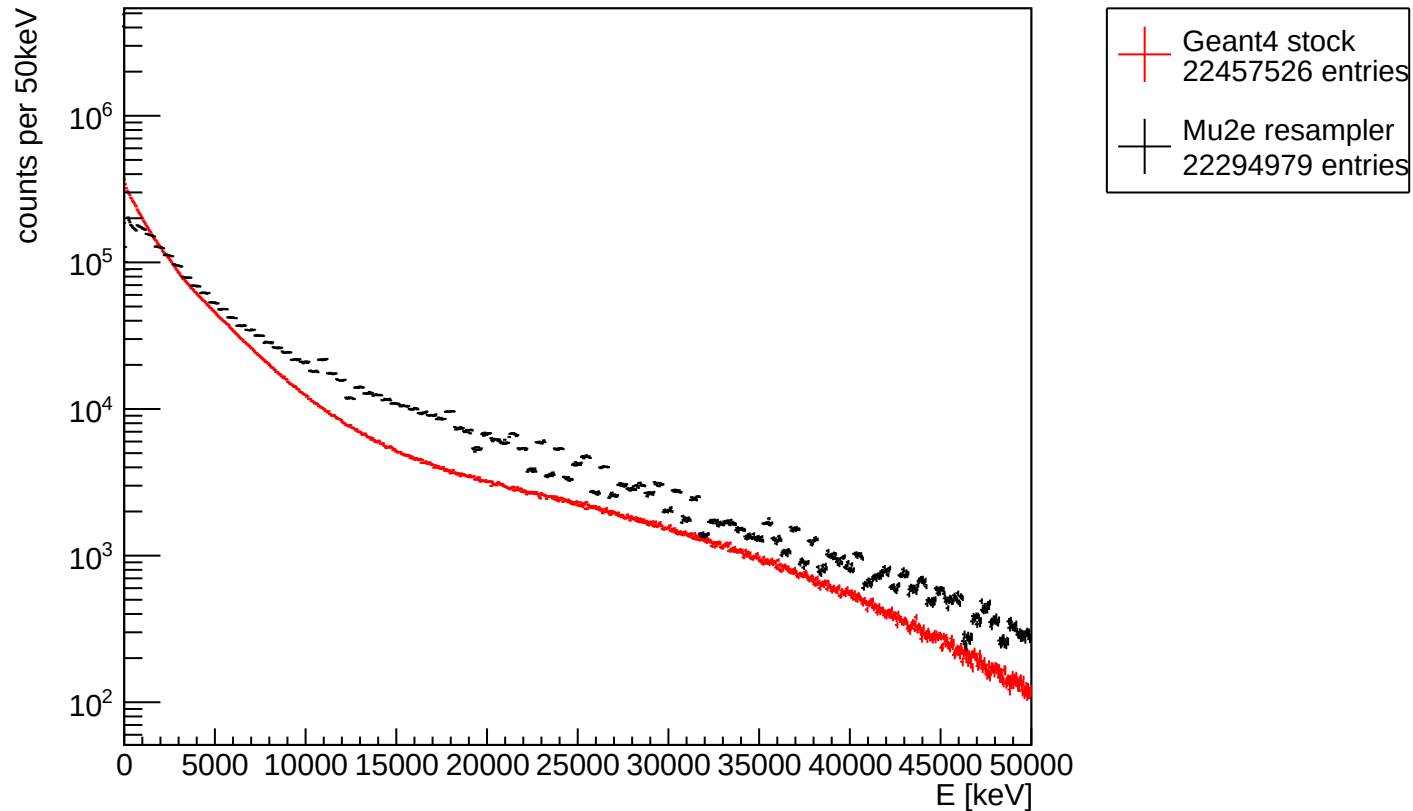
Entries	2.414217e+07
Mean	3438
Std Dev	5907
Underflow	0
Overflow	1.043e+04
Integral	2.245e+07

Mu2e mu resampler: neutron kin. energy

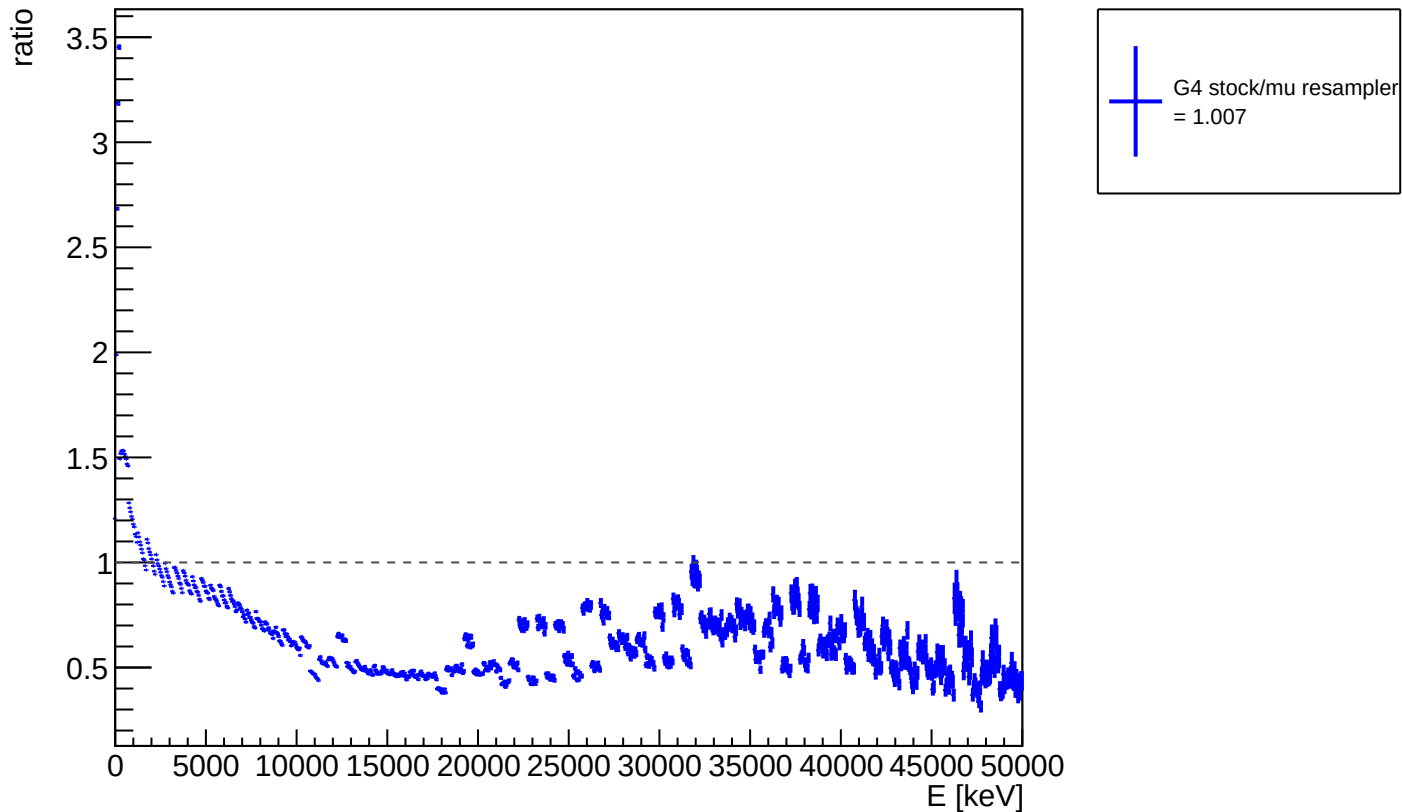


Entries	2.209432e+07
Mean	5149
Std Dev	7300
Underflow	0
Overflow	5.412e+04
Integral	2.224e+07

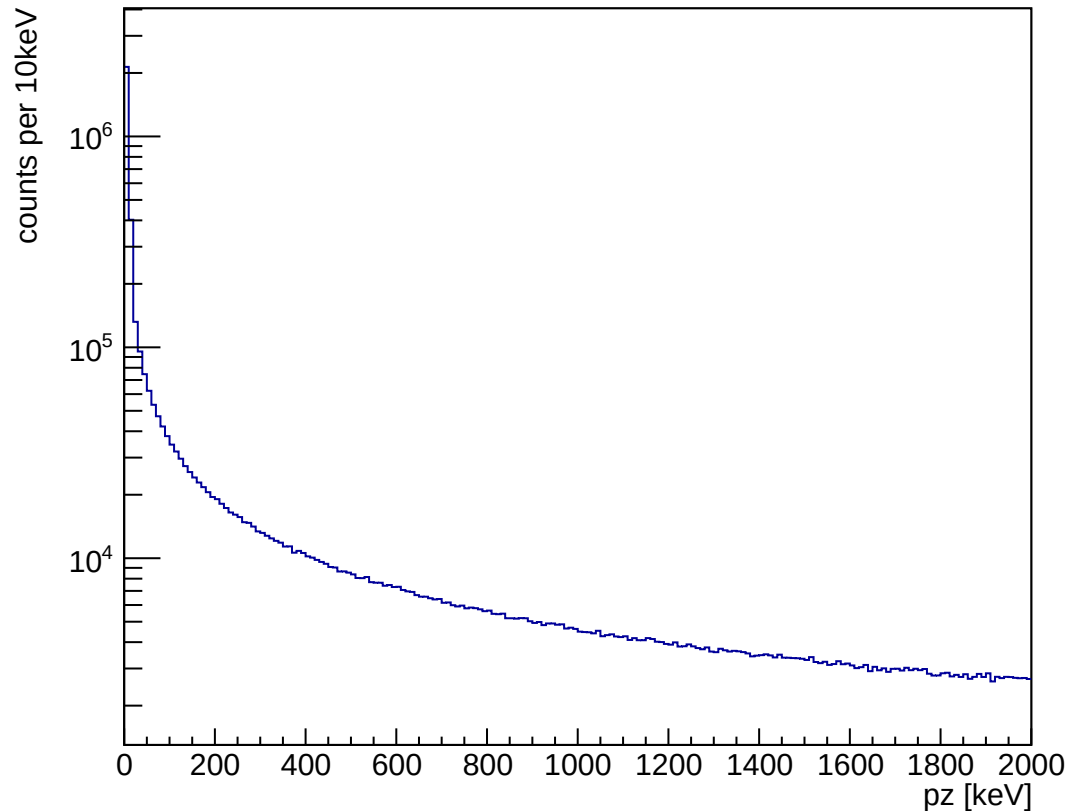
neutron kin. energy



ratio of G4 stock / muon resampler: neutron kin. energy

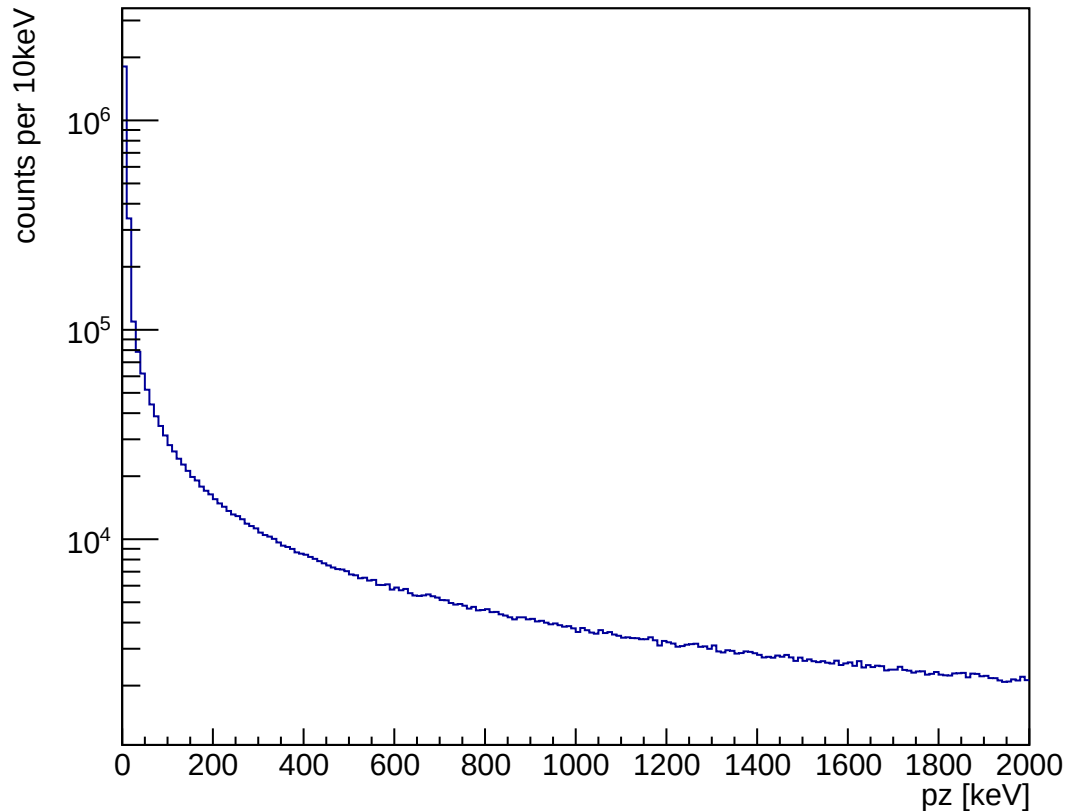


Stock G4: neutron momentum pz zoomed



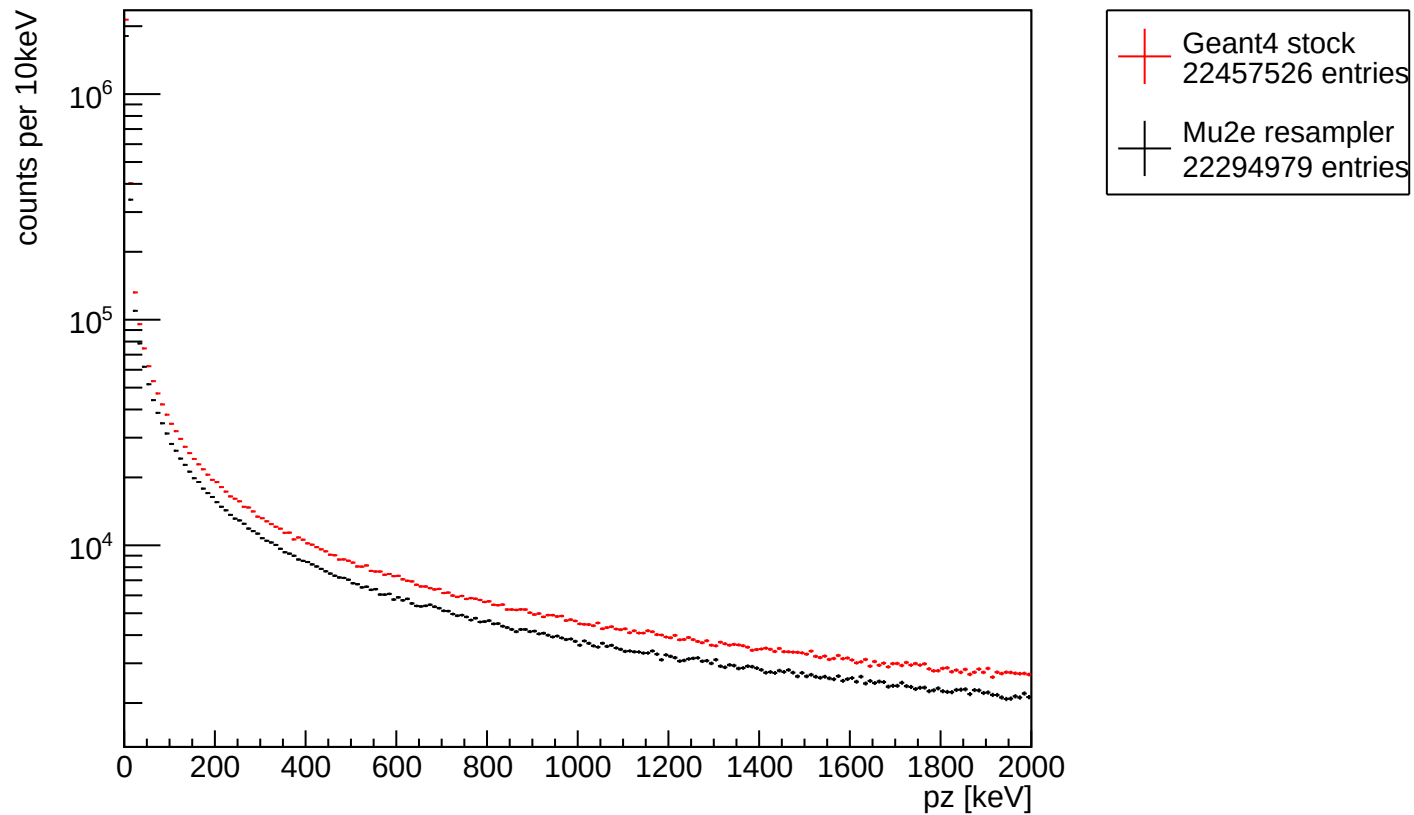
Entries	2.414217e+07
Mean	210.7
Std Dev	420.2
Underflow	0
Overflow	1.81e+07
Integral	4.357e+06

Mu2e mu resampler: neutron momentum pz zoomed

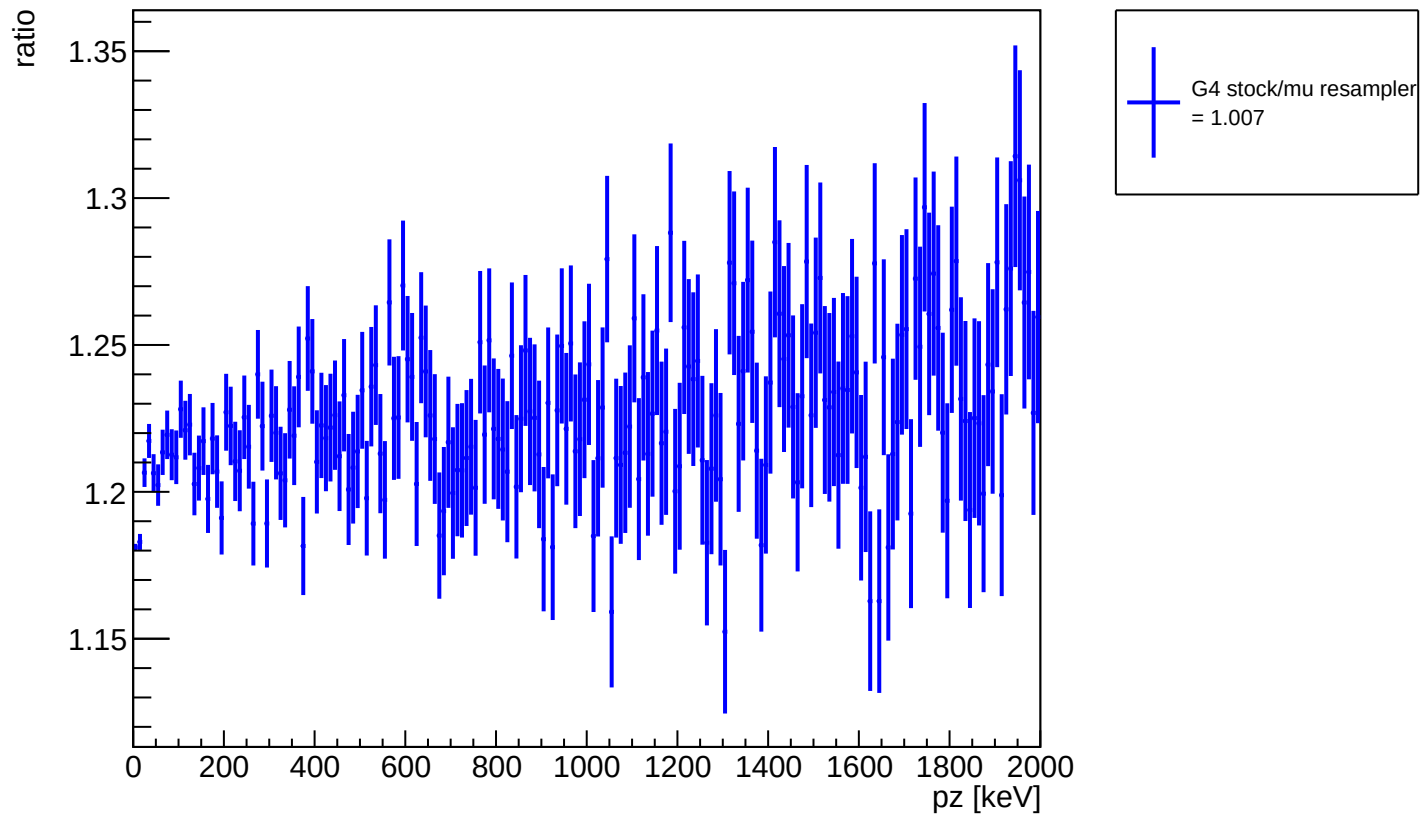


Entries	2.209432e+07
Mean	205.5
Std Dev	415.1
Underflow	0
Overflow	1.865e+07
Integral	3.642e+06

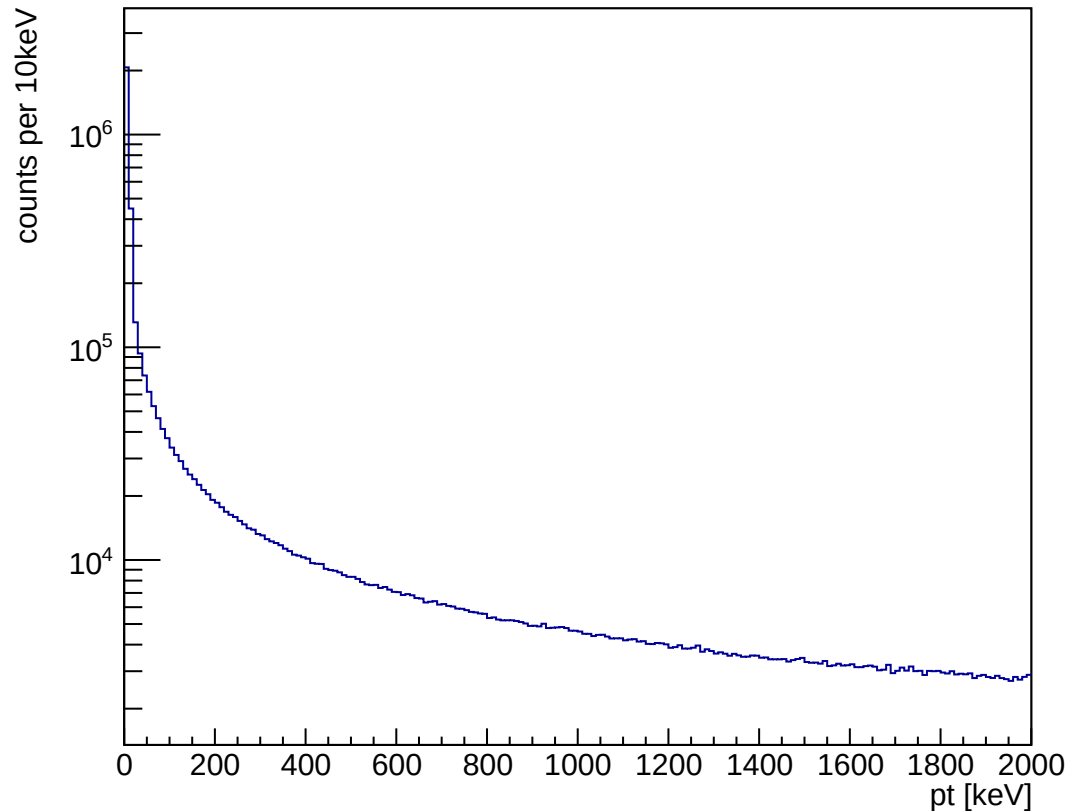
neutron momentum pz zoomed



ratio of G4 stock / muon resampler: neutron momentum pz zoomed

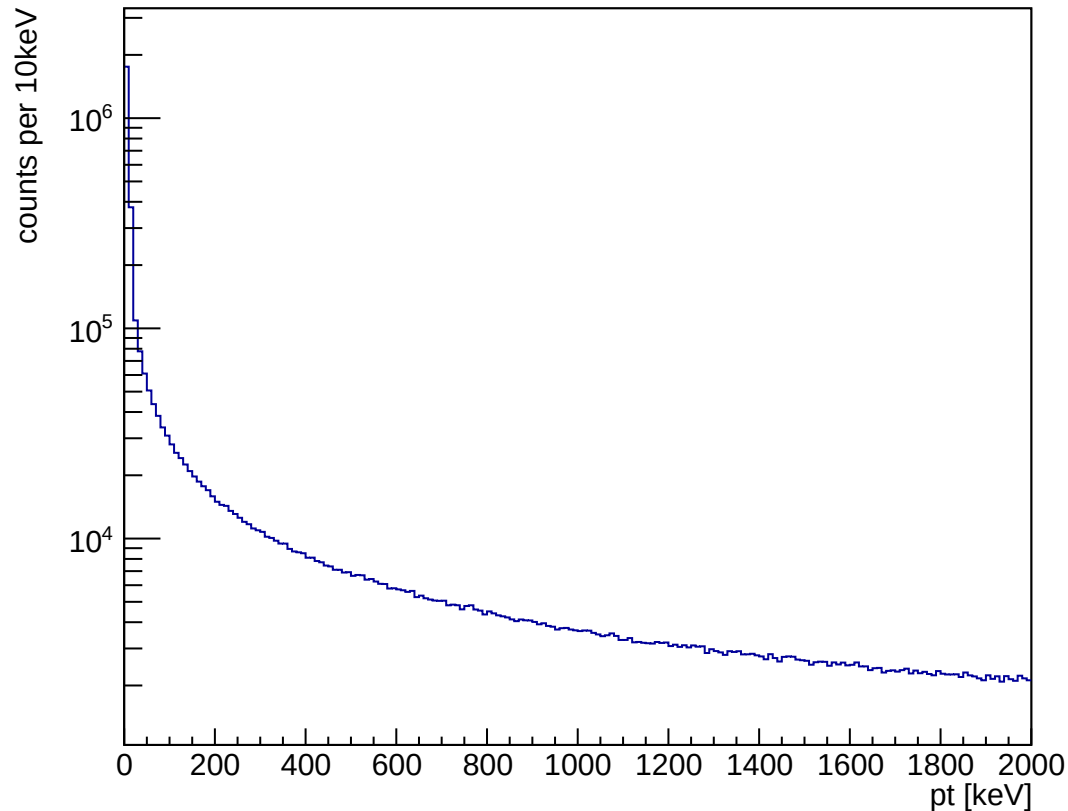


Stock G4: neutron transverse momentum pT zoomed



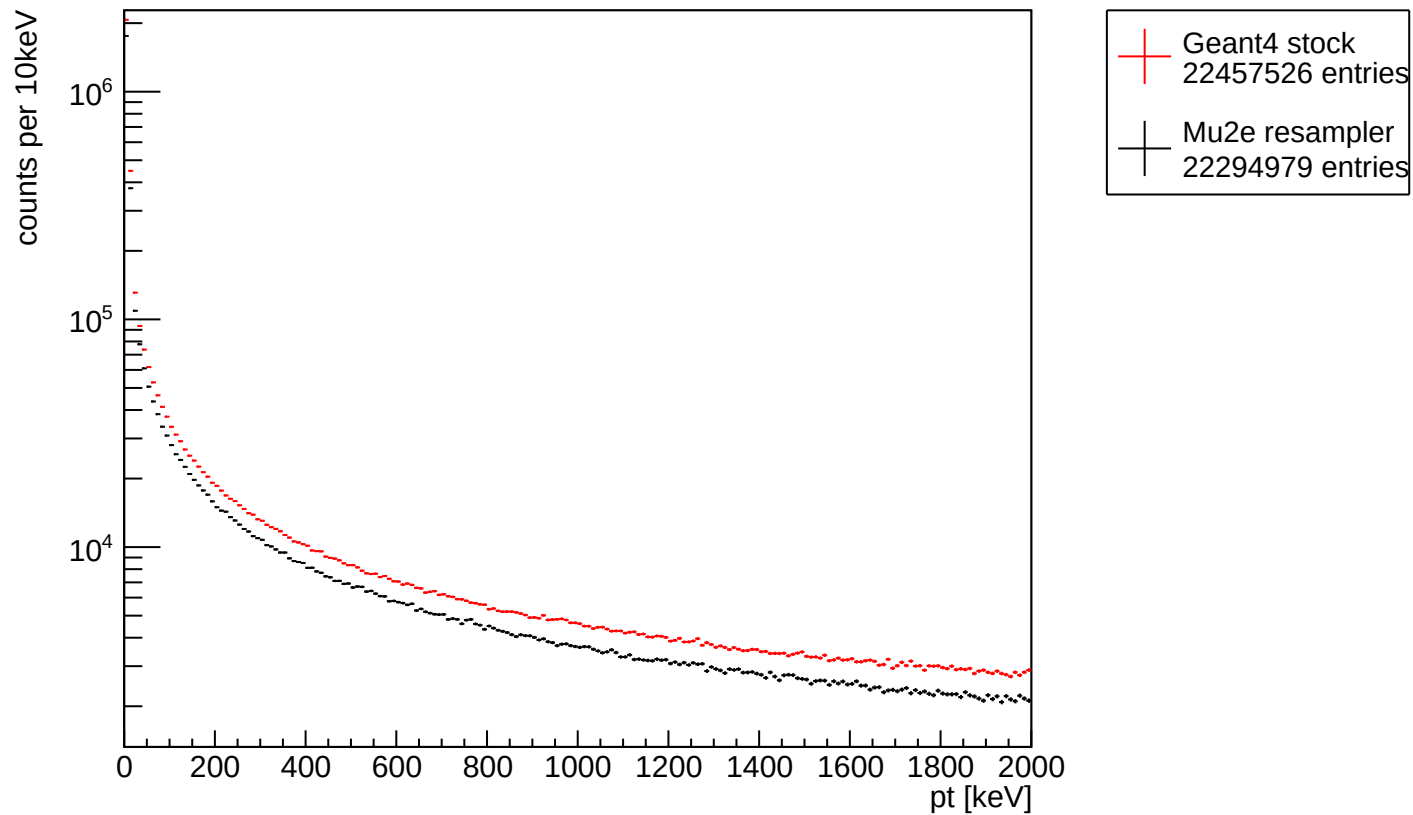
Entries	2.414217e+07
Mean	213.3
Std Dev	424.8
Underflow	0
Overflow	1.814e+07
Integral	4.319e+06

Mu2e mu resampler: neutron transverse momentum pT zoomed

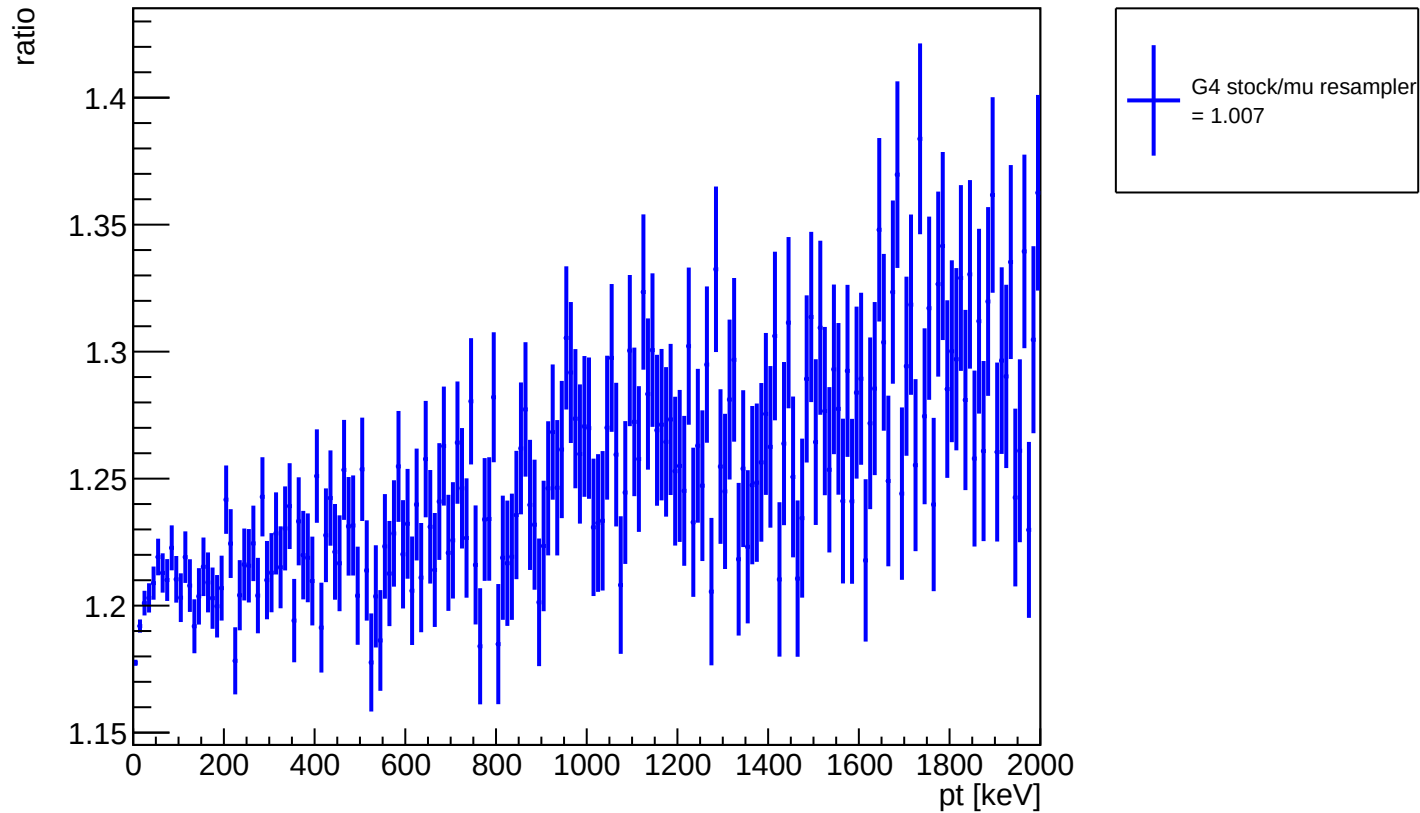


Entries	2.209432e+07
Mean	204.1
Std Dev	414.3
Underflow	0
Overflow	1.869e+07
Integral	3.602e+06

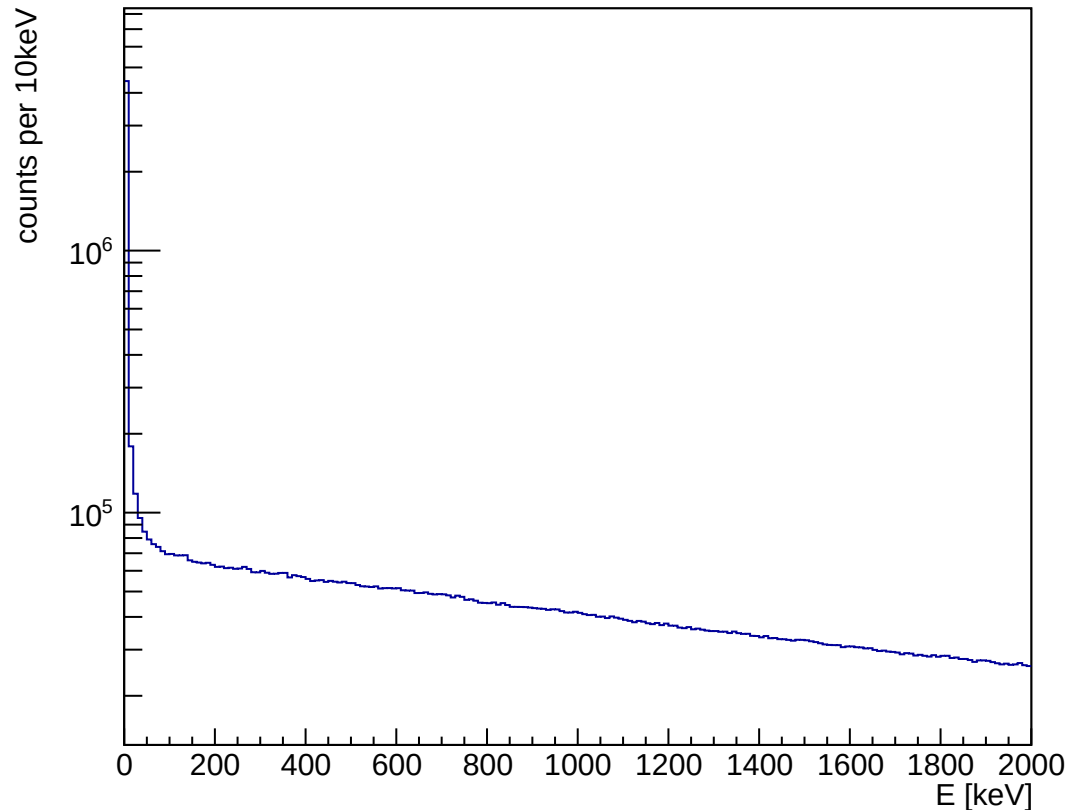
neutron transverse momentum pT zoomed



ratio of G4 stock / muon resampler: neutron transverse momentum pT zoomed

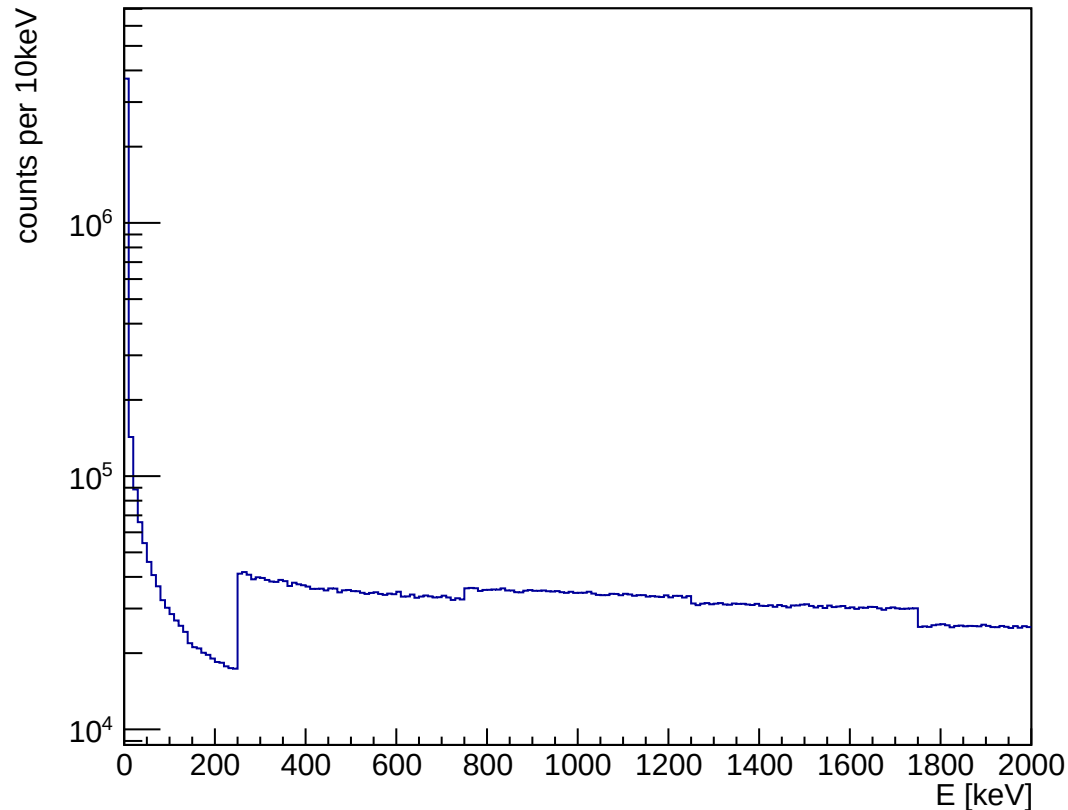


Stock G4: neutron kin. energy zoomed



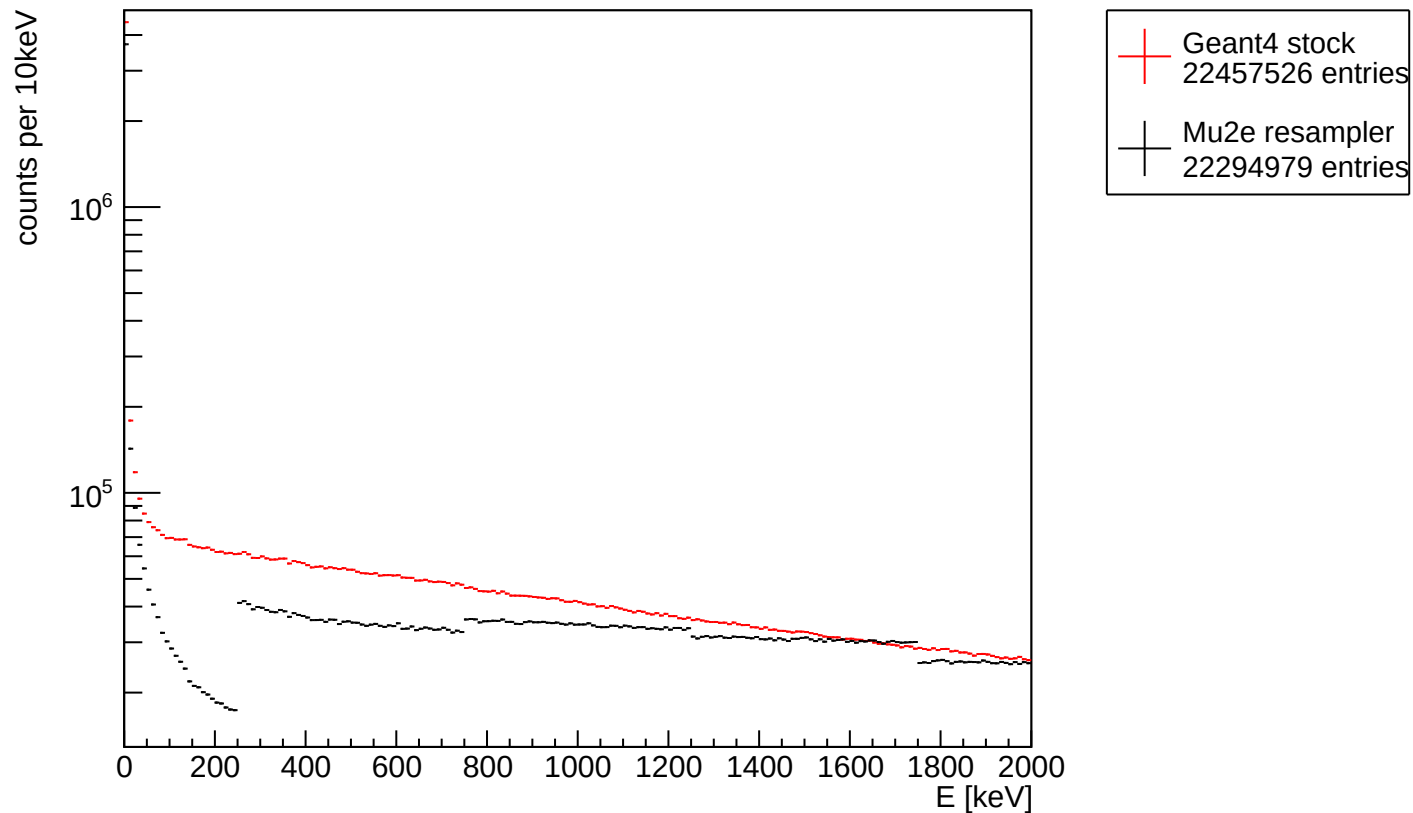
Entries	2.414217e+07
Mean	545.5
Std Dev	604.5
Underflow	0
Overflow	9.148e+06
Integral	1.331e+07

Mu2e mu resampler: neutron kin. energy zoomed

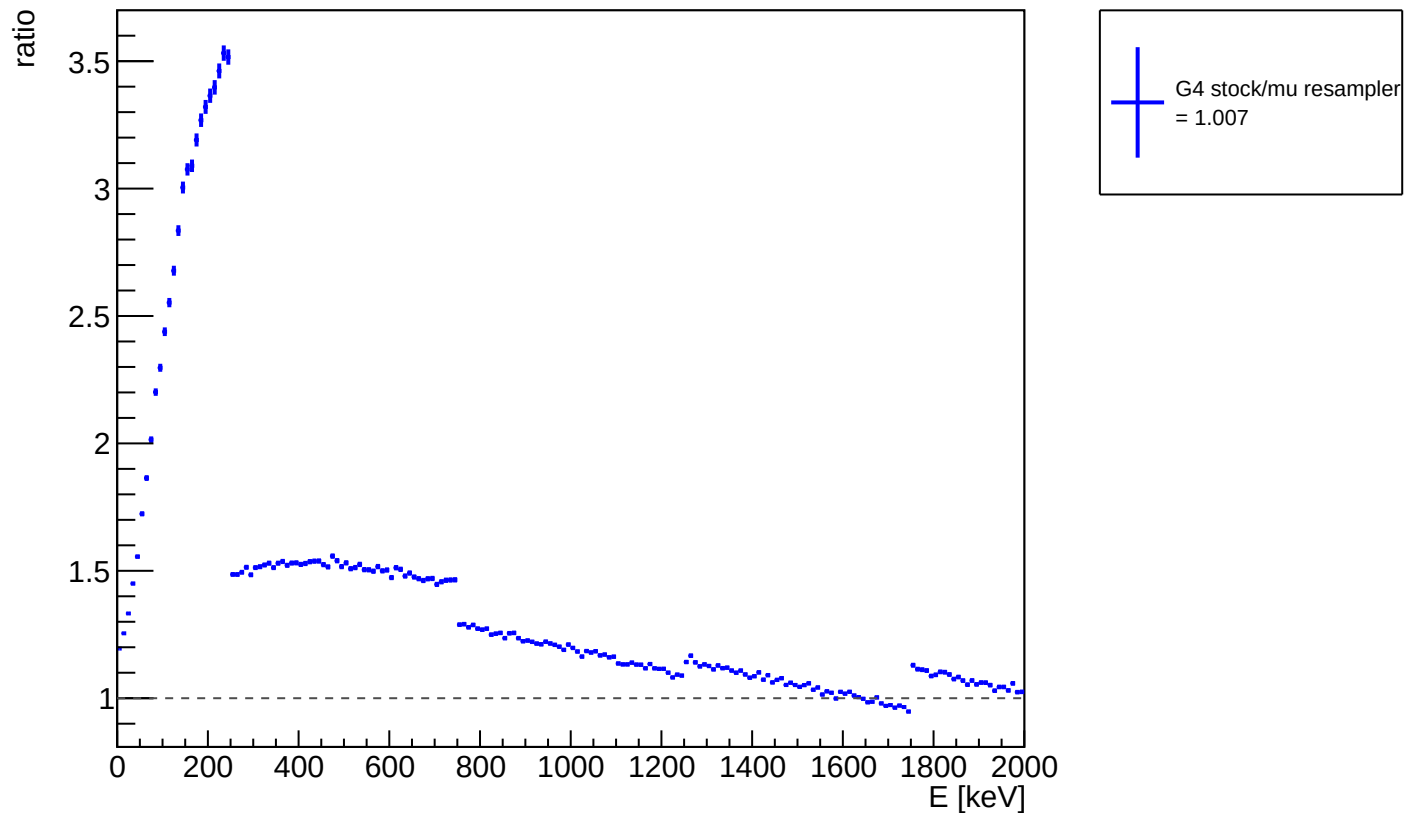


Entries	2.209432e+07
Mean	602
Std Dev	642.8
Underflow	0
Overflow	1.204e+07
Integral	1.026e+07

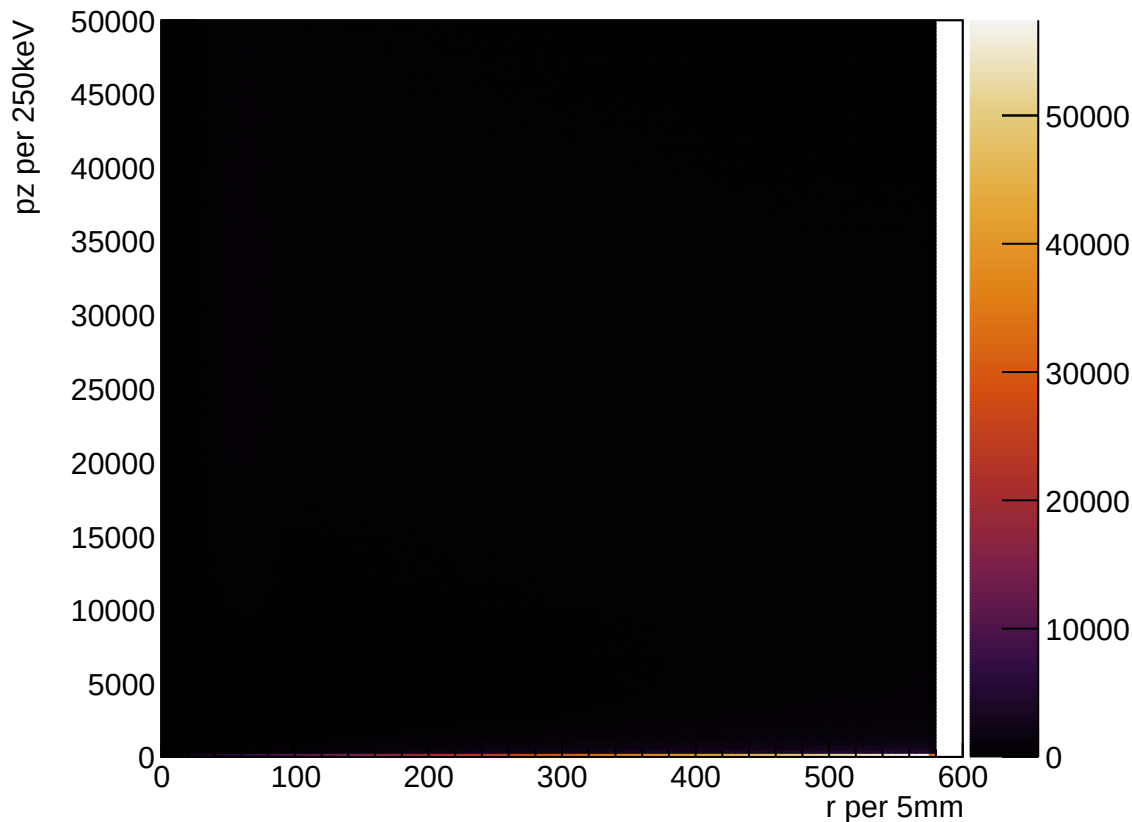
neutron kin. energy zoomed



ratio of G4 stock / muon resampler: neutron kin. energy zoomed

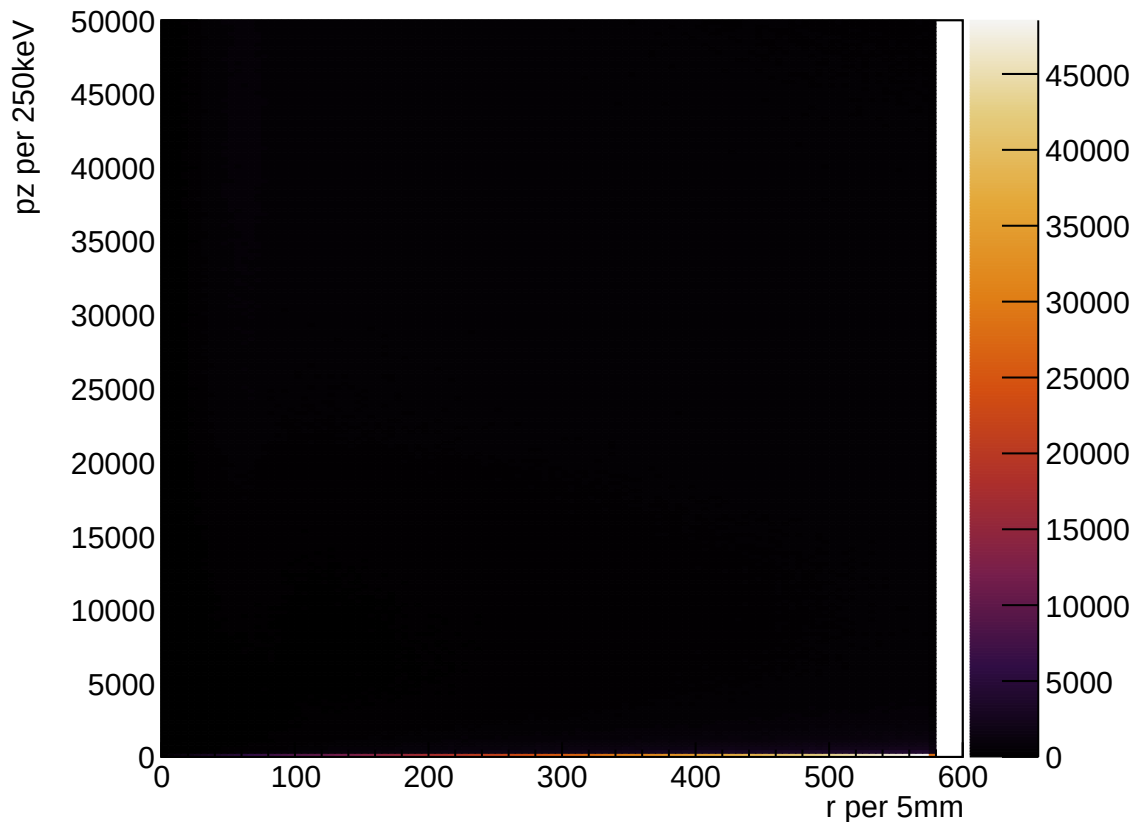


Stock G4: neutron pz vs r



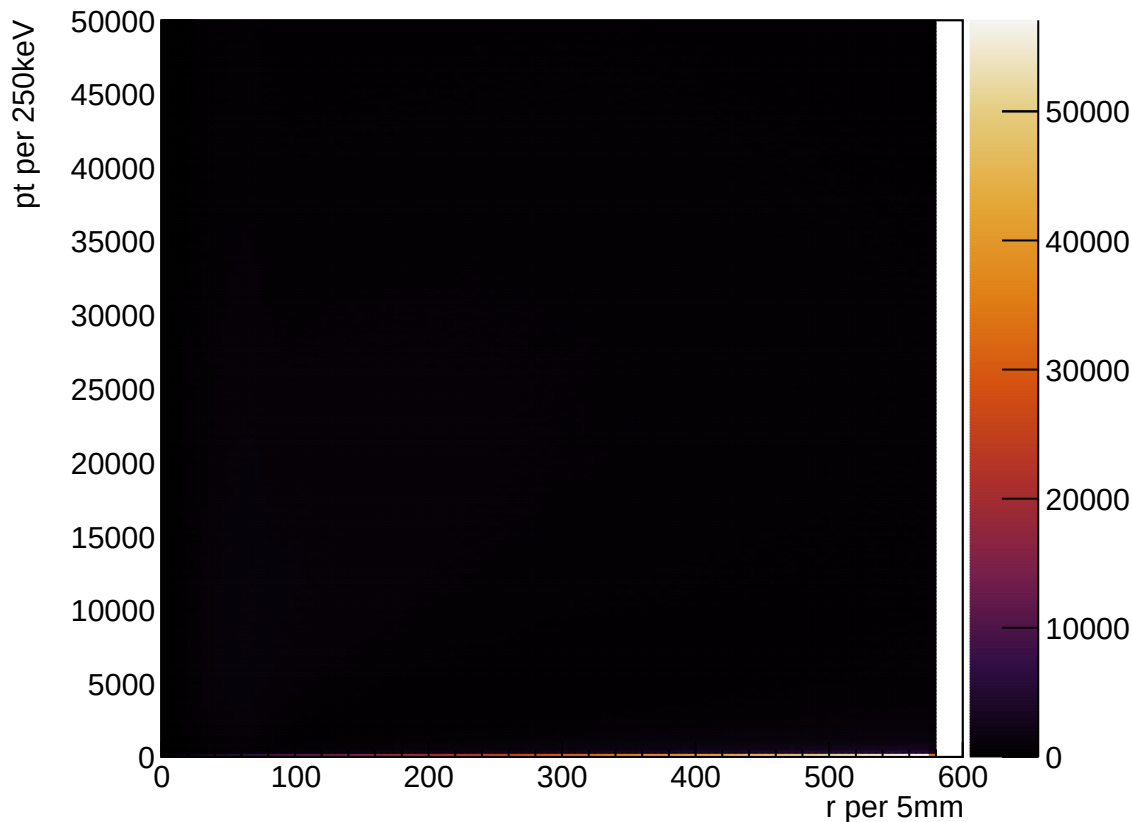
Entries	2.414217e+07		
Mean x	325.5		
Mean y	1.846e+04		
Std Dev x	160.1		
Std Dev y	1.619e+04		
Integral	1.494e+07		
0	7.51381e+06	0	0
0	1.49437e+07	0	0
0	0	0	0

Mu2e mu resampler: neutron pz vs r



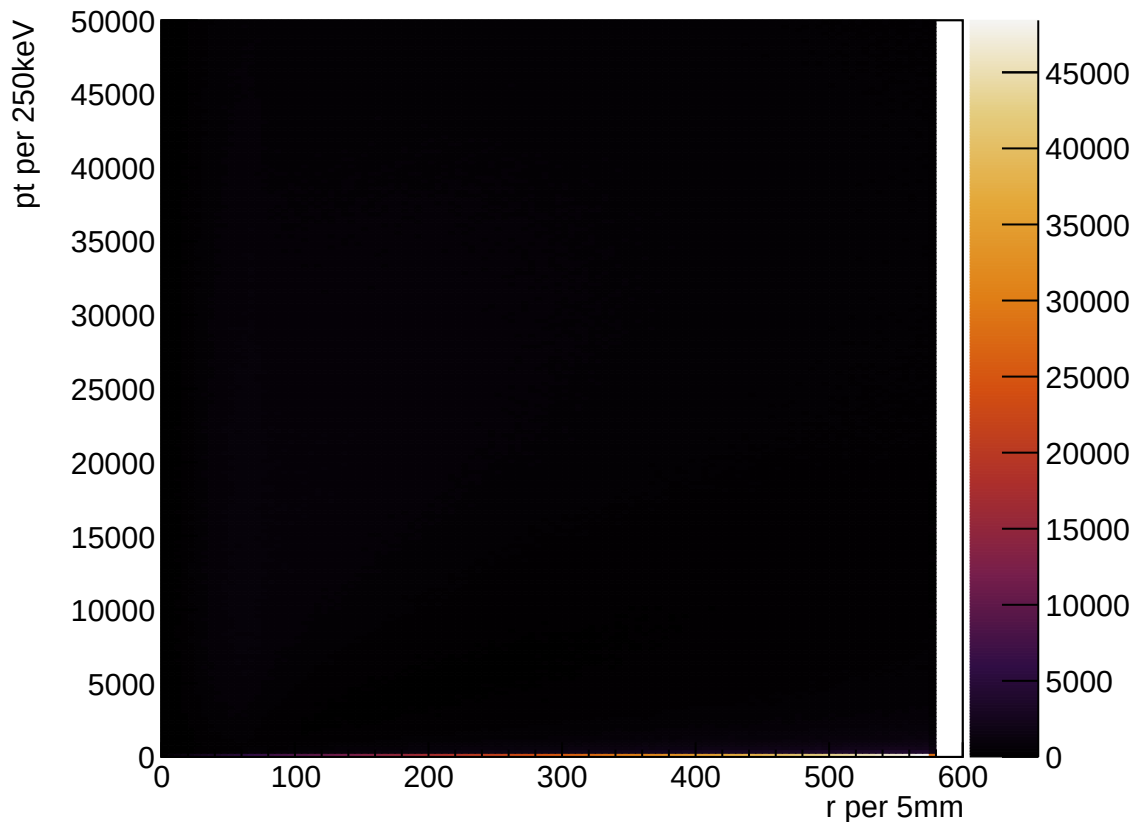
Entries	2.209432e+07		
Mean x	332.1		
Mean y	1.975e+04		
Std Dev x	158.9		
Std Dev y	1.695e+04		
Integral	1.219e+07		
0	1.01001e+07	0	
0	1.21949e+07	0	
0	0	0	

Stock G4: neutron pt vs r



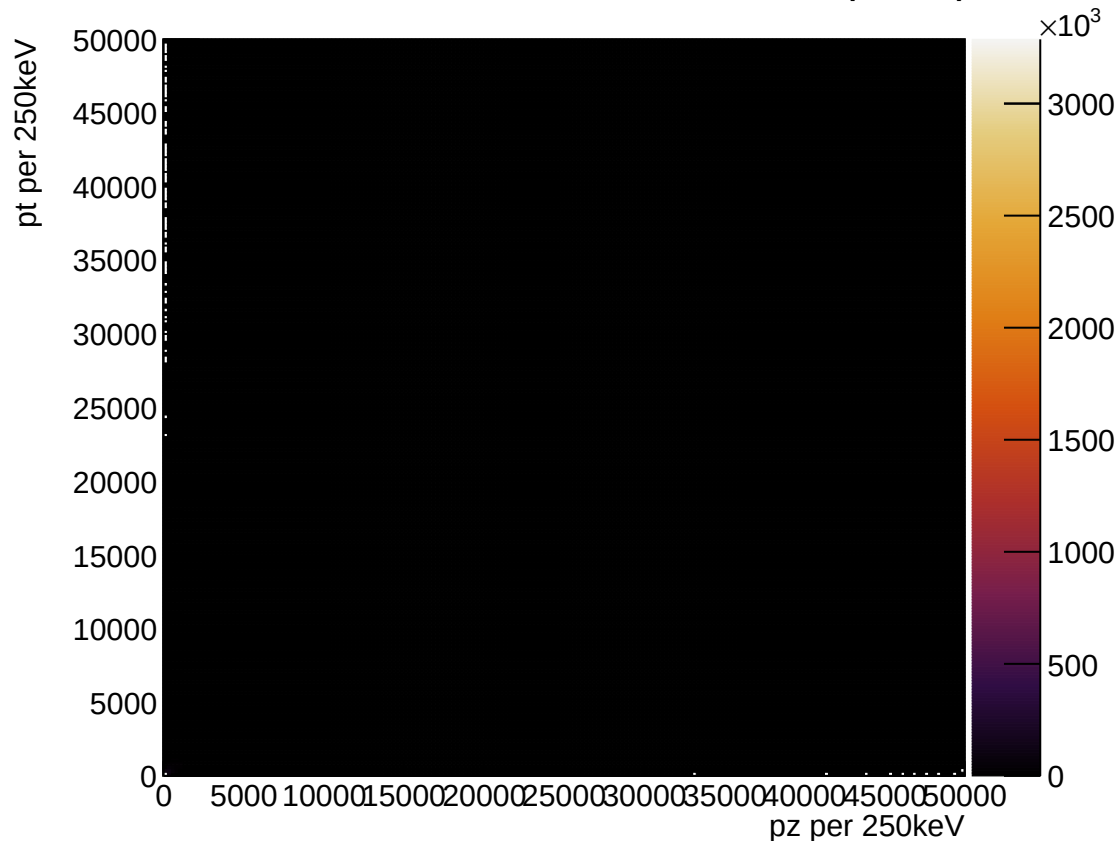
Entries	2.414217e+07		
Mean x	303.3		
Mean y	1.838e+04		
Std Dev x	162.7		
Std Dev y	1.566e+04		
Integral	1.628e+07		
0	6.17275e+06	0	0
0	1.62848e+07	0	0
0	0	0	0

Mu2e mu resampler: neutron pt vs r



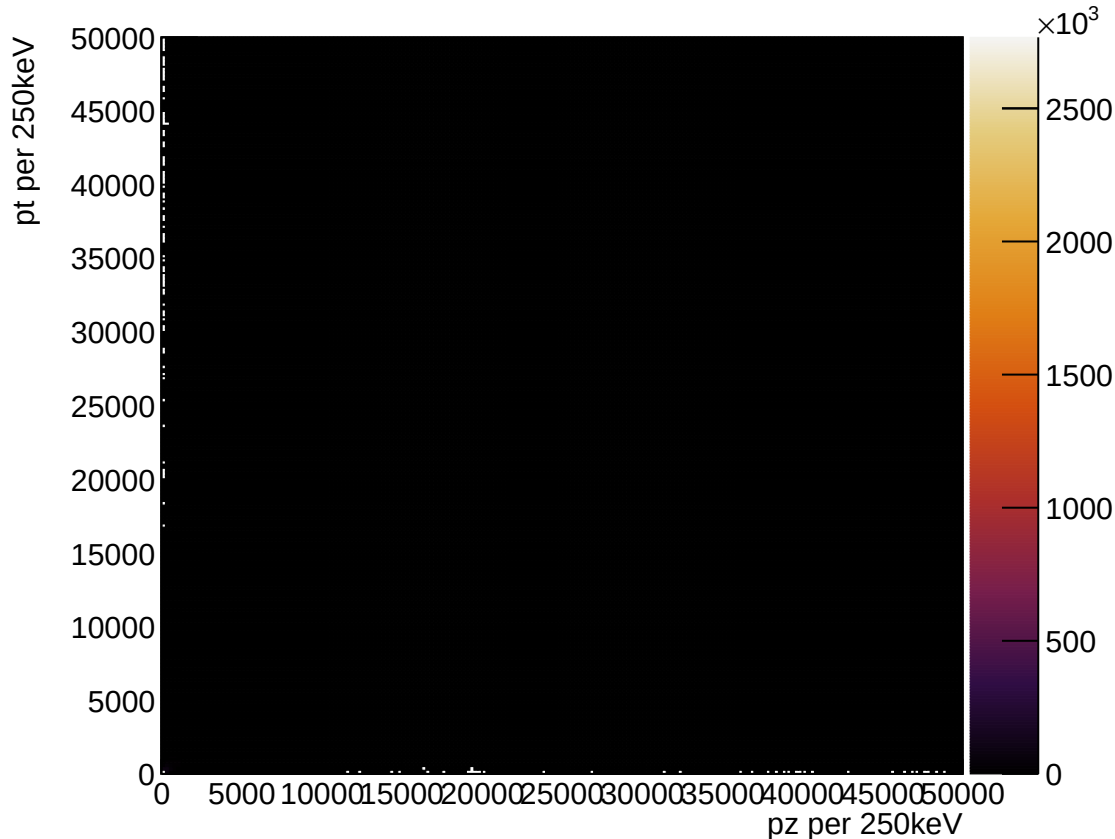
Entries	2.209432e+07		
Mean x	300.2		
Mean y	2.021e+04		
Std Dev x	163.5		
Std Dev y	1.634e+04		
Integral	1.374e+07		
0	8.5562e+06	0	
0	1.37388e+07	0	
0	0	0	

Stock G4: neutron pt vs pz



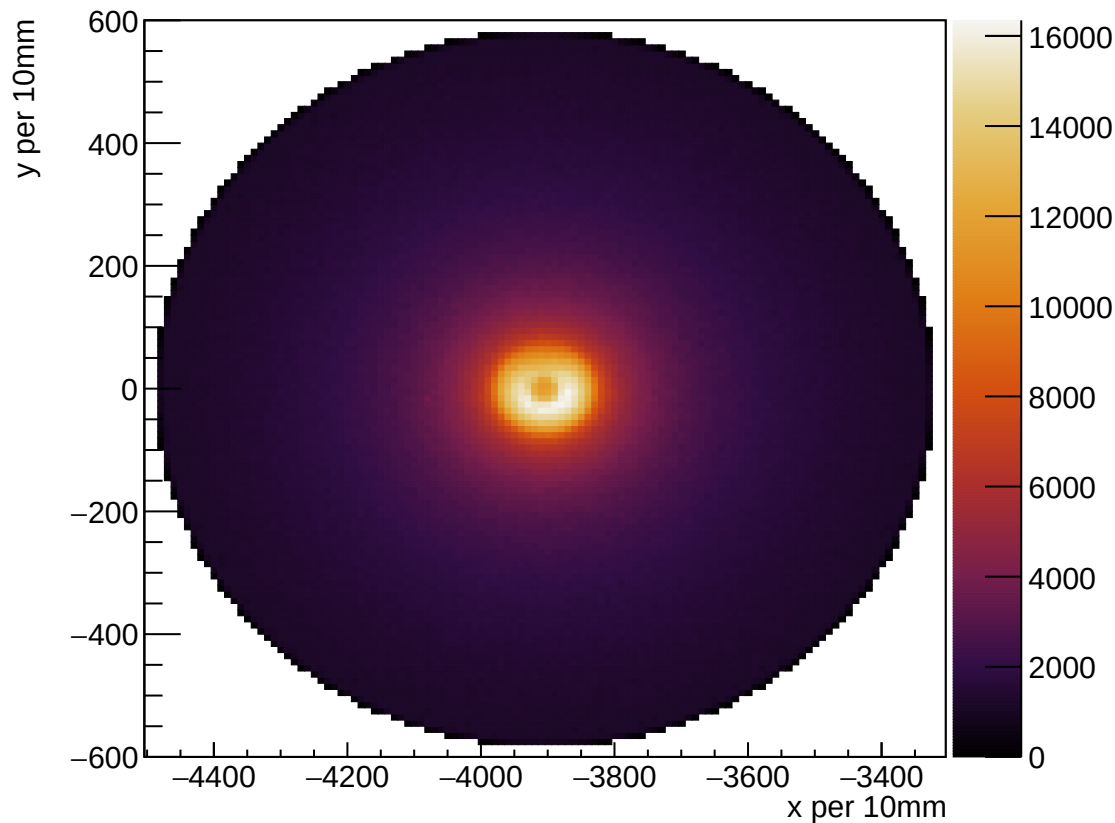
Entries	2.414217e+07	
Mean x	1.641e+04	
Mean y	1.504e+04	
Std Dev x	1.581e+04	
Std Dev y	1.479e+04	
Integral	1.287e+07	
0	2.073e+06	4.09975e+06
0	1.28707e+07	1.1405e+06
0	0	0

Mu2e mu resampler: neutron pt vs pz



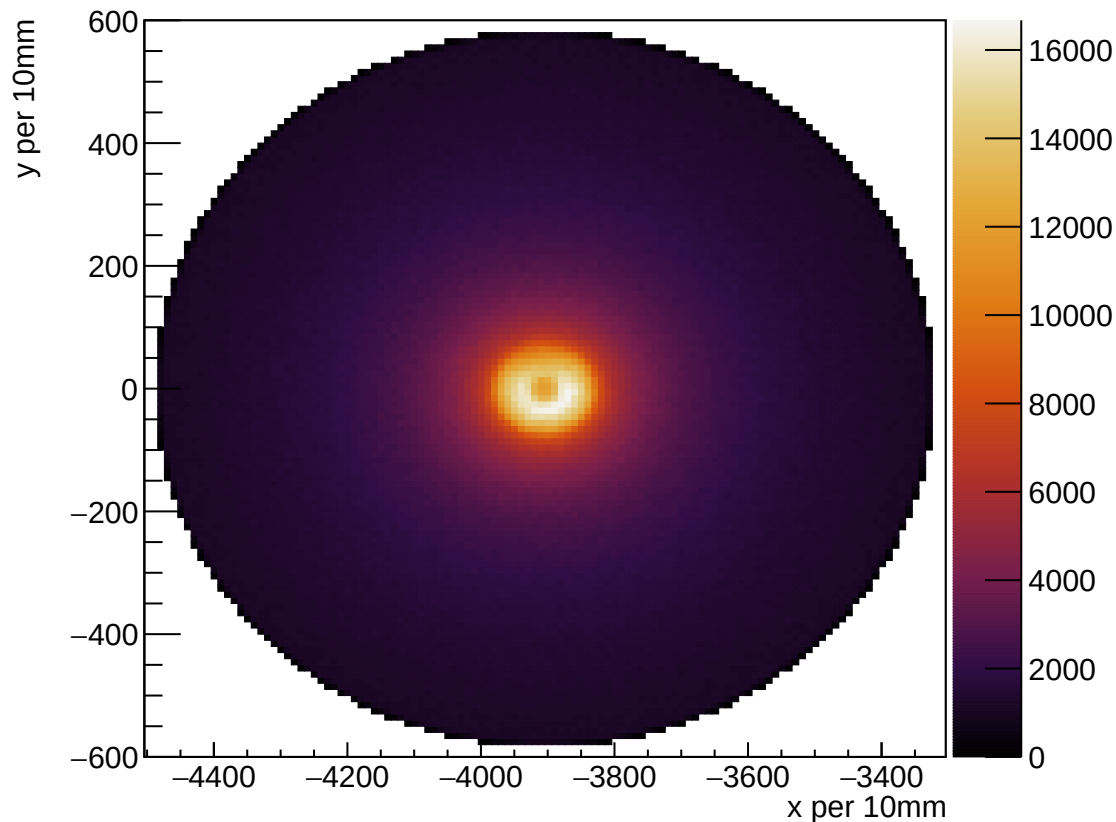
Entries	2.209432e+07	
Mean x	1.694e+04	
Mean y	1.58e+04	
Std Dev x	1.672e+04	
Std Dev y	1.572e+04	
Integral	9.85e+06	
0	2.34501e+06	6.1119e+06
0	9.84987e+05	8.8891e+06
0	0	0

Stock G4: neutron y vs x



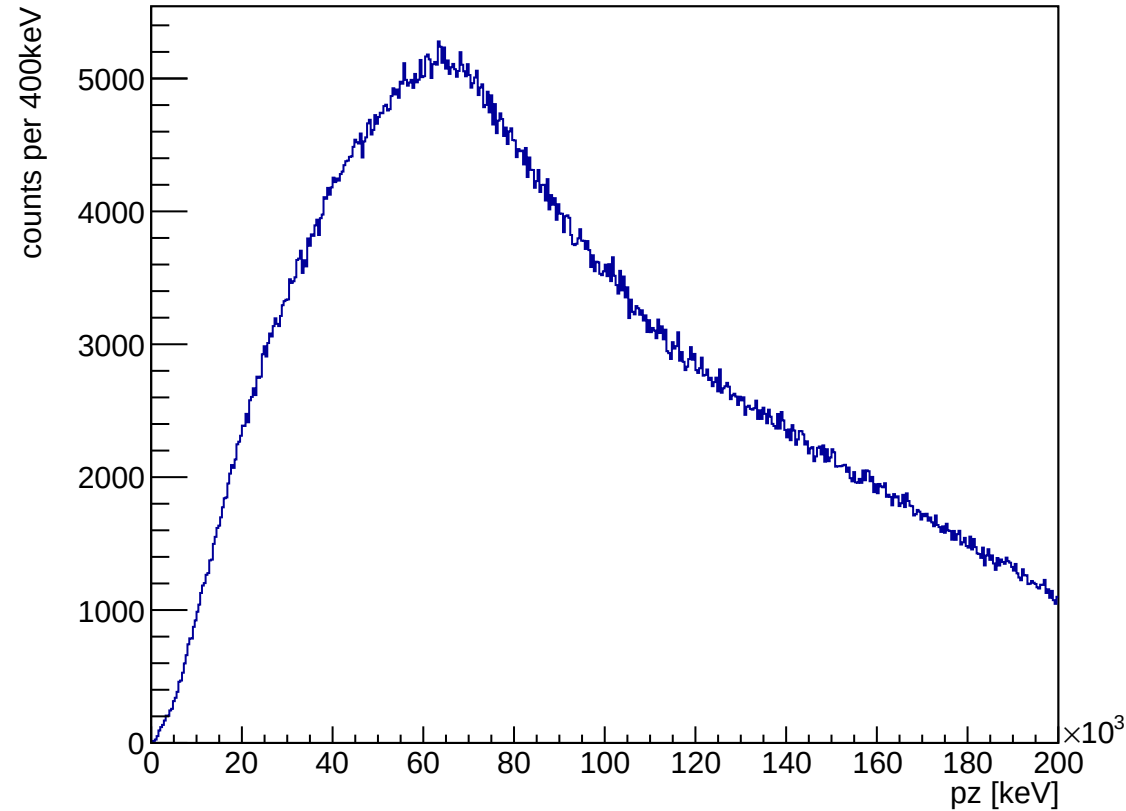
Entries	2.414217e+07		
Mean x	-3903		
Mean y	-0.3705		
Std Dev x	242.6		
Std Dev y	242.7		
Integral	2.246e+07		
0	0	0	
0	2.24575e+07	0	
0	0	0	

Mu2e mu resampler: neutron y vs x



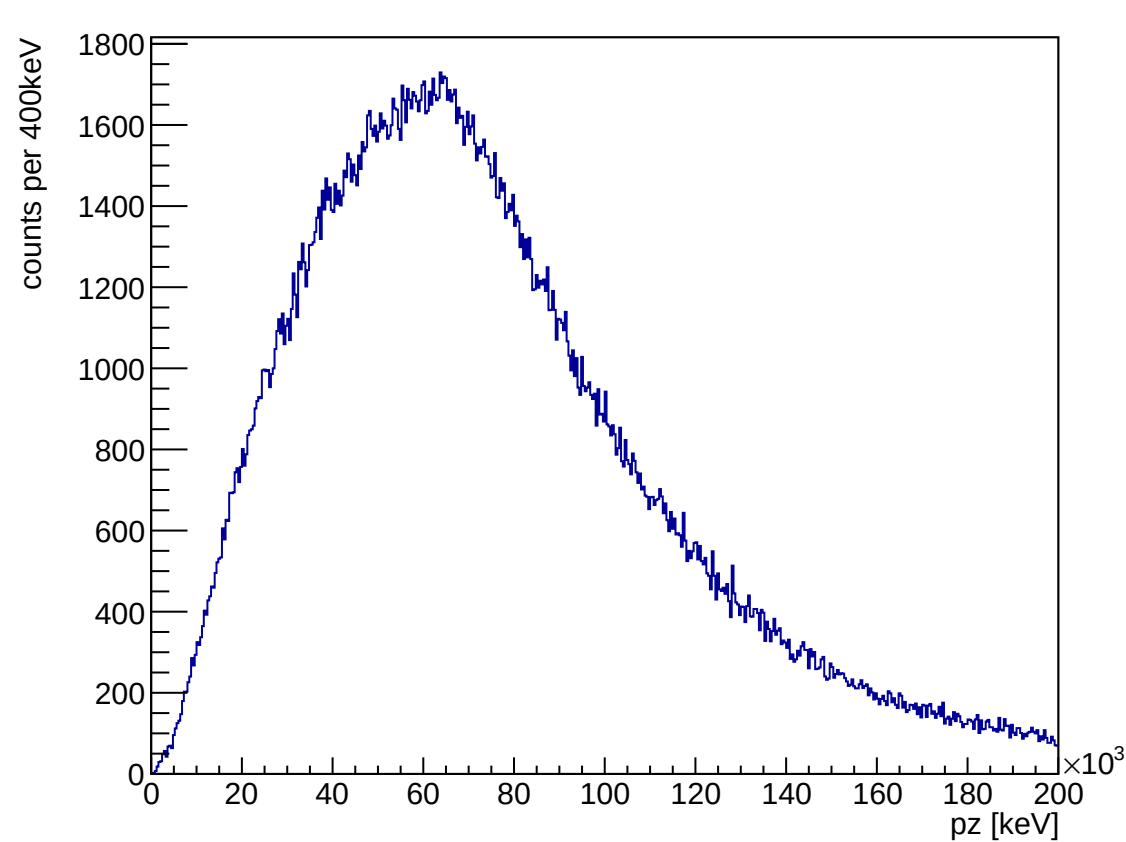
Entries	2.209432e+07		
Mean x	-3903		
Mean y	-0.4154		
Std Dev x	240.6		
Std Dev y	240.6		
Integral	2.229e+07		
0	0	0	
0	2.2295e+07	0	
0	0	0	

Stock G4: proton momentum pz



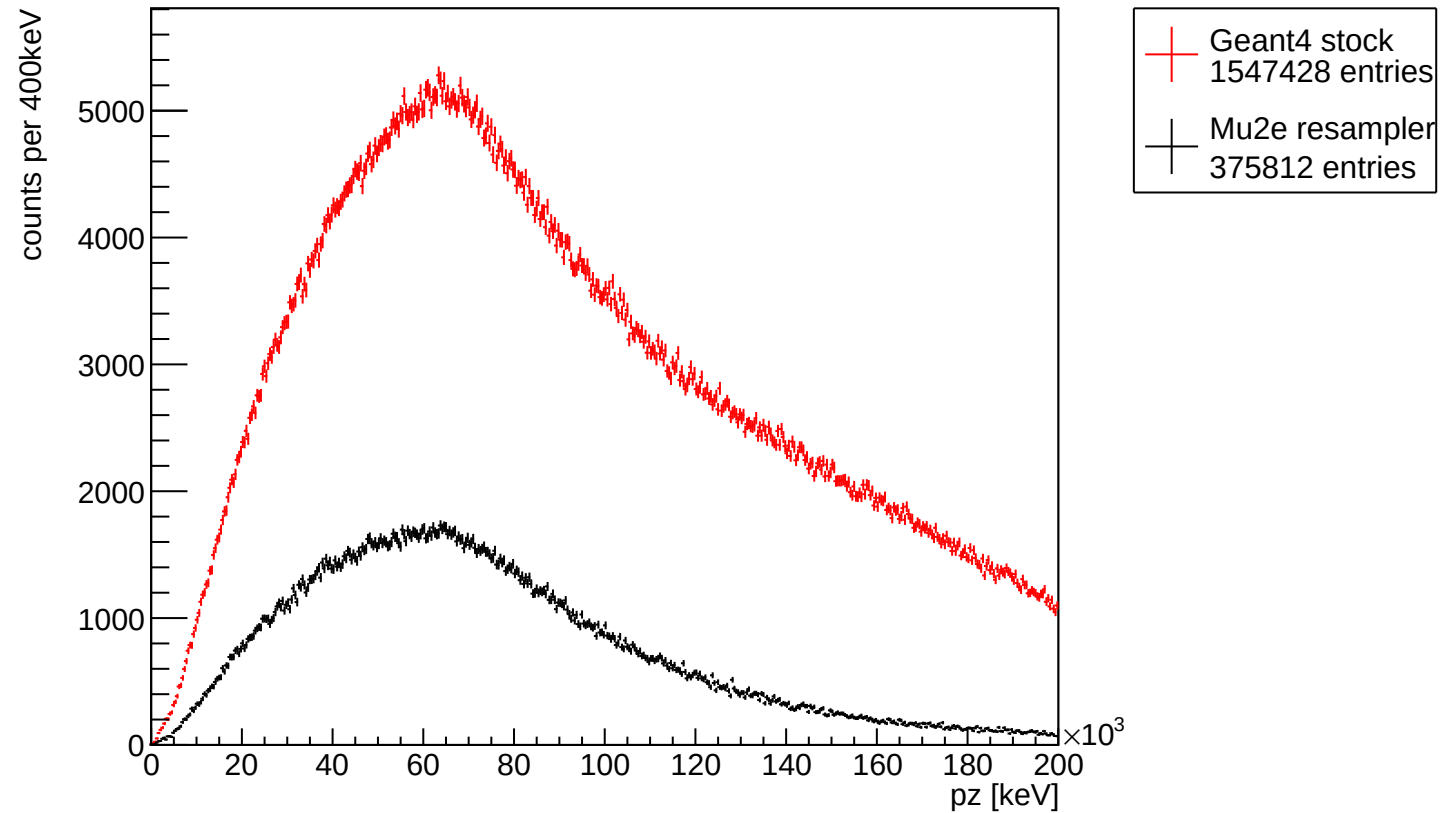
Entries	1663508
Mean	8.991e+04
Std Dev	4.724e+04
Underflow	0
Overflow	1.02e+05
Integral	1.445e+06

Mu2e mu resampler: proton momentum pz

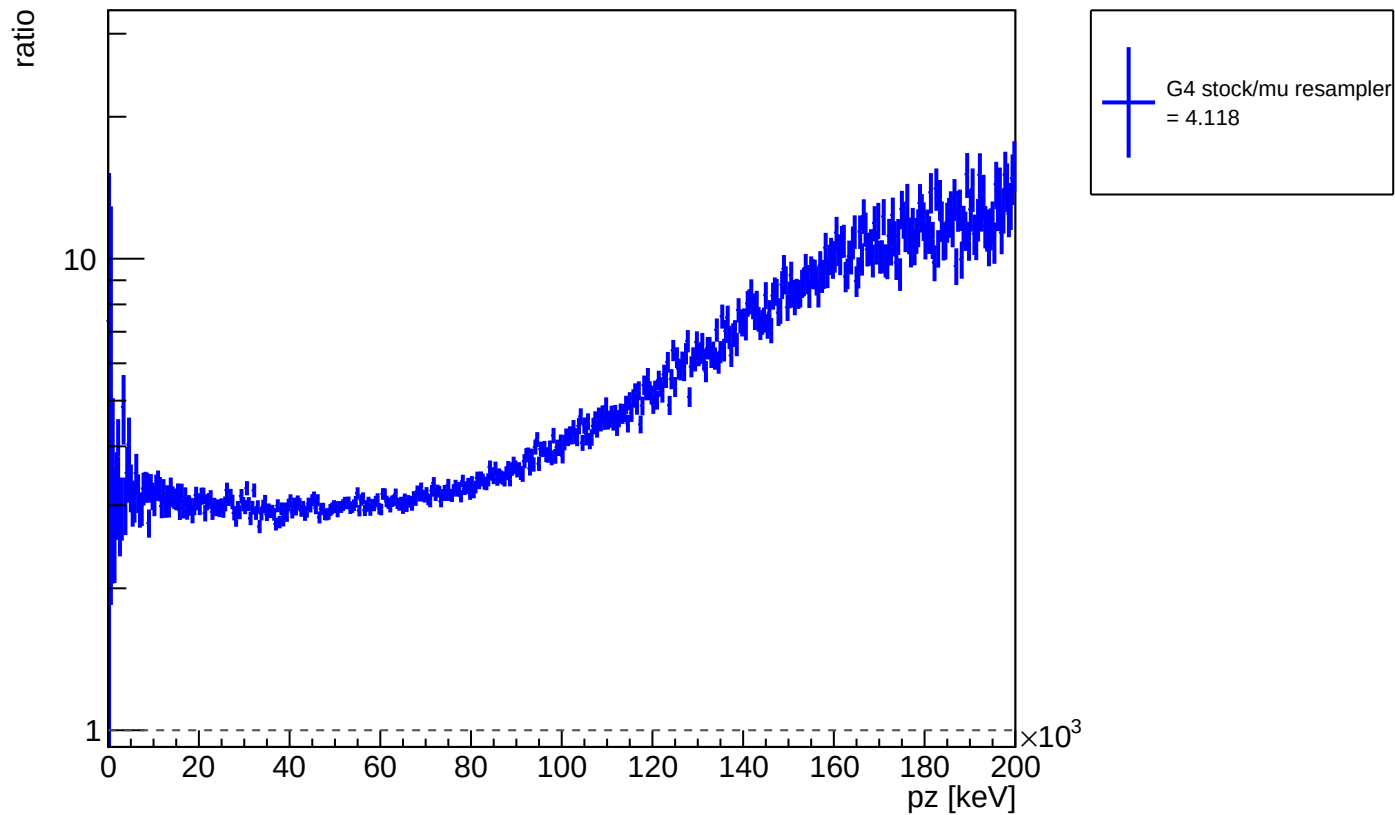


Entries	372430
Mean	7.393e+04
Std Dev	3.891e+04
Underflow	0
Overflow	8183
Integral	3.676e+05

proton momentum pz

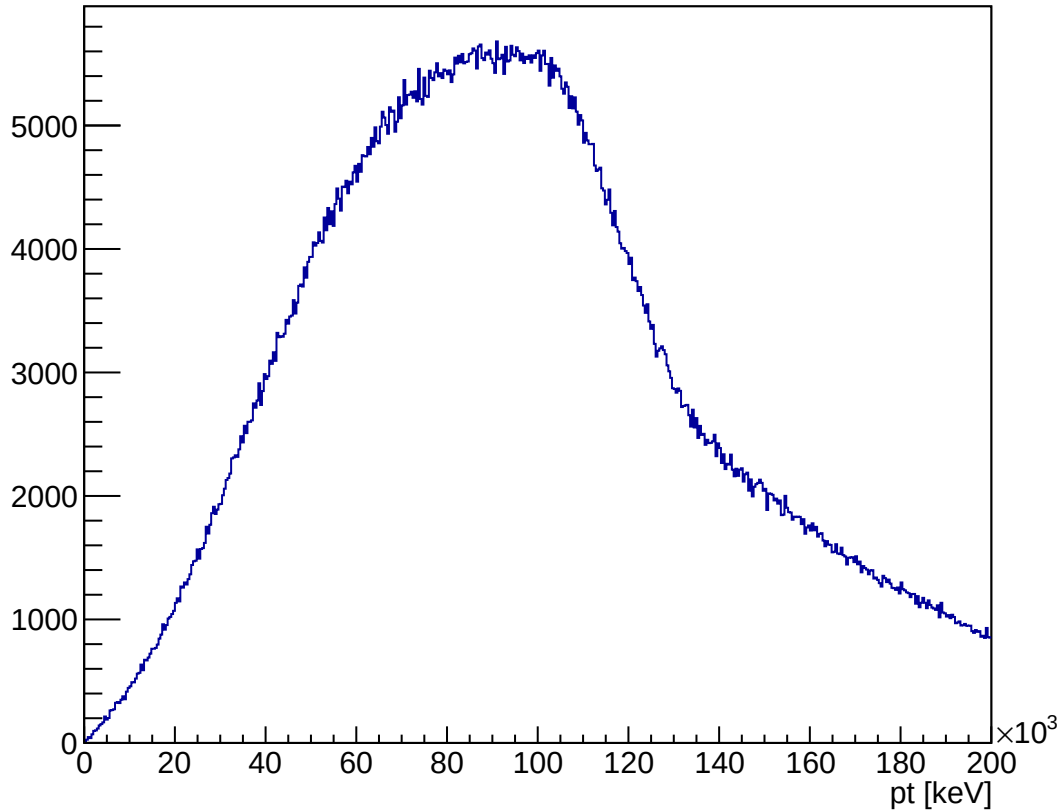


ratio of G4 stock / muon resampler: proton momentum pz



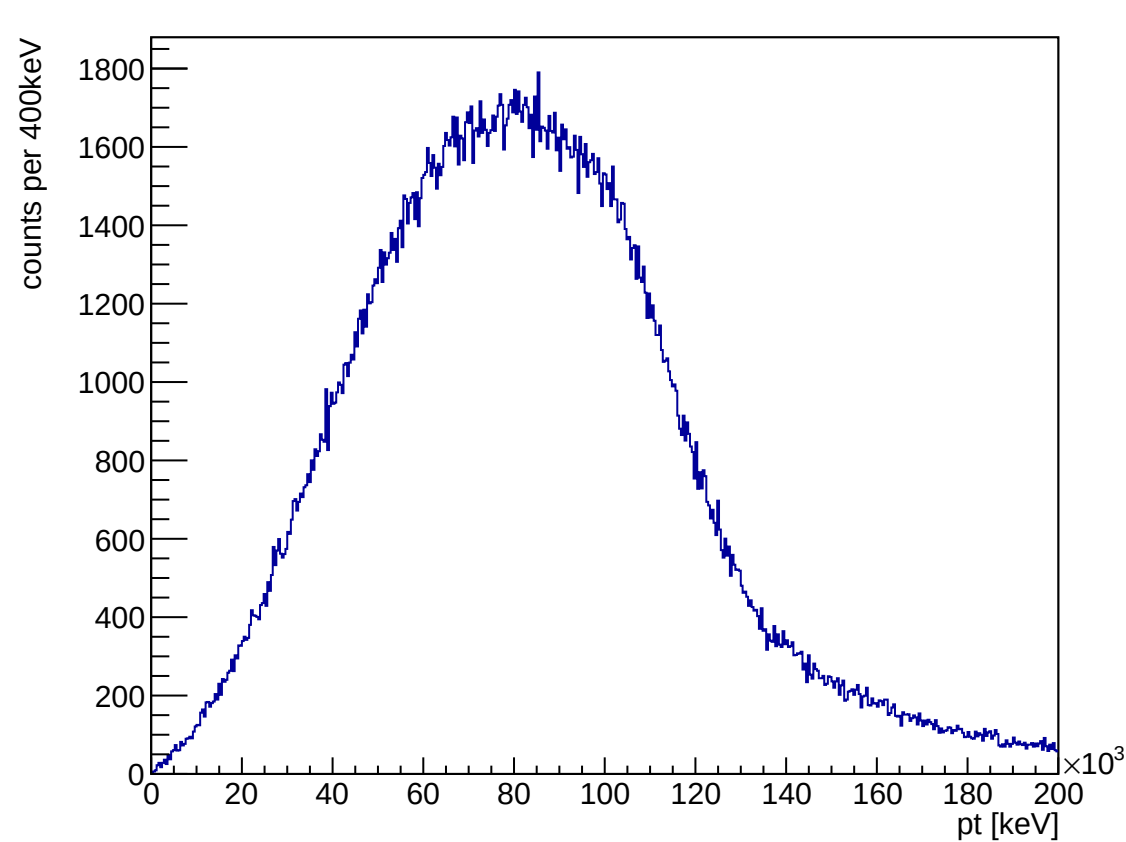
Stock G4: proton transverse momentum pT

counts per 400keV



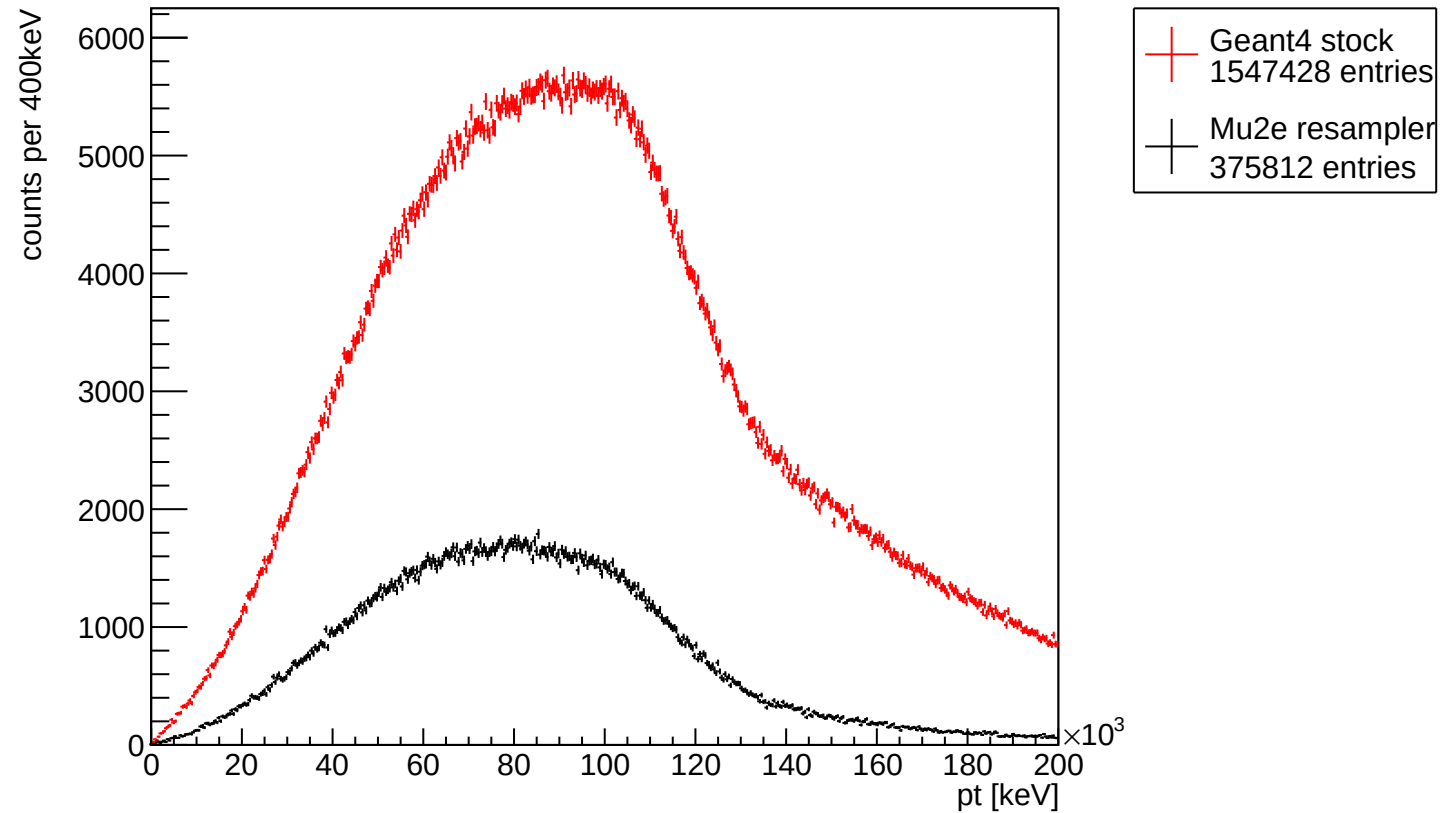
Entries	1663508
Mean	9.5e+04
Std Dev	4.161e+04
Underflow	0
Overflow	7.513e+04
Integral	1.472e+06

Mu2e mu resampler: proton transverse momentum pT

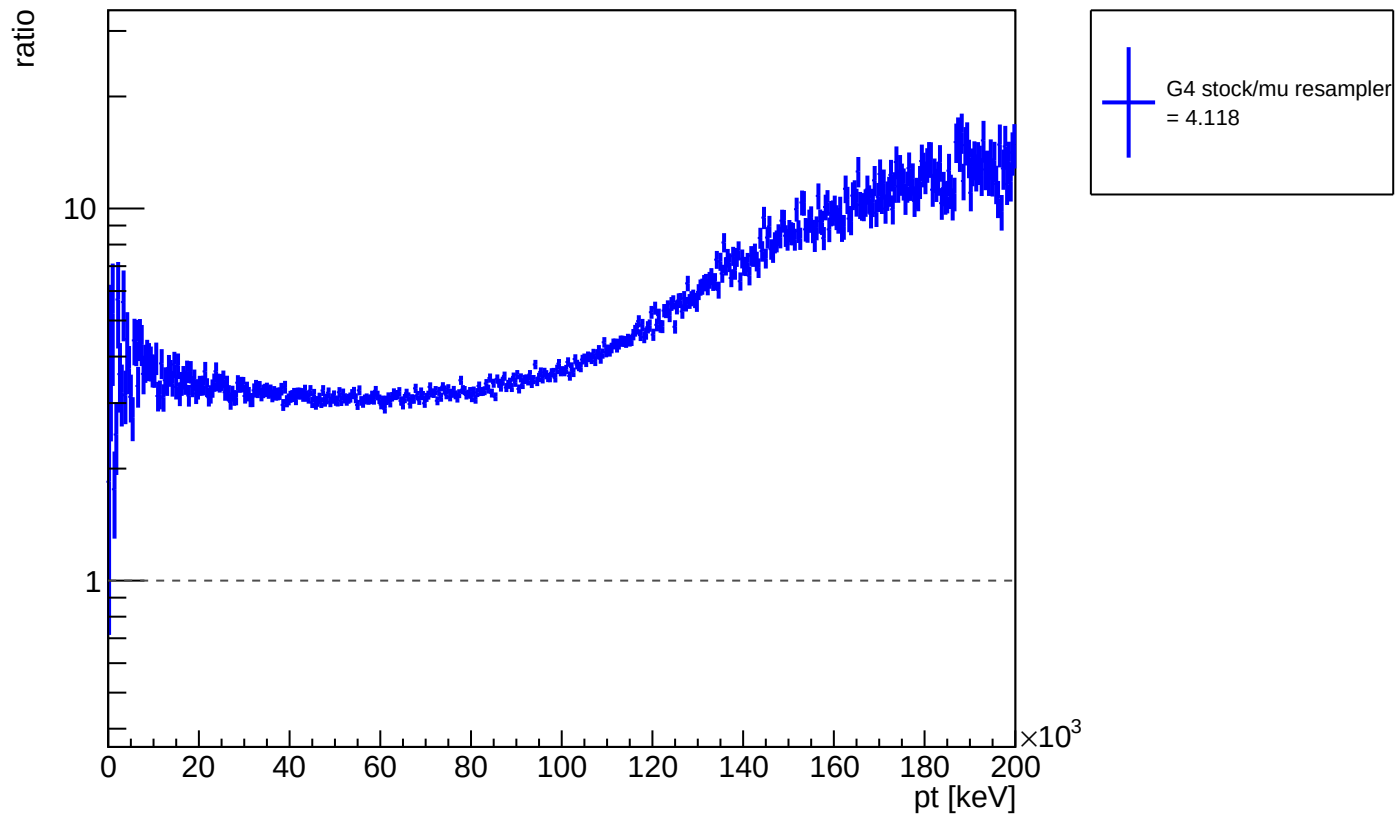


Entries	372430
Mean	8.289e+04
Std Dev	3.489e+04
Underflow	0
Overflow	5692
Integral	3.701e+05

proton transverse momentum pT

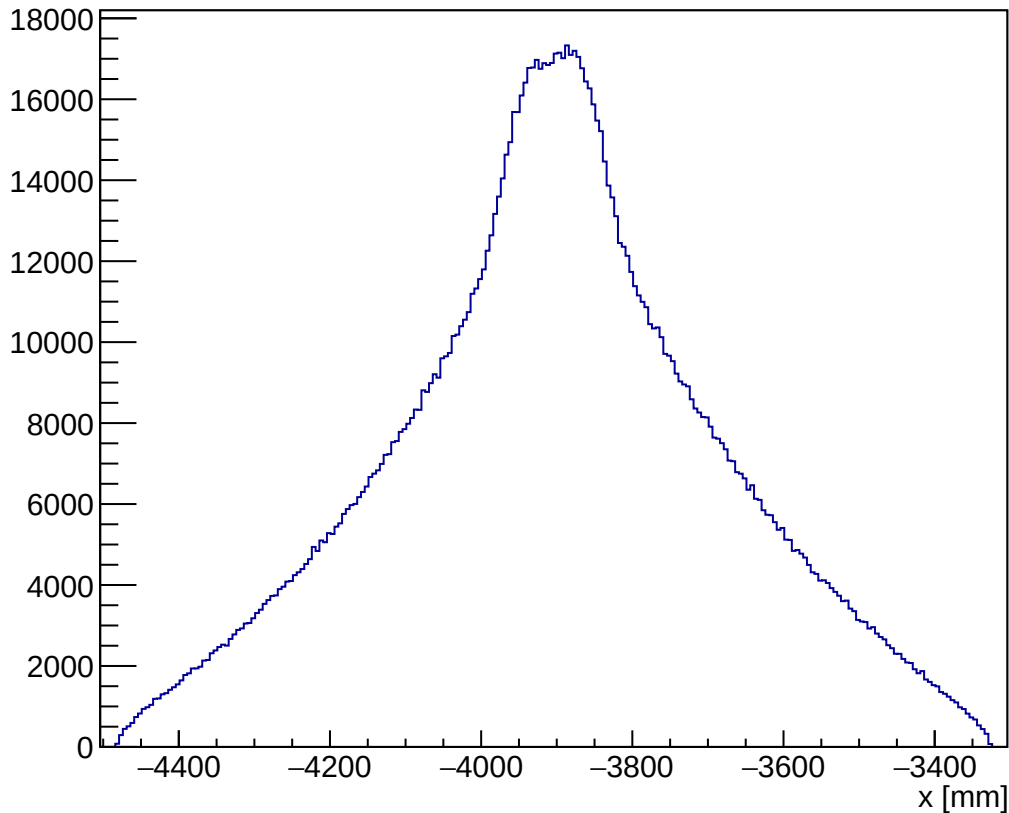


ratio of G4 stock / muon resampler: proton transverse momentum pT



Stock G4: proton x position

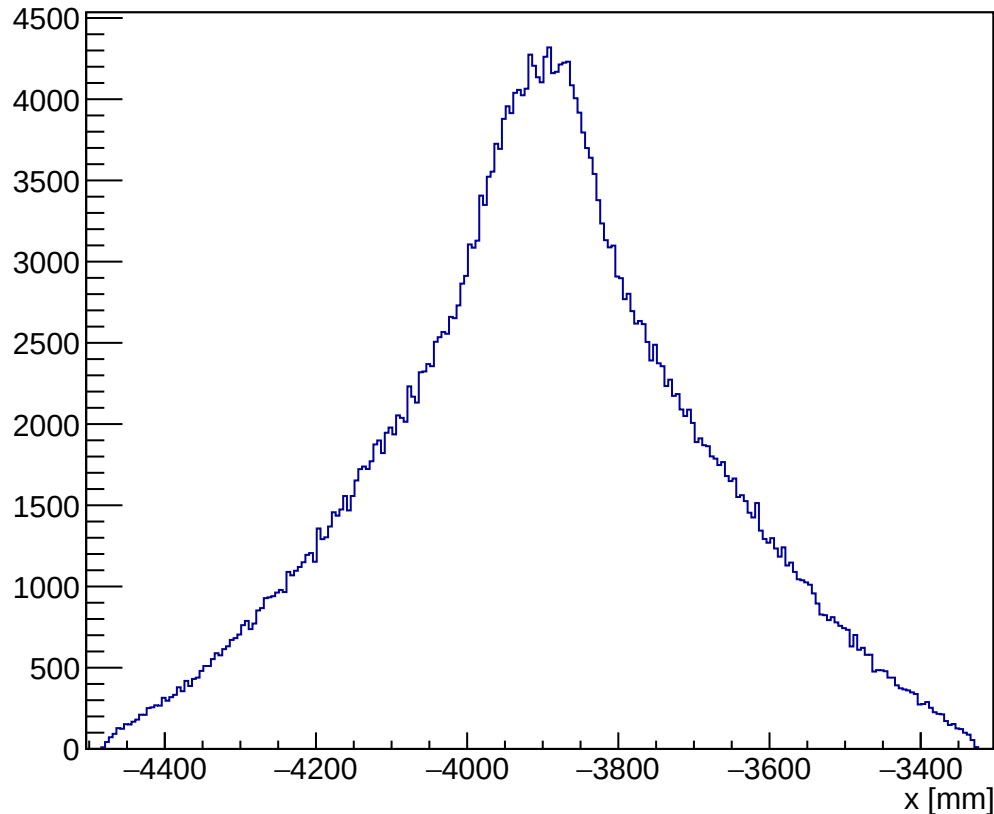
counts per 5mm



Entries	1663508
Mean	-3901
Std Dev	218.3
Underflow	0
Overflow	0
Integral	1.547e+06

Mu2e mu resampler: proton x position

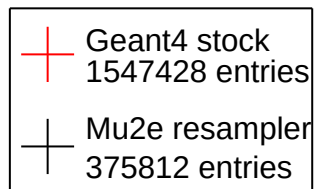
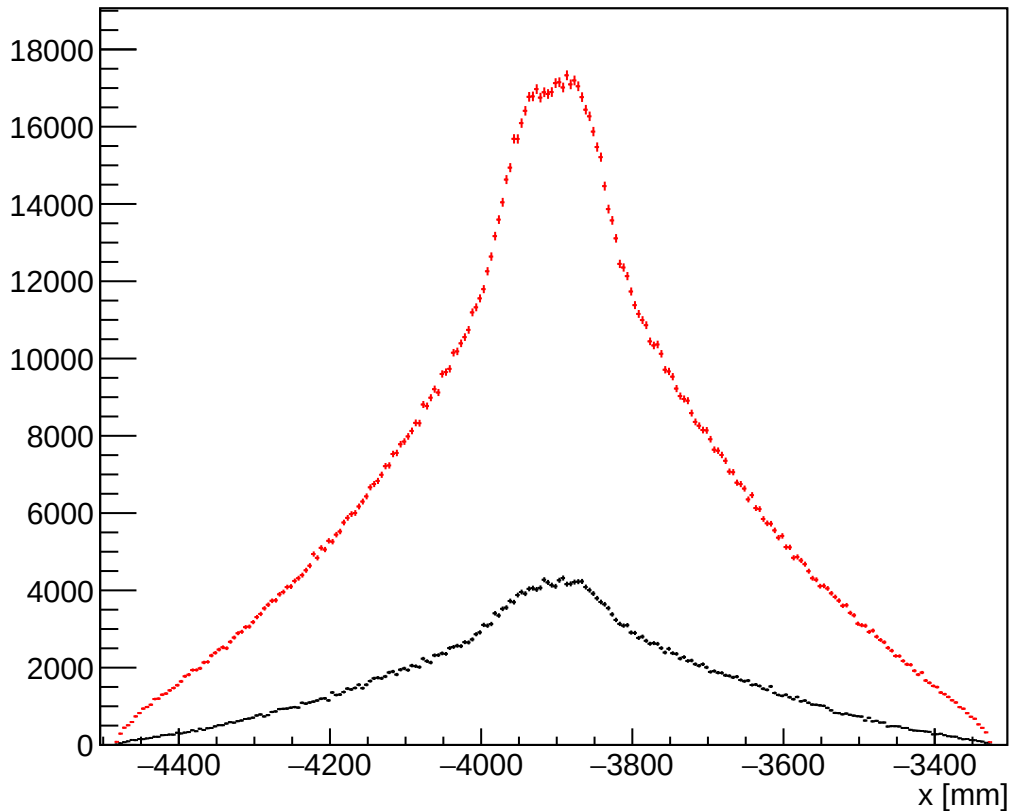
counts per 5mm



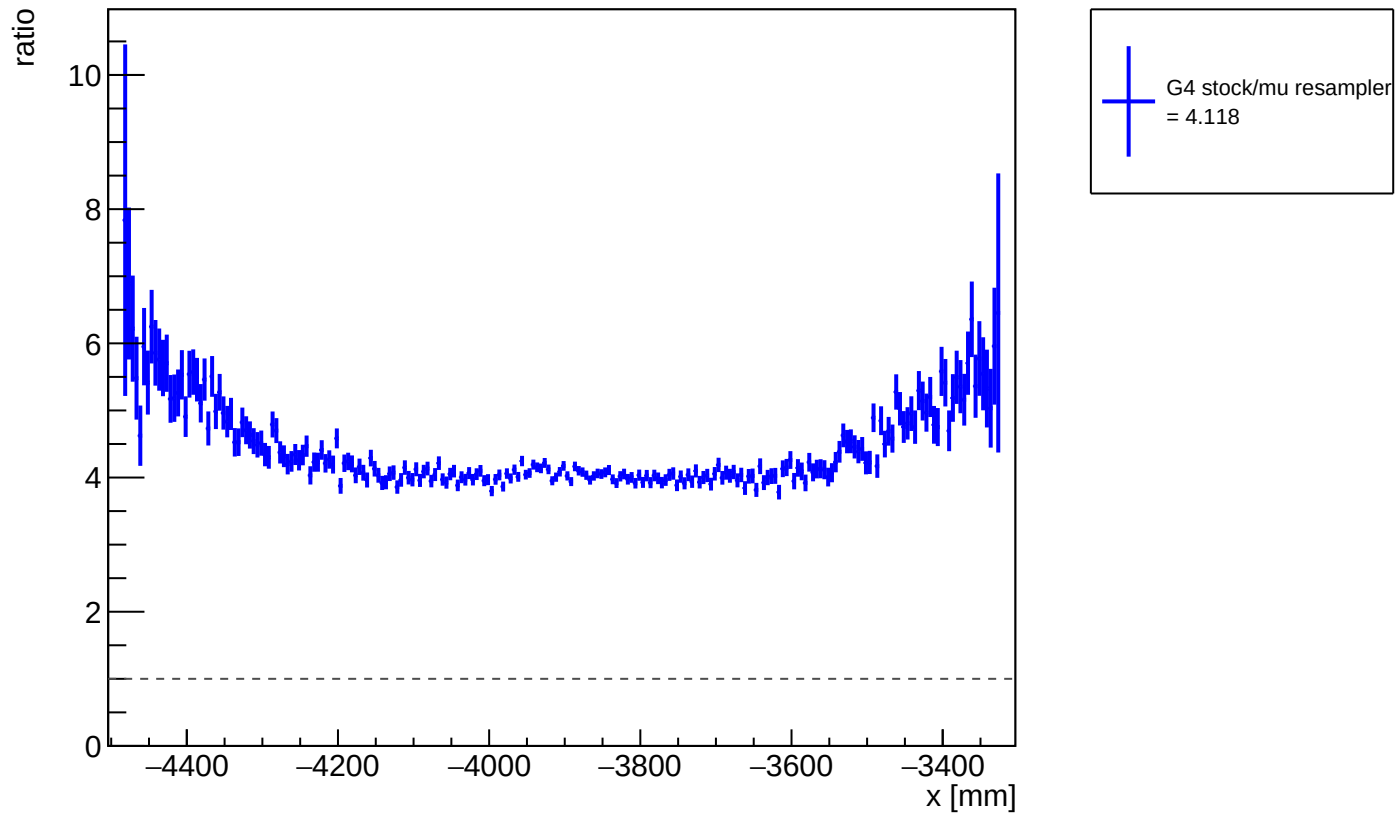
Entries	372430
Mean	-3900
Std Dev	210.7
Underflow	0
Overflow	0
Integral	3.758e+05

proton x position

counts per 5mm

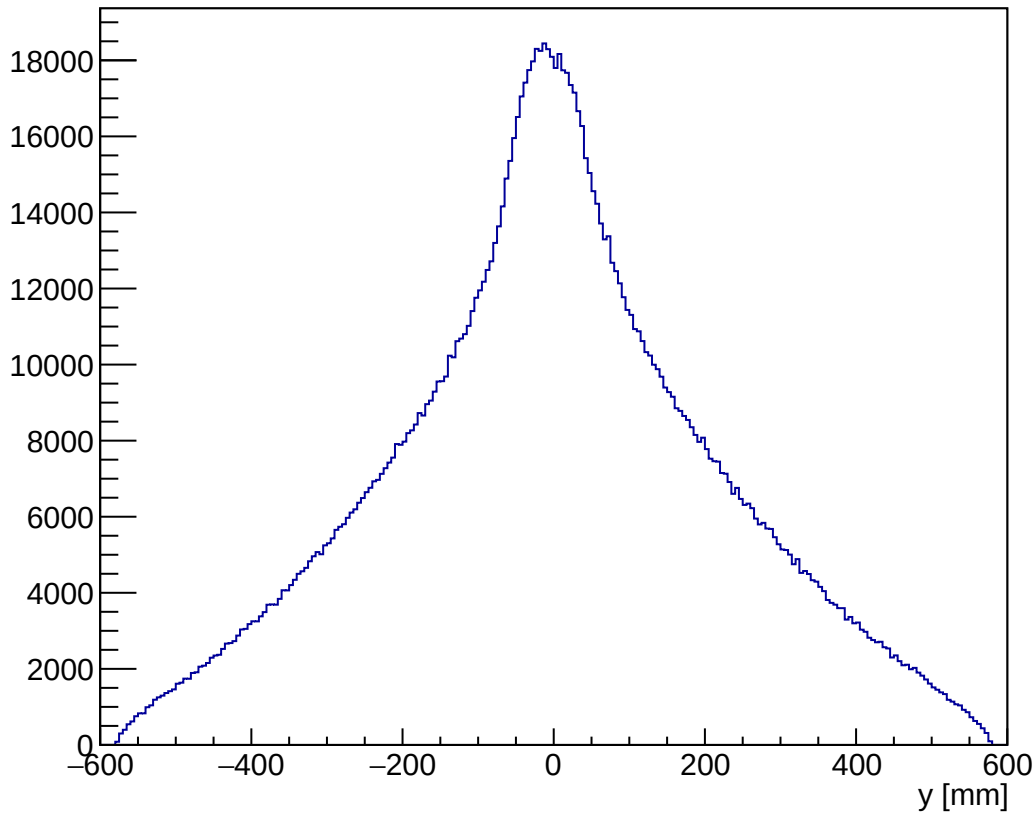


ratio of G4 stock / muon resampler: proton x position



Stock G4: proton y position

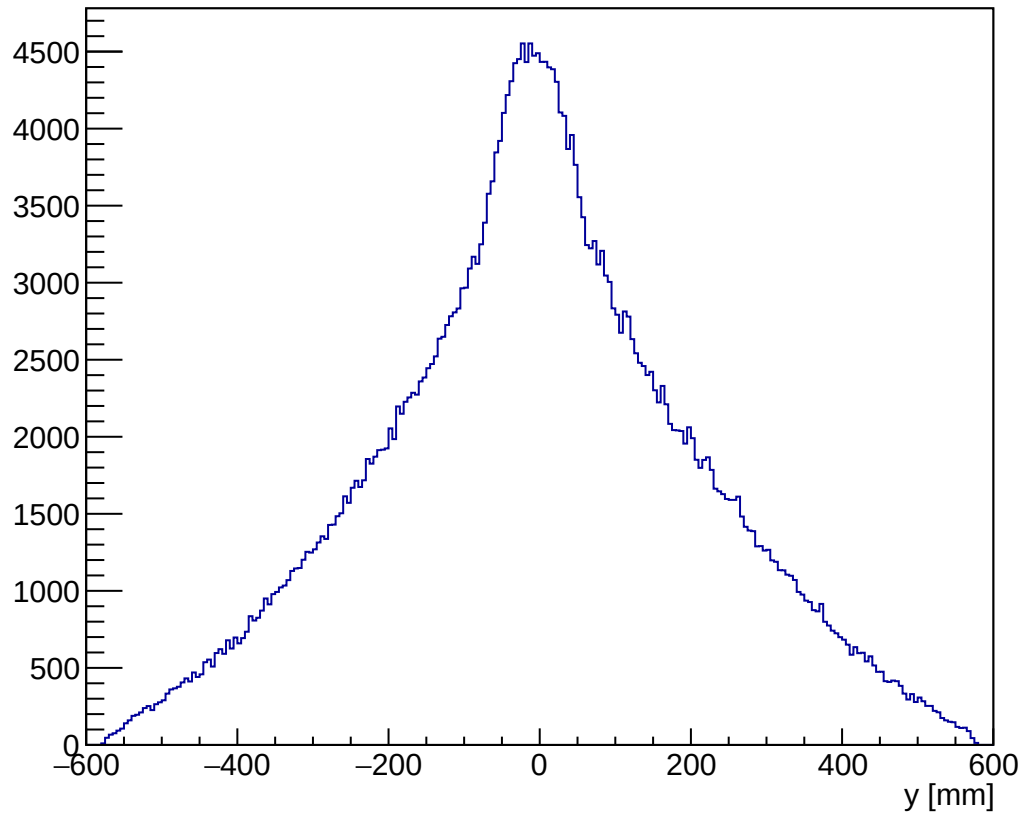
counts per 5mm



Entries	1663508
Mean	-0.7733
Std Dev	217.8
Underflow	0
Overflow	0
Integral	1.547e+06

Mu2e mu resampler: proton y position

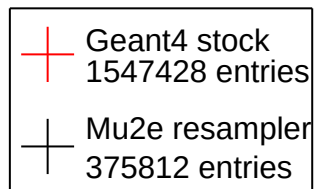
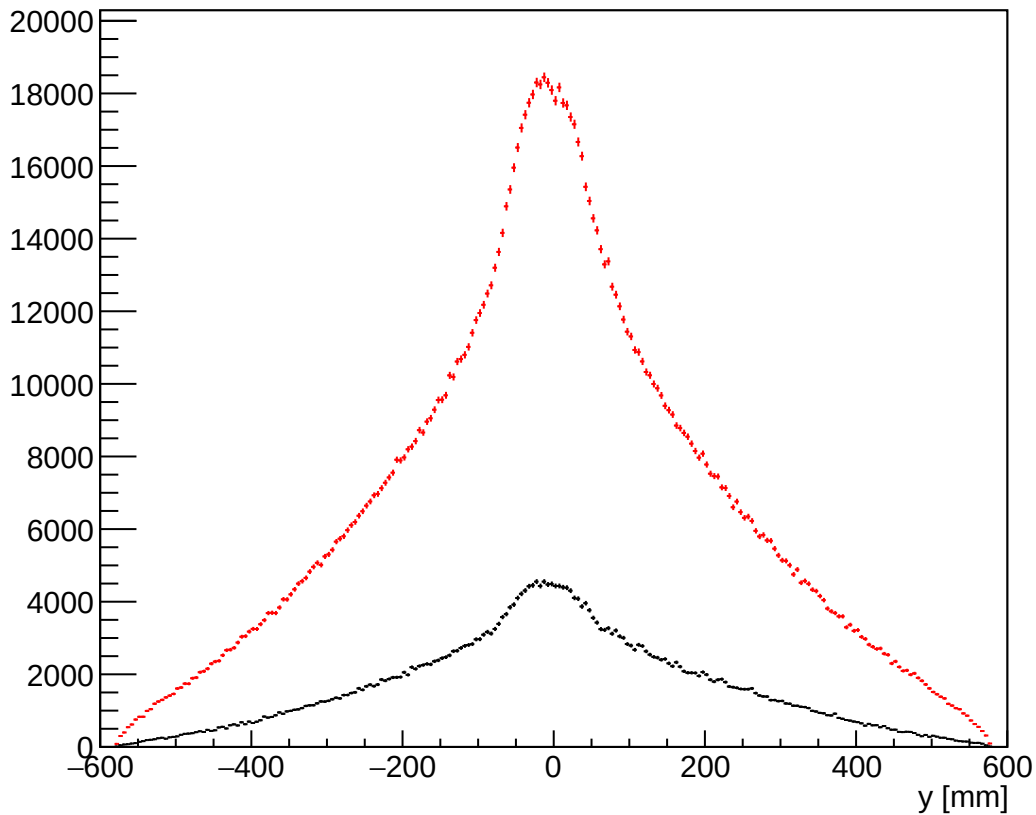
counts per 5mm



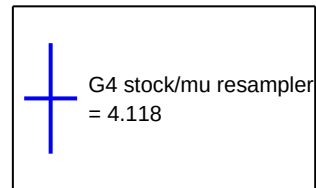
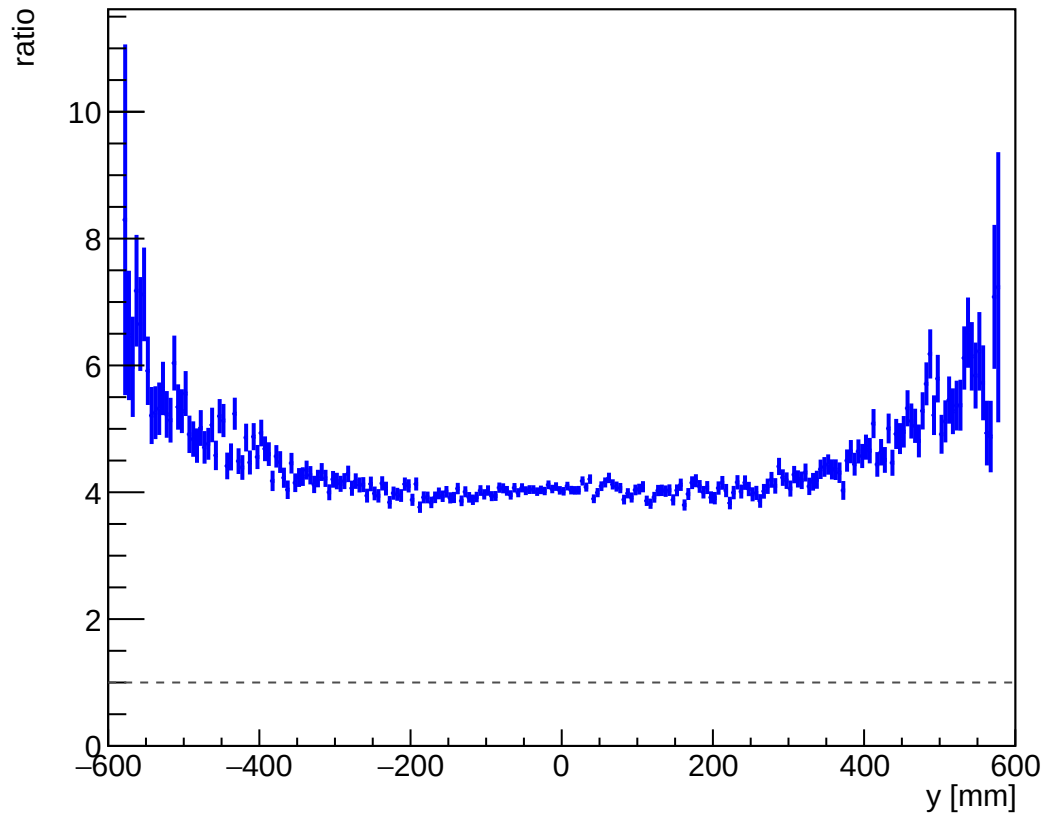
Entries	372430
Mean	-1.614
Std Dev	209.4
Underflow	0
Overflow	0
Integral	3.758e+05

proton y position

counts per 5mm

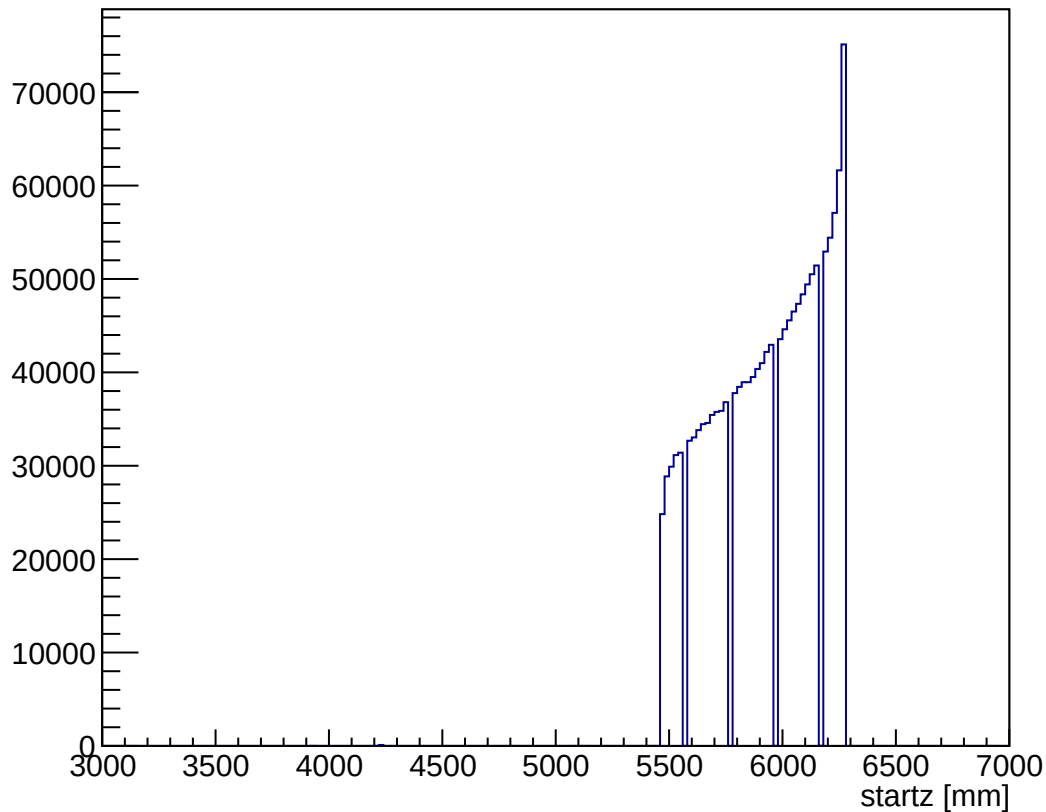


ratio of G4 stock / muon resampler: proton y position



Stock G4: proton start z

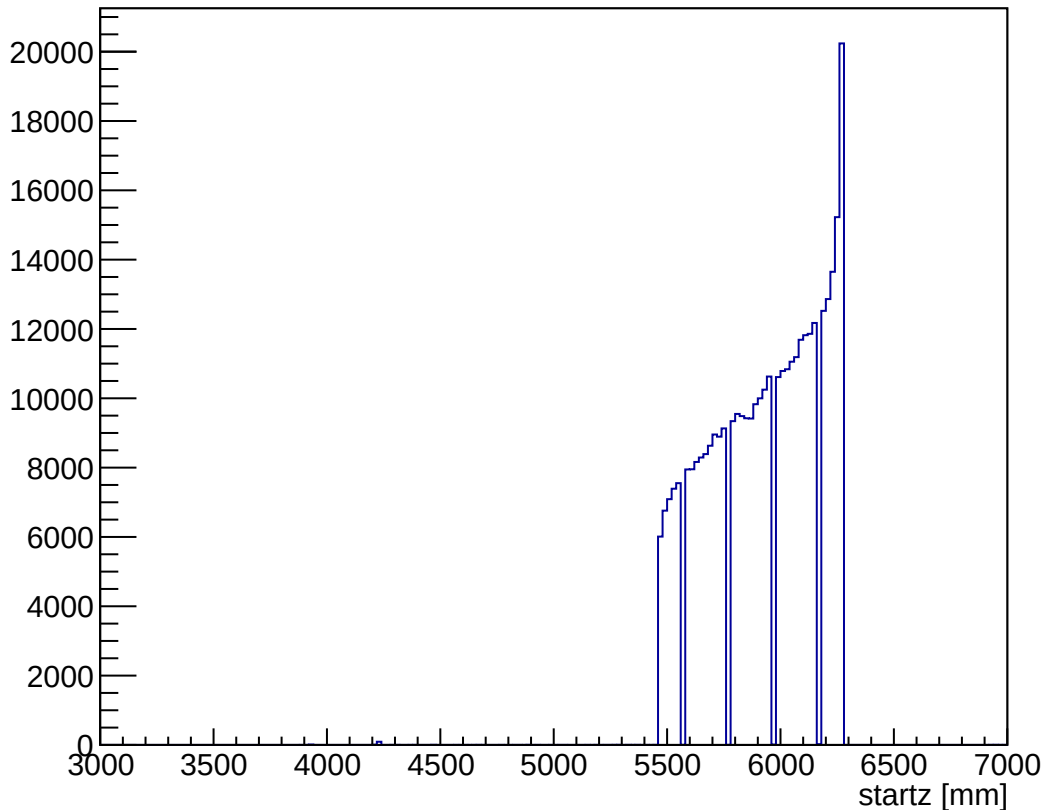
counts per 20mm



Entries	1663508
Mean	5924
Std Dev	237.2
Underflow	0
Overflow	7.442
Integral	1.547e+06

Mu2e mu resampler: proton start z

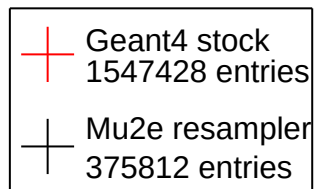
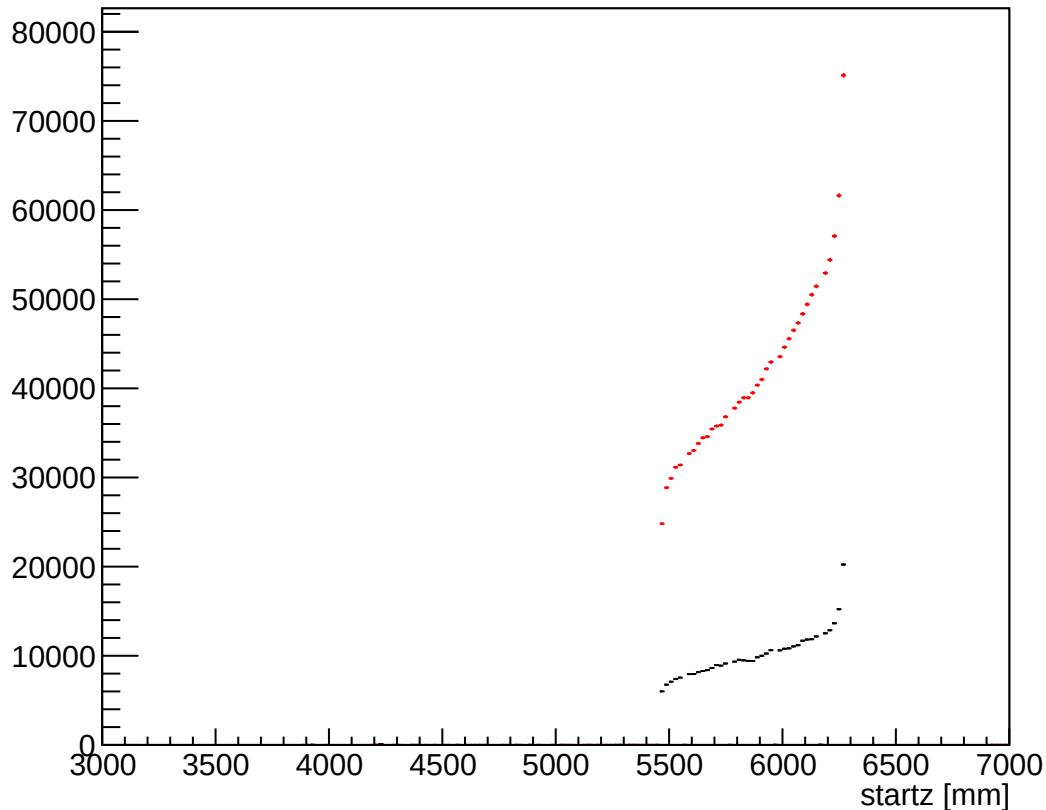
counts per 20mm



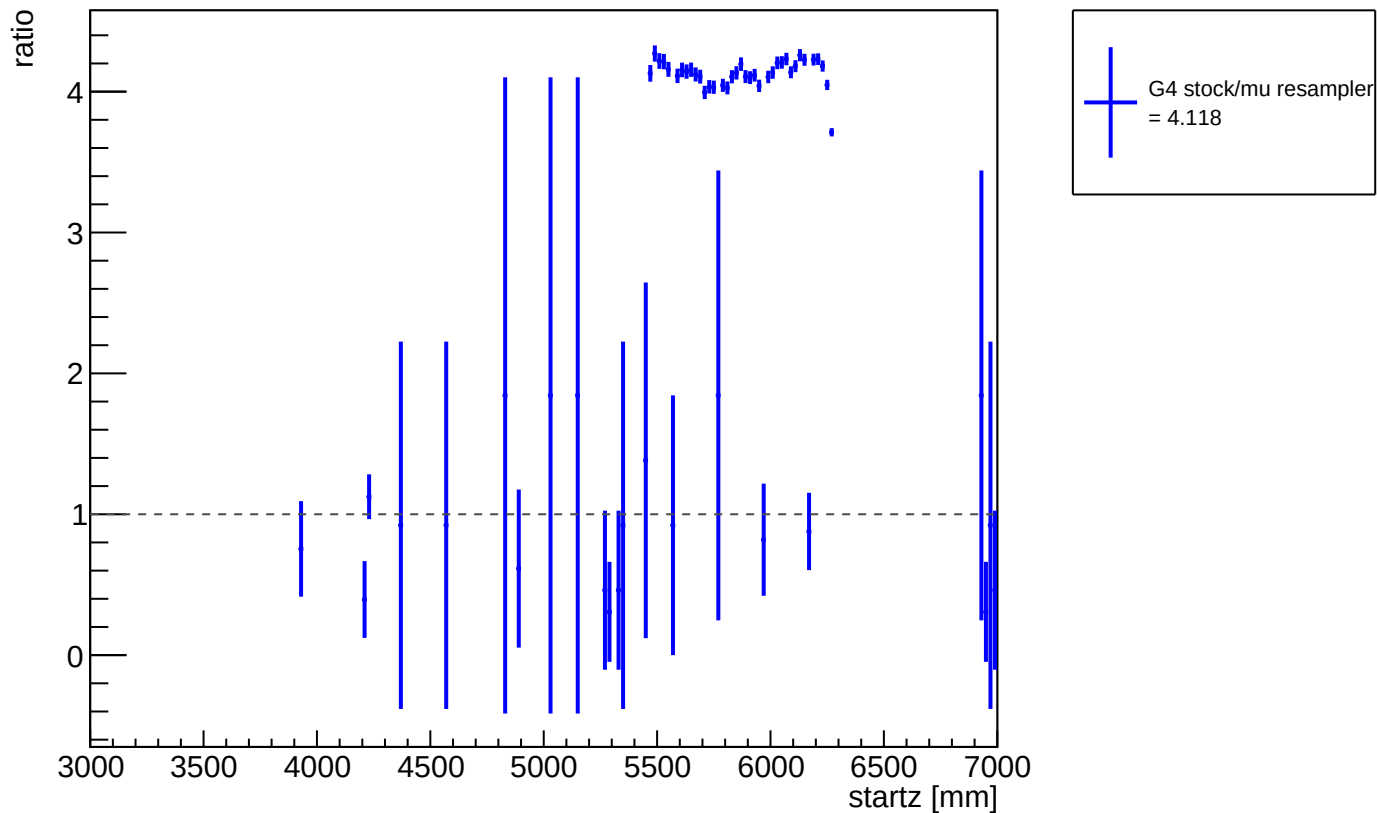
Entries	372430
Mean	5925
Std Dev	239.1
Underflow	0
Overflow	9.082
Integral	3.758e+05

proton start z

counts per 20mm

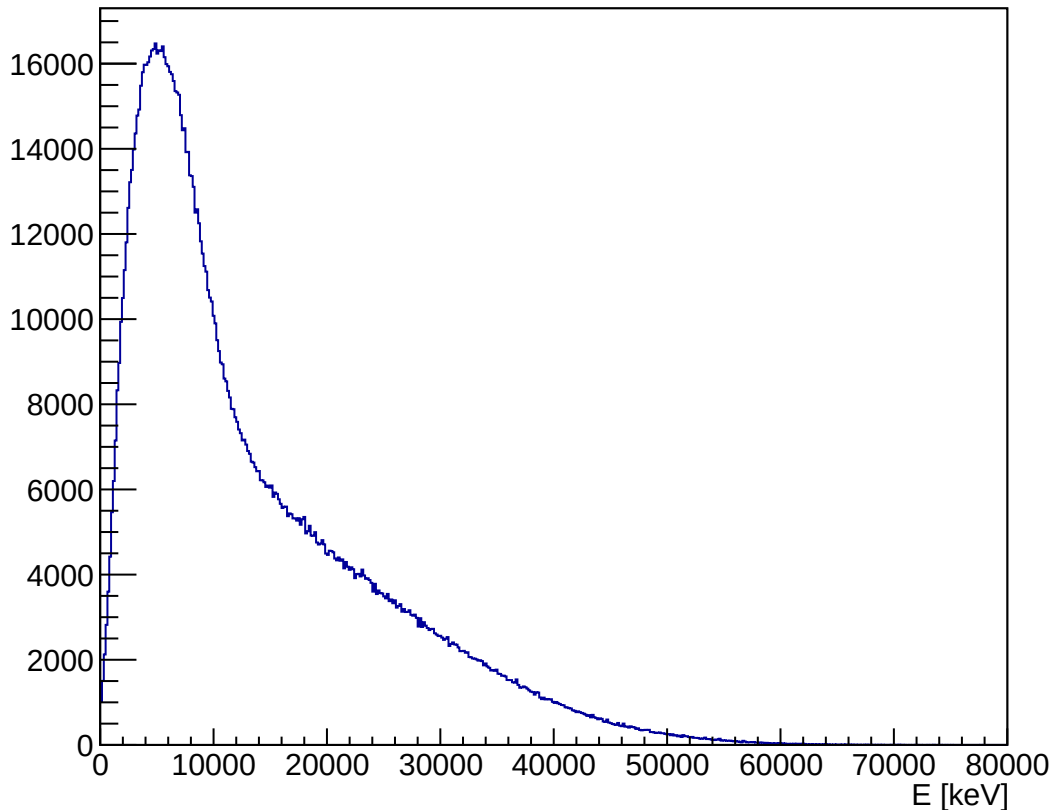


ratio of G4 stock / muon resampler: proton start z



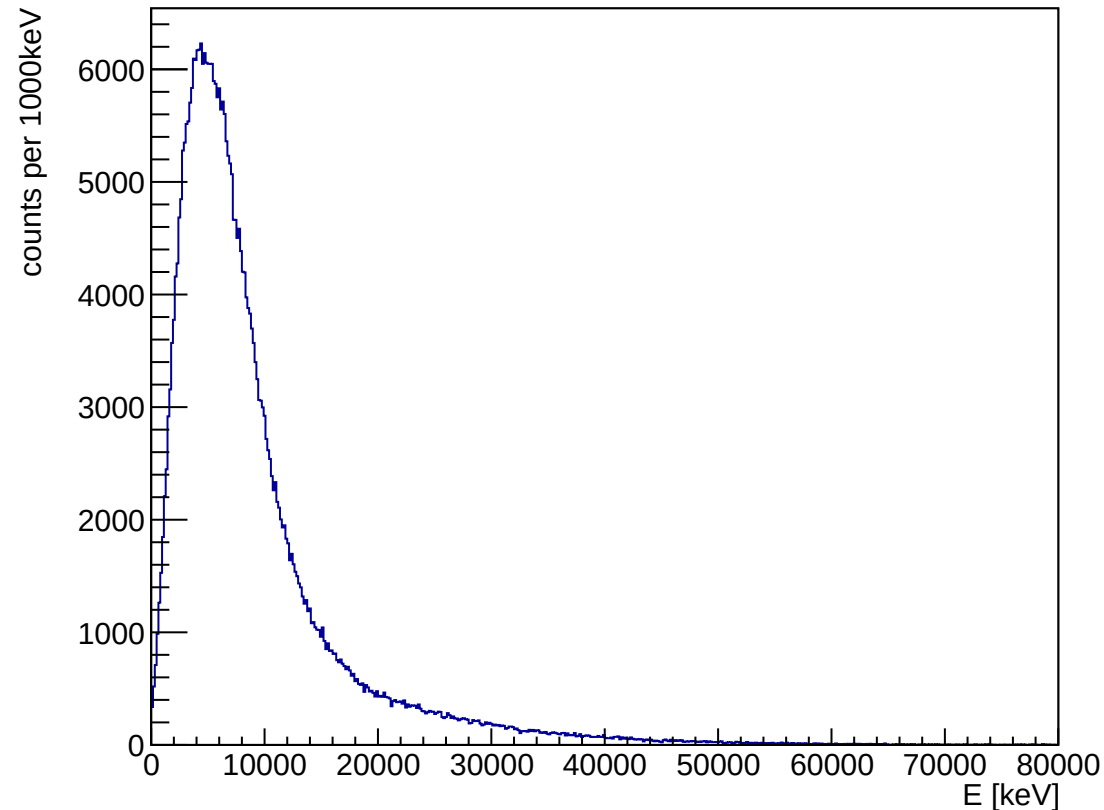
Stock G4: proton kin. energy

counts per 1000keV



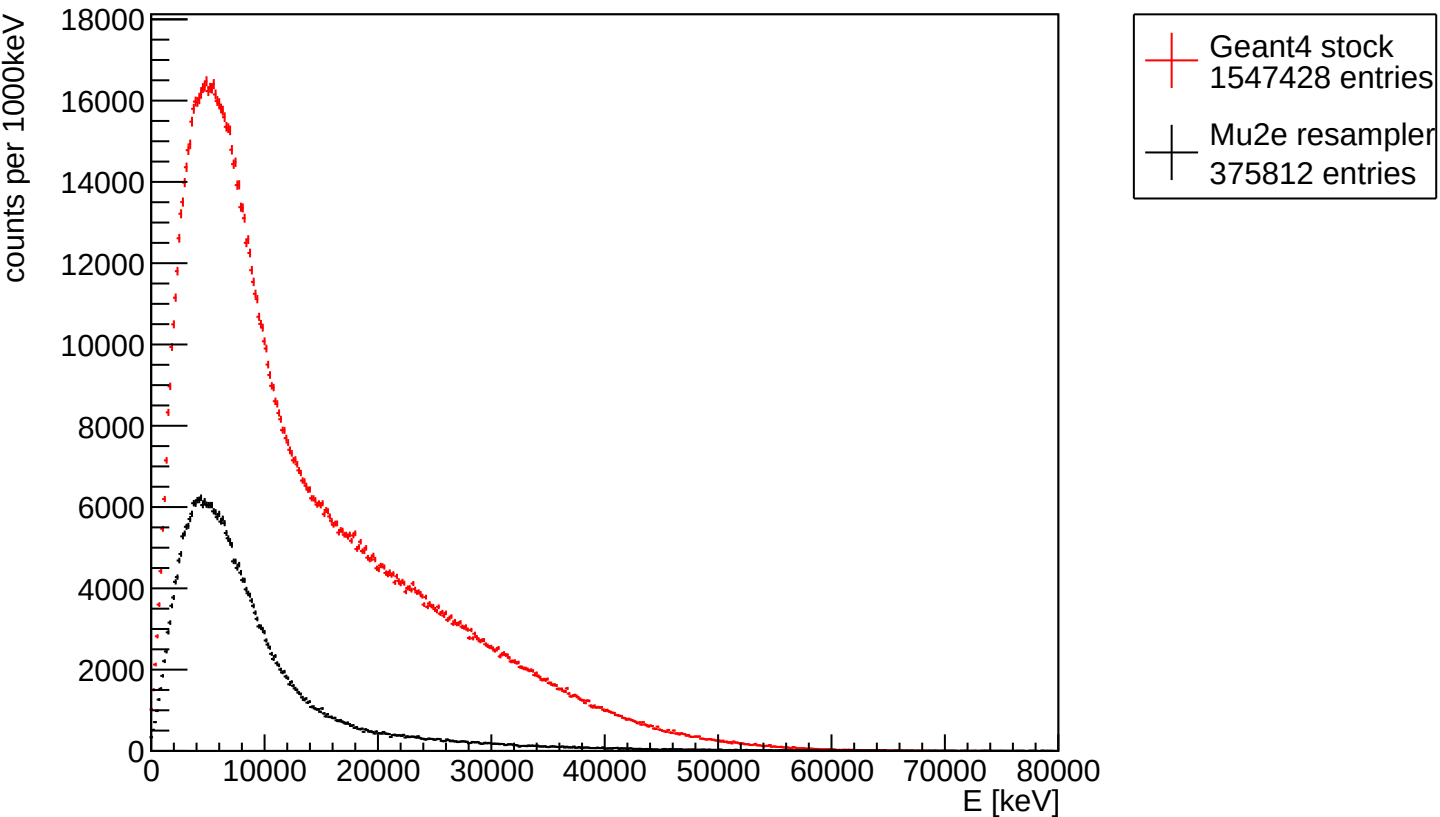
Entries	1663508
Mean	1.366e+04
Std Dev	1.083e+04
Underflow	0
Overflow	0
Integral	1.547e+06

Mu2e mu resampler: proton kin. energy

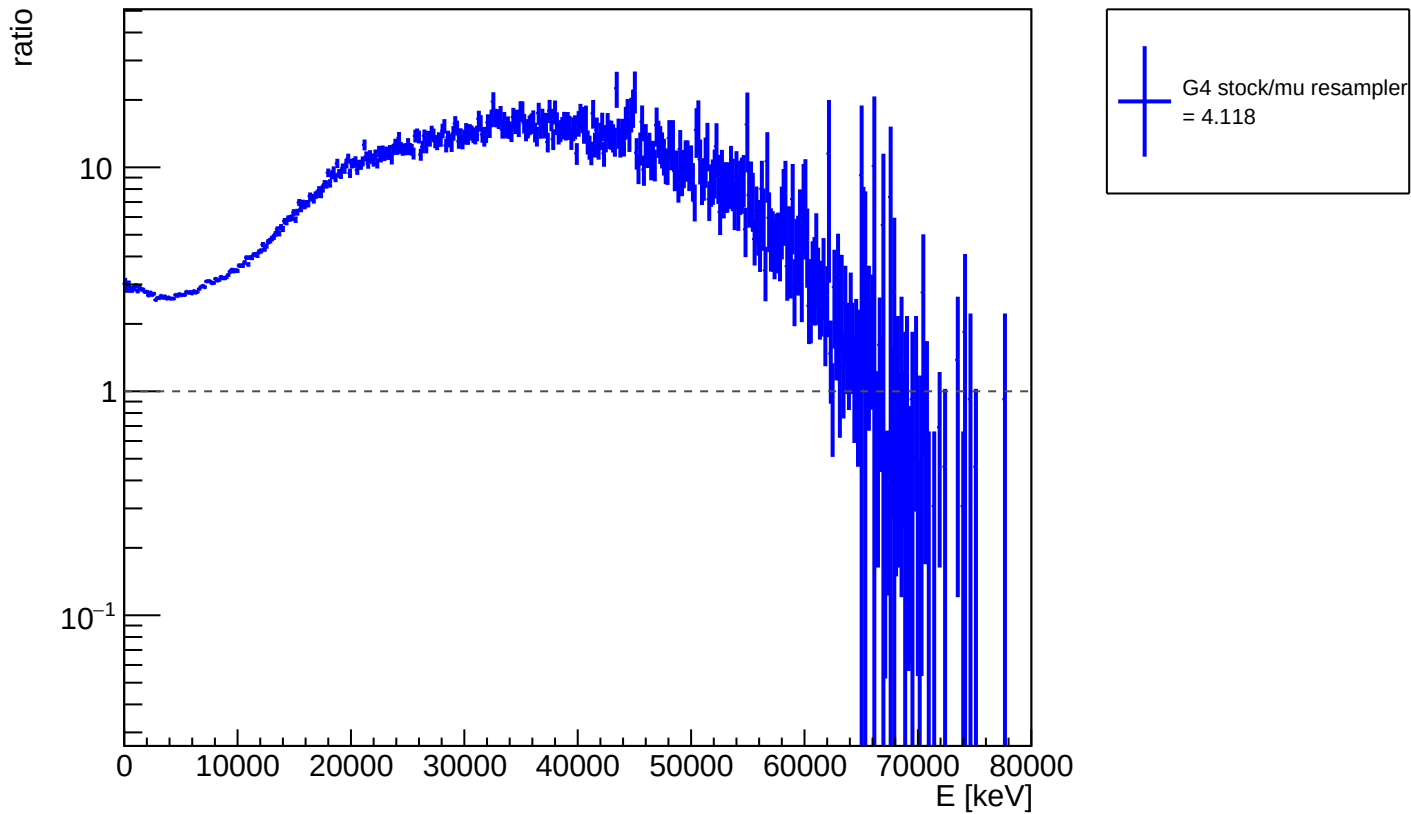


Entries	372430
Mean	8885
Std Dev	7866
Underflow	0
Overflow	59.54
Integral	3.758e+05

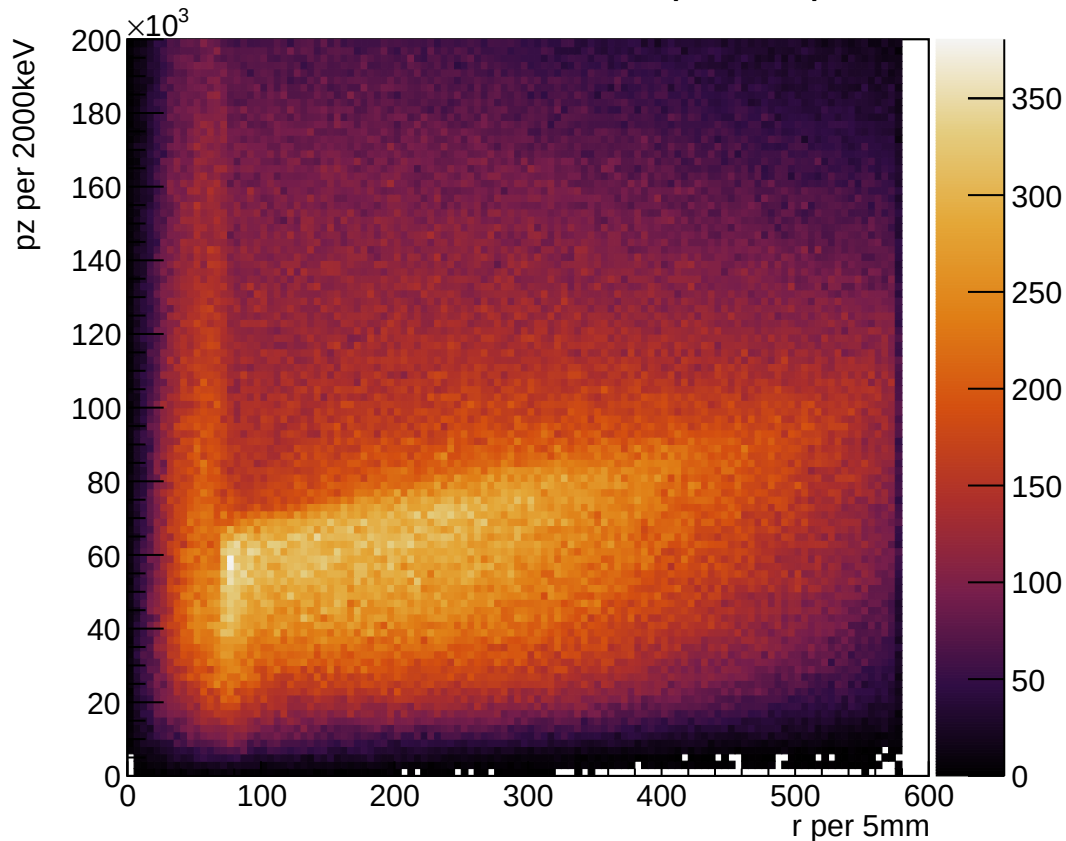
proton kin. energy



ratio of G4 stock / muon resampler: proton kin. energy

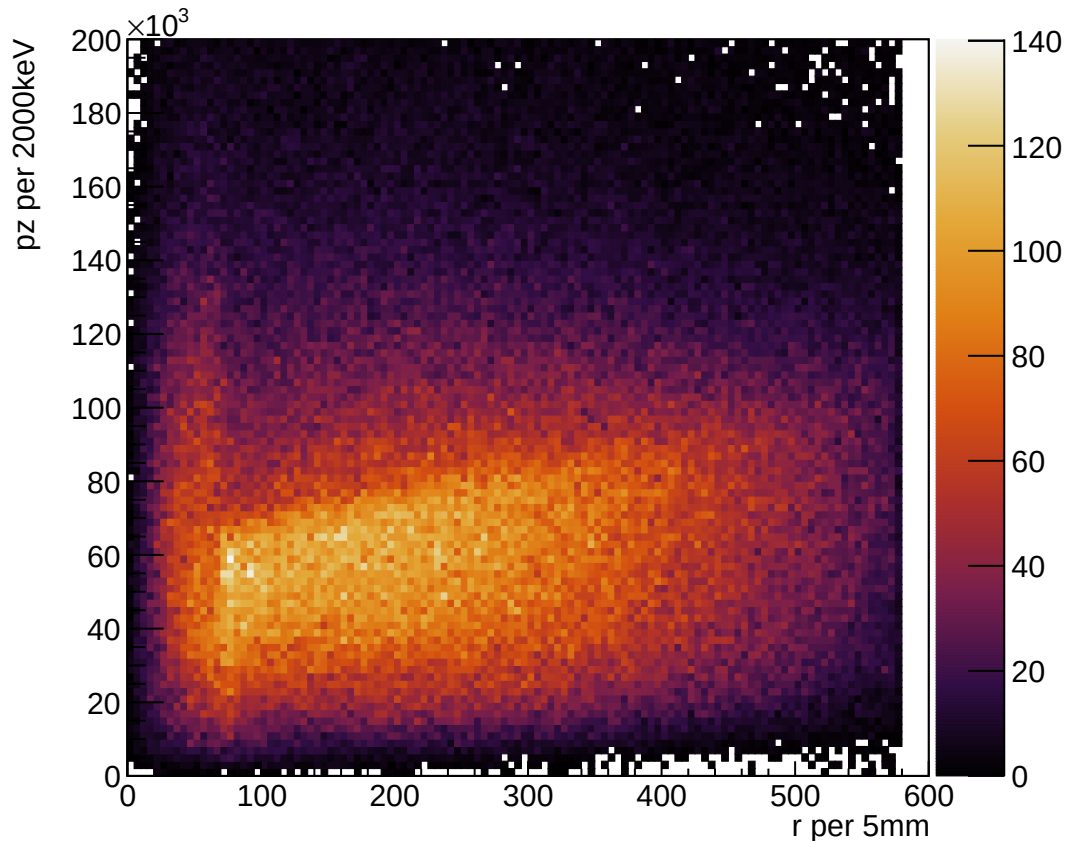


Stock G4: proton pz vs r



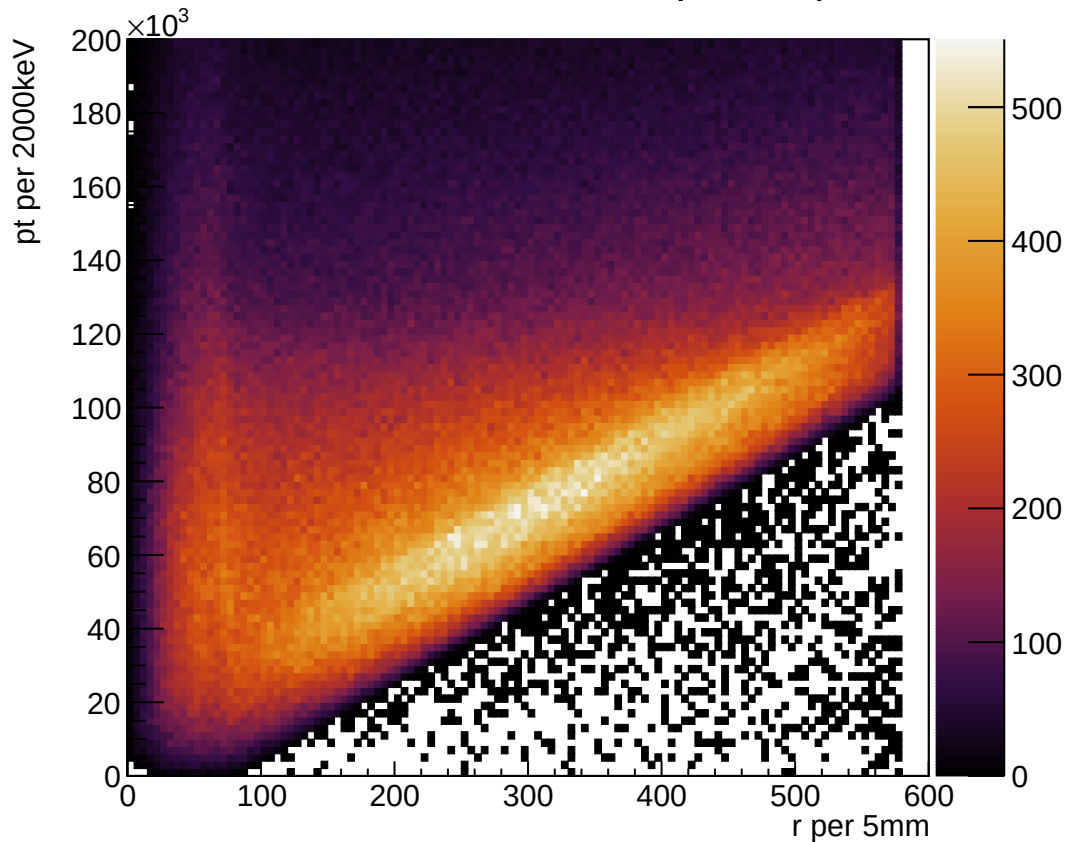
Entries	1663508	
Mean x	272.1	
Mean y	8.991e+04	
Std Dev x	152	
Std Dev y	4.724e+04	
Integral	1.445e+06	
0	102041	0
0	1.44539e+06	0
0	0	0

Mu2e mu resampler: proton pz vs r



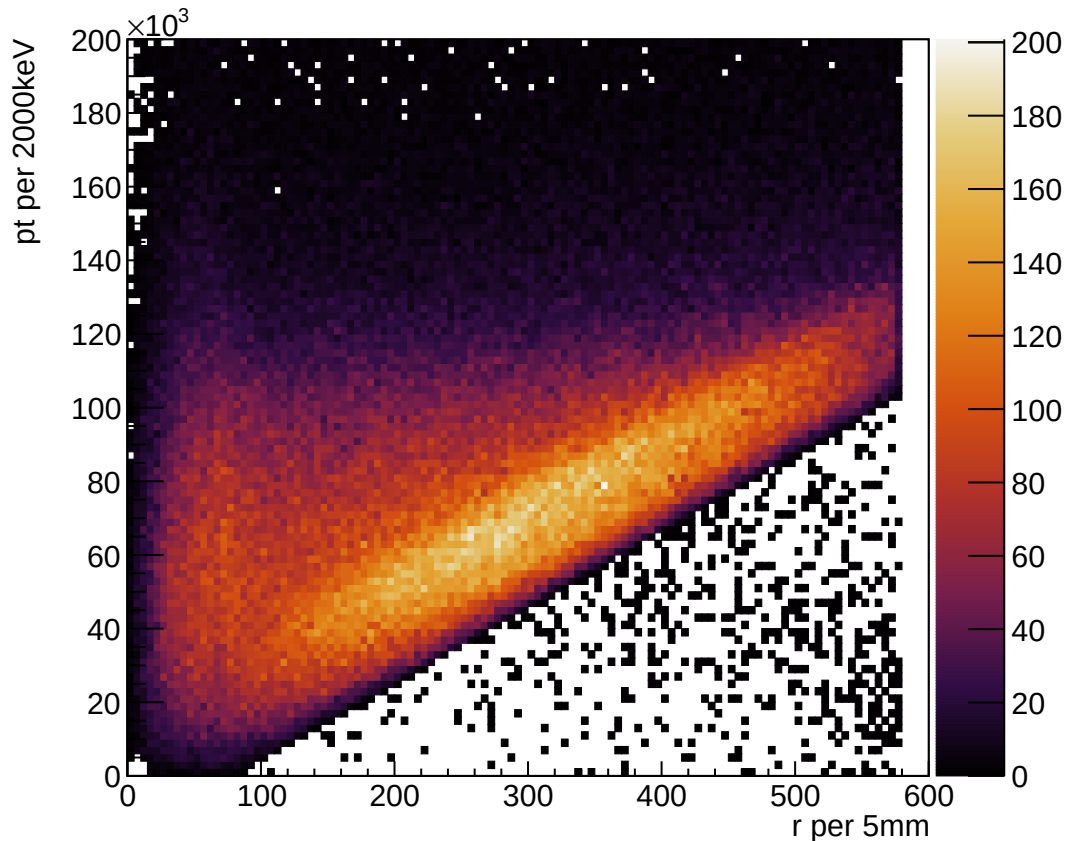
Entries	372430	
Mean x	260	
Mean y	7.393e+04	
Std Dev x	145.3	
Std Dev y	3.891e+04	
Integral	3.676e+05	
0	8182.64	0
0	367630	0
0	0	0

Stock G4: proton pt vs r



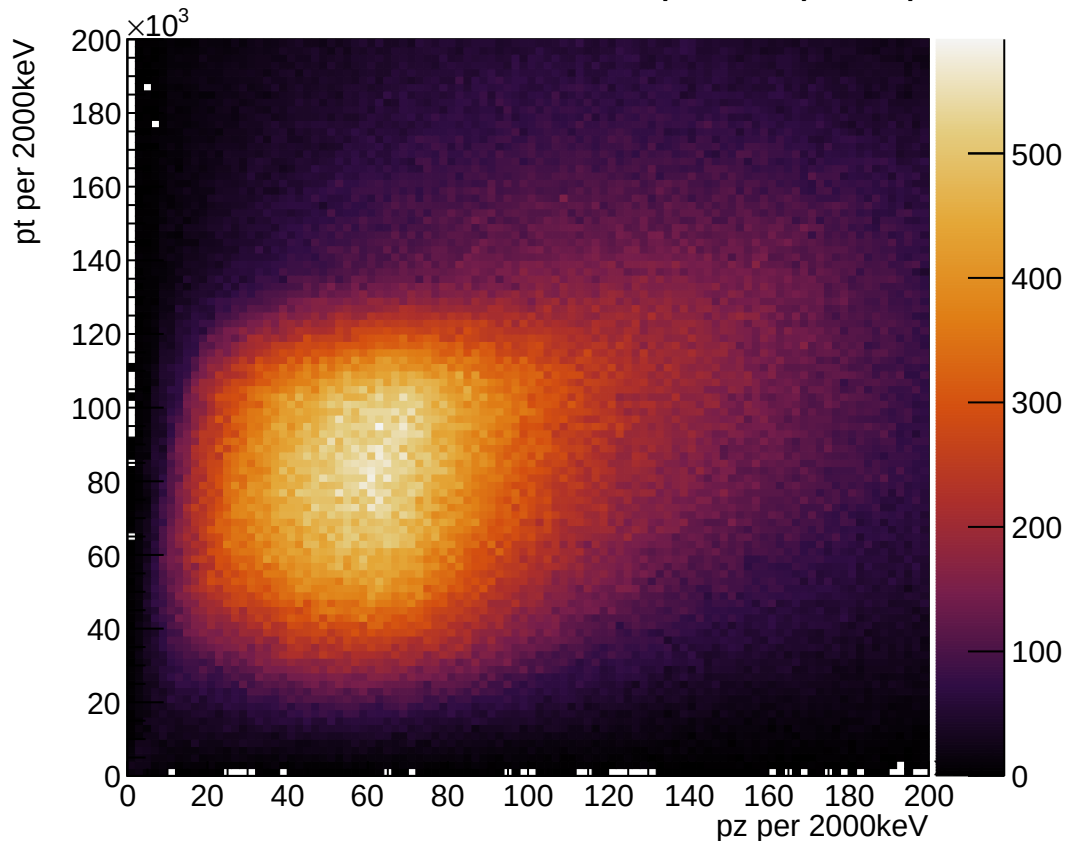
Entries	1663508	
Mean x	266.5	
Mean y	9.5e+04	
Std Dev x	150.7	
Std Dev y	4.161e+04	
Integral	1.472e+06	
0	75132	0
0	1.4723e+06	0
0	0	0

Mu2e mu resampler: proton pt vs r



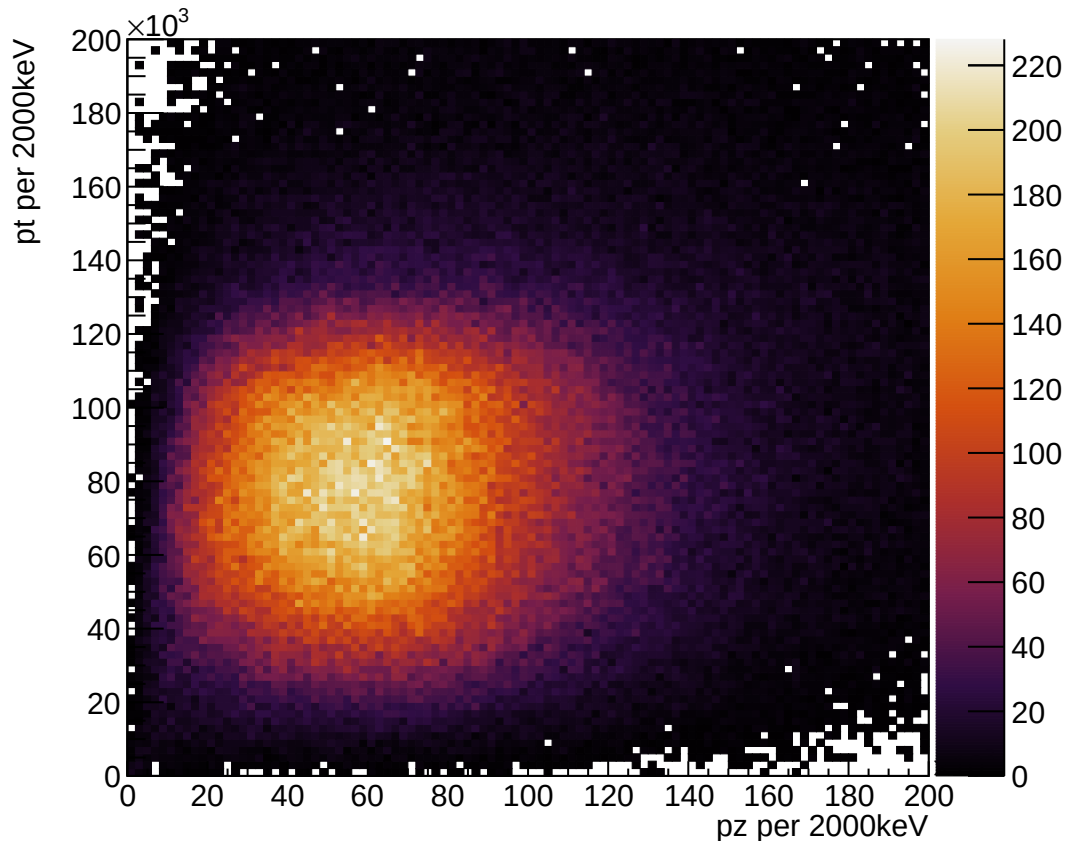
Entries	372430	
Mean x	258.3	
Mean y	8.289e+04	
Std Dev x	144.9	
Std Dev y	3.489e+04	
Integral	3.701e+05	
0	5692.23	0
0	370120	0
0	0	0

Stock G4: proton pt vs pz



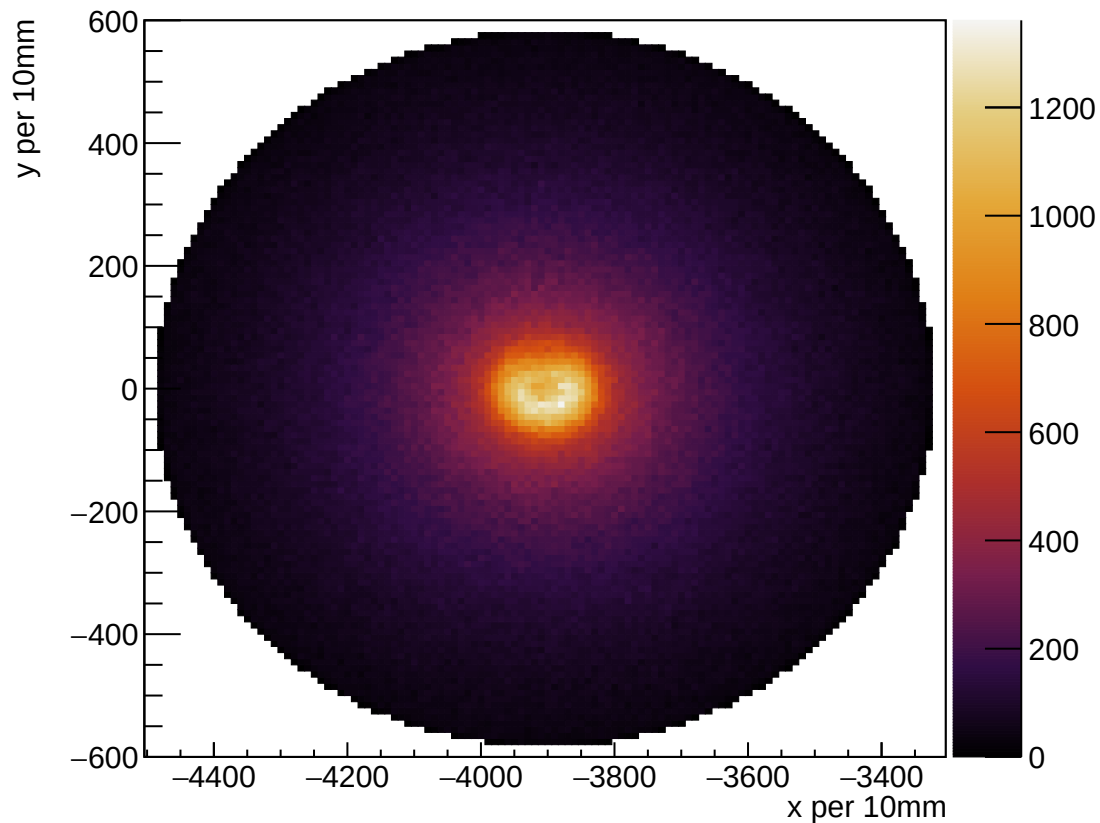
Entries	1663508	
Mean x	8.912e+04	
Mean y	9.411e+04	
Std Dev x	4.717e+04	
Std Dev y	4.124e+04	
Integral	1.374e+06	
0	70899.5	4232.5
0	1.37449e+06	97808.9
0	0	0

Mu2e mu resampler: proton pt vs pz



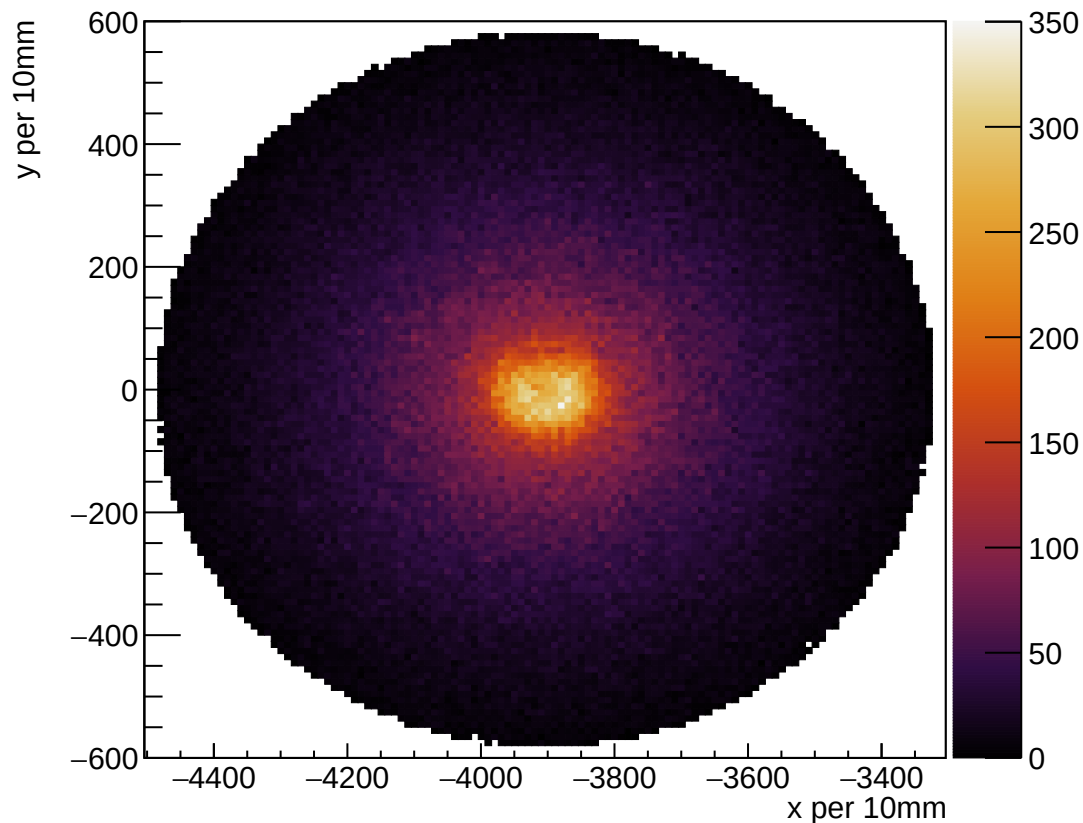
Entries	372430	
Mean x	7.348e+04	
Mean y	8.237e+04	
Std Dev x	3.858e+04	
Std Dev y	3.443e+04	
Integral	3.626e+05	
0	5047.43	644.803
0	362582	7537.84
0	0	0

Stock G4: proton y vs x



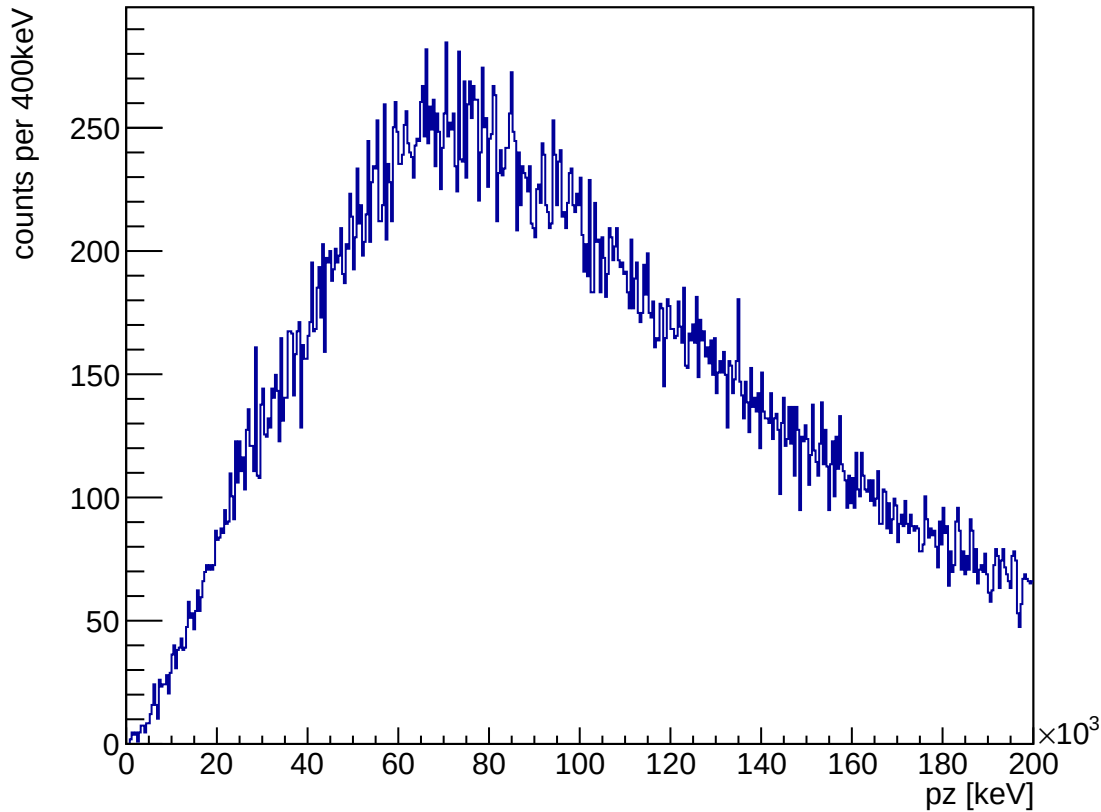
Entries	1663508		
Mean x	-3901		
Mean y	-0.7733		
Std Dev x	218.3		
Std Dev y	217.8		
Integral	1.547e+06		
0	0	0	
0	1.54743e+06	0	
0	0	0	

Mu2e mu resampler: proton y vs x



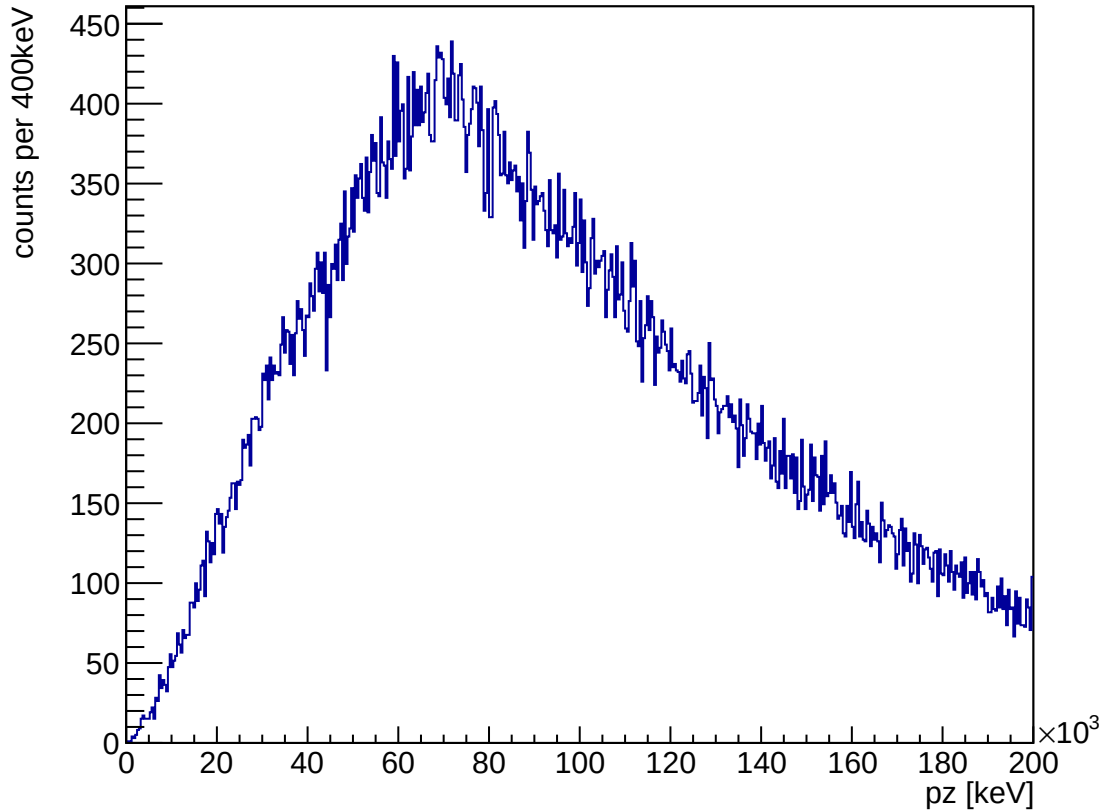
Entries	372430	
Mean x	-3900	
Mean y	-1.614	
Std Dev x	210.7	
Std Dev y	209.4	
Integral	3.758e+05	
0	0	0
0	375812	0
0	0	0

Stock G4: D momentum pz



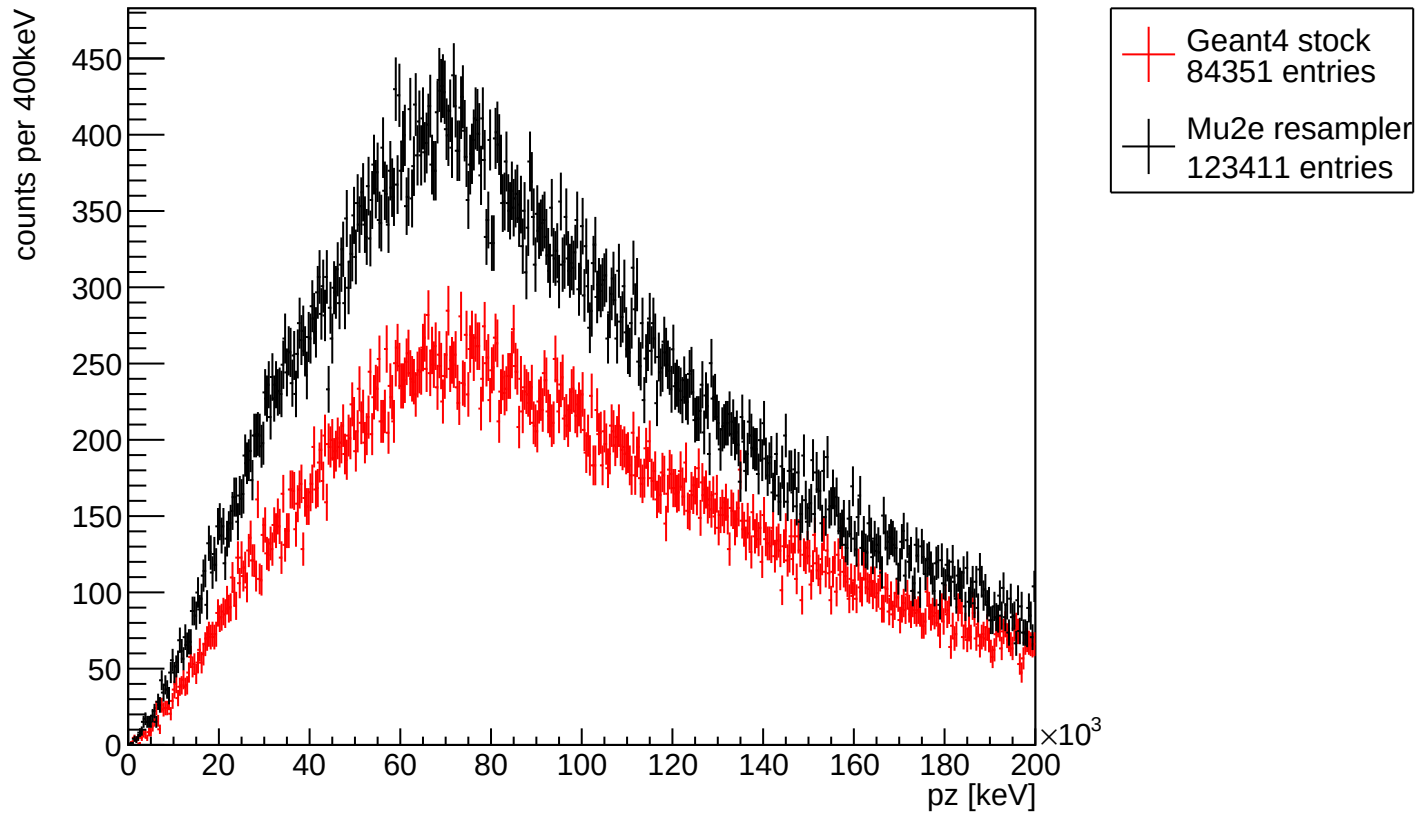
Entries	90679
Mean	9.569e+04
Std Dev	4.59e+04
Underflow	0
Overflow	1.03e+04
Integral	7.405e+04

Mu2e mu resampler: D momentum pz

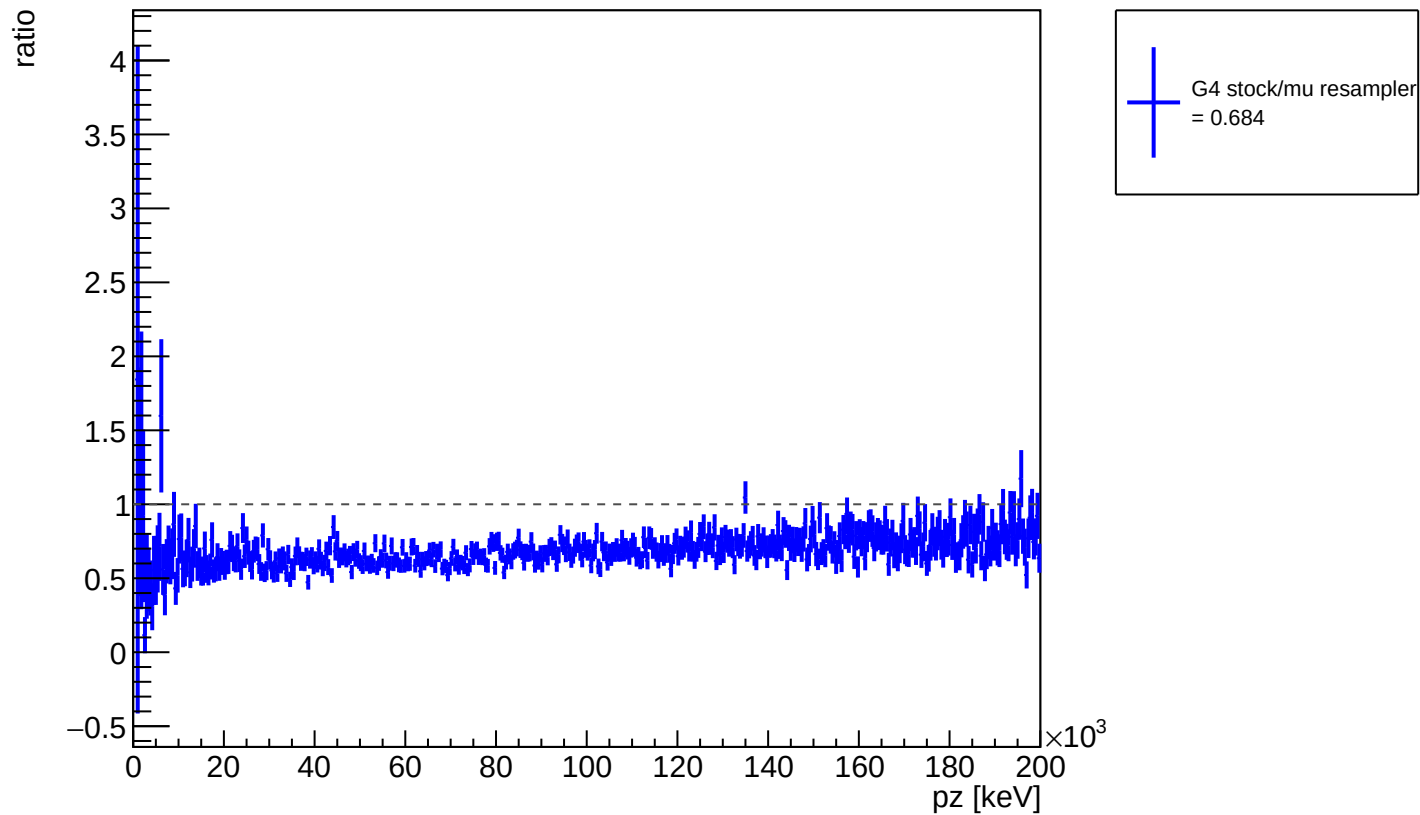


Entries	122300
Mean	9.262e+04
Std Dev	4.522e+04
Underflow	0
Overflow	1.311e+04
Integral	1.103e+05

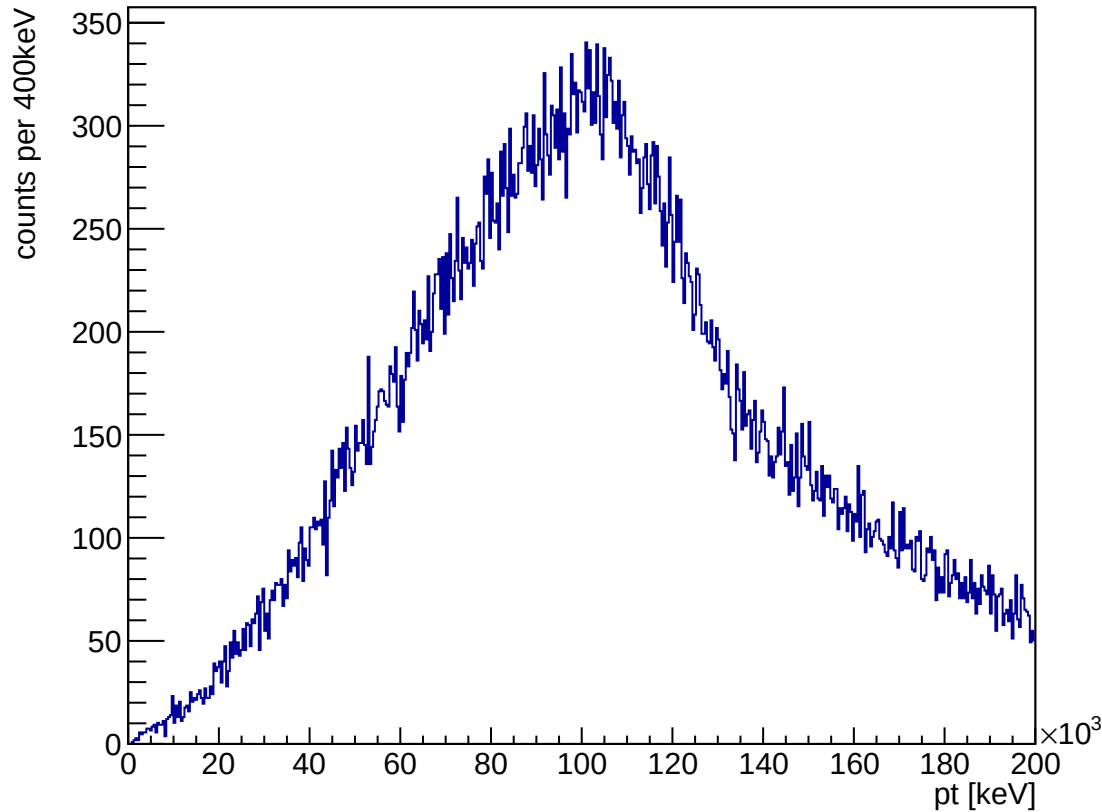
D momentum pz



ratio of G4 stock / muon resampler: D momentum pz

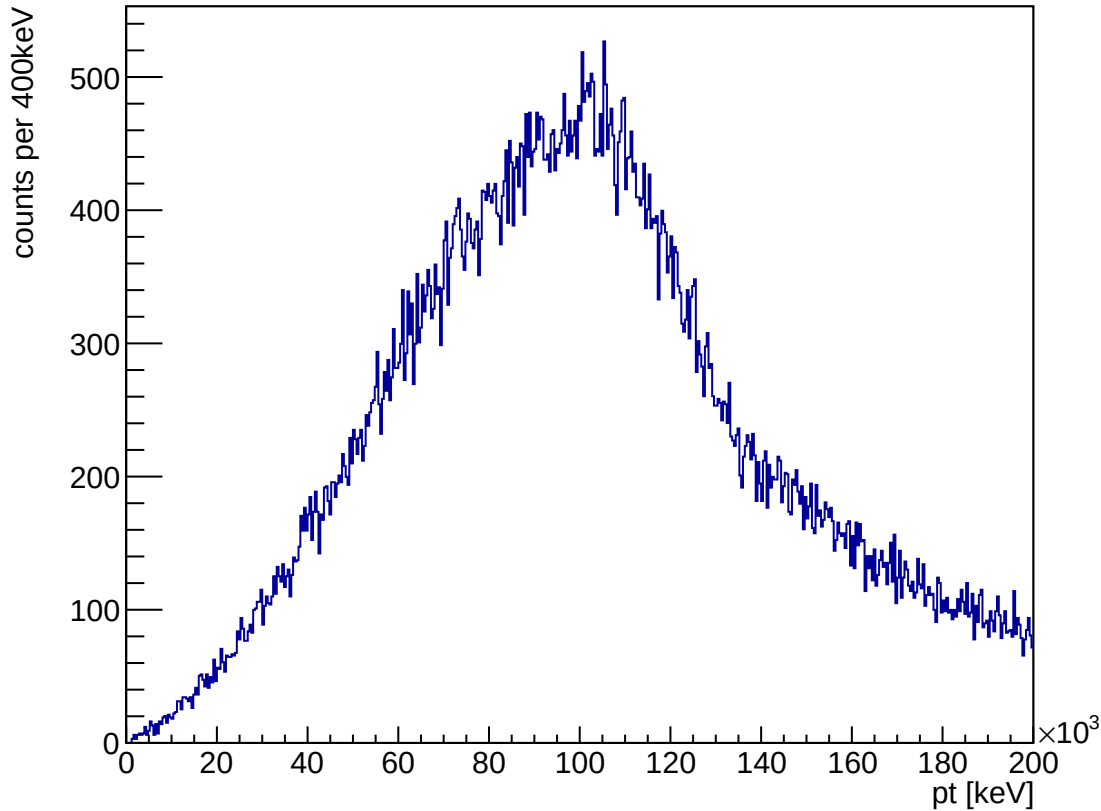


Stock G4: D transverse momentum pT



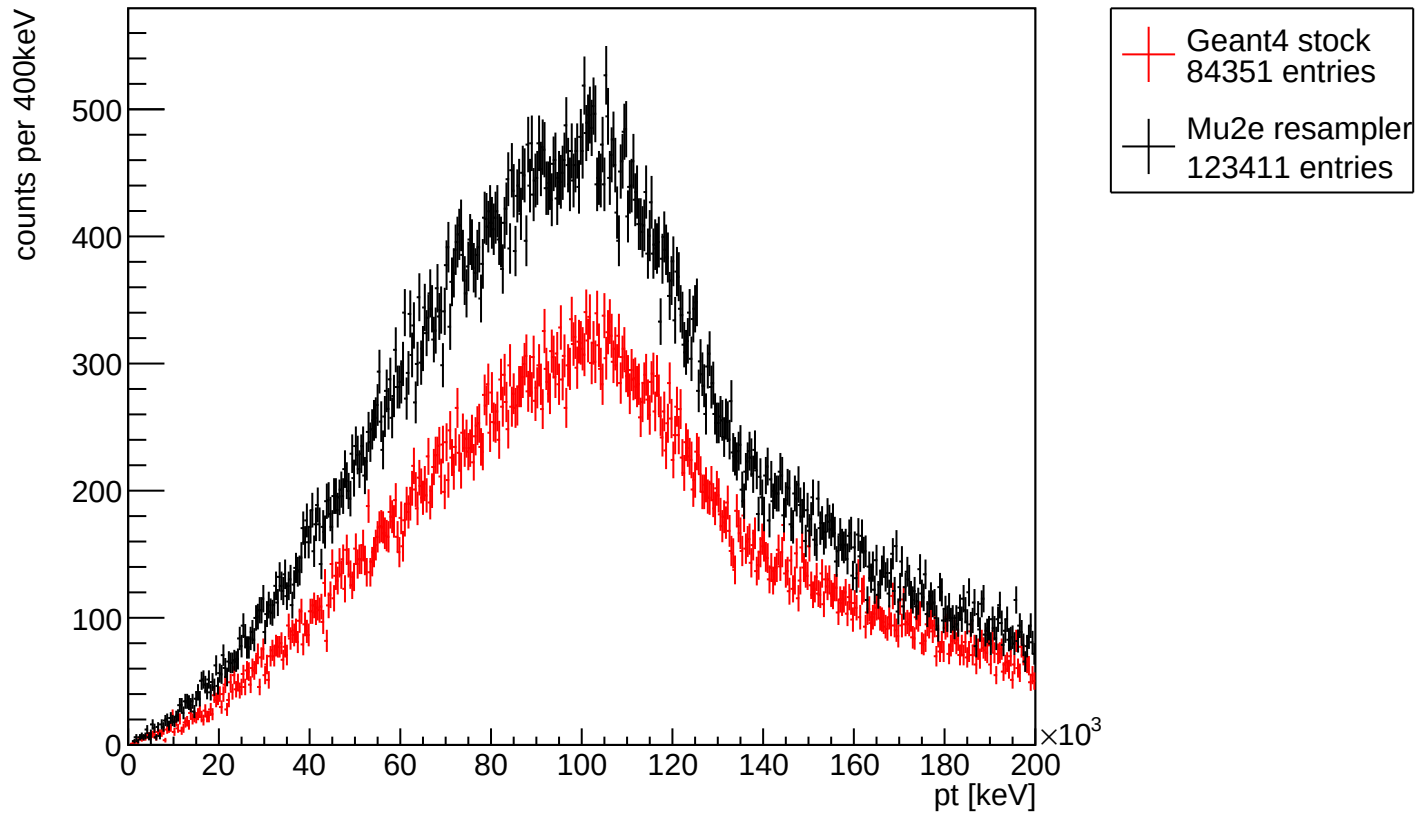
Entries	90679
Mean	1.039e+05
Std Dev	4.101e+04
Underflow	0
Overflow	8723
Integral	7.563e+04

Mu2e mu resampler: D transverse momentum pT

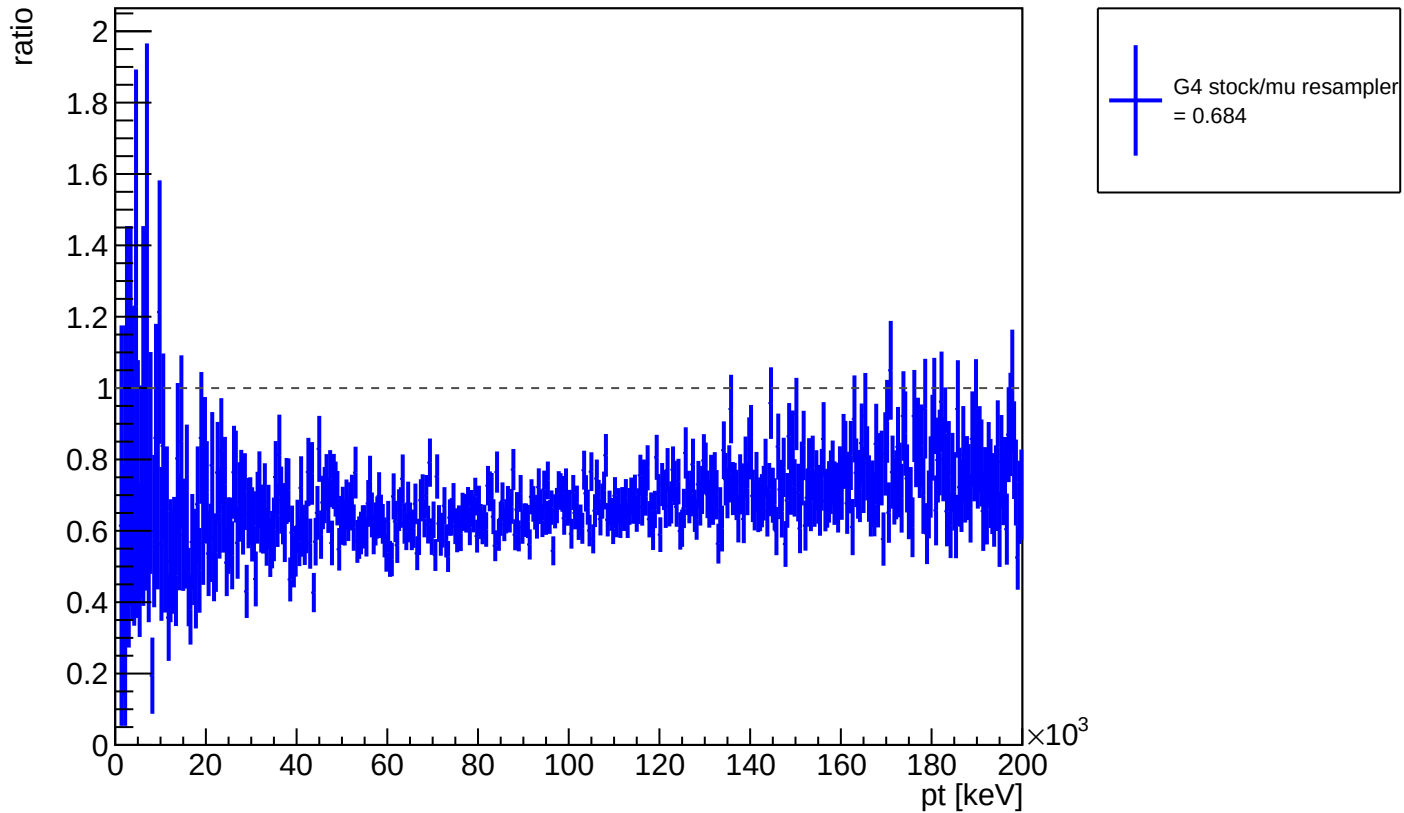


Entries	122300
Mean	1.016e+05
Std Dev	4.048e+04
Underflow	0
Overflow	1.097e+04
Integral	1.124e+05

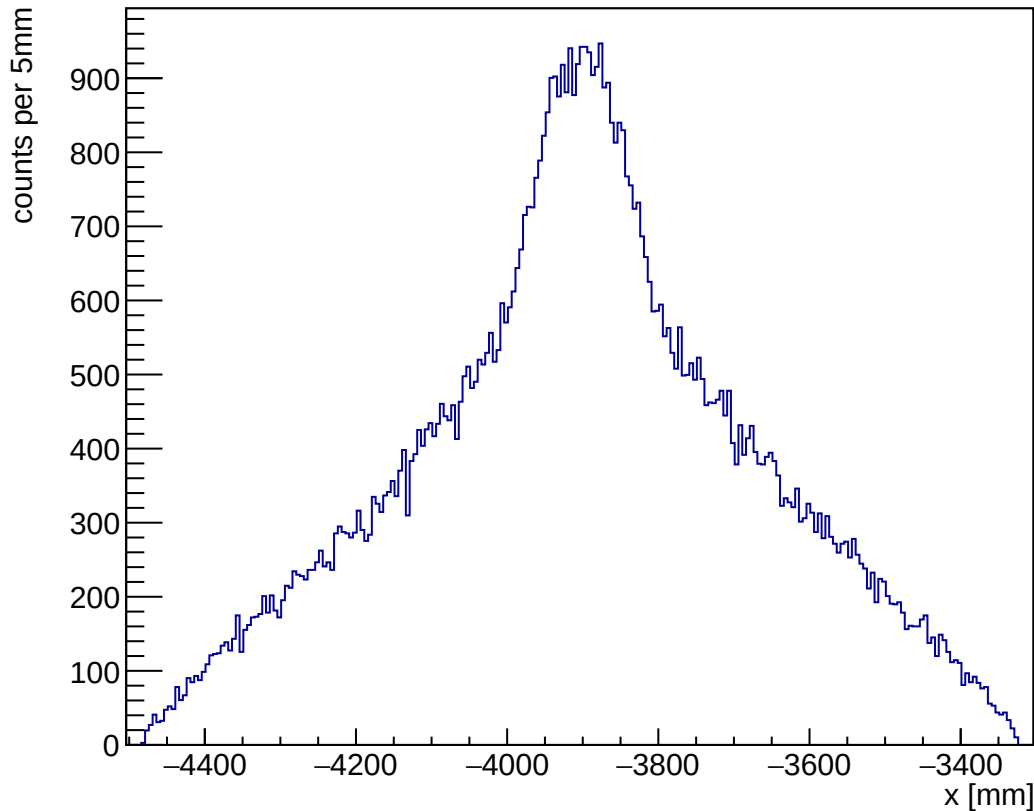
D transverse momentum pT



ratio of G4 stock / muon resampler: D transverse momentum pT



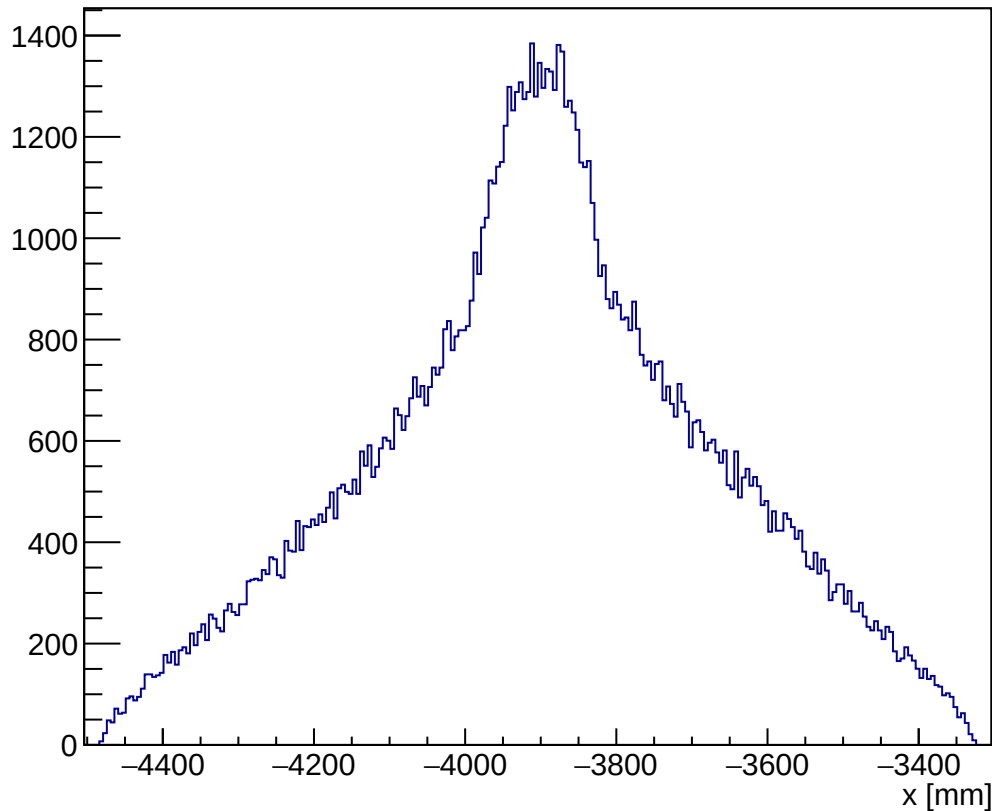
Stock G4: D x position



Entries	90679
Mean	-3898
Std Dev	228.7
Underflow	0
Overflow	0
Integral	8.435e+04

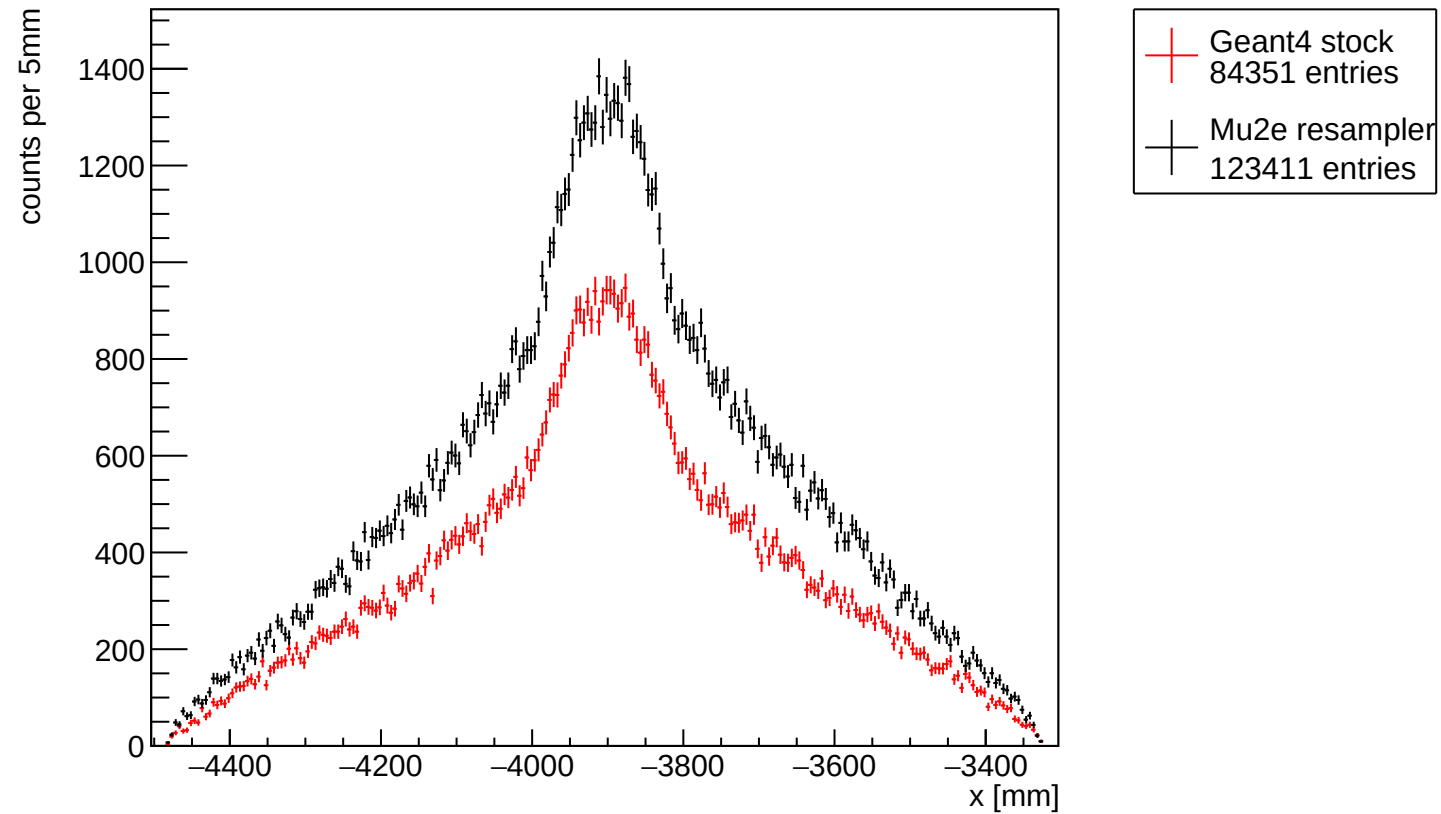
Mu2e mu resampler: D x position

counts per 5mm

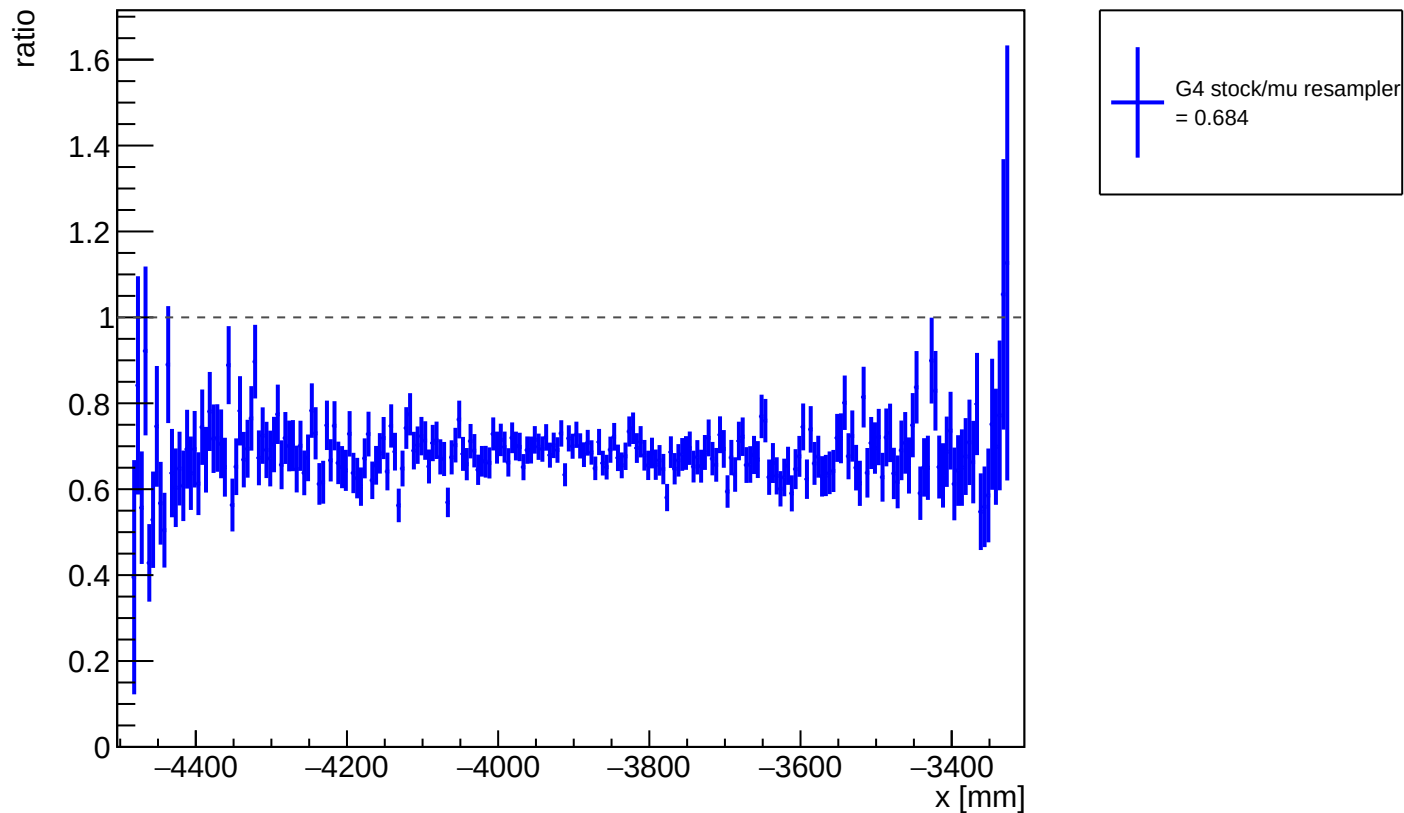


Entries	122300
Mean	-3896
Std Dev	228.9
Underflow	0
Overflow	0
Integral	1.234e+05

D x position

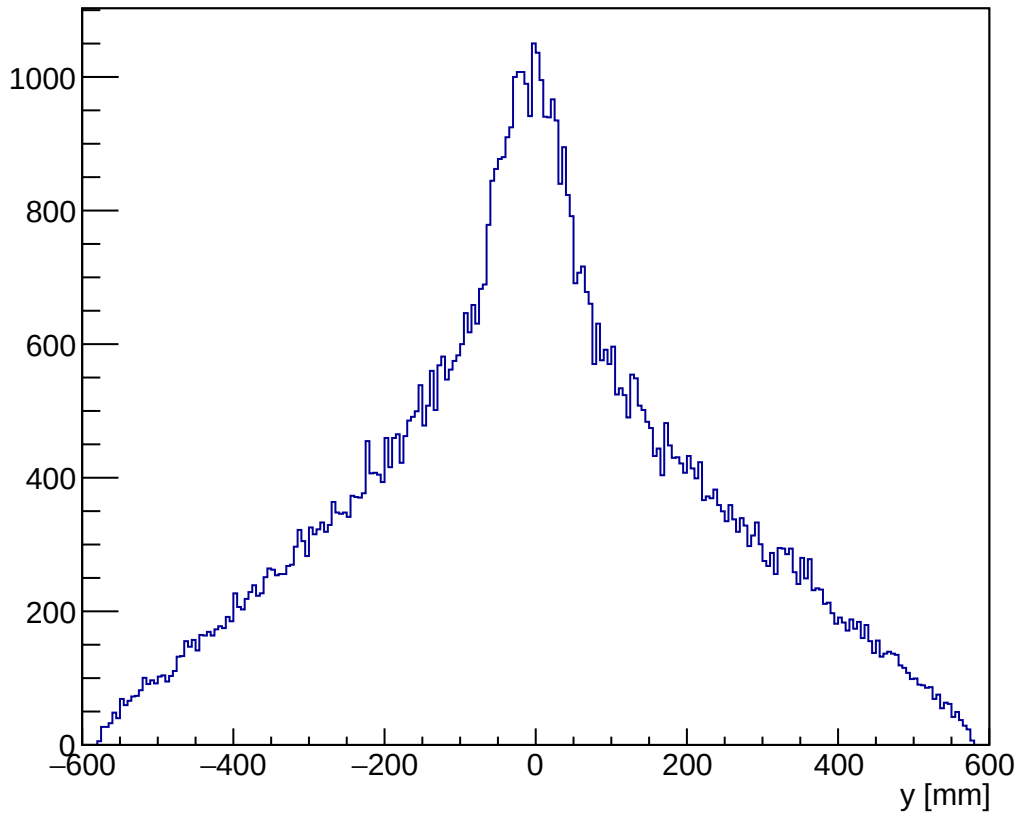


ratio of G4 stock / muon resampler: D x position



Stock G4: D y position

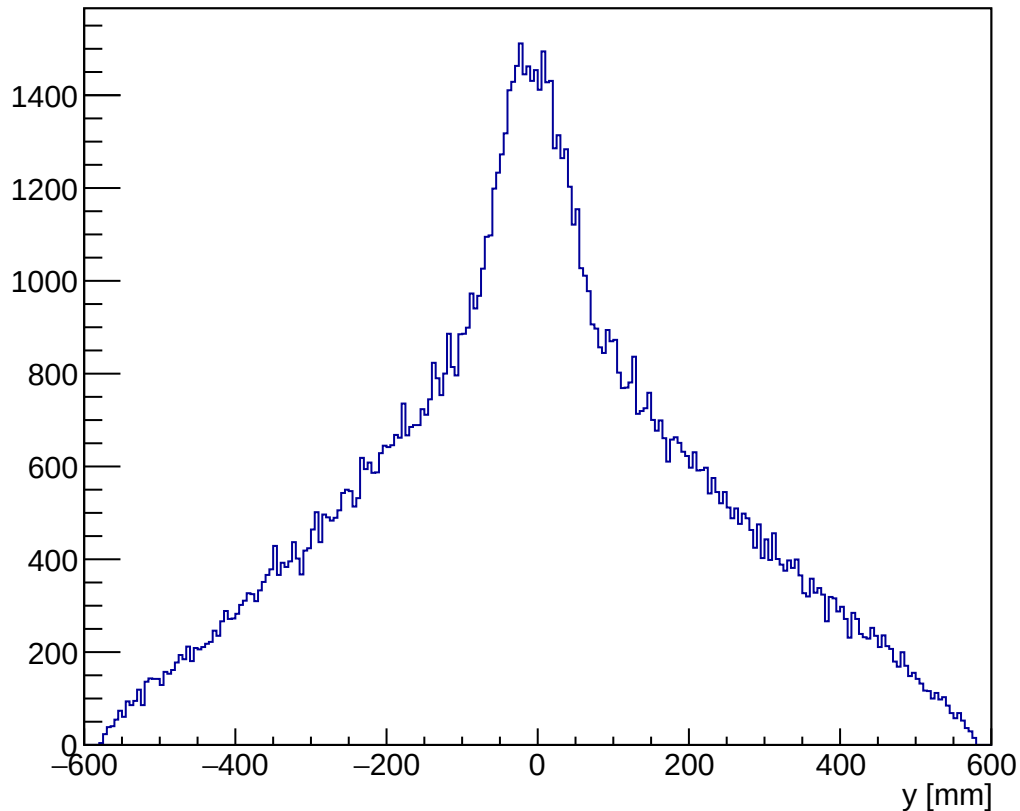
counts per 5mm



Entries	90679
Mean	-1.773
Std Dev	227.9
Underflow	0
Overflow	0
Integral	8.435e+04

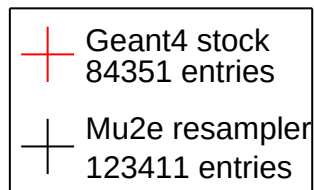
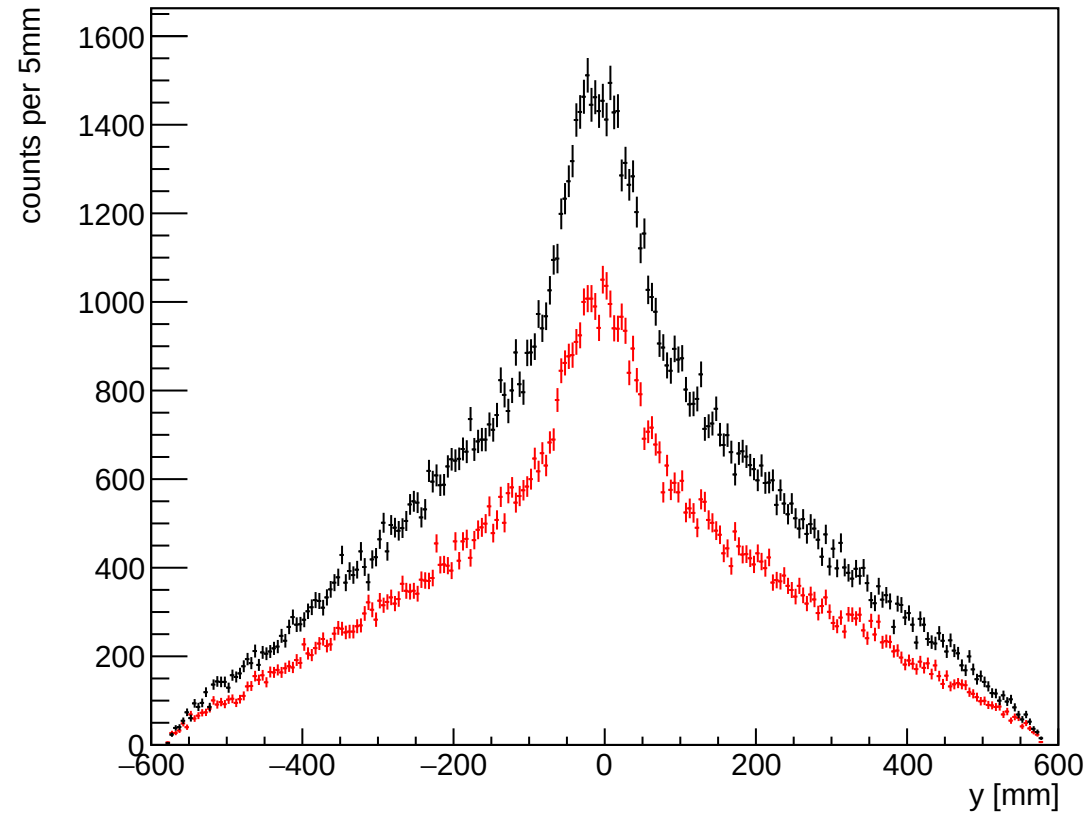
Mu2e mu resampler: D y position

counts per 5mm

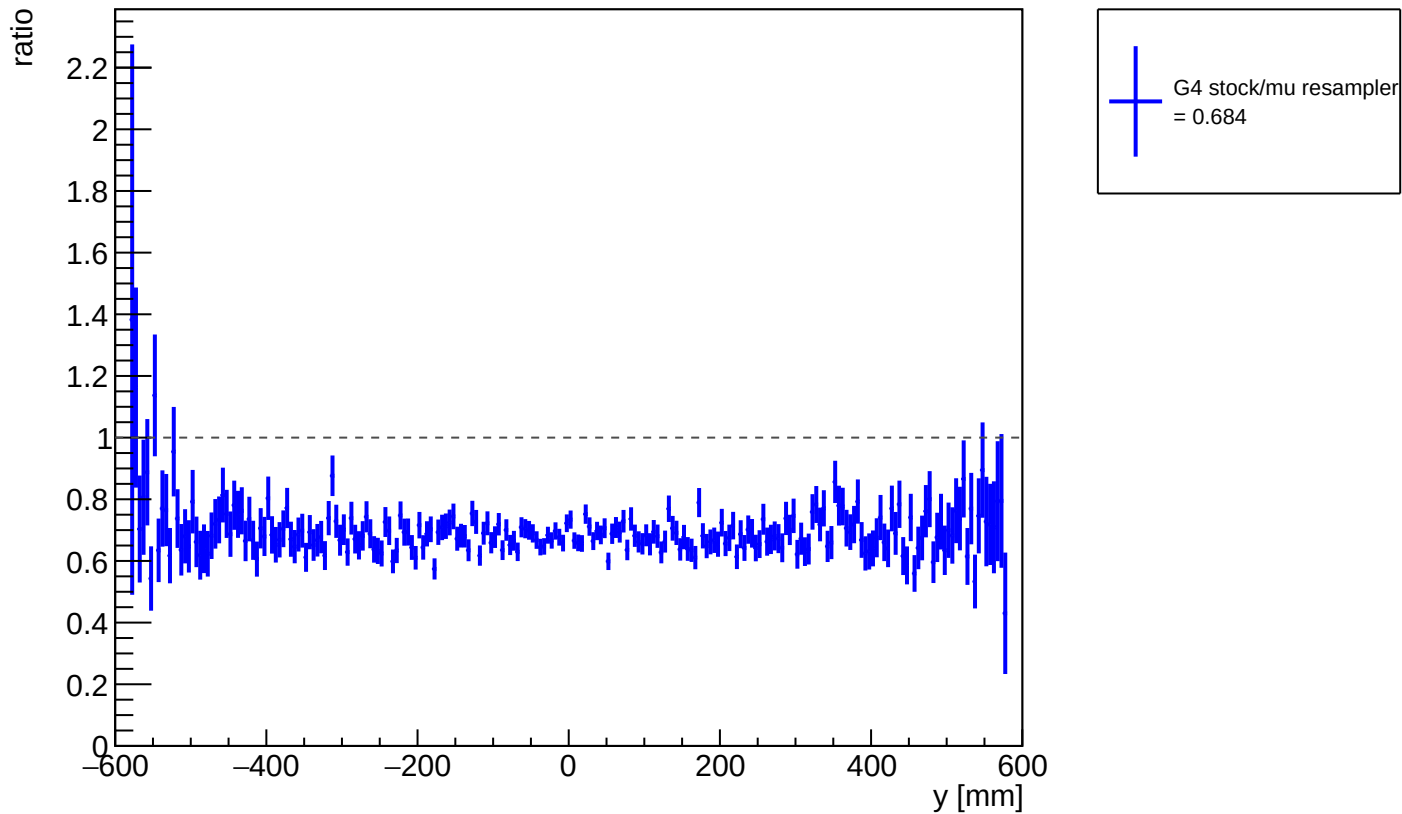


Entries	122300
Mean	-1.519
Std Dev	226.3
Underflow	0
Overflow	0
Integral	1.234e+05

D y position

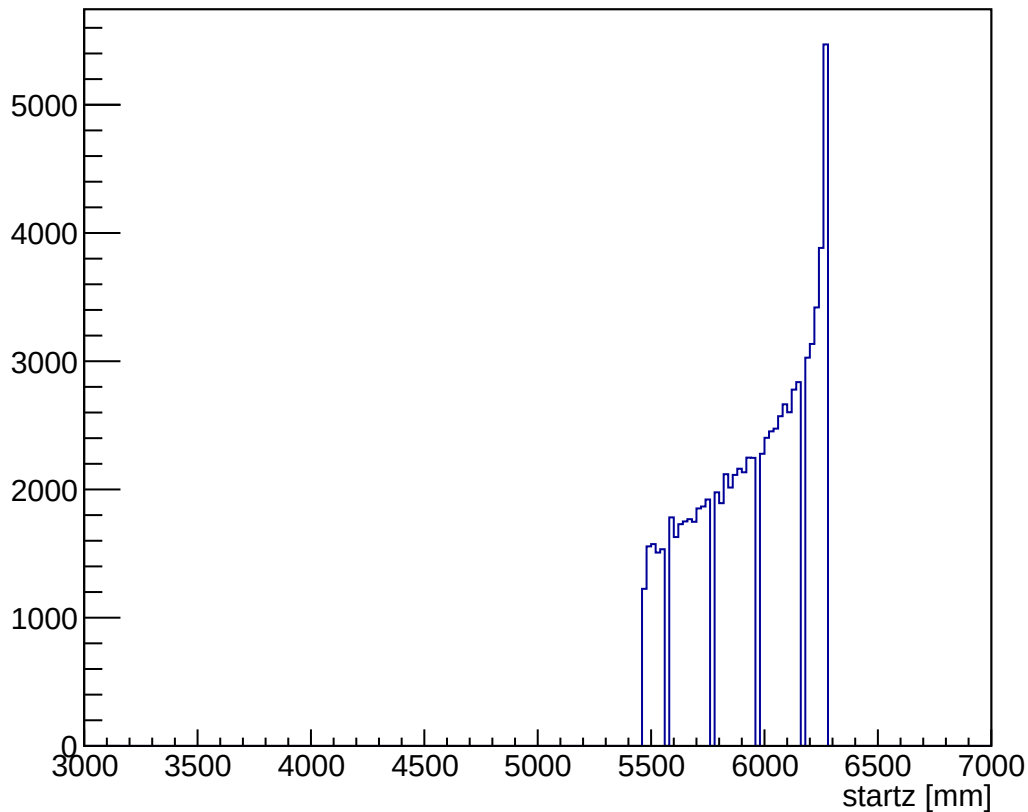


ratio of G4 stock / muon resampler: D y position



Stock G4: D start z

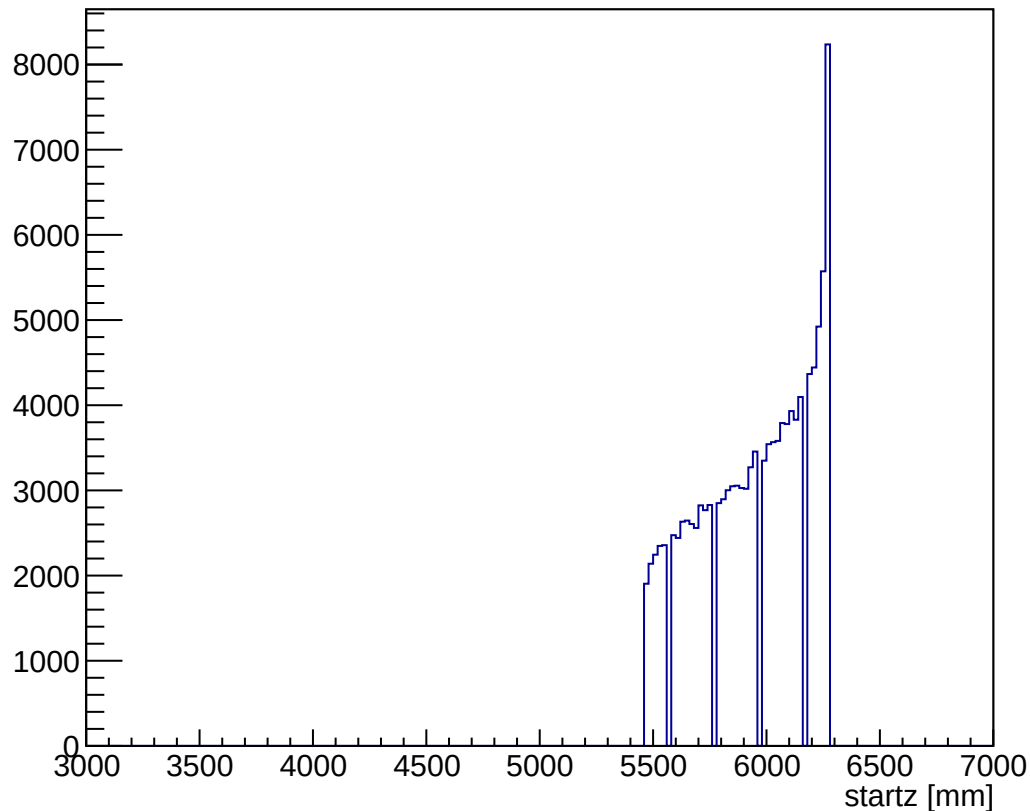
counts per 20mm



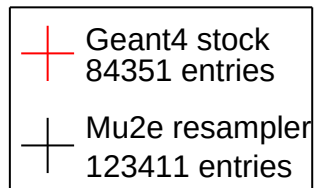
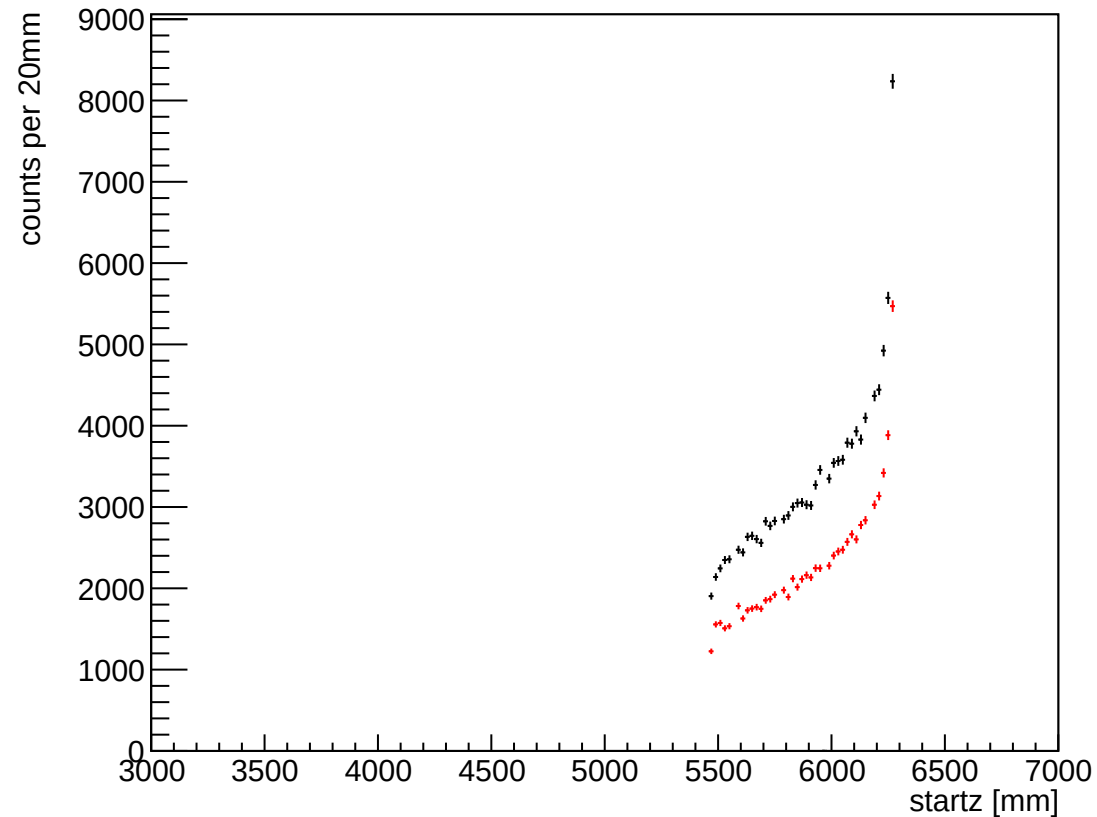
Entries	90679
Mean	5940
Std Dev	238.9
Underflow	0
Overflow	0
Integral	8.435e+04

Mu2e mu resampler: D start z

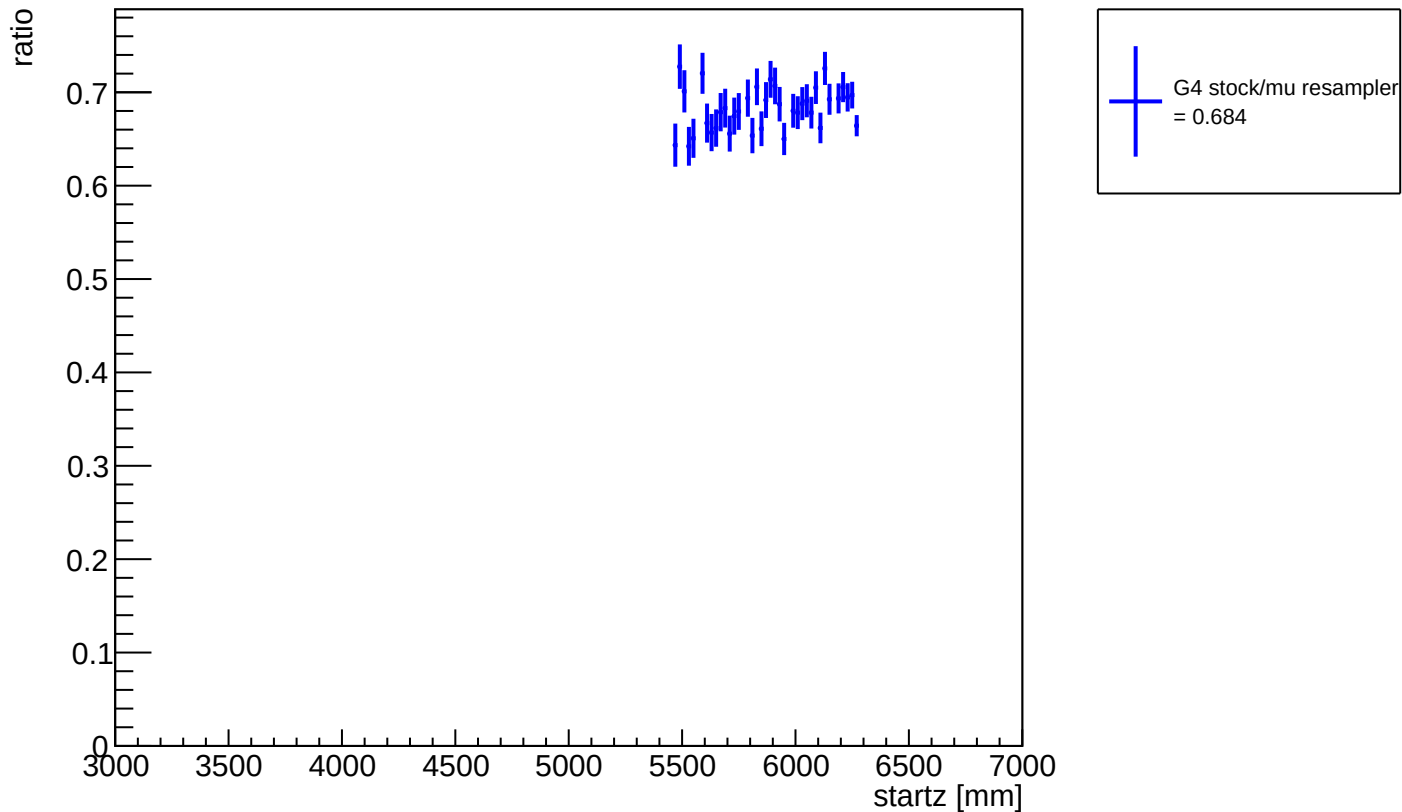
counts per 20mm



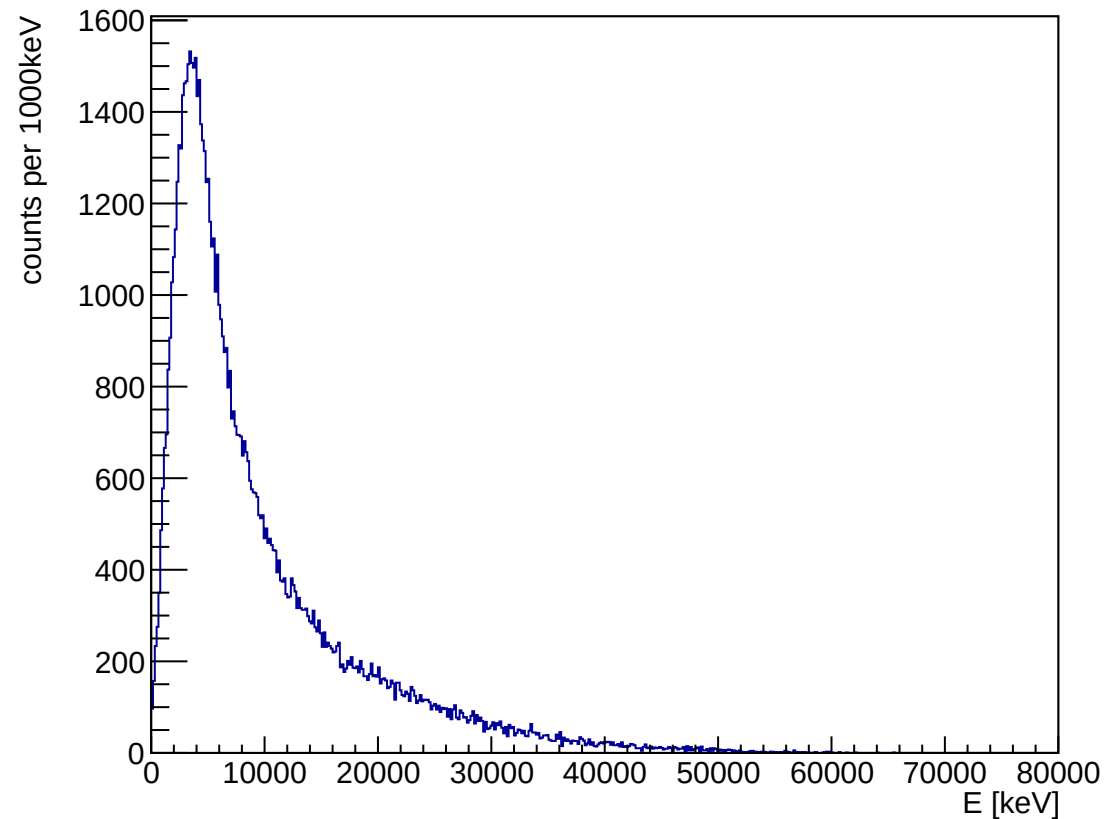
D start z



ratio of G4 stock / muon resampler: D start z

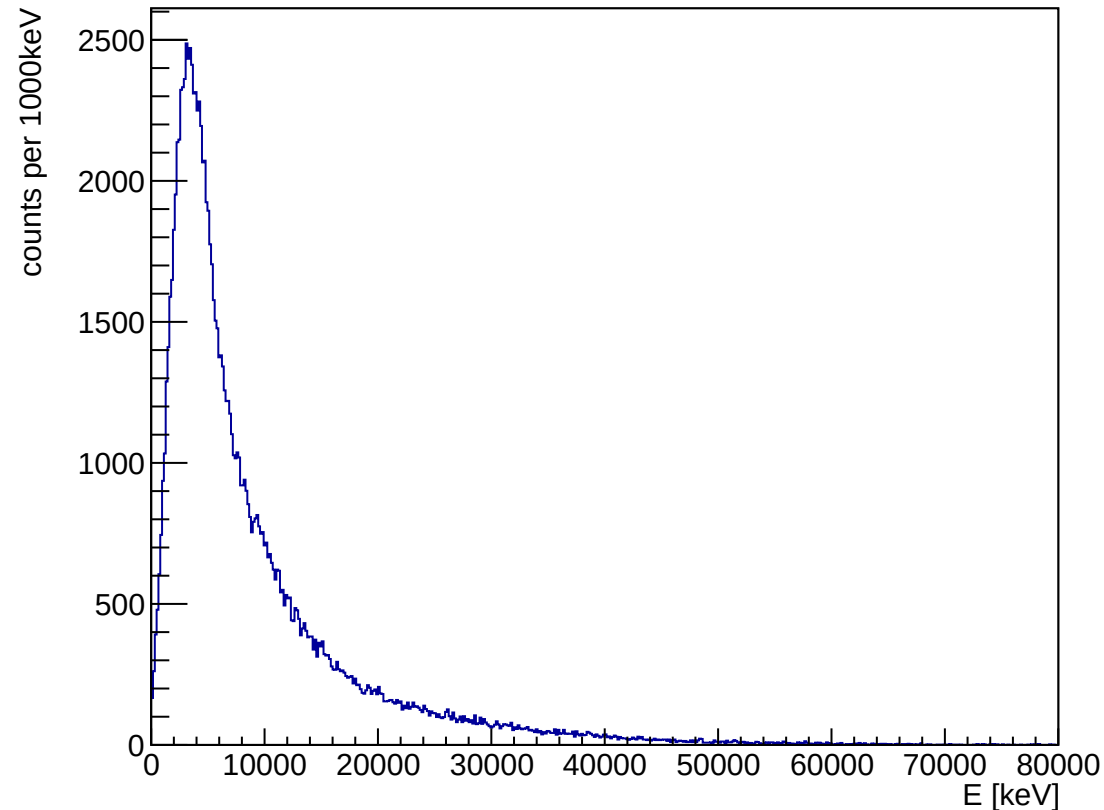


Stock G4: D kin. energy



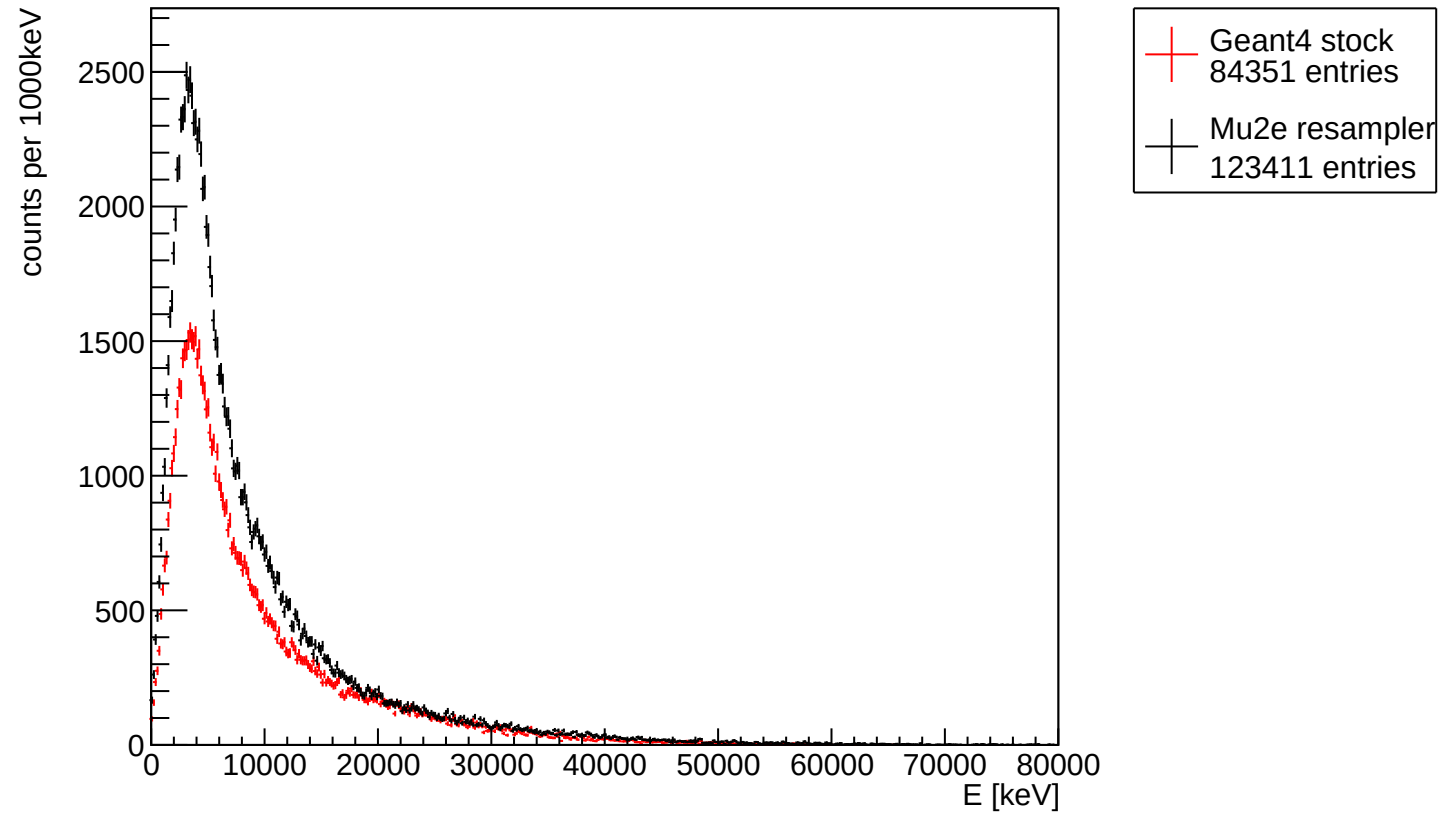
Entries	90679
Mean	9478
Std Dev	8581
Underflow	0
Overflow	0
Integral	8.435e+04

Mu2e mu resampler: D kin. energy

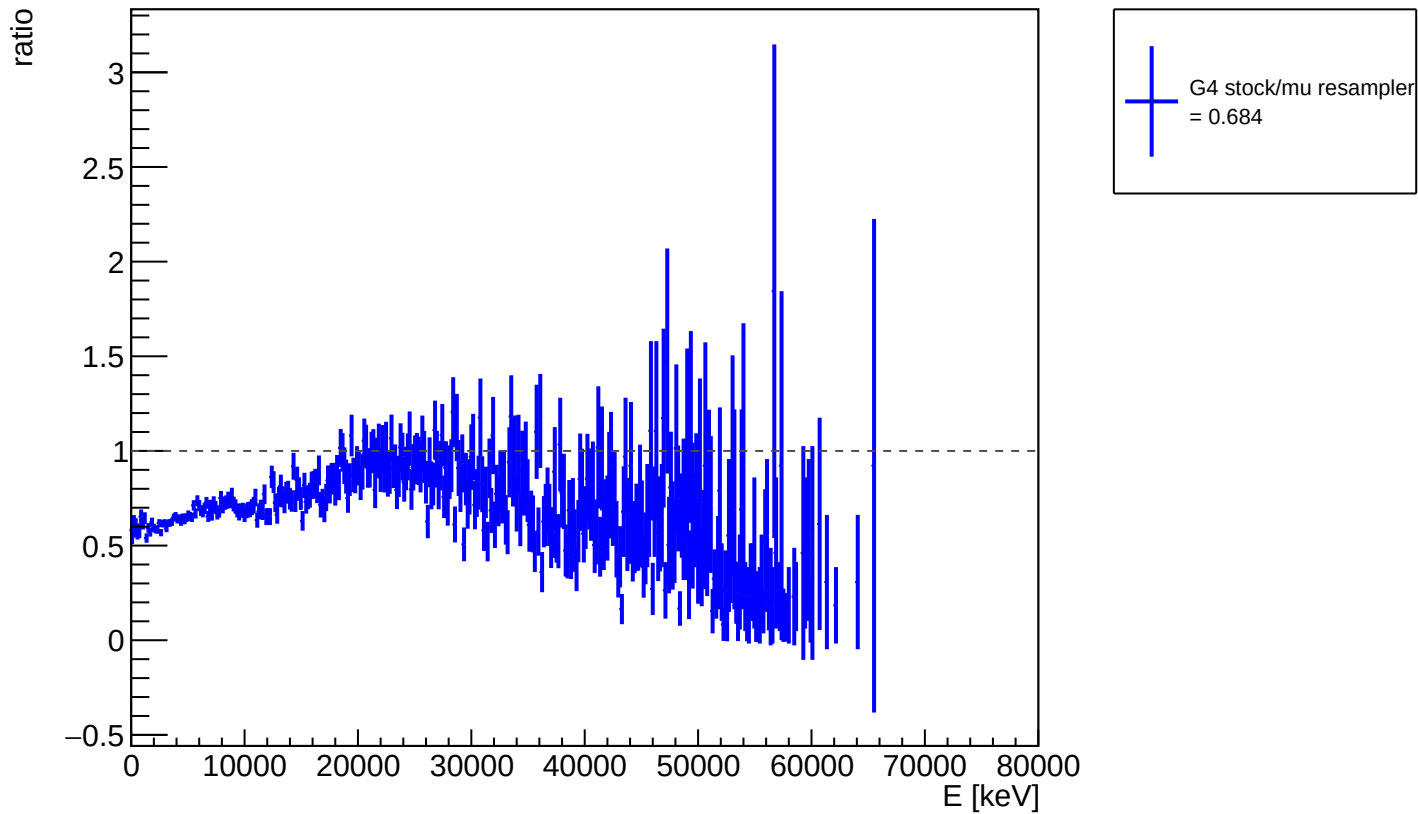


Entries	122300
Mean	8924
Std Dev	8879
Underflow	0
Overflow	37.34
Integral	1.234e+05

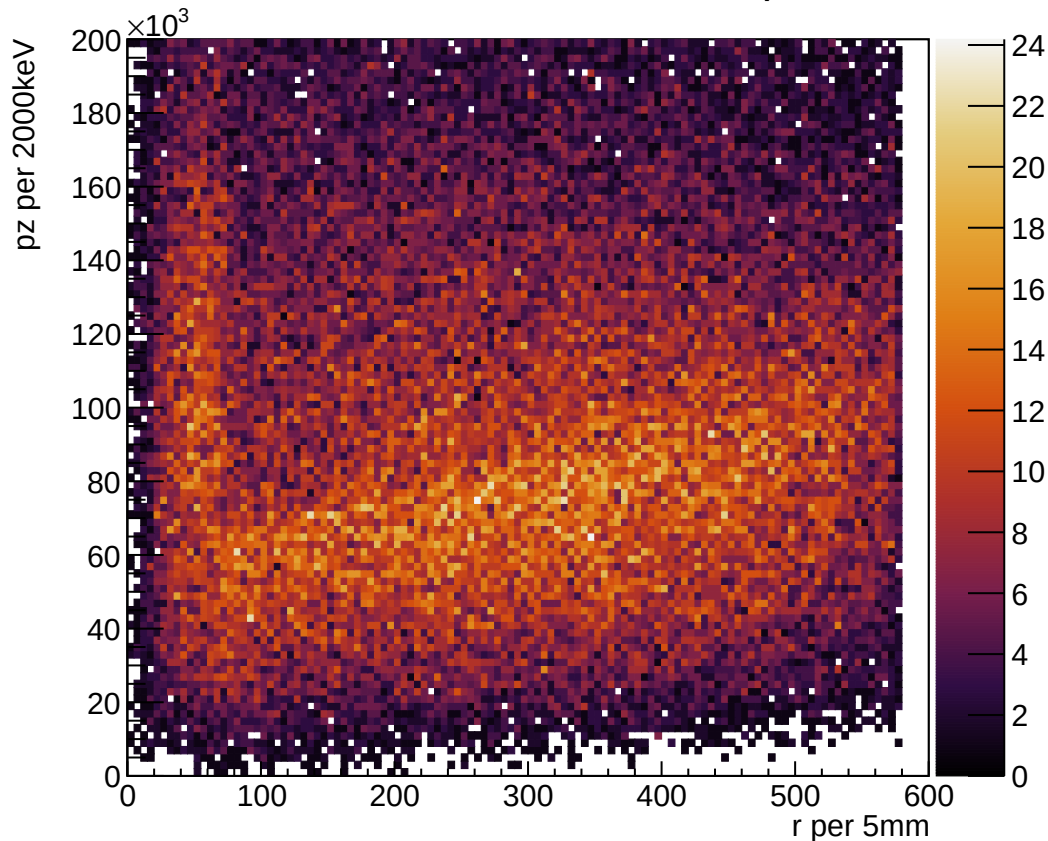
D kin. energy



ratio of G4 stock / muon resampler: D kin. energy

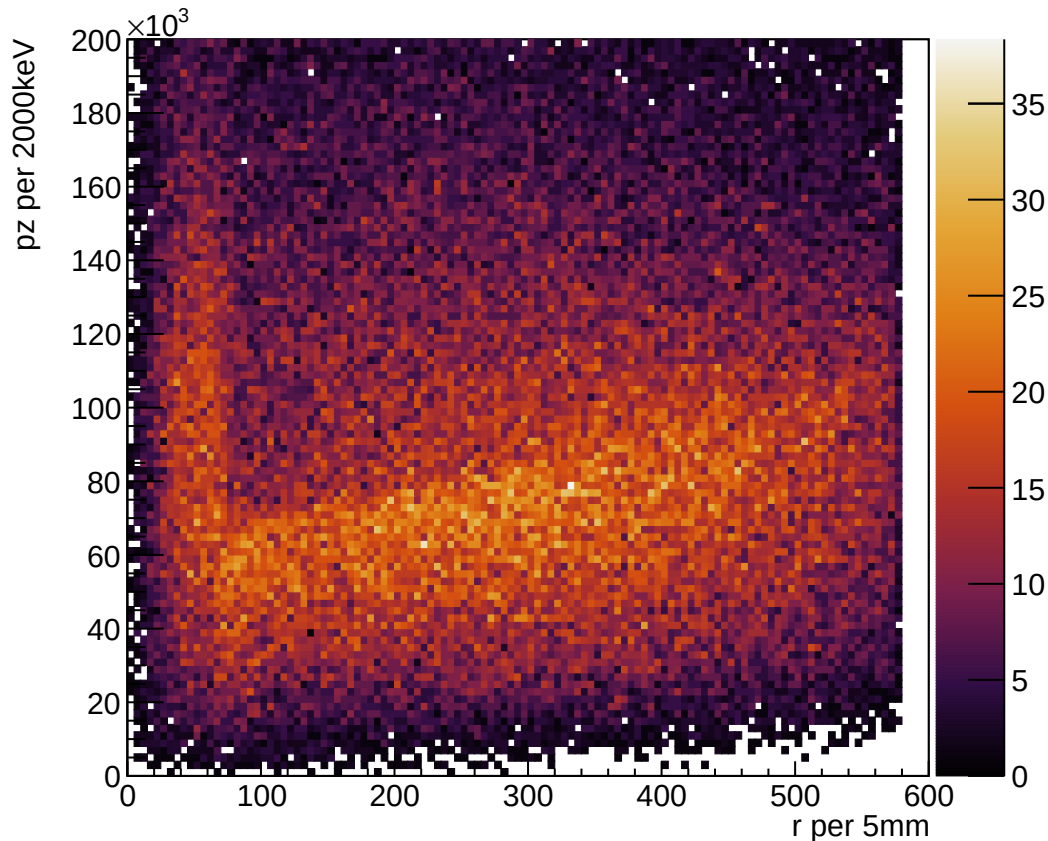


Stock G4: D pz vs r



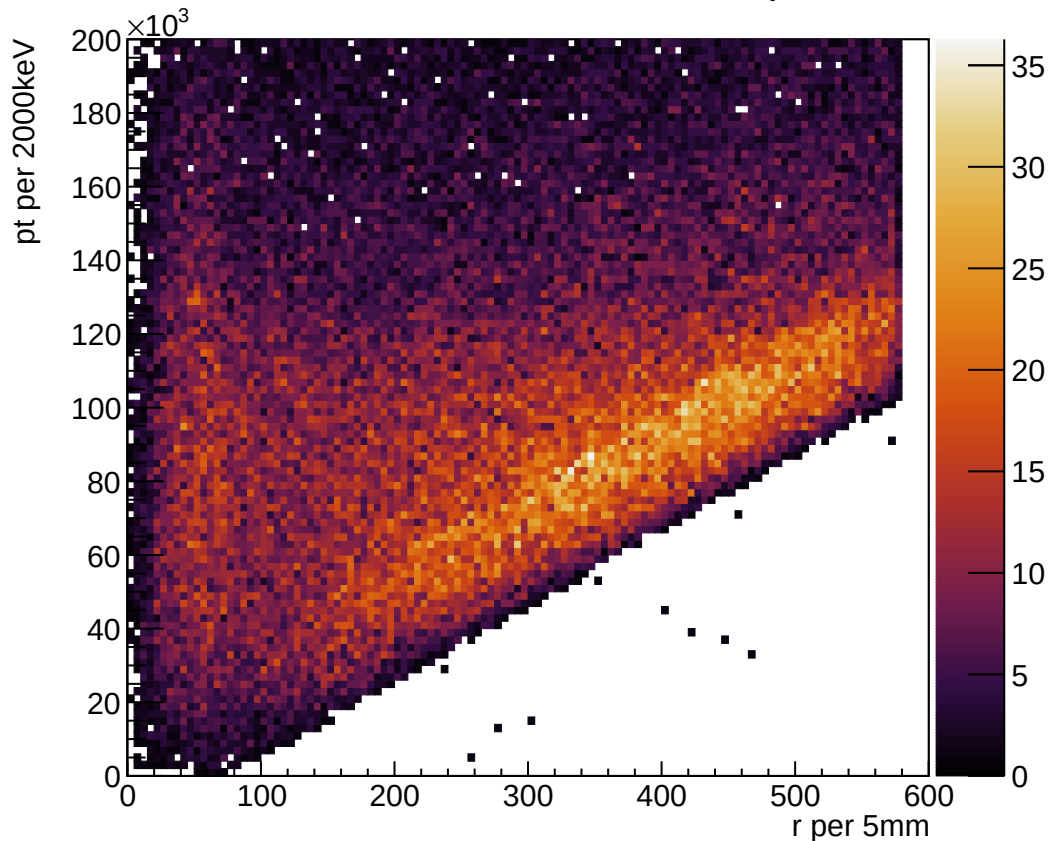
Entries	90679	
Mean x	287.3	
Mean y	9.569e+04	
Std Dev x	156.9	
Std Dev y	4.59e+04	
Integral	7.405e+04	
0	10303.1	0
0	74048.3	0
0	0	0

Mu2e mu resampler: D pz vs r



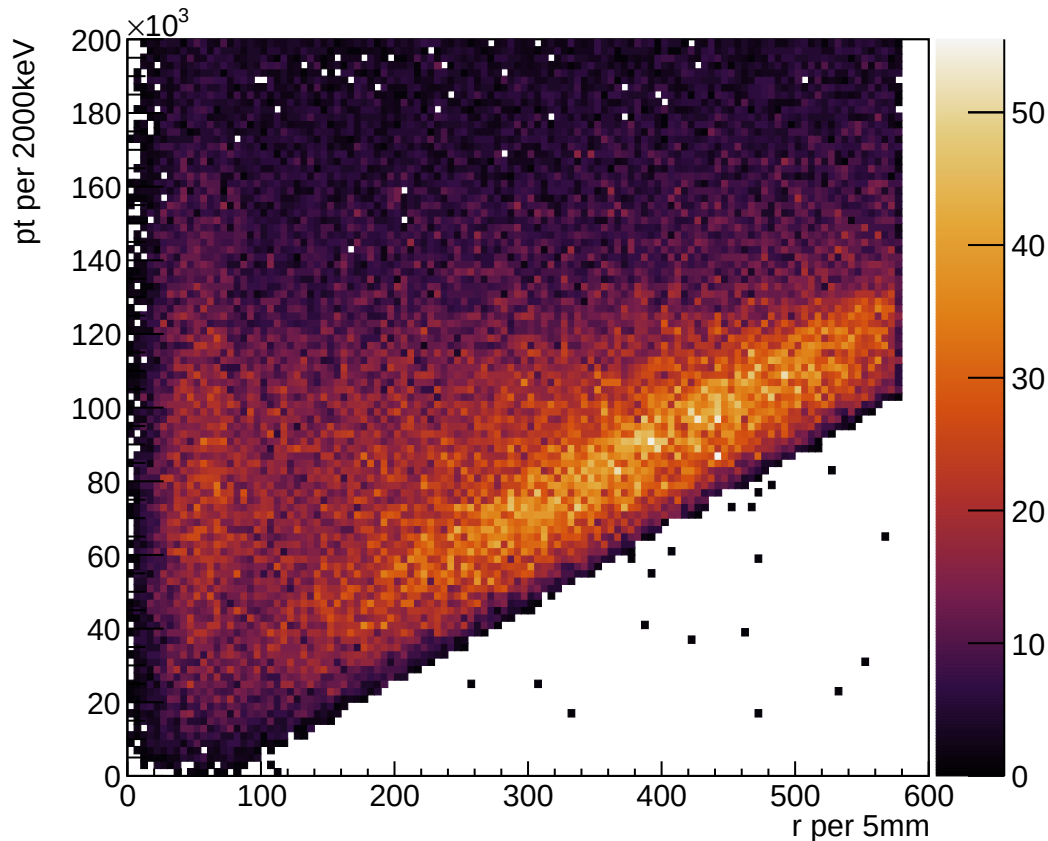
Entries	122300	
Mean x	285.8	
Mean y	9.262e+04	
Std Dev x	155.8	
Std Dev y	4.522e+04	
Integral	1.103e+05	
0	13111	0
0	110300	0
0	0	0

Stock G4: D pt vs r



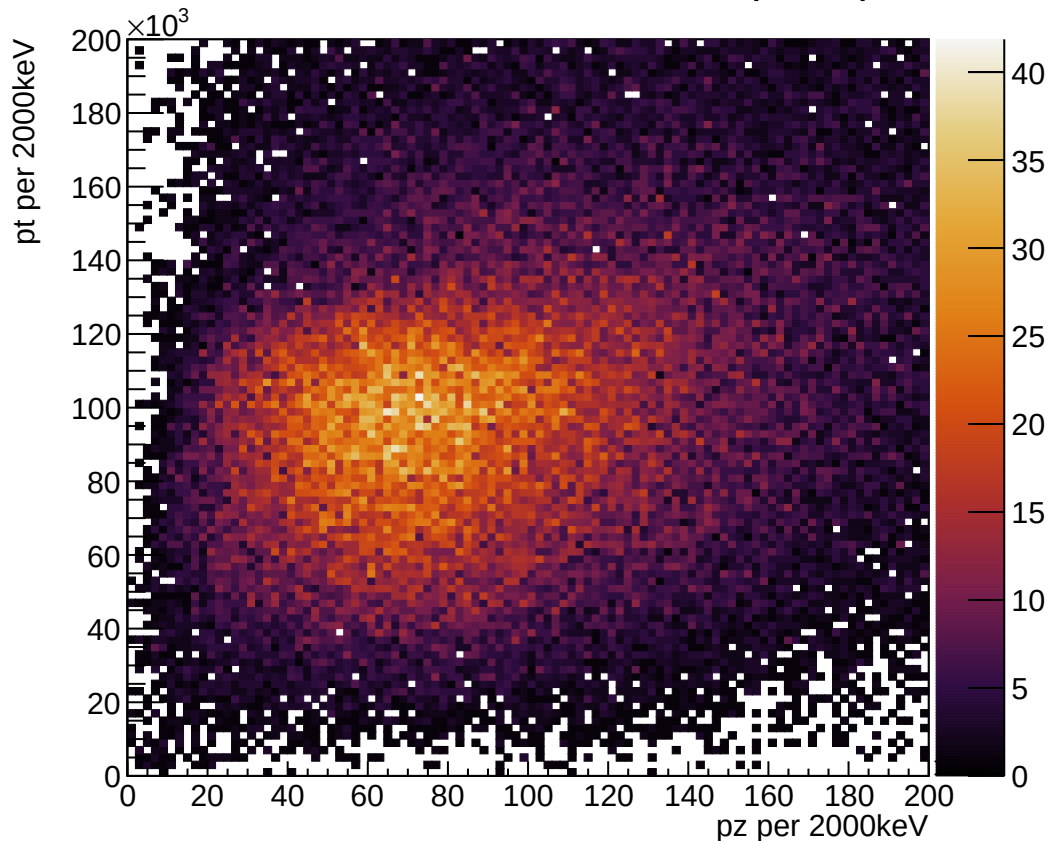
Entries	90679	
Mean x	279.6	
Mean y	1.039e+05	
Std Dev x	155.4	
Std Dev y	4.101e+04	
Integral	7.563e+04	
0	8722.67	0
0	75628.7	0
0	0	0

Mu2e mu resampler: D pt vs r



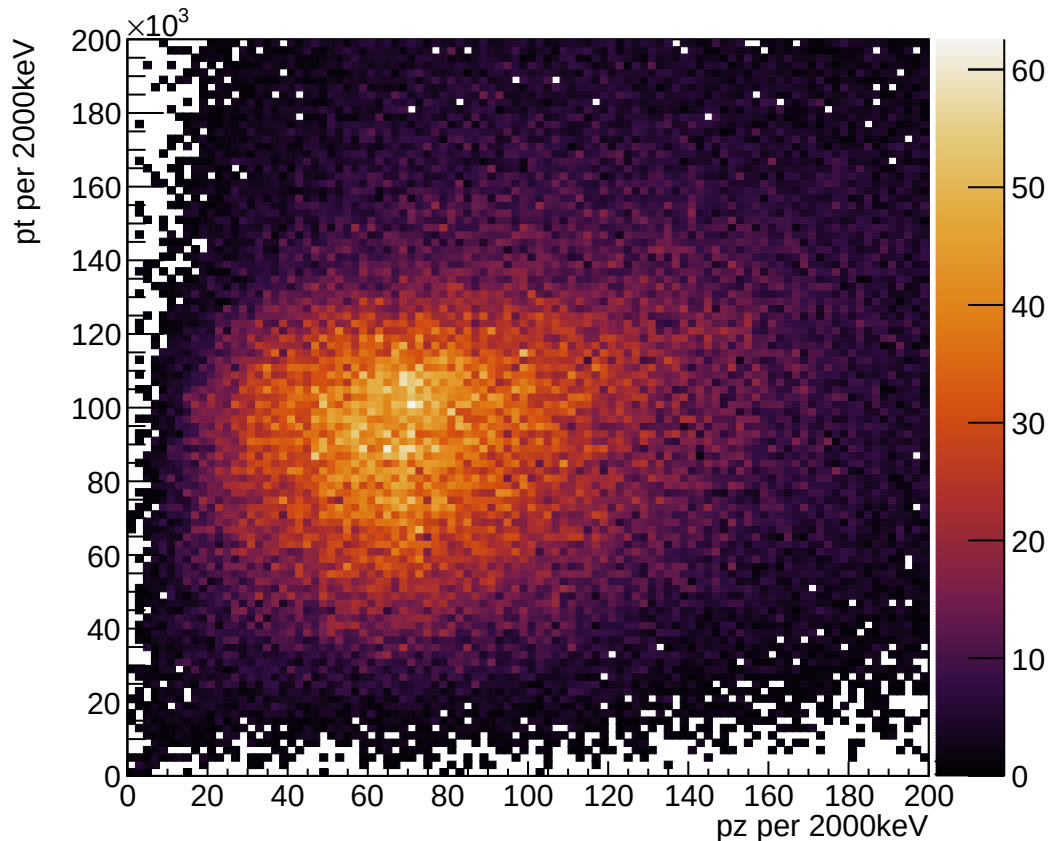
Entries	122300	
Mean x	279.7	
Mean y	1.016e+05	
Std Dev x	154.6	
Std Dev y	4.048e+04	
Integral	1.124e+05	
0	10971.7	0
0	112439	0
0	0	0

Stock G4: D pt vs pz



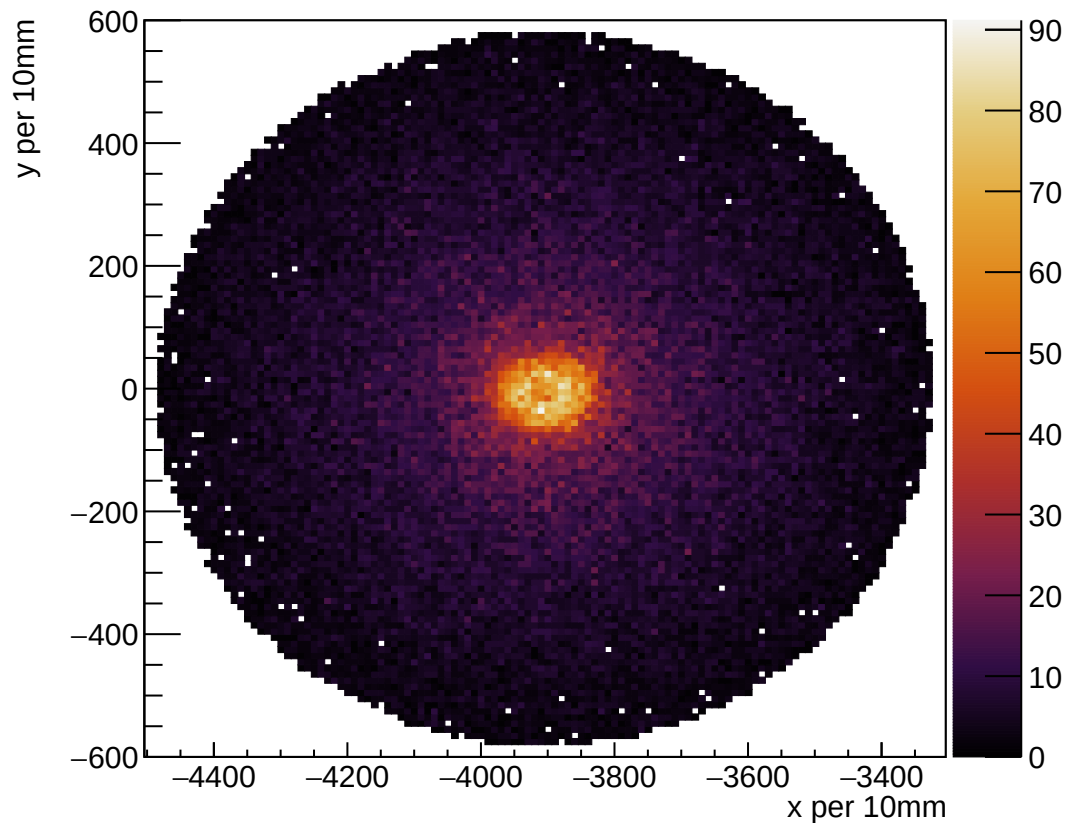
Entries	90679	
Mean x	9.353e+04	
Mean y	1.017e+05	
Std Dev x	4.517e+04	
Std Dev y	3.99e+04	
Integral	6.778e+04	
0	6270.61	2452.06
0	67777.7	7851.05
0	0	0

Mu2e mu resampler: D pt vs pz



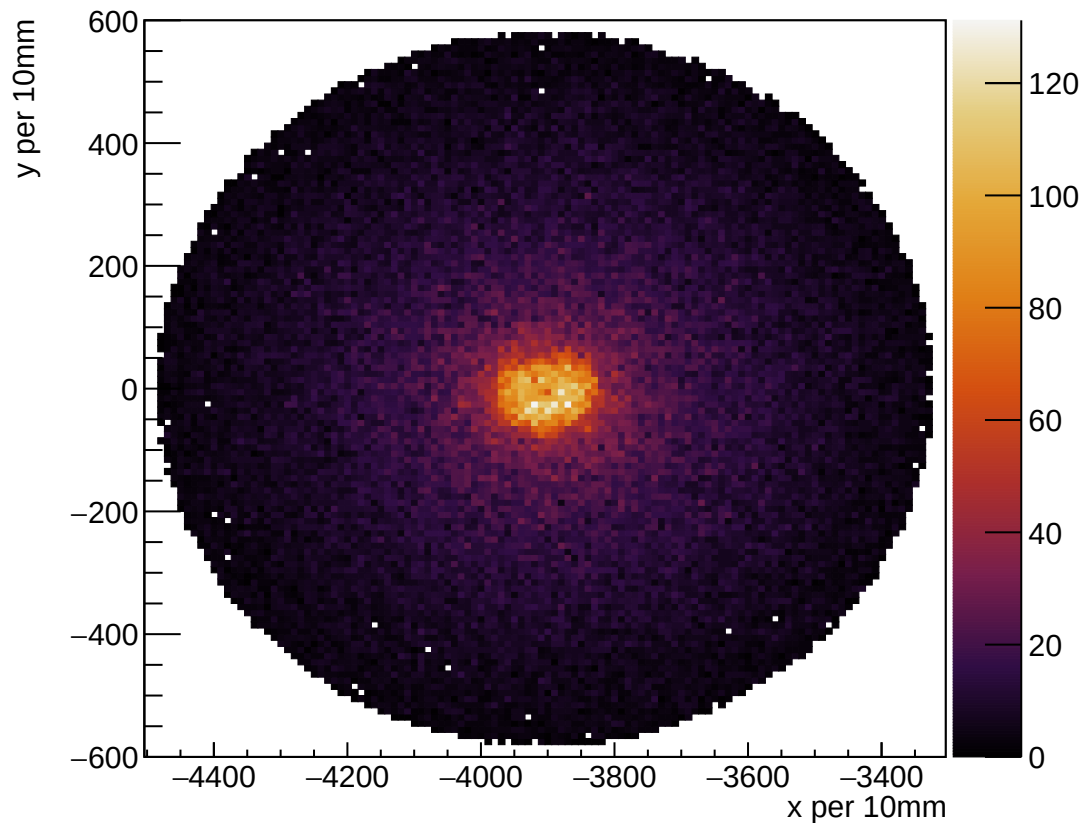
Entries	122300	
Mean x	9.094e+04	
Mean y	9.957e+04	
Std Dev x	4.458e+04	
Std Dev y	3.941e+04	
Integral	1.029e+05	
0	7438.95	3532.8
0	102861	9578.2
0	0	0

Stock G4: D y vs x



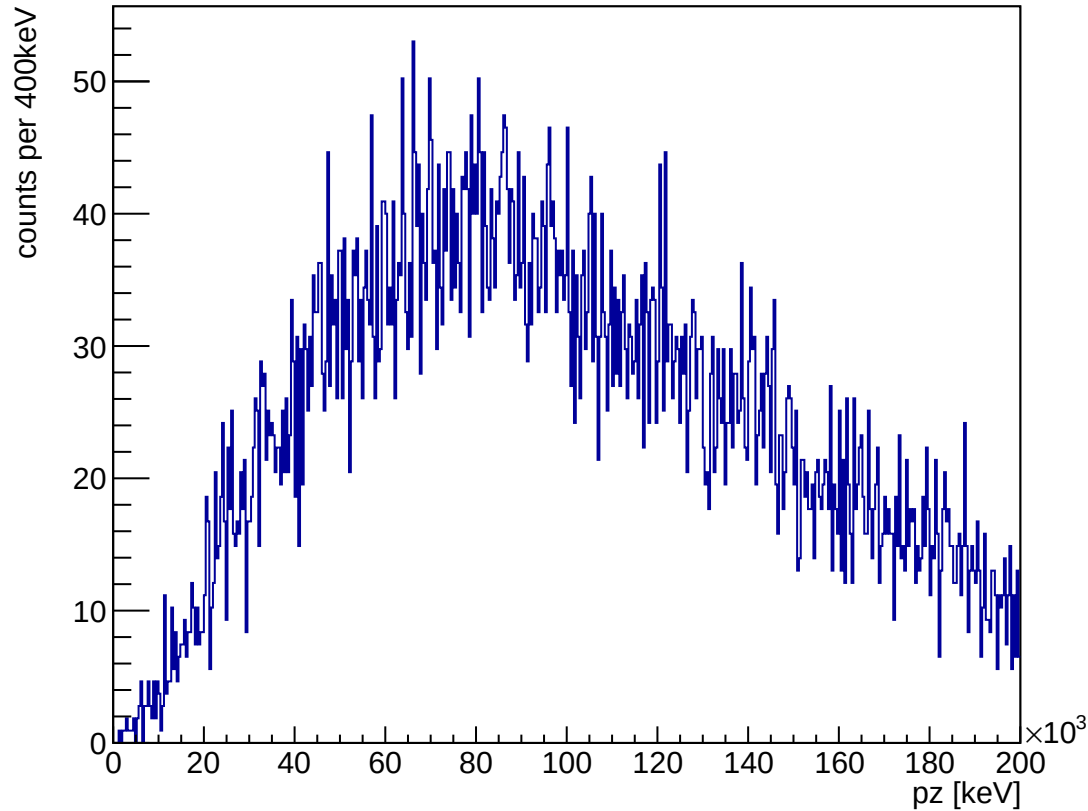
Entries	90679	
Mean x	-3898	
Mean y	-1.773	
Std Dev x	228.7	
Std Dev y	227.9	
Integral	8.435e+04	
0	0	0
0	84351.4	0
0	0	0

Mu2e mu resampler: D y vs x



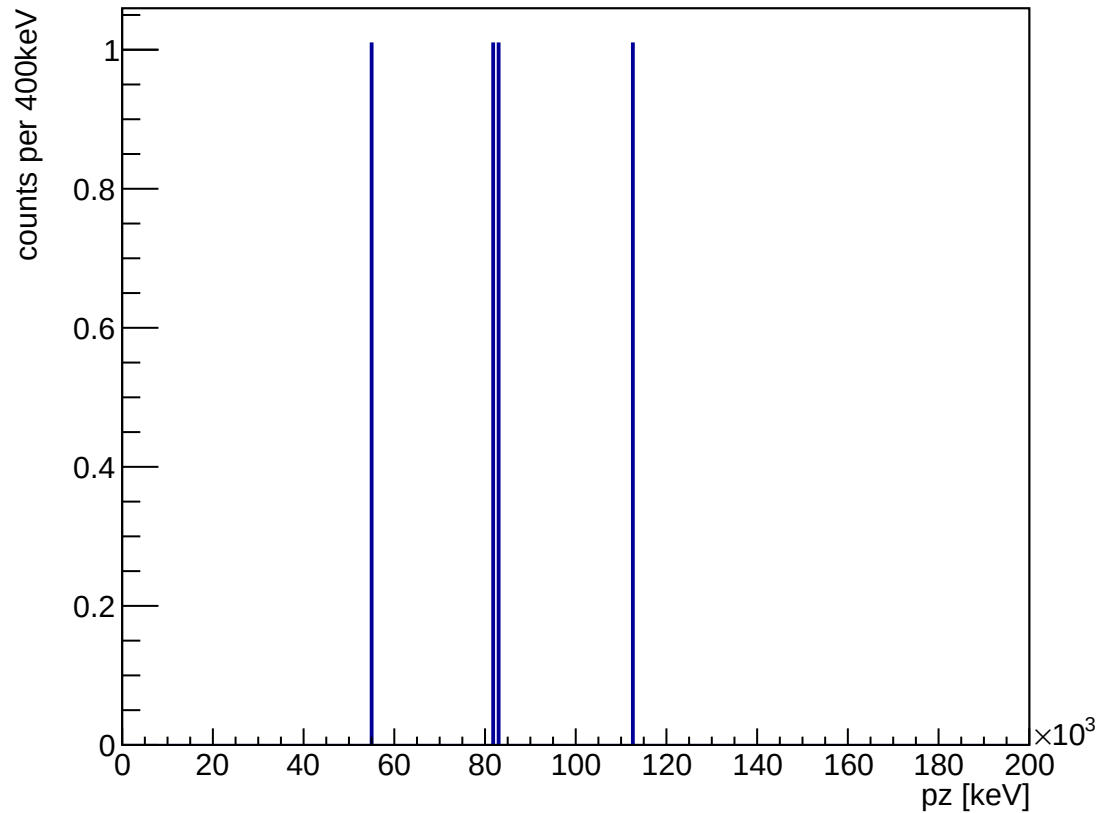
Entries	122300	
Mean x	-3896	
Mean y	-1.519	
Std Dev x	228.9	
Std Dev y	226.3	
Integral	1.234e+05	
0	0	0
0	123411	0
0	0	0

Stock G4: T momentum pz



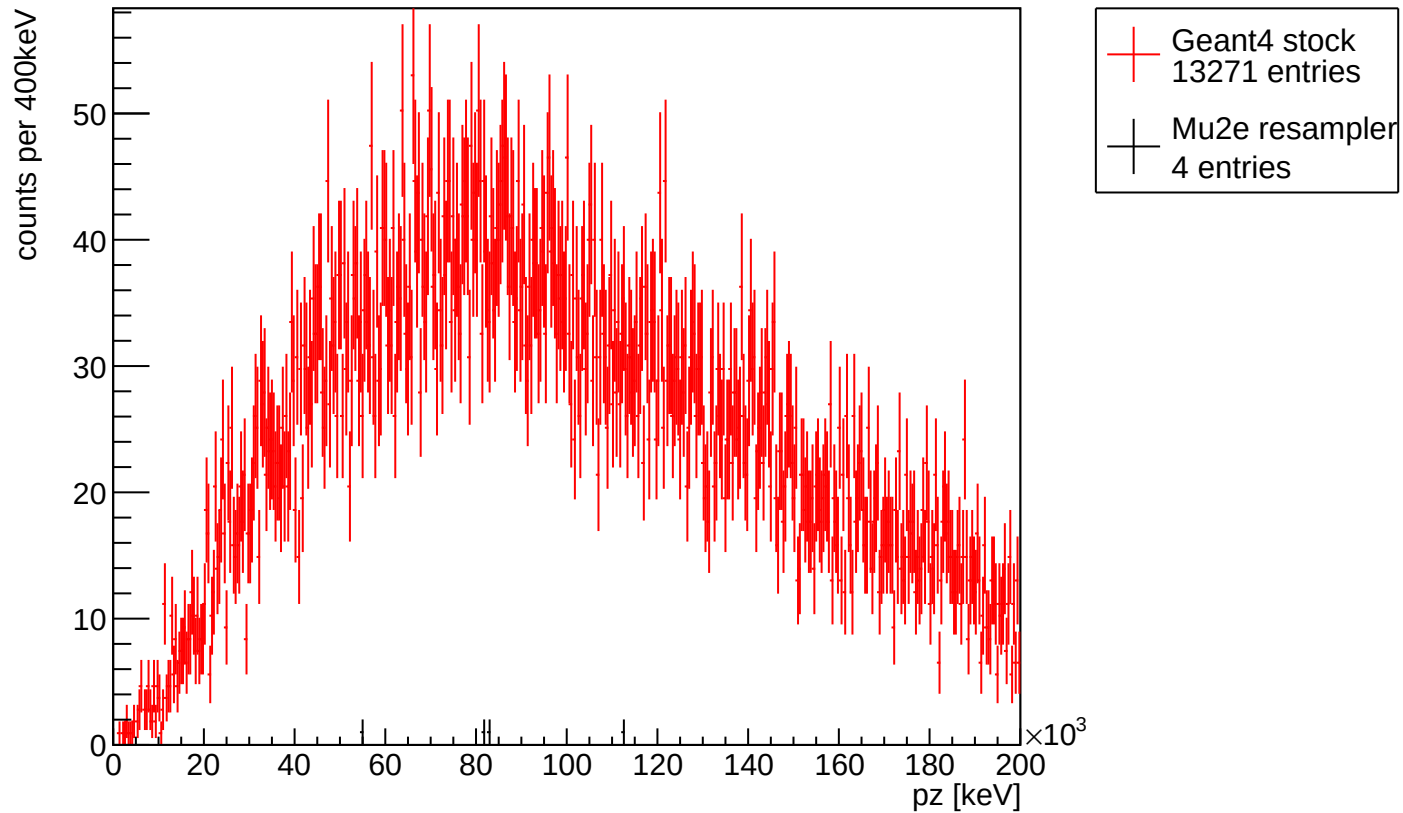
Entries	14266
Mean	9.877e+04
Std Dev	4.596e+04
Underflow	0
Overflow	1051
Integral	1.222e+04

Mu2e mu resampler: T momentum pz

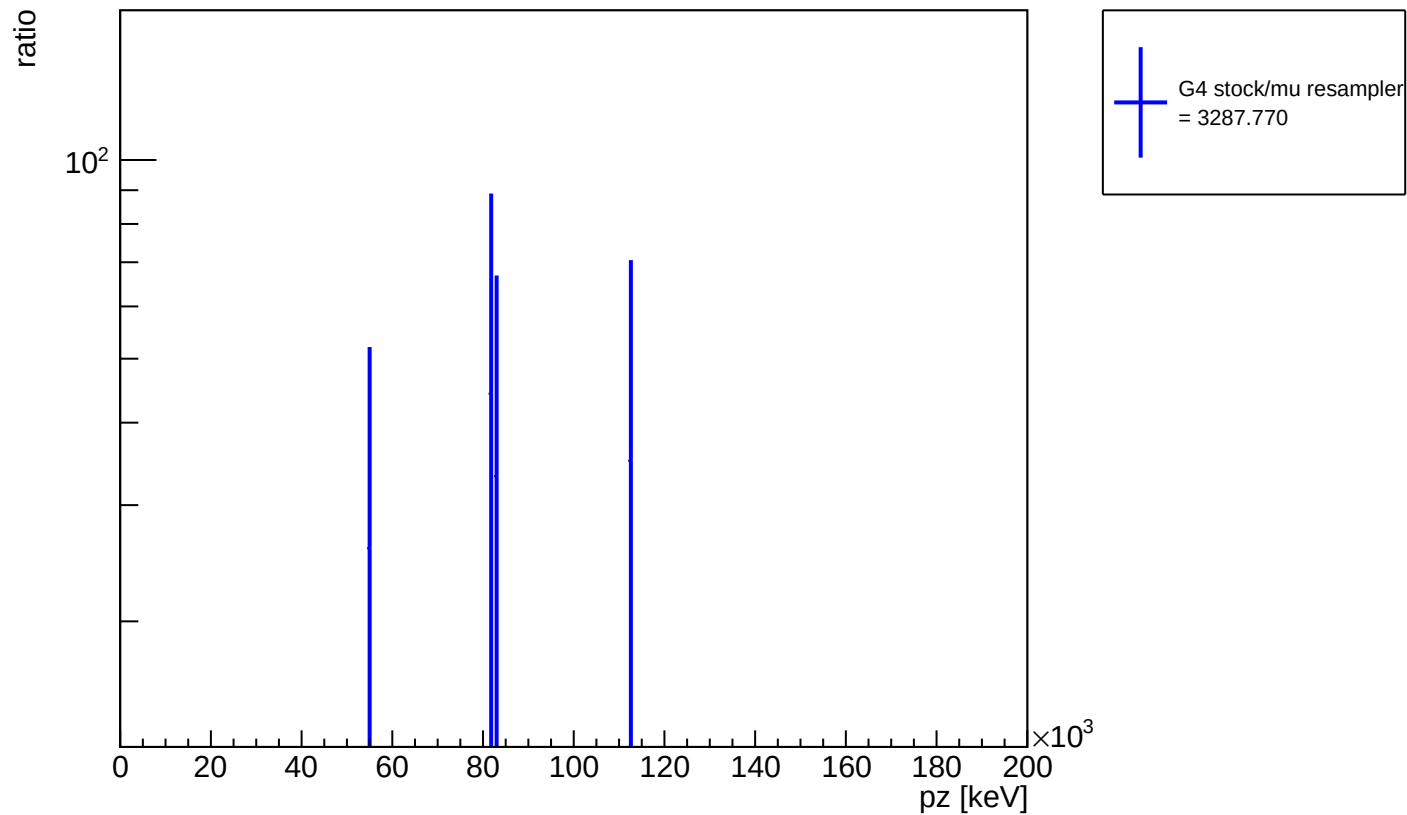


Entries	4
Mean	8.301e+04
Std Dev	2.041e+04
Underflow	0
Overflow	0
Integral	4.036

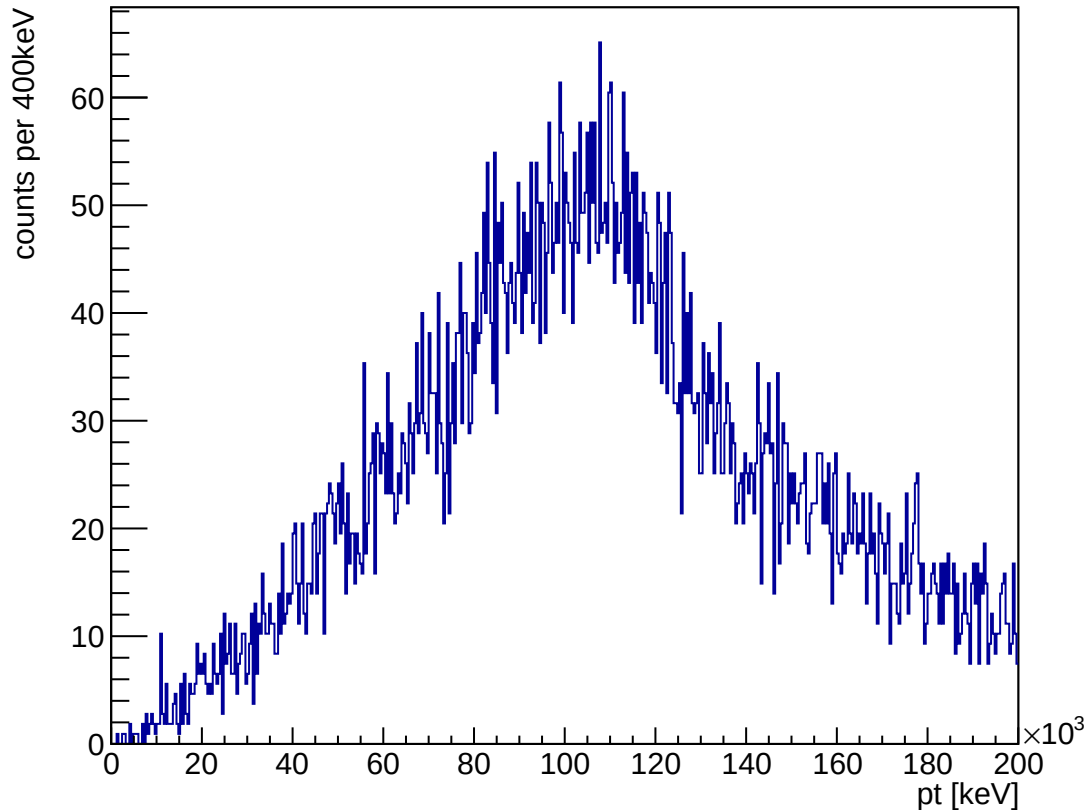
T momentum pz



ratio of G4 stock / muon resampler: T momentum pz

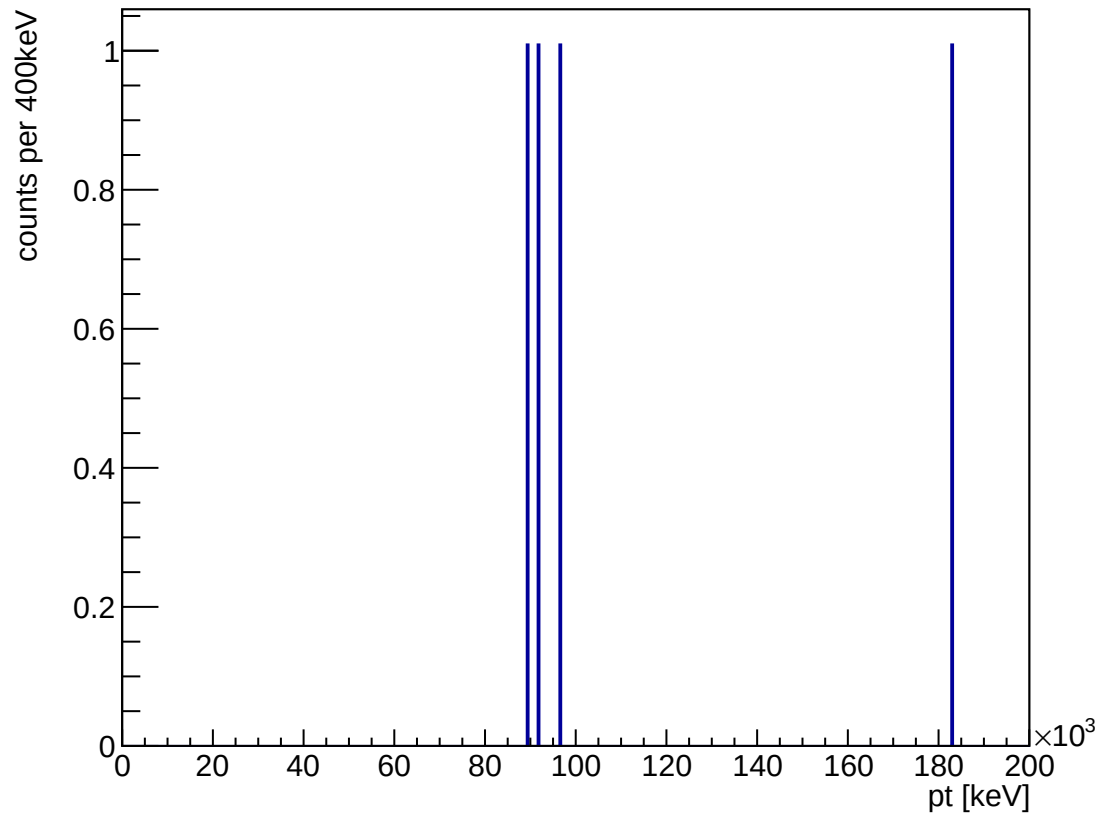


Stock G4: T transverse momentum pT



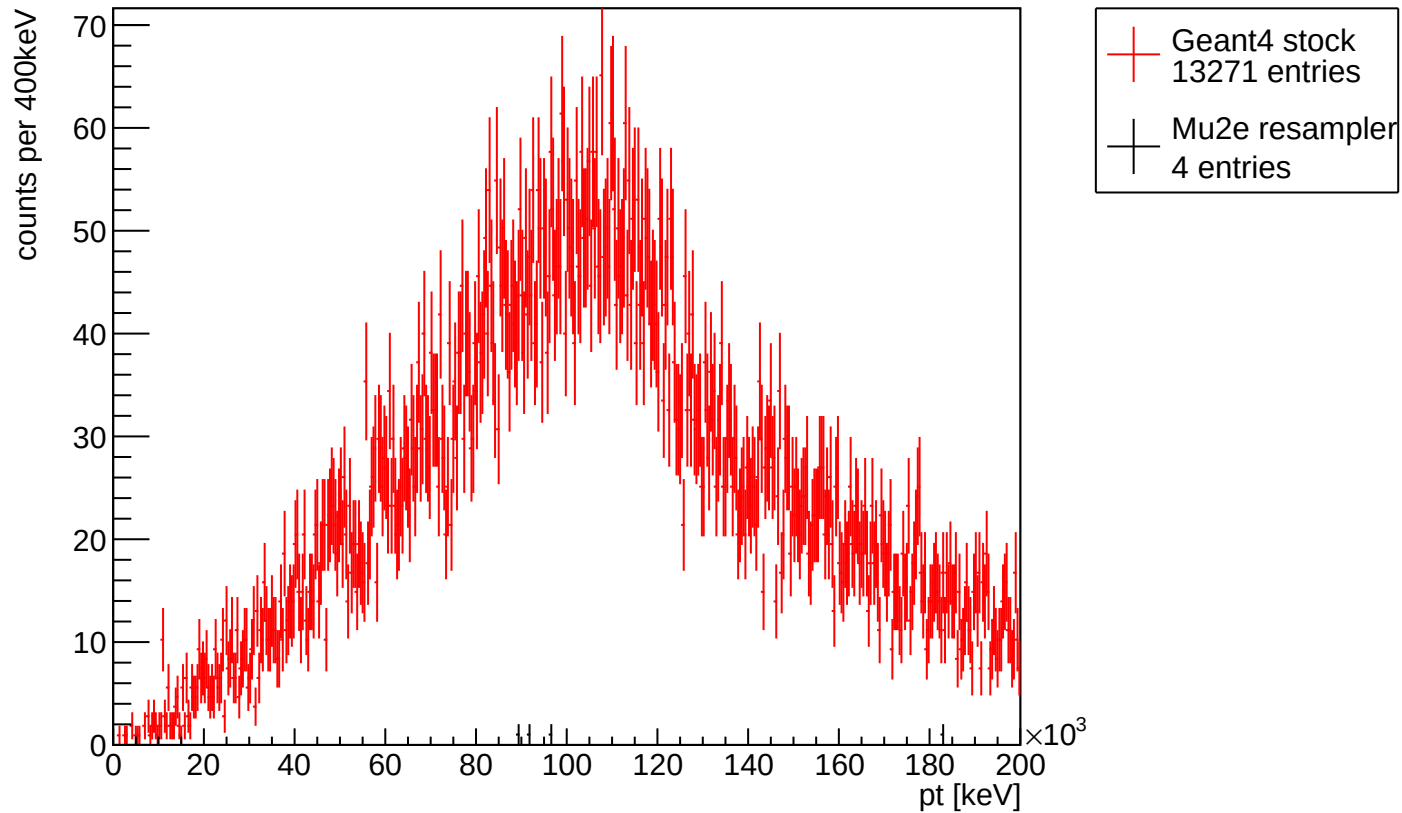
Entries	14266
Mean	1.076e+05
Std Dev	4.126e+04
Underflow	0
Overflow	1090
Integral	1.218e+04

Mu2e mu resampler: T transverse momentum pT

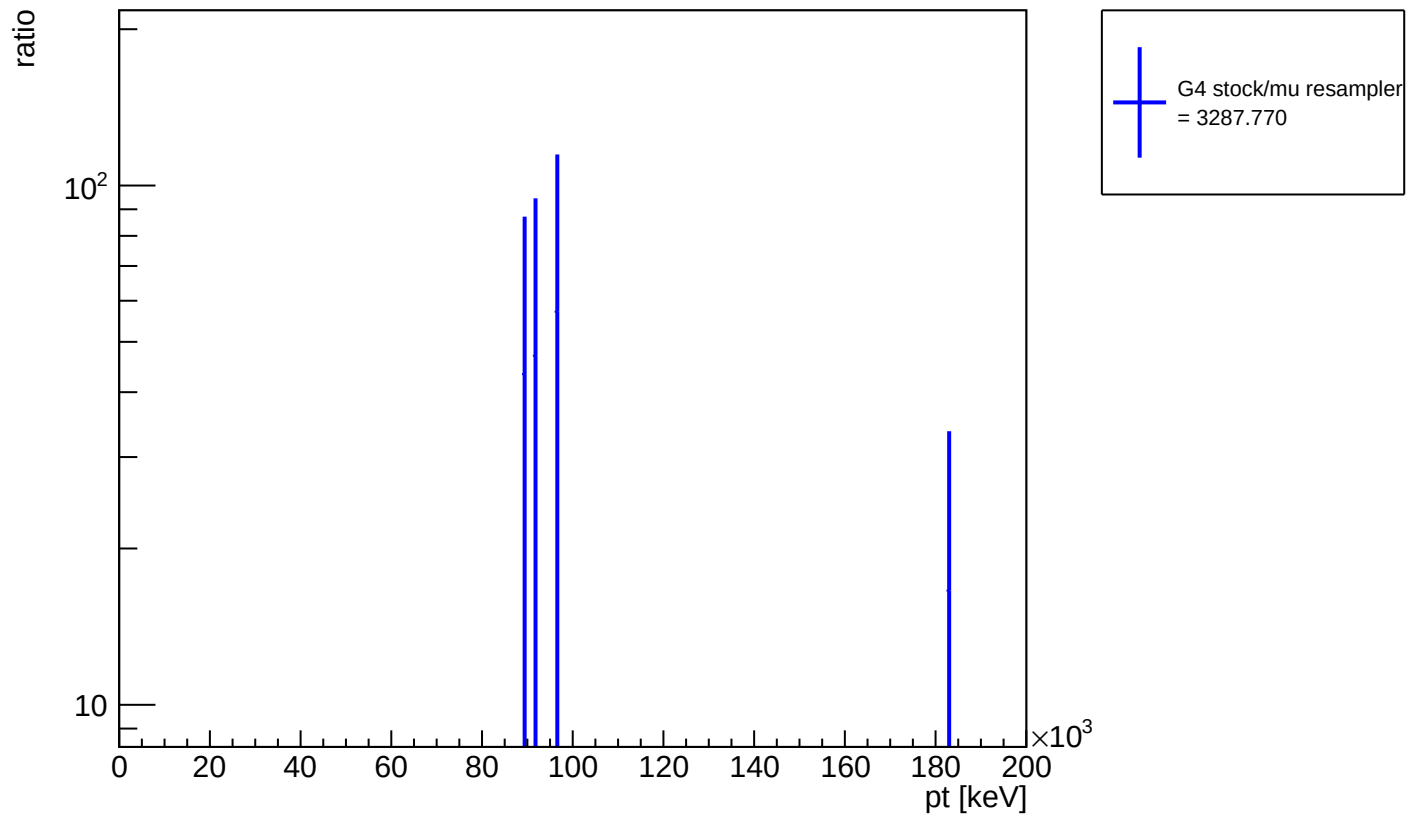


Entries	4
Mean	1.152e+05
Std Dev	3.925e+04
Underflow	0
Overflow	0
Integral	4.036

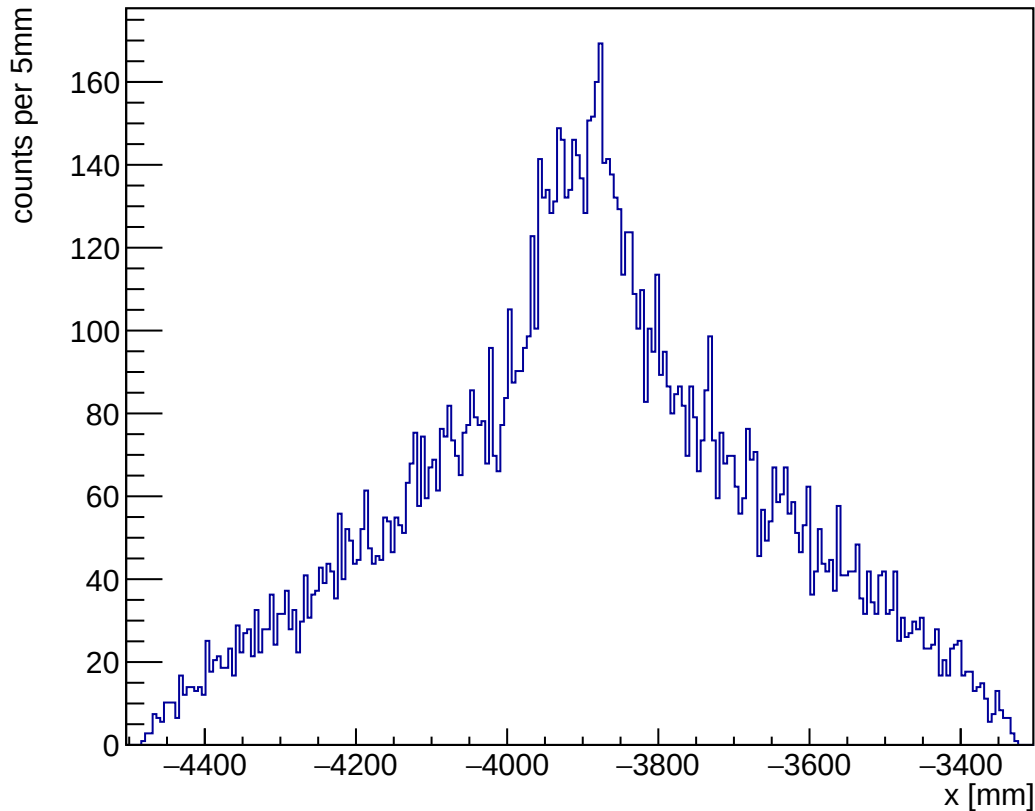
T transverse momentum pT



ratio of G4 stock / muon resampler: T transverse momentum pT

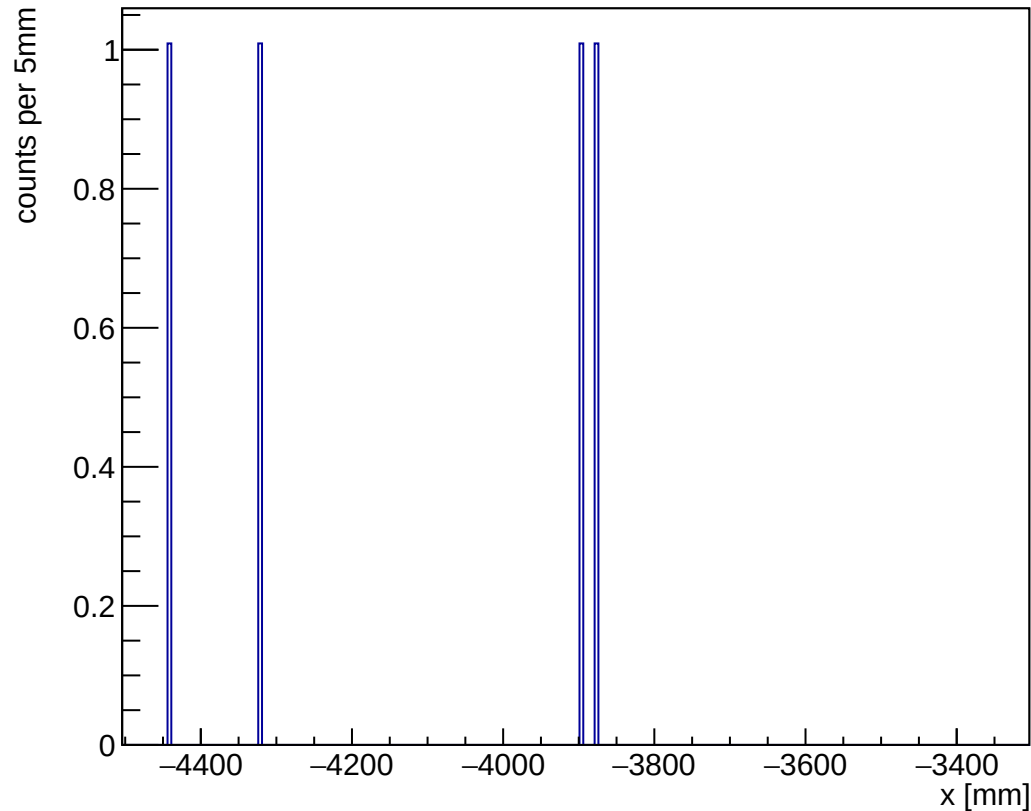


Stock G4: T x position



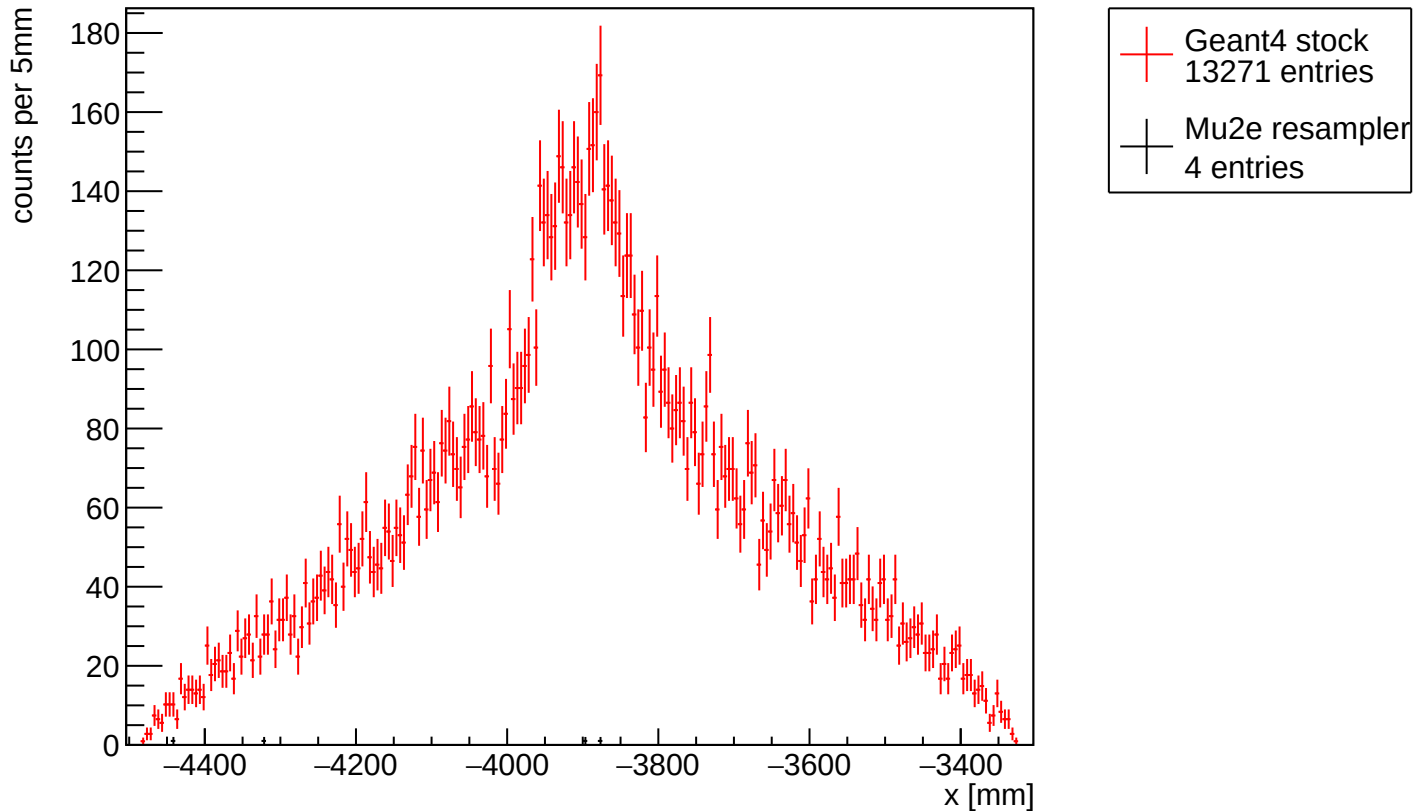
Entries	14266
Mean	-3894
Std Dev	231.9
Underflow	0
Overflow	0
Integral	1.327e+04

Mu2e mu resampler: T x position

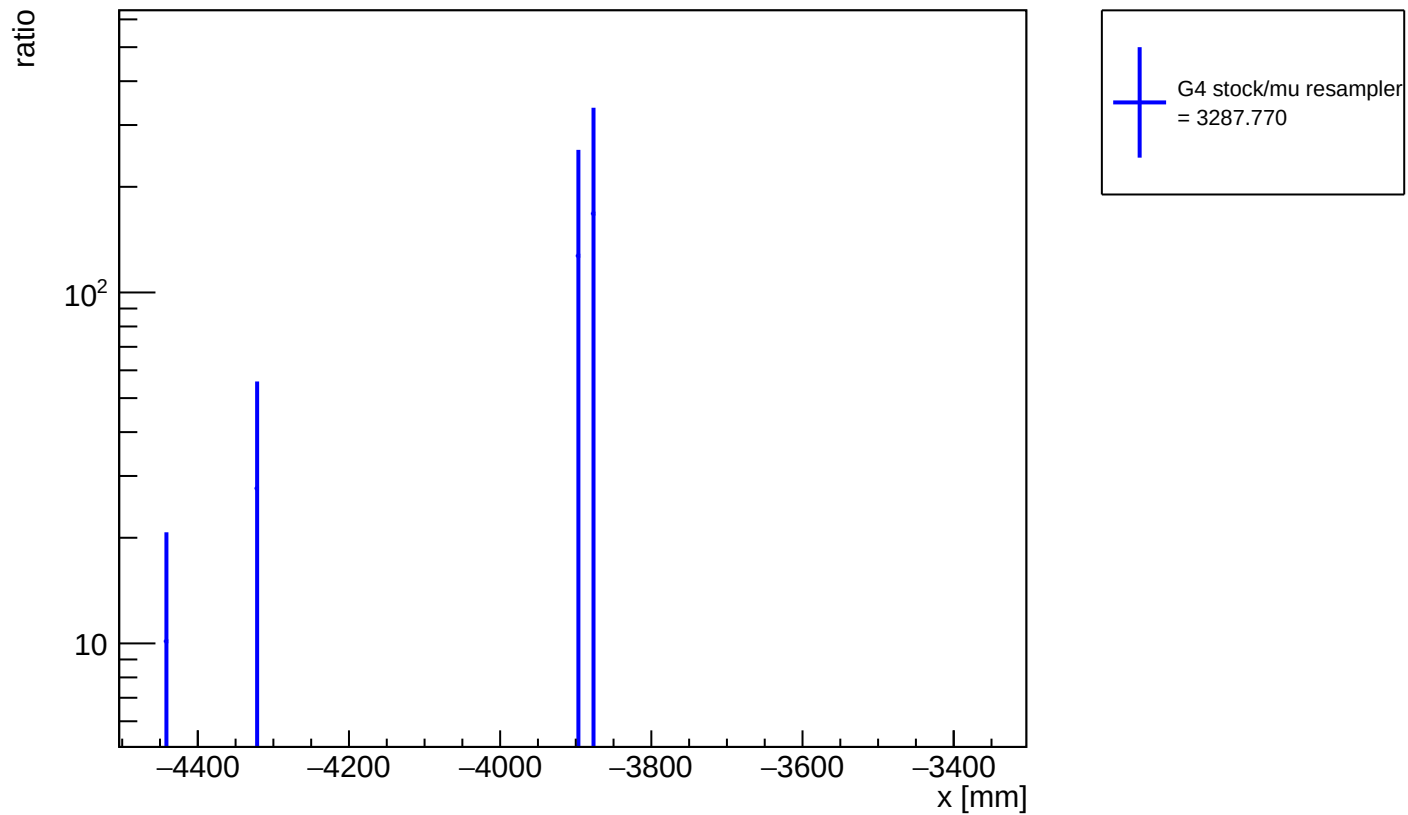


Entries	4
Mean	-4133
Std Dev	250.8
Underflow	0
Overflow	0
Integral	4.036

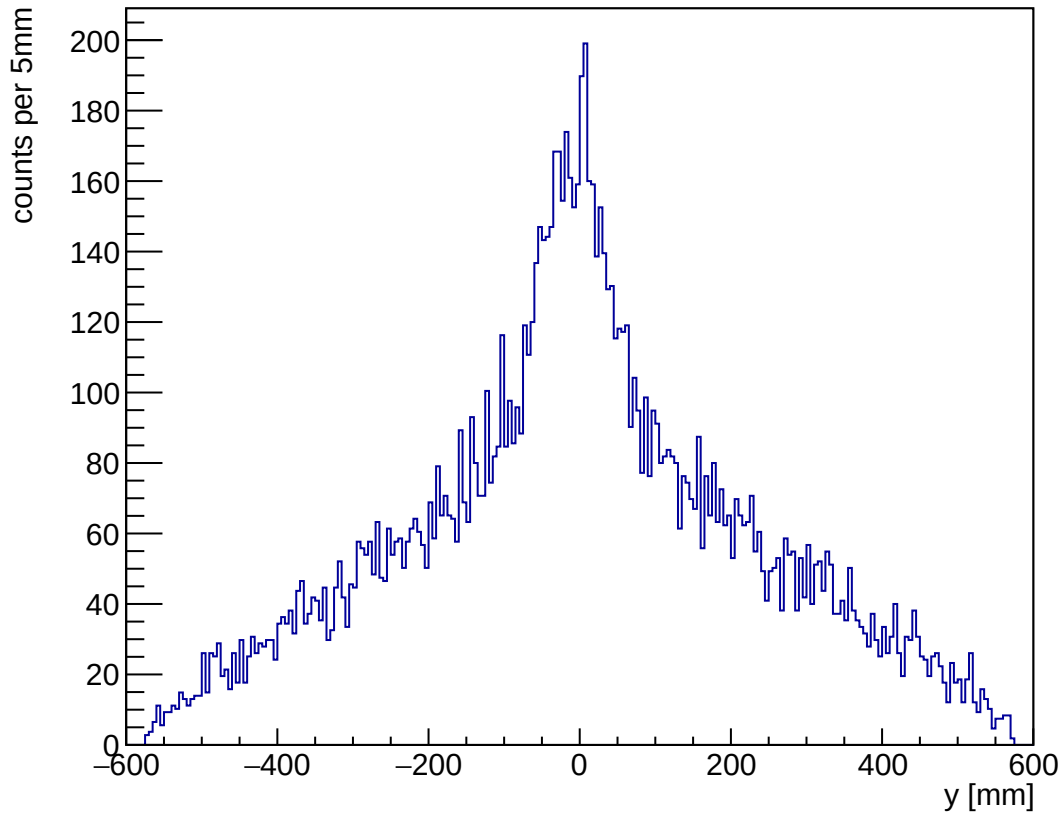
T x position



ratio of G4 stock / muon resampler: T x position

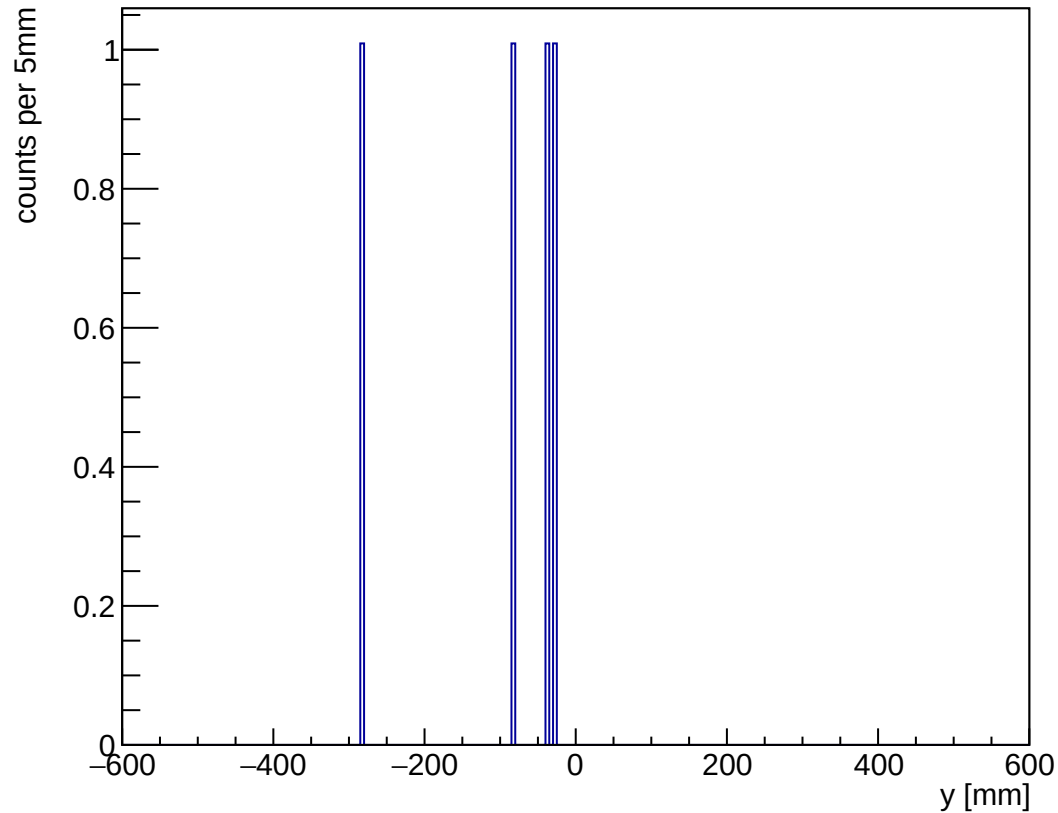


Stock G4: T y position



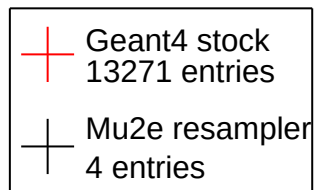
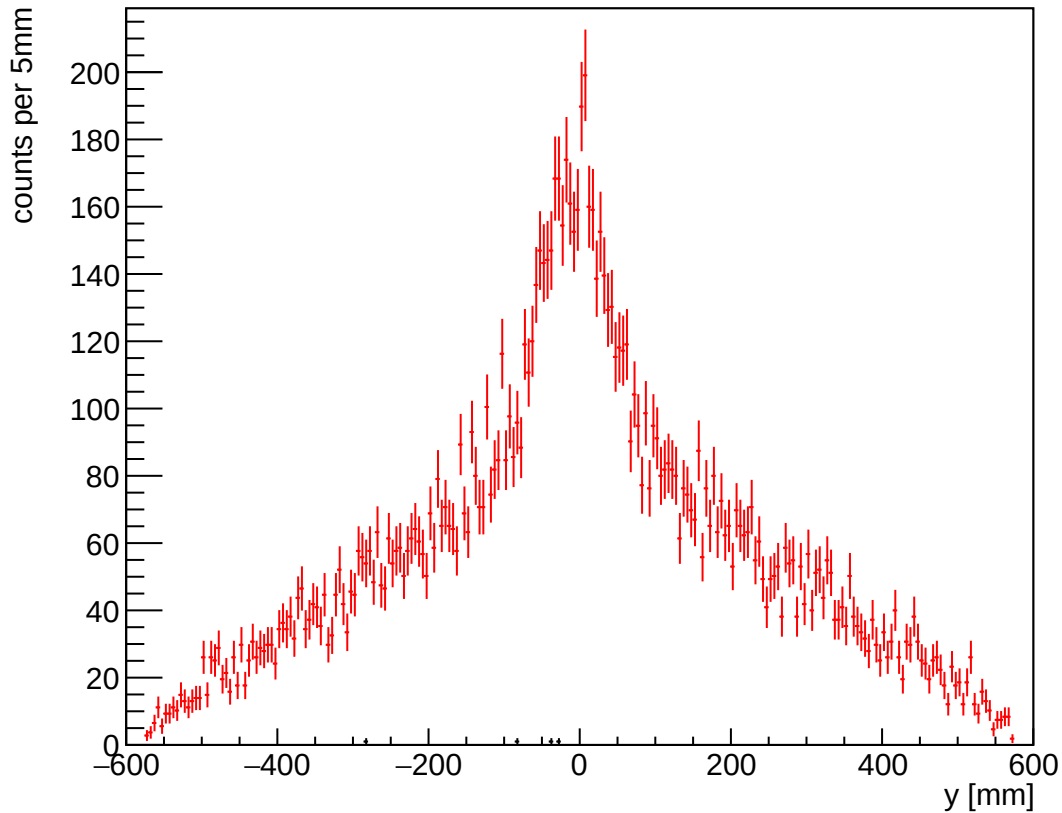
Entries	14266
Mean	0.4615
Std Dev	230.1
Underflow	0
Overflow	0
Integral	1.327e+04

Mu2e mu resampler: T y position

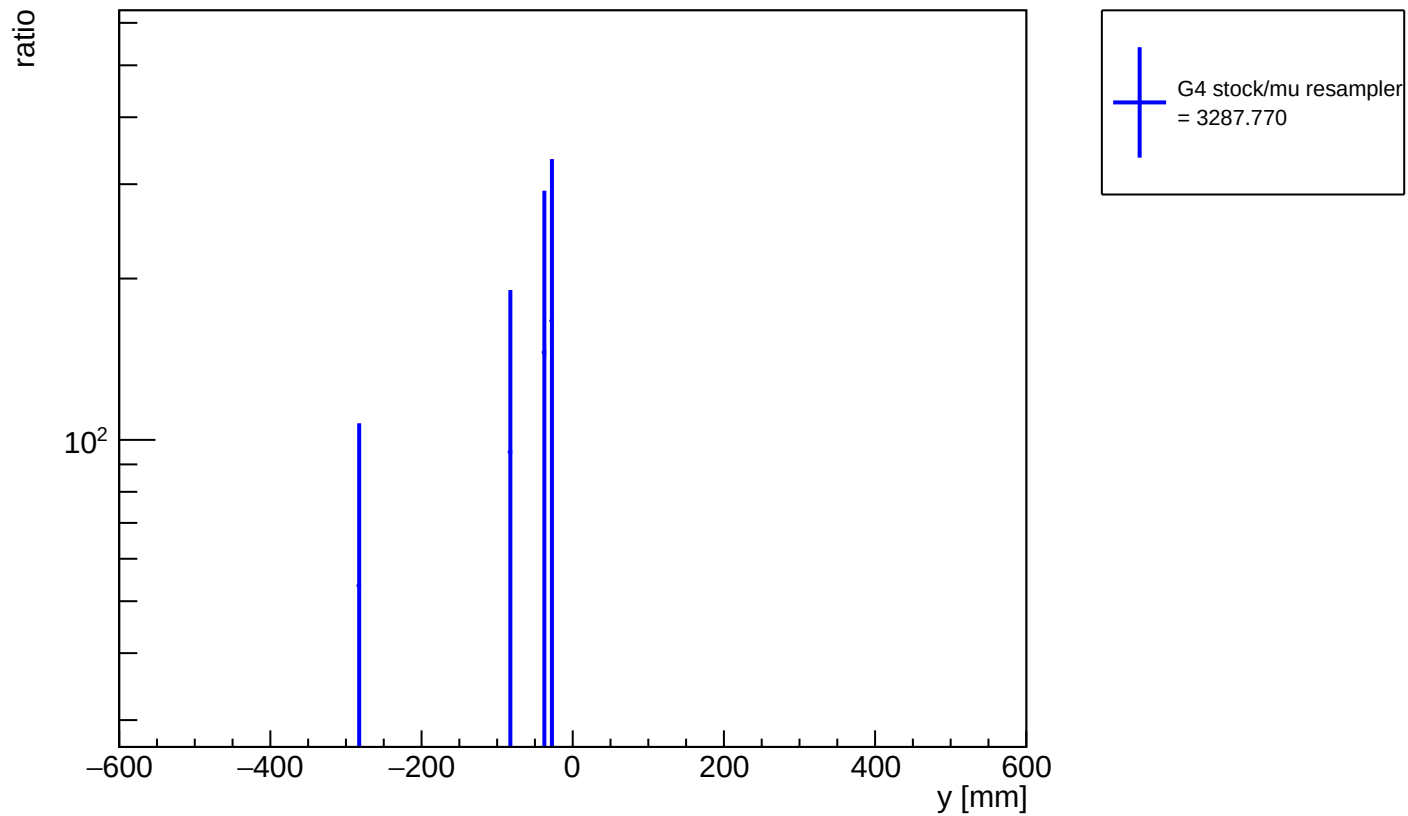


Entries	4
Mean	-106.7
Std Dev	103.4
Underflow	0
Overflow	0
Integral	4.036

T y position

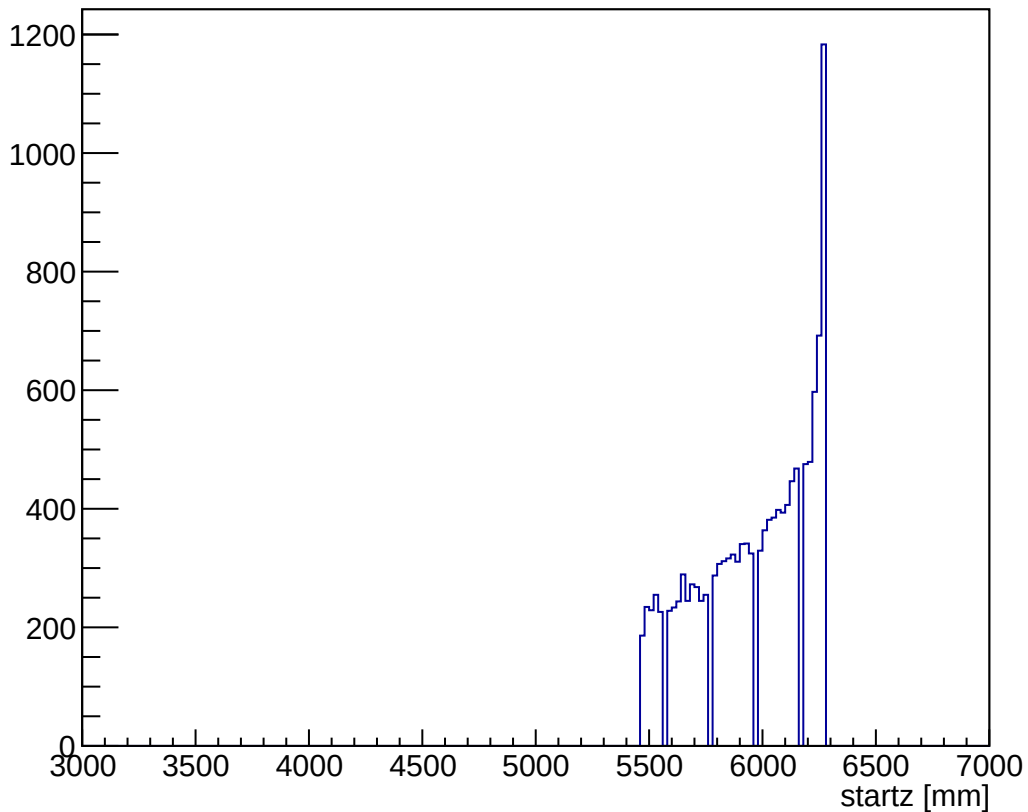


ratio of G4 stock / muon resampler: T y position



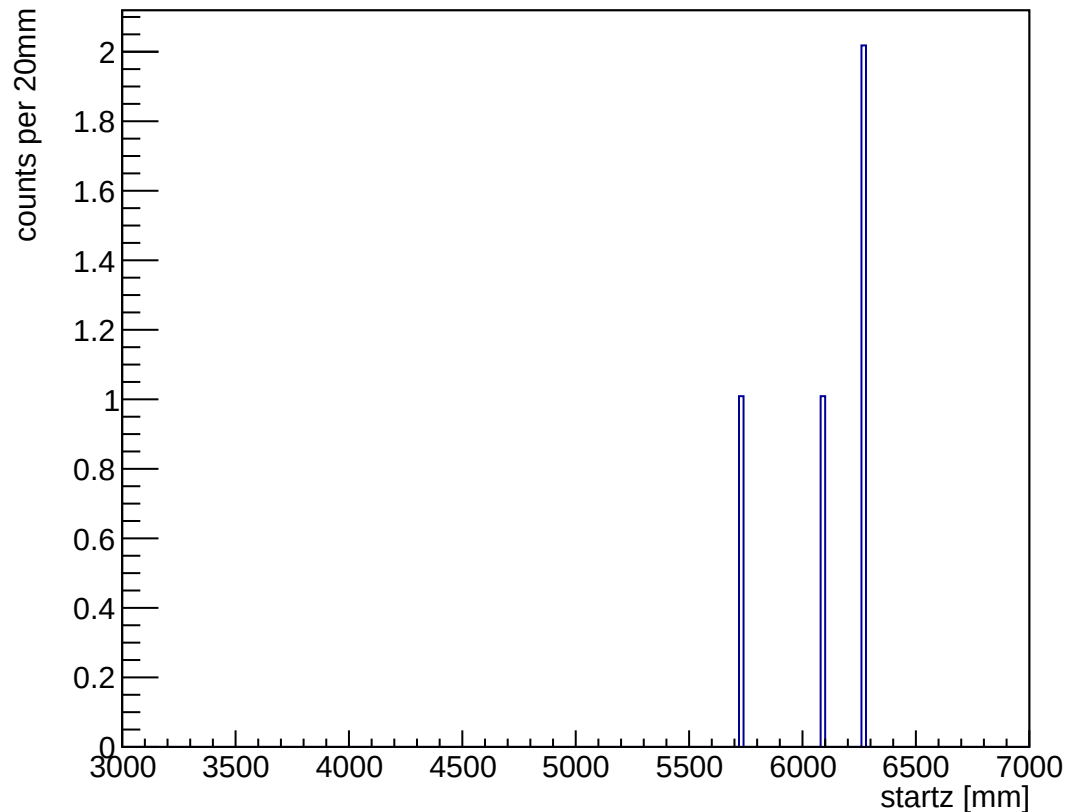
Stock G4: T start z

counts per 20mm



Entries	14266
Mean	5957
Std Dev	241.7
Underflow	0
Overflow	0
Integral	1.327e+04

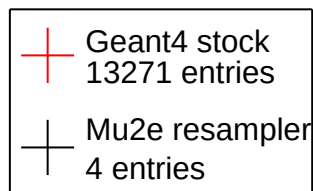
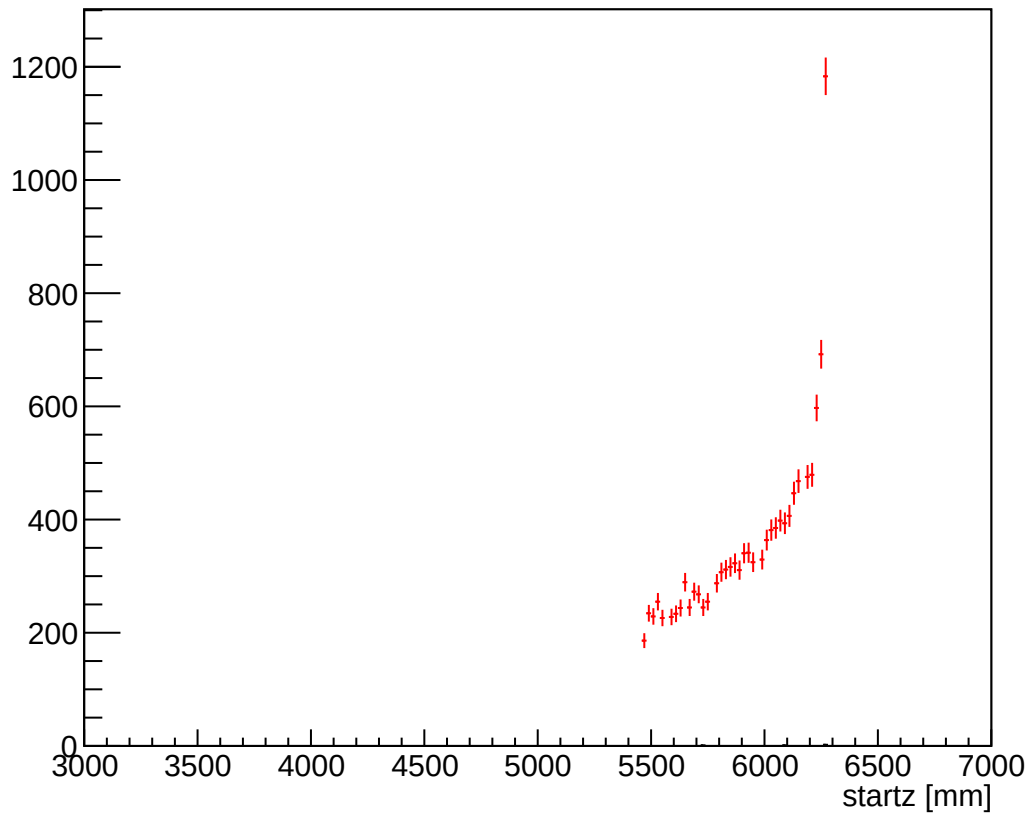
Mu2e mu resampler: T start z



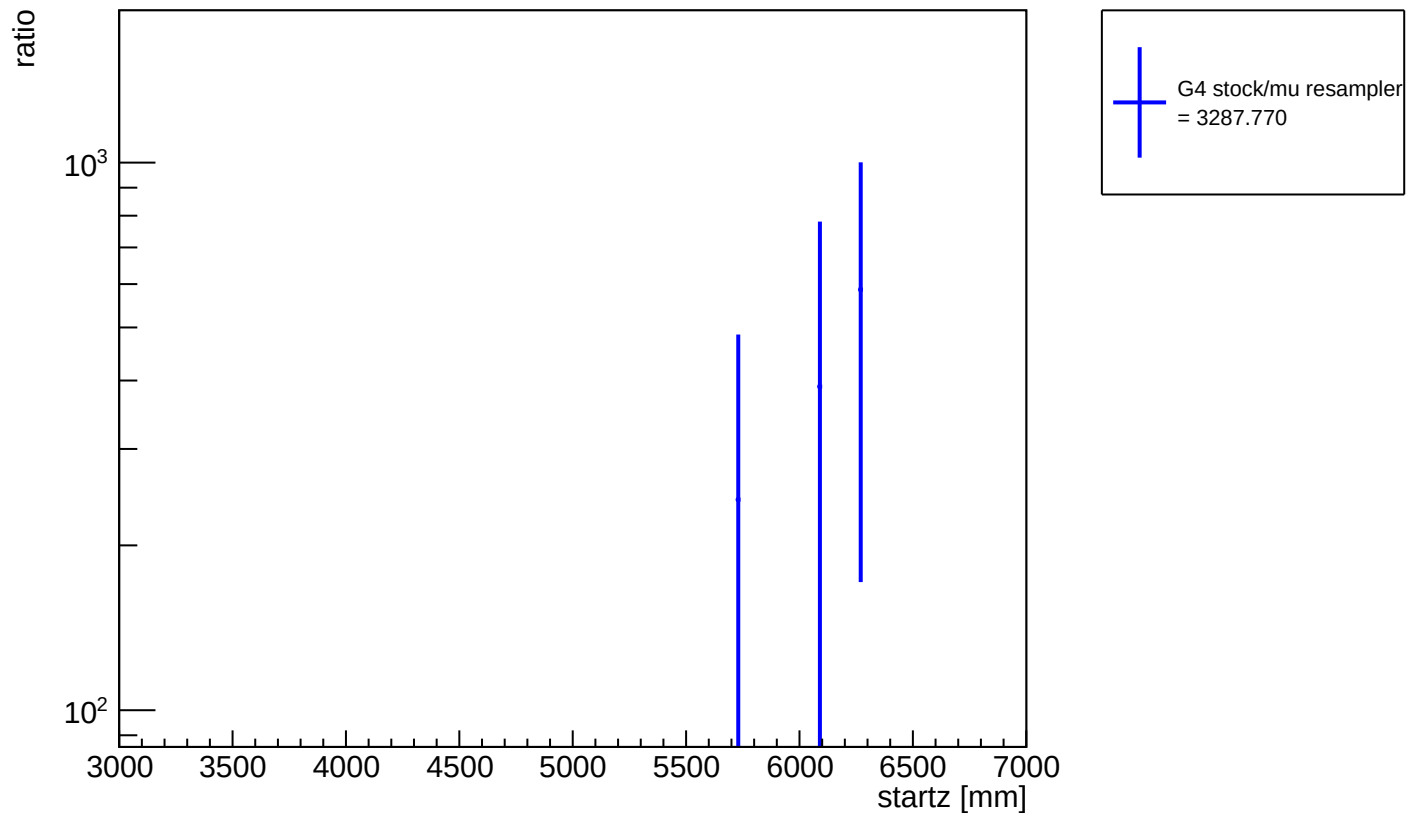
Entries	4
Mean	6093
Std Dev	217.7
Underflow	0
Overflow	0
Integral	4.036

T start z

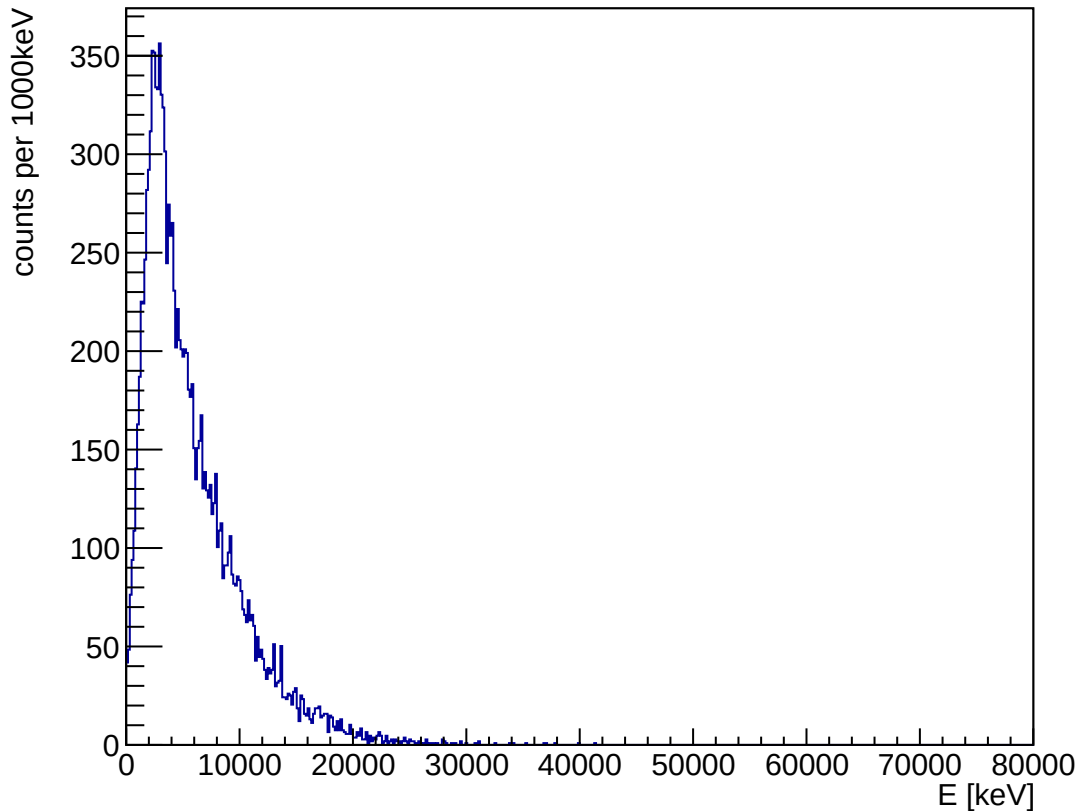
counts per 20mm



ratio of G4 stock / muon resampler: T start z

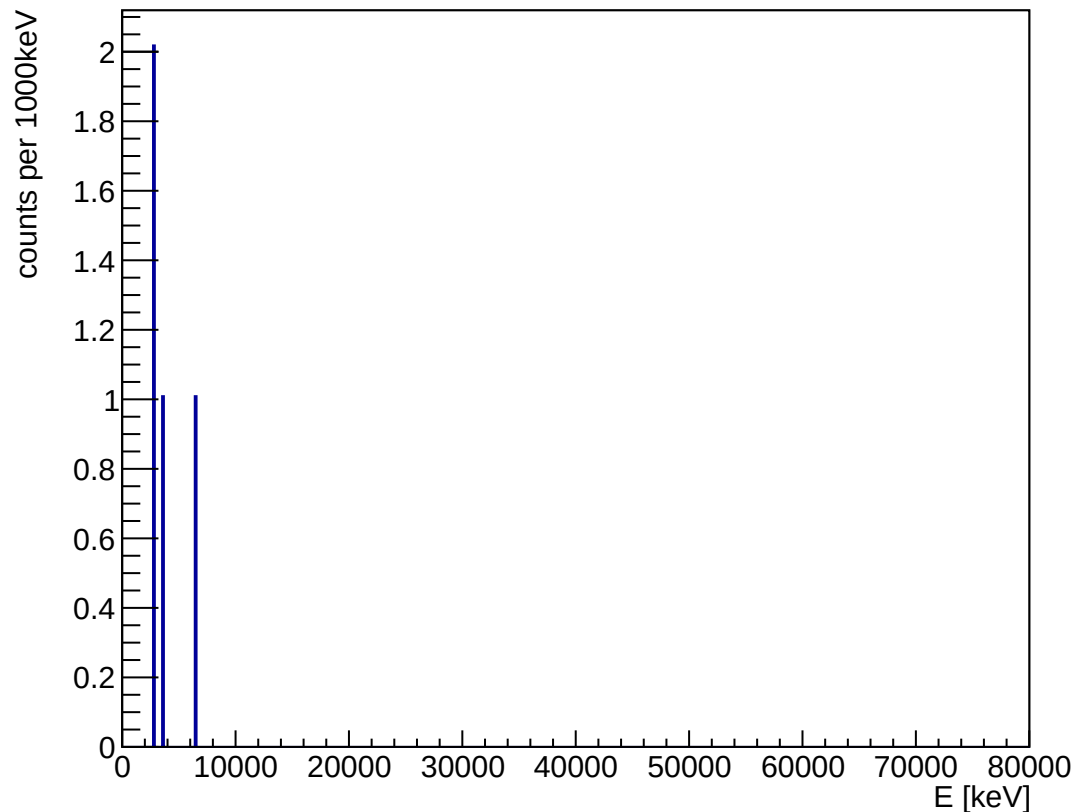


Stock G4: T kin. energy



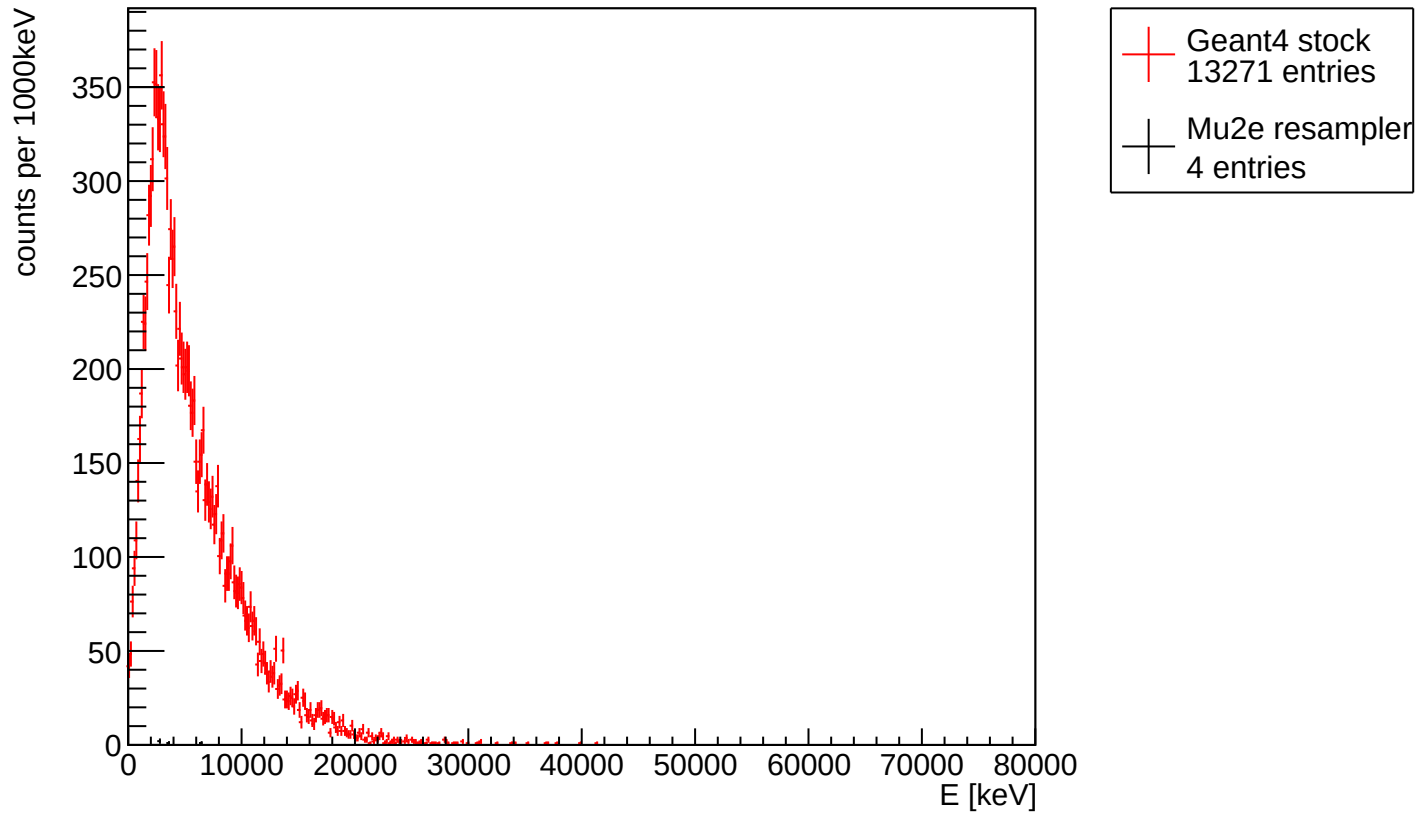
Entries	14266
Mean	5745
Std Dev	4349
Underflow	0
Overflow	0
Integral	1.327e+04

Mu2e mu resampler: T kin. energy

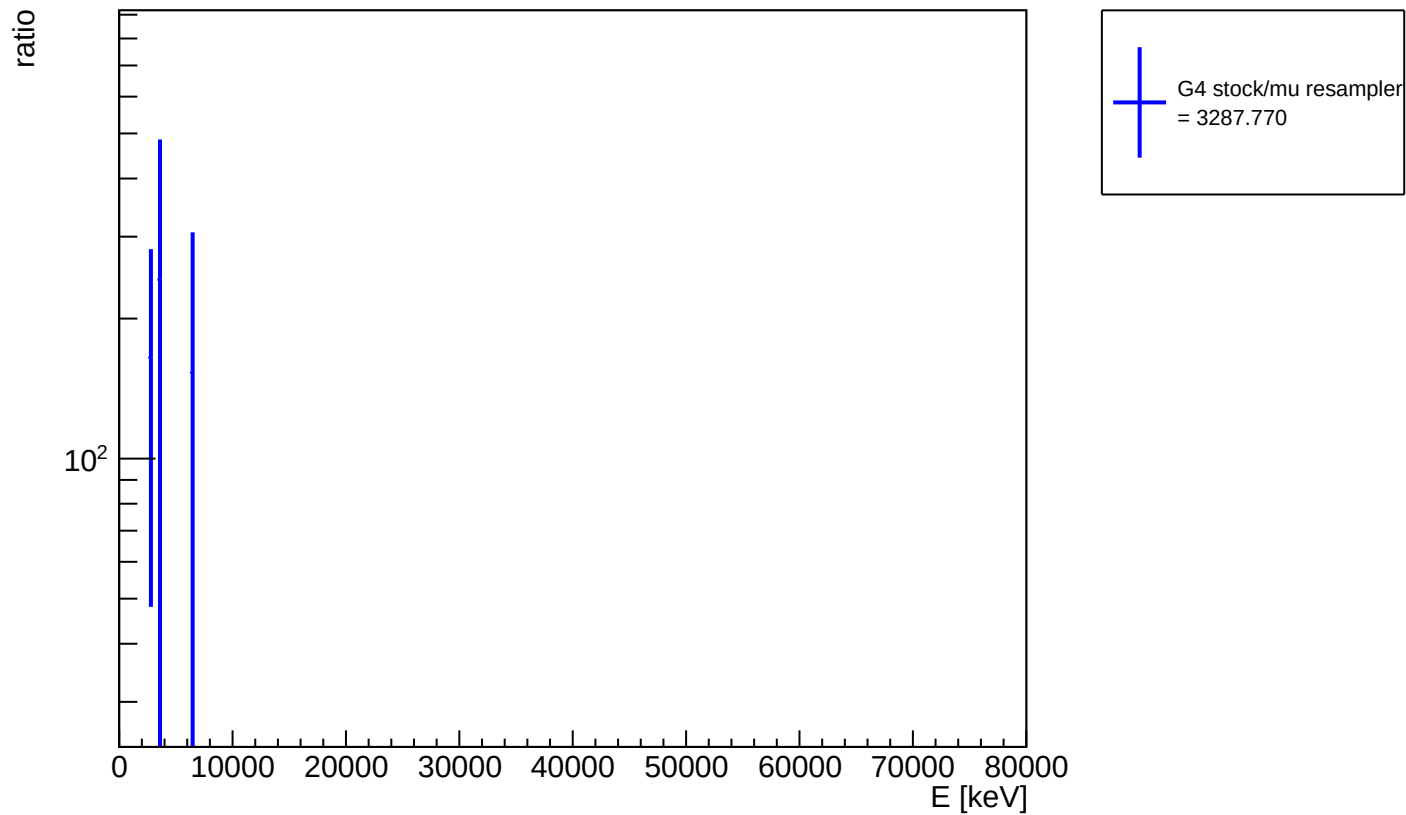


Entries	4
Mean	3933
Std Dev	1521
Underflow	0
Overflow	0
Integral	4.036

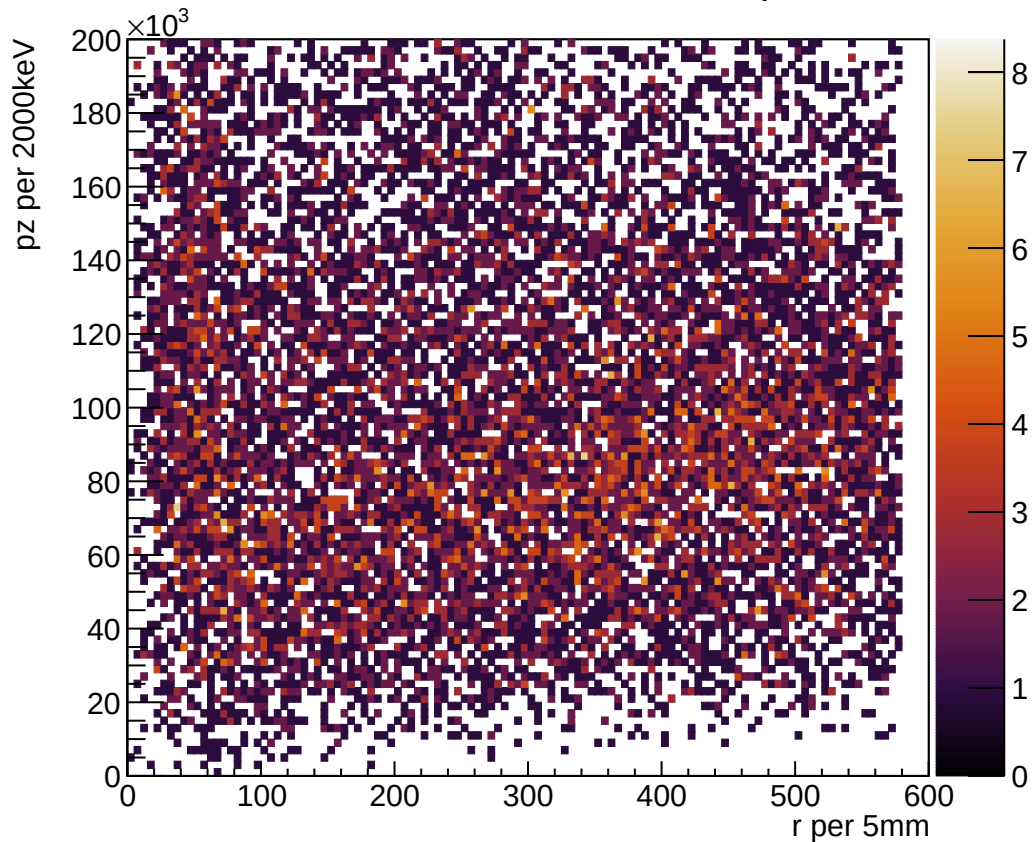
T kin. energy



ratio of G4 stock / muon resampler: T kin. energy

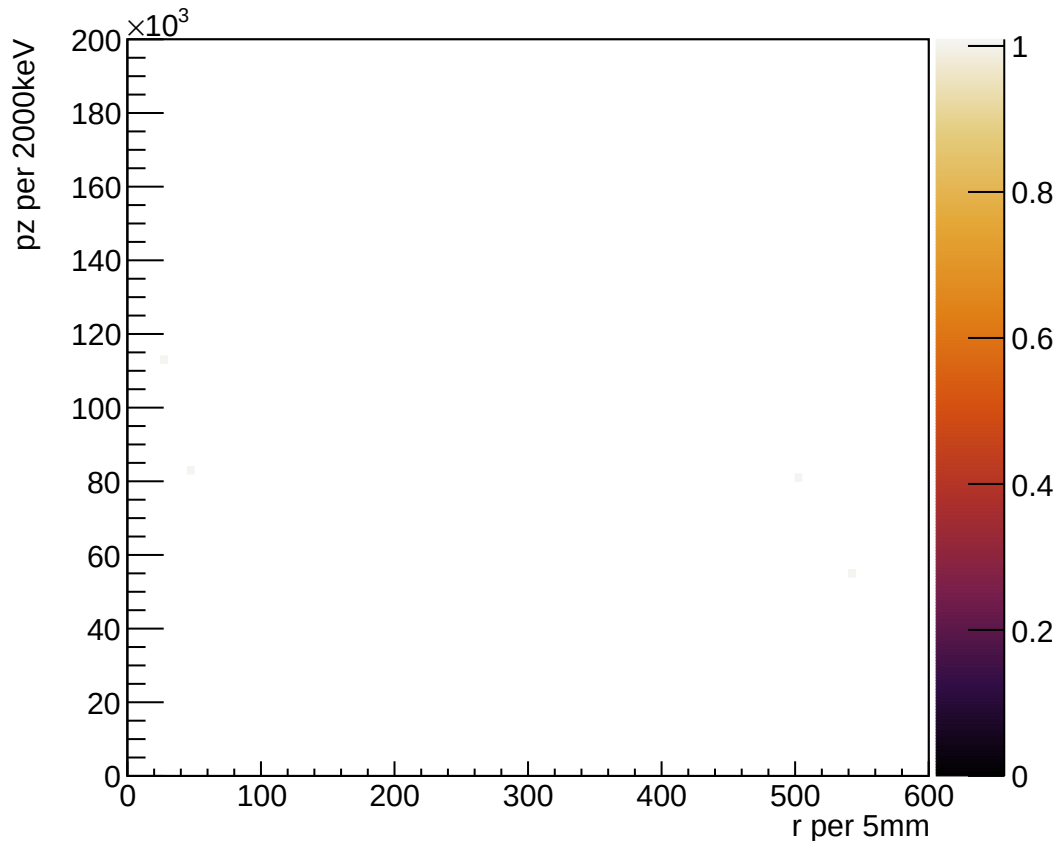


Stock G4: T pz vs r



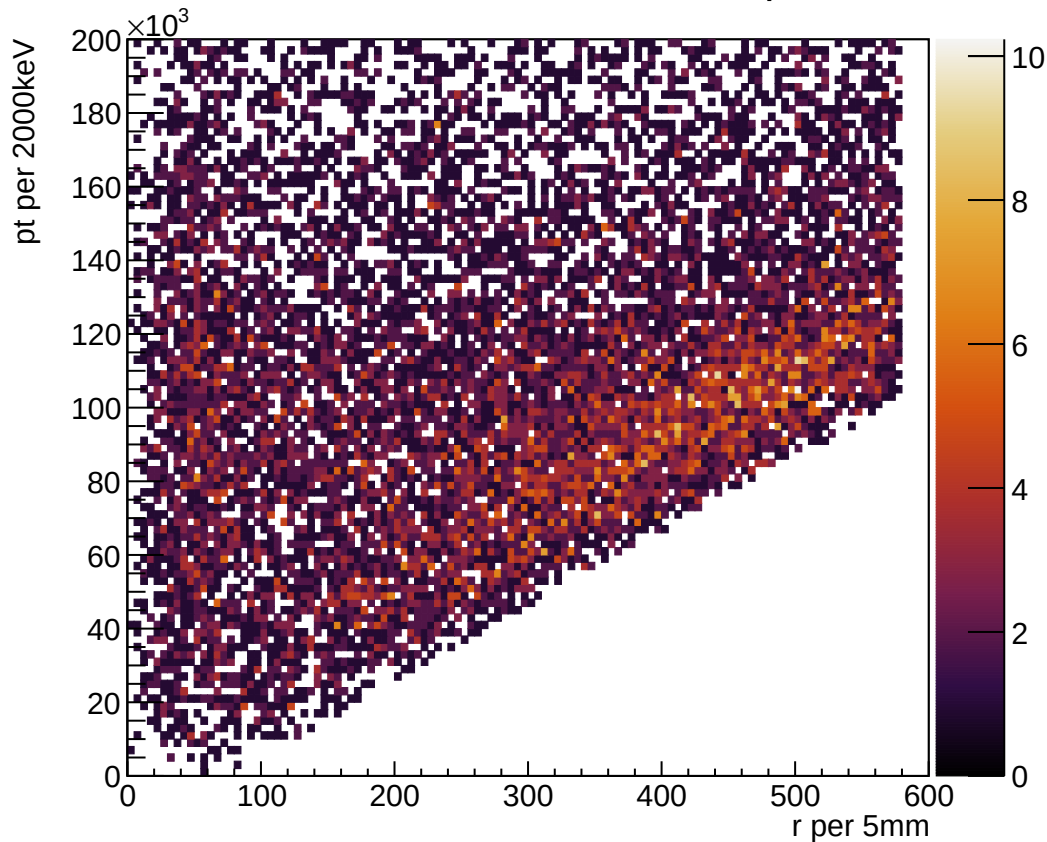
Entries	14266	
Mean x	287.9	
Mean y	9.877e+04	
Std Dev x	160.8	
Std Dev y	4.596e+04	
Integral	1.222e+04	
0	1051.15	0
0	12219.4	0
0	0	0

Mu2e mu resampler: T pz vs r



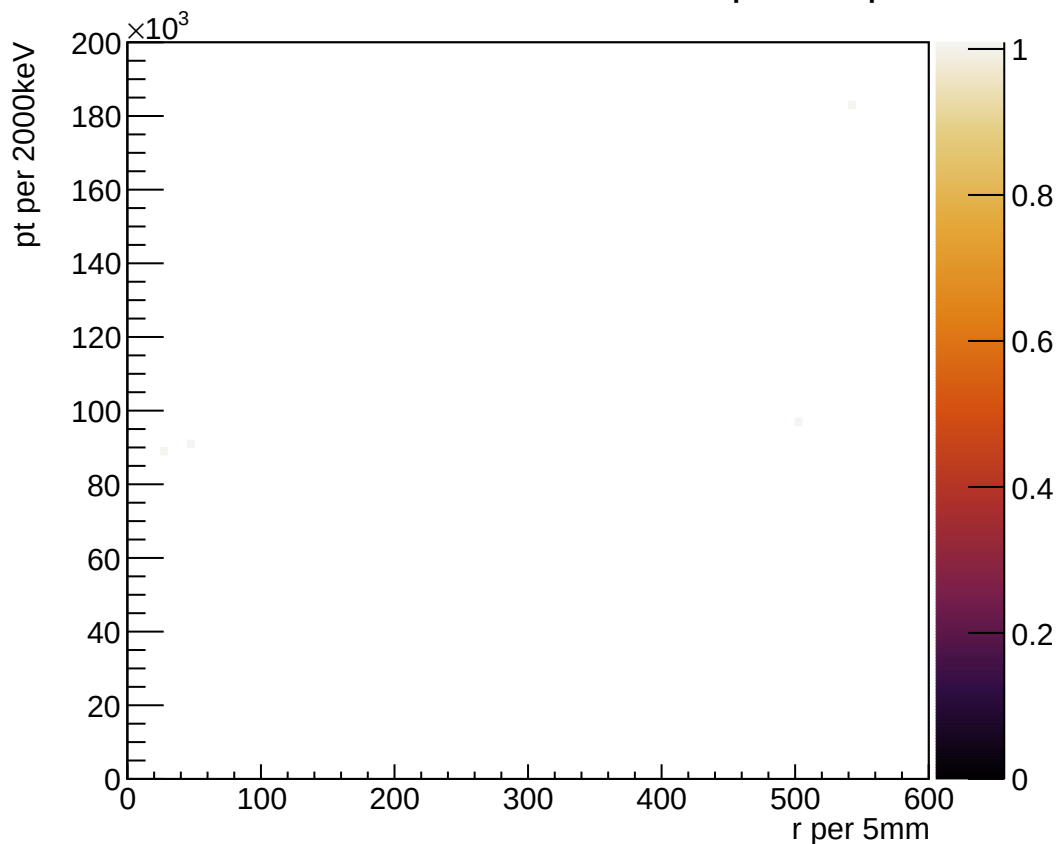
Entries	4	
Mean x	279.9	
Mean y	8.301e+04	
Std Dev x	243	
Std Dev y	2.041e+04	
Integral	4.036	
0	0	0
0	4.03633	0
0	0	0

Stock G4: T pt vs r



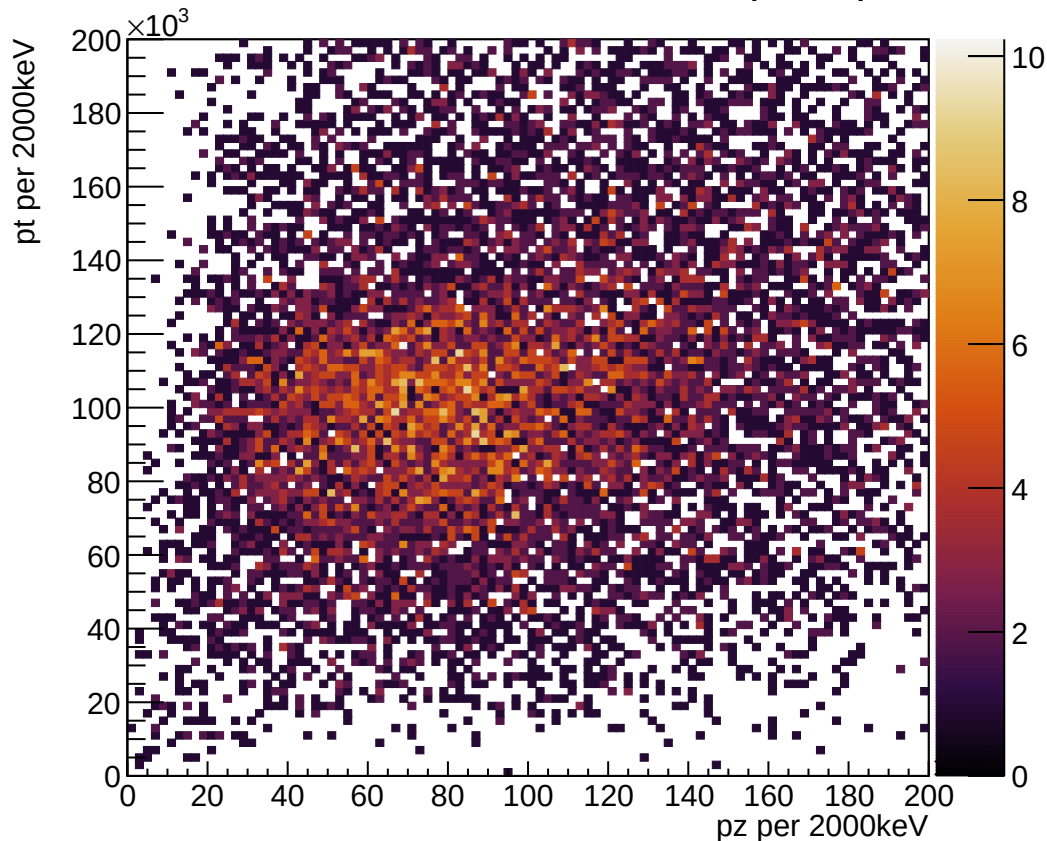
Entries	14266	
Mean x	283.7	
Mean y	1.076e+05	
Std Dev x	159.7	
Std Dev y	4.126e+04	
Integral	1.218e+04	
0	1090.22	0
0	12180.3	0
0	0	0

Mu2e mu resampler: T pt vs r



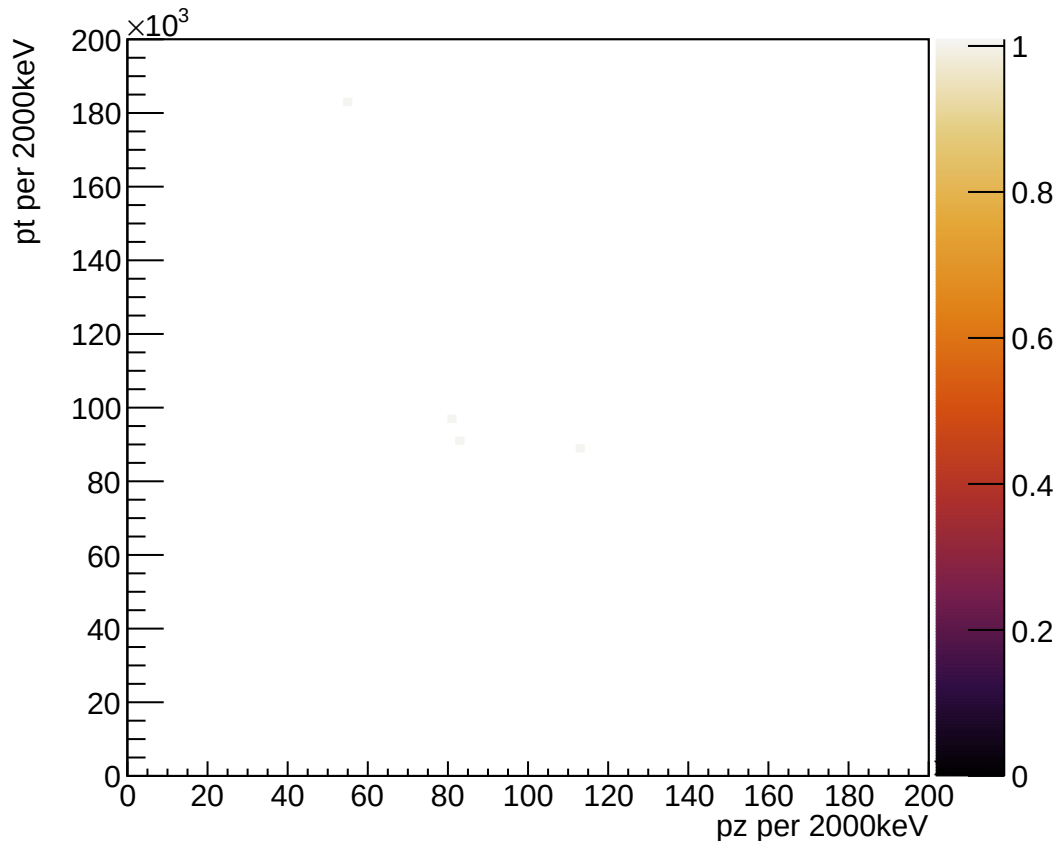
Entries	4	
Mean x	279.9	
Mean y	1.152e+05	
Std Dev x	243	
Std Dev y	3.925e+04	
Integral	4.036	
0	0	0
0	4.03633	0
0	0	0

Stock G4: T pt vs pz



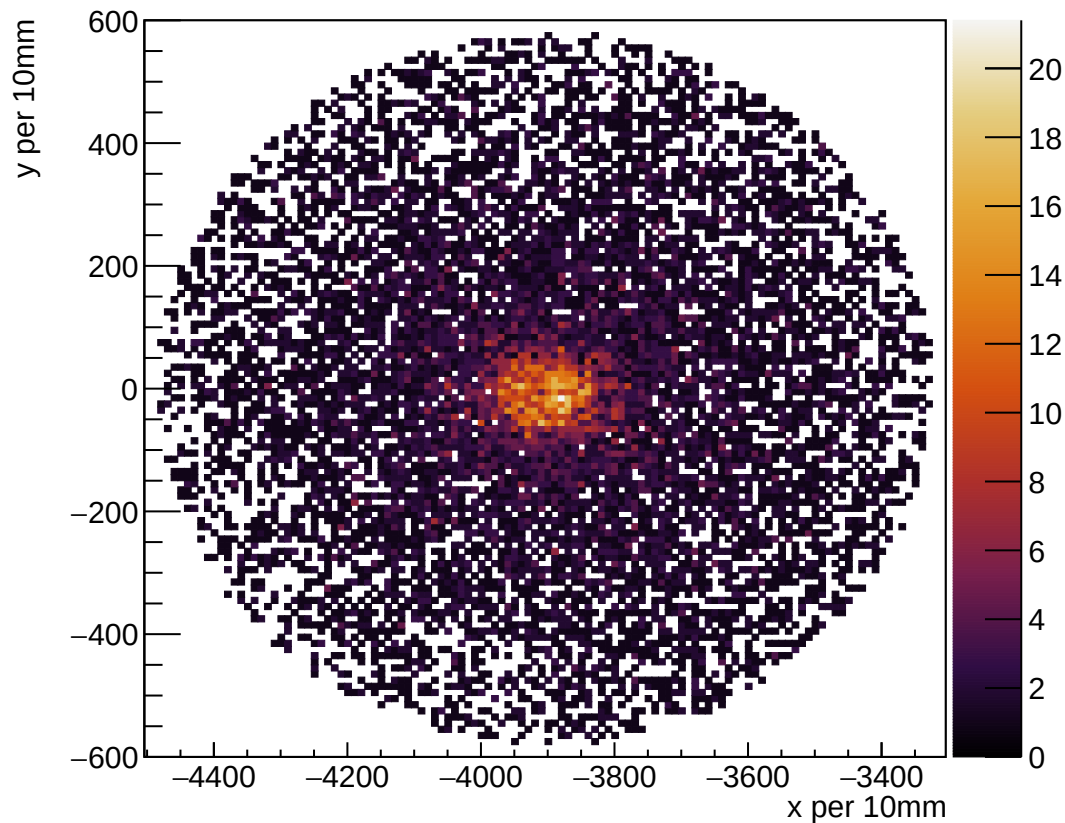
Entries	14266	
Mean x	9.776e+04	
Mean y	1.066e+05	
Std Dev x	4.582e+04	
Std Dev y	4.083e+04	
Integral	1.129e+04	
0	928.359	161.858
0	11291	889.29
0	0	0

Mu2e mu resampler: T pt vs pz



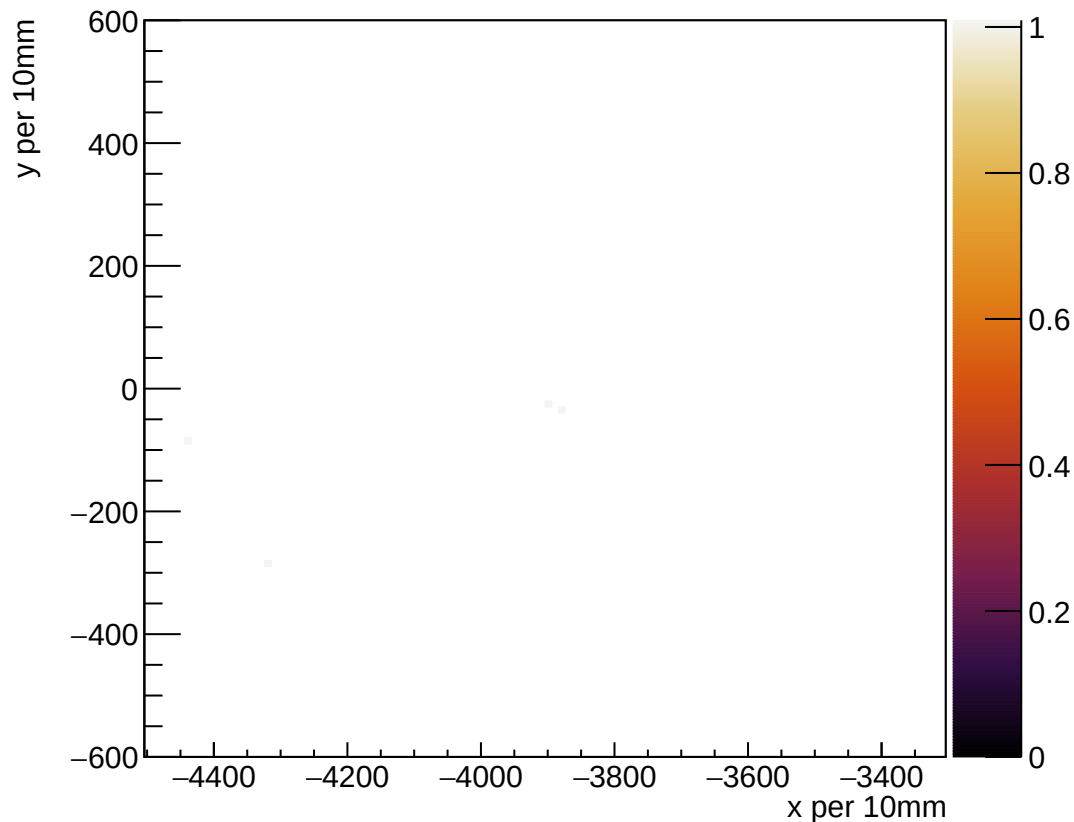
Entries		4
Mean x	8.301e+04	
Mean y	1.152e+05	
Std Dev x	2.041e+04	
Std Dev y	3.925e+04	
Integral		4.036
0	0	0
0	4.03633	0
0	0	0

Stock G4: T y vs x



Entries	14266		
Mean x	-3894		
Mean y	0.4615		
Std Dev x	231.9		
Std Dev y	230.1		
Integral	1.327e+04		
0	0	0	
0	13270.5	0	
0	0	0	

Mu2e mu resampler: T y vs x



Entries	4		
Mean x	-4133		
Mean y	-106.7		
Std Dev x	250.8		
Std Dev y	103.4		
Integral	4.036		
0	0	0	
0	4.03633	0	
0	0	0	